



Technical Documentation Version 9.6

System Control Table (SCT)



**Center for Advanced Decision Support
for Water and Environmental Systems
UNIVERSITY OF COLORADO BOULDER**

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Chapter 1

SCT Overview

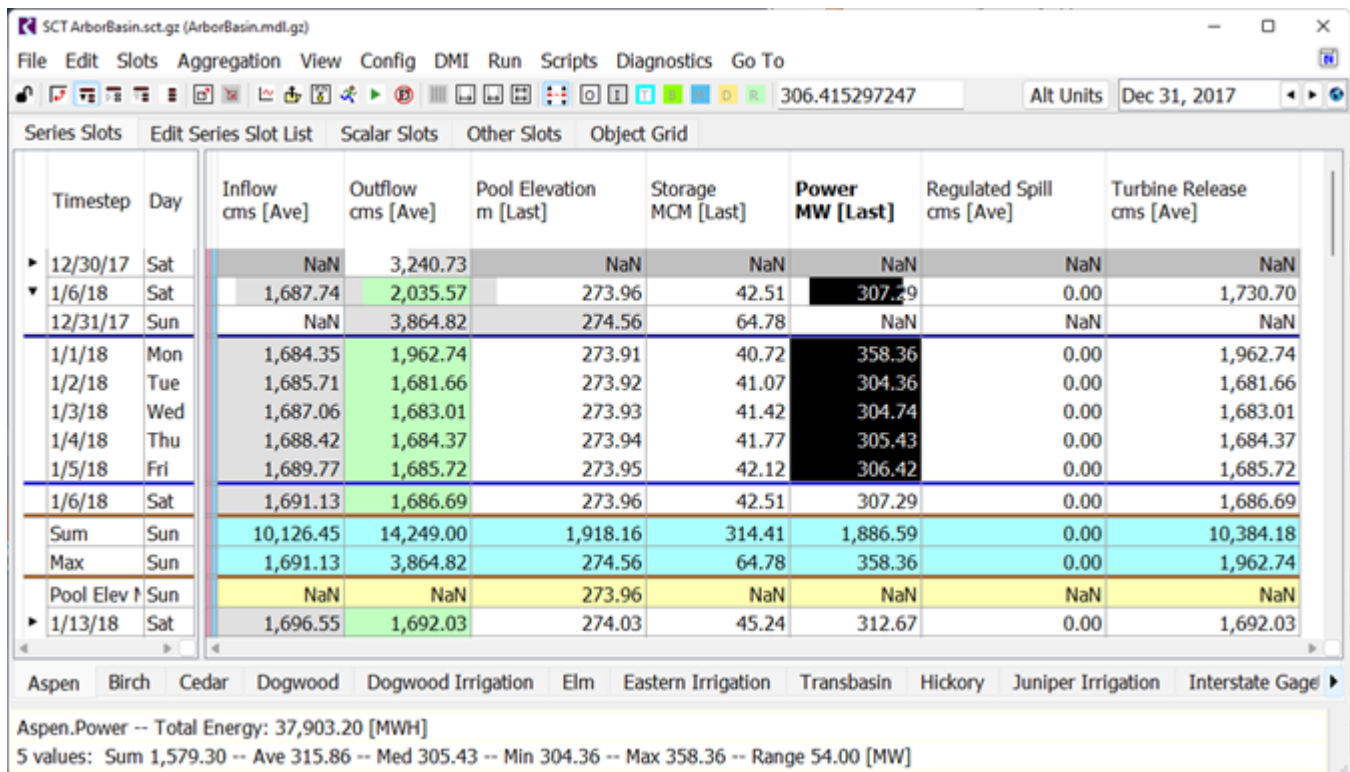
Topics

- [About the SCT, page 1](#)
- [SCT Features, page 2](#)

About the SCT

The RiverWare System Control Table (SCT) is a customizable user interface to view and edit data, make runs, investigate results, and access other applicable utilities. [Figure 1.1](#) is a sample SCT, displaying series data in a spreadsheet-like view. In this view, each timeseries slot is a column, and each row represents a timestep or aggregation of many timesteps. The orientation of the slots and time axes can be easily swapped.

Figure 1.1 Screenshot of a sample SCT



Chapter 1

SCT Overview

The SCT is highly configurable; nearly everything can be changed to meet the modeling needs, including the following:

- Look and feel of the SCT, such as the colors, fonts, metadata, and orientation of the data
- Content displayed, including the slots, the order, and the sheets on which they are displayed.

The slots on the SCT are organized by the type of data. Series data is shown on the Series Slots tab and can be further broken into different sheets. The Edit Series Slot List tab is used to specify and order the series slots. Scalar data can be listed on the Scalar Slots tab or shown in a grid of objects on the Object Grid tab. Table and periodic data is listed on the Other Slots tab, providing easy access to open these slots in their own dialog boxes.

The SCT is a powerful tool to increase productivity to set up data, run a simulation, and analyze the results. You can create an SCT for a five-minute task and never save it, or configure an SCT that is used everyday to operate a model. Many users open RiverWare, load their model and SCT, and use the SCT to operate the model not using the workspace, run control, or slot dialogs at all.

The SCT provides access to many of the features needed for operations such as the Data Management Interface, Scripts, Diagnostics, Run Analysis, and Output Devices. Also, you can open multiple SCTs to provide different views of the same data in a model. Since SCTs are a separate file, a single SCT can be used with multiple models of the same basin, providing a single interface.

SCT Features

Many functions normally associated with running a model are available from the SCT, allowing you to use the SCT as a primary RiverWare interface. Following are some of the operations you can perform directly from the SCT.

- Open, run, and save models.
- View any times series data in the model, in any order.
- View the data in time-aggregated form.
- Edit and manipulate series slot data by entering values directly, adjusting them by a constant or percentage, interpolating, and copy and paste.
- Change series slot flags (such as Input, Output, and Target) using SCT toolbar buttons, keyboard shortcuts, and other utilities.
- Execute model runs.
- Enable and disable dispatching of certain objects.
- Plot slot values.
- Display the Diagnostics windows and information for debugging and simulation analysis.
- Access Data Management Interfaces (DMIs) to import and export data from the model.
- Run any scripts defined in the model.

Following are some of the ways you can customize the SCT to fit your needs.

- Configure multiple SCTs for the same model. Each SCT can include any slots, allowing you to view different areas of your basin or view the data in different ways.

- Open more than one SCT at a time and easily switch from one SCT to another.
- Create an SCT on the fly for a quick task.
- Define the included slots and the order in which they appear.
- Organize slots into logical groups and even sheets, for ease of viewing and management.
- Create a similar view of the same named slots on different objects.
- View timesteps vertically or horizontally and easily switch between the views.
- Aggregate groups of timesteps (for example, aggregate seven daily values into one weekly value) and specify whether to display only summary values, or summary and details.
- For series data, define Color Alerts that automatically change background or text color of a cell as the slot value meets specified conditions.
- Configure the look and feel of the SCT, including: font type, size, and color; divider colors; how missing values are shown; and how flags and priorities are shown.

Chapter 1

SCT Overview

Chapter 2

SCT Basics

Topics

- [Data in the SCT, page 5](#)
- [Multiple SCTs, page 5](#)
- [SCT Layout, page 6](#)
- [Time Series Data, page 15](#)
- [Scalar Data on the Object Grid, page 31](#)
- [SCT Configuration, page 33](#)
- [Accessing Other Utilities From the SCT, page 34](#)

Data in the SCT

The SCT is a window or view into the model. Although you can change model data and start simulation runs through an SCT, the SCT is only a *representation* of the underlying model data. When using the SCT to change slot values, keep the following principles in mind.

- No slot values are saved with the SCT.
- When you save an SCT, the SCT *configuration* is saved, not the slot values.
- To retain changes made to slot values, you must also save the model.

The time range displayed by the SCT matches the run control configuration of the model, plus a user-defined number of pre- and post-simulation timesteps.

Related topics

- [Manipulating Slot Values, page 67](#)

Multiple SCTs

You can use multiple SCTs in a single RiverWare session. This allows you to show distinct views of the same RiverWare model.

An SCT can be used with multiple RiverWare models as long as the object and slot names are exactly the same in the two models.

Related topics

- [Getting Started With the SCT](#), page 37

SCT Layout

The SCT is a separate RiverWare window that you can open, resize, and position to suit your workflow. [Figure 2.1](#) is an illustration of the main sections of the SCT.

Topics

- [General layout](#), page 6
- [Series Slots tab](#), page 7
- [Edit Series Slot List tab](#), page 10
- [Scalar Slots tab](#), page 11
- [Other Slots tab](#), page 12
- [Object Grid tab](#), page 13

General layout

[Figure 2.1](#) illustrates the general layout of the SCT.

The SCT file name is in the top border of the window. The model file name is shown in parenthesis.

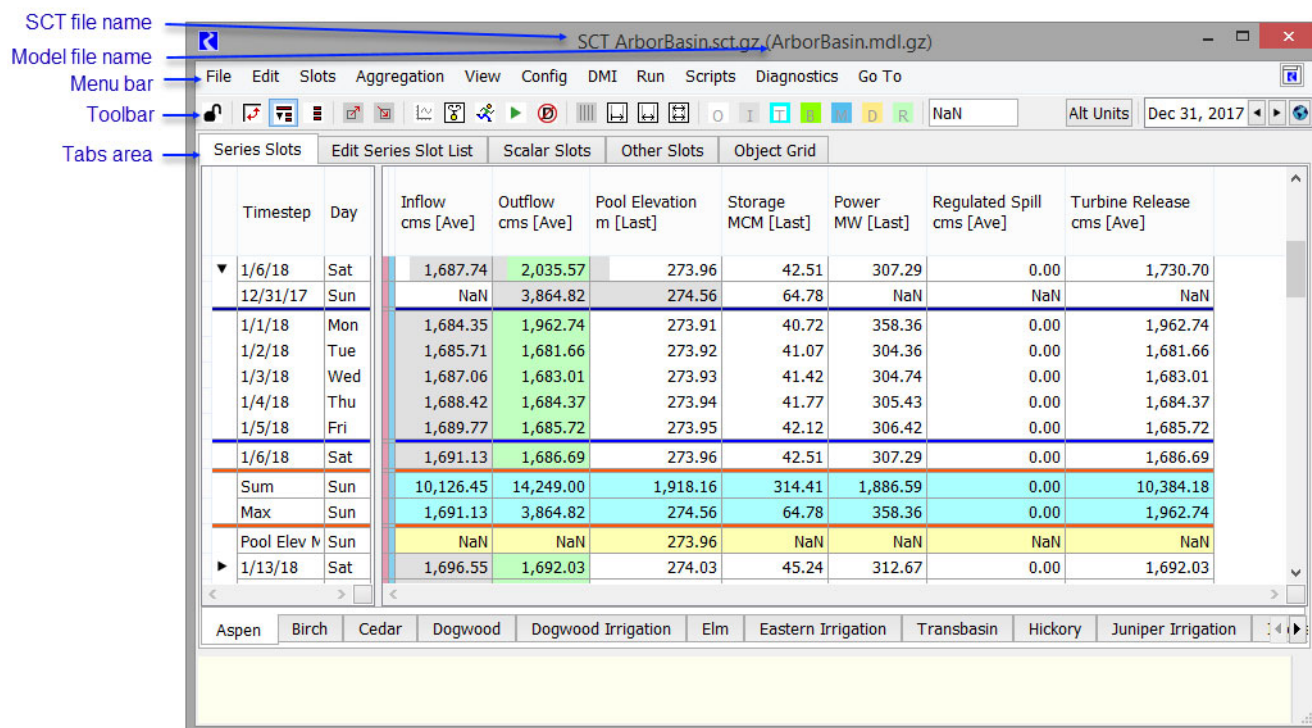
The menu bar provides access to most SCT functions. Some functions are available from context-sensitive shortcut menus within the SCT.

The toolbar provides quick access to common SCT menu functions. You can customize the toolbar to include the buttons you use most frequently and hide the ones you do not need. User-configurable buttons include actions and cell value type flags.

The tabs area includes five tabs that are used to view, configure, and edit different types of data.

You can also selectively show or hide the Diagnostics Output area in the SCT. See [Diagnostics](#), page 35 for more information.

Figure 2.1 SCT layout



Series Slots tab

This tab displays series slots, including AggSeries and table series slots, on physical simulation objects and accounting slots. It allows you to display series slots and manipulate their values.

Series data is displayed in a spreadsheet-like format, with timesteps on one axis and slots on the other. Each cell displays a value for the slot at that timestep. You can easily switch the orientation of the axes so the timesteps are listed either vertically, down the page, or horizontally, across the page. See [Figure 2.2](#) and [Figure 2.3](#) for illustrations.

Chapter 2 SCT Basics

Figure 2.2 Series Slots tab—Vertical timestep orientation

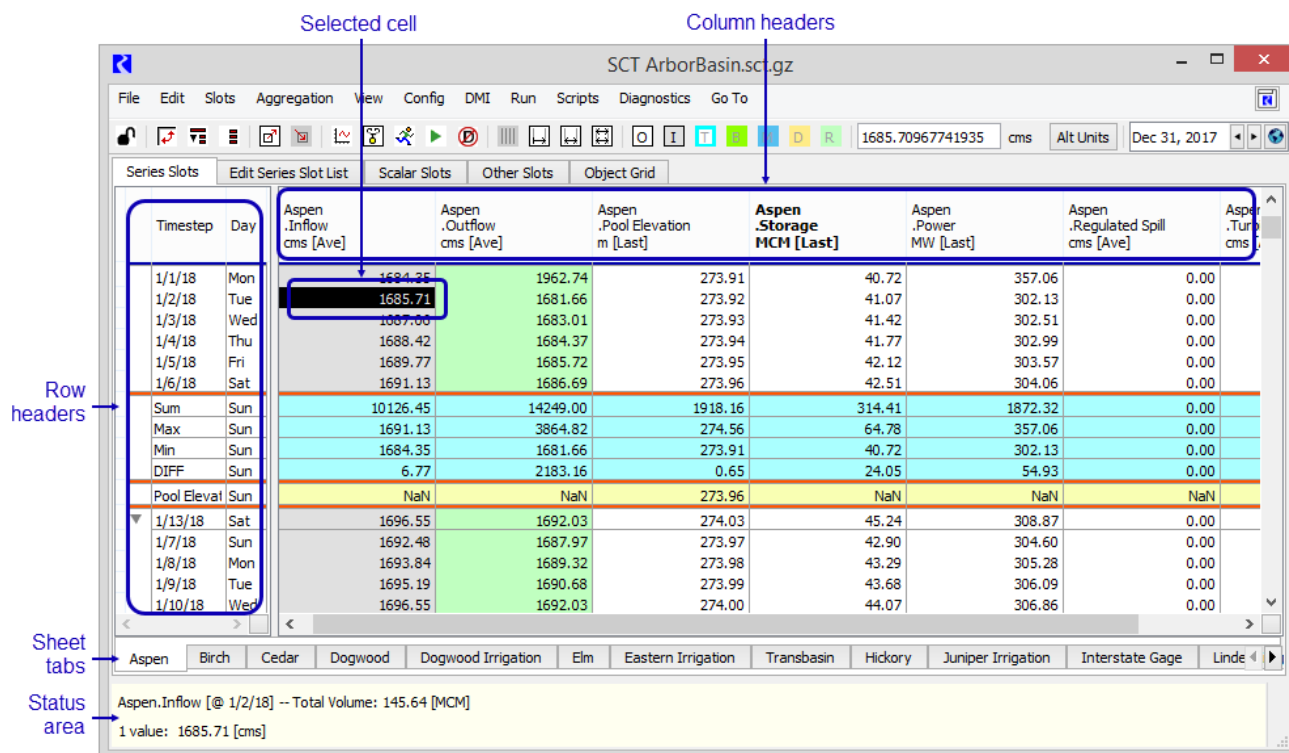
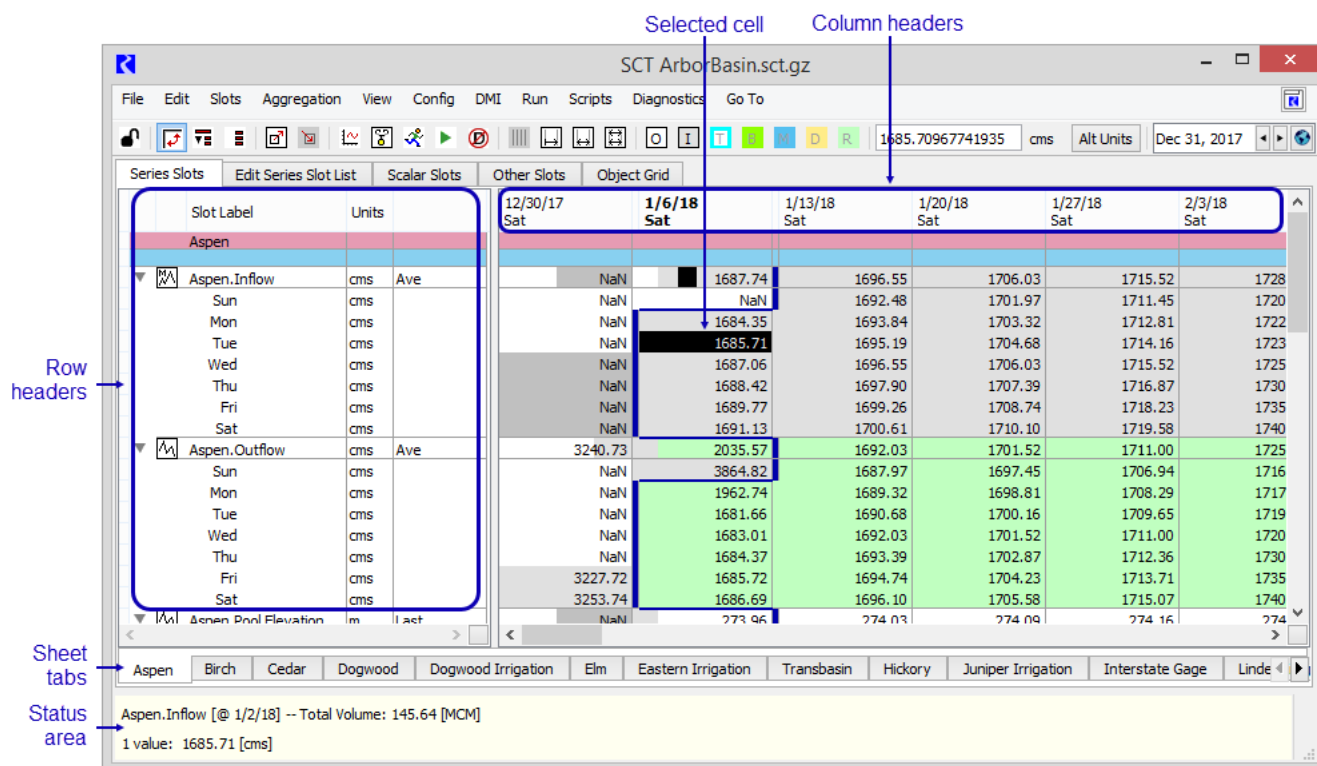


Figure 2.3 Series Slots tab—Horizontal timestep orientation



The background color of each cell typically identifies the *Flag*, which indicates how the value in the cell was calculated or assigned. For example, there is one color for Input (I) values and another for Output (O). You can optionally customize these flag background colors. Alternatively, you can create Color Alerts that change colors as the values meet specified conditions.

Optionally, you can aggregate timestep values according to a variety of user-selected summary functions, including average, median, minimum, maximum, first, and last. After defining an aggregation interval for the entire SCT, you can assign summary functions by slot or by unit type.

The Status area, at the bottom of the Series Slots tab, alternates between the following displays.

- Selection summary information. Most of the time, the Status area displays detail on selected cells, including the name or number of slots selected, and basic timestep statistics, such as Sum, Average, Median, Min, Max, Range, and Difference.
- Run status. When a simulation run is started, the Status area displays the run status, if the Show Run Status setting is enabled in the SCT configuration. This area also displays object dispatch information for the run; an icon displays the most limiting object dispatching condition in the run and the number of objects that meet this condition.

Related topics

- [Time Series Data](#), page 15
- [Timestep orientation](#), page 16
- [Aggregation](#), page 21

Chapter 2 SCT Basics

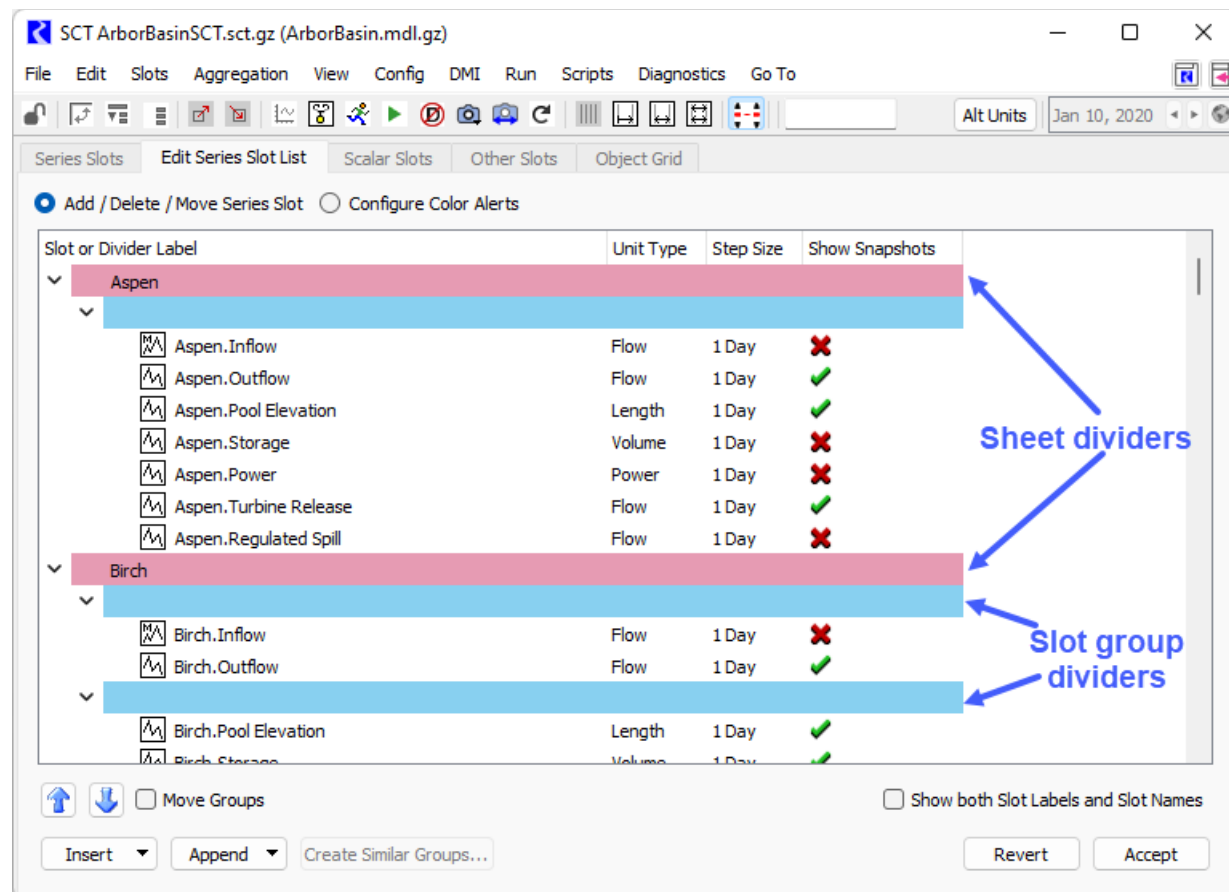
- [Flags and cell colors](#), page 26
- [Define custom colors for highlights and color alerts](#), page 191
- [Configuring Color Alerts](#), page 150
- [Adding Series Slots](#), page 46
- [Manipulating Slot Values](#), page 67

Edit Series Slot List tab

This tab lists all series slots, slot group dividers, and sheet dividers that are displayed on the Series Slots tab. It allows you to organize these slots and configure slot value threshold alerts.

Note: You must unlock the SCT configuration to go to this tab.

Figure 2.4 Edit Series Slot List tab



You can perform the following actions on this tab.

- Add, delete, and reorder slots.
- Organize slots into user-defined logical slot groups by adding and arranging slot group dividers.

- Organize objects and slots into multiple sheets by adding and arranging sheet dividers. Each sheet is accessible from a tab at the bottom of the Series Slots tab.
- Copy the structure of a selected slot group or sheet to other, similar objects.
- Configure whether to show snapshot slots for each slot, dependent on the setting to show recent snapshots.
- Configure custom, labeled colors for cell text and background.
- Configure Color Alerts that automatically apply a custom color when slot values meet a specified condition.

Related topics

- [Edit Series Slot List tab details](#), page 278
- [Sheet dividers and slot dividers](#), page 19
- [Adding Series Slots](#), page 46
- [Organizing Series Slots](#), page 53
- [Configuring Color Alerts](#), page 150

Scalar Slots tab

This tab displays scalar and 1x1 table slots. It provides a single convenient location for you to track, organize, and edit these slots.

Note: 1x1 table slots were used for scalar values before scalar slots were implemented.

Each slot appears on a separate row. The object icon identifies the object type, and the slot icon identifies the slot type. The current value and unit type for the slot are also displayed.

Chapter 2

SCT Basics

Figure 2.5 Scalar Slots tab

Object	Slot	Value	Units
Dogwood	Diversion Min Elevation	171.00	m
Aspen	Minimum Power Elevation	235.00	m
Birch	Minimum Power Elevation	207.26	m
Dogwood	Minimum Power Elevation	161.54	m
Elm	Minimum Power Elevation	146.76	m
Elm	Pool Elevation License Min	147.00	m
Aspen	Pool Elevation Max	276.00	m
Birch	Pool Elevation Max	235.00	m
Elm	Pool Elevation Max	149.00	m
Hickory	Pool Elevation Max	140.00	m
Aspen	Pool Elevation Min	273.00	m
Birch	Pool Elevation Min	250.50	m
Elm	Pool Elevation Min	147.00	m
Hickory	Pool Elevation Min	133.00	m
Hickory	Pool Elevation Min to Supplement Irrigation	135.00	m
Elm	Pool Elevation Target	148.50	m

You can perform the following actions on this tab:

- Add, delete, sort, and reorder scalar slots.
- Set slot values directly; alternatively, you can open slots in the Slot Viewer and edit values there.

Related topics

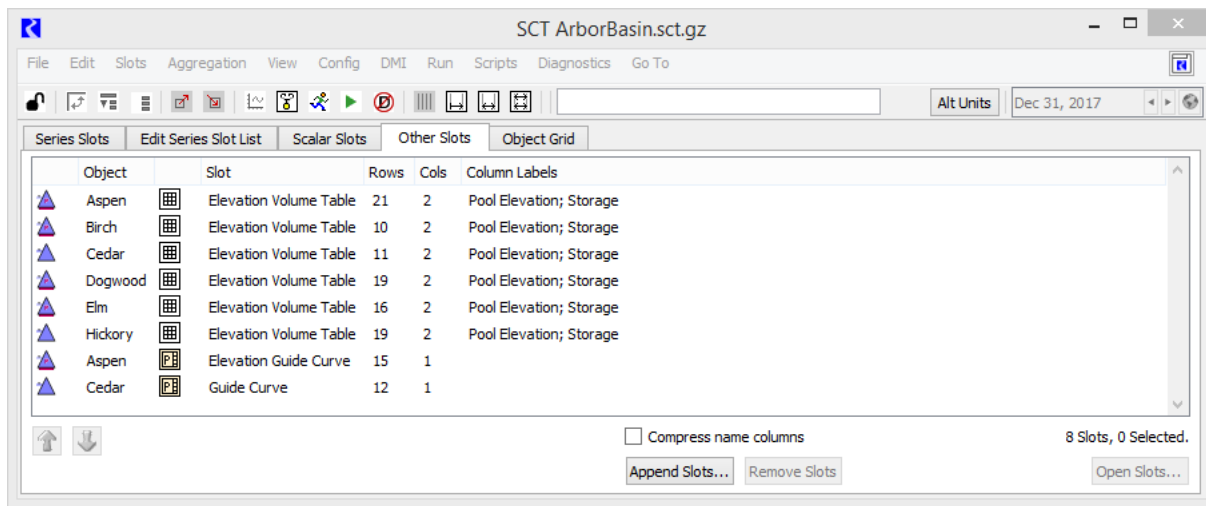
- [Scalar Slots tab details](#), page 281
- [Organizing Scalar and Table Slots](#), page 213

Other Slots tab

This tab displays a list of table slots, periodic slots, list slots, and statistical table slots. It provides a single convenient location for you to track and organize these slots.

Each slot appears on a separate row in the list. The object icon identifies the object type, and the slot icon identifies the slot type. The number of rows and columns and the column labels are also displayed.

Figure 2.6 Other Slots tab



You can perform the following actions on this tab:

- Add, delete, sort, and reorder slots.
- Open slots in separate Slot Viewer windows. While you cannot edit slot values directly on this tab, you can edit values in the Slot Viewer.

Related topics

- [Other Slots tab details](#), page 283
- [Organizing Scalar and Table Slots](#), page 213

Object Grid tab

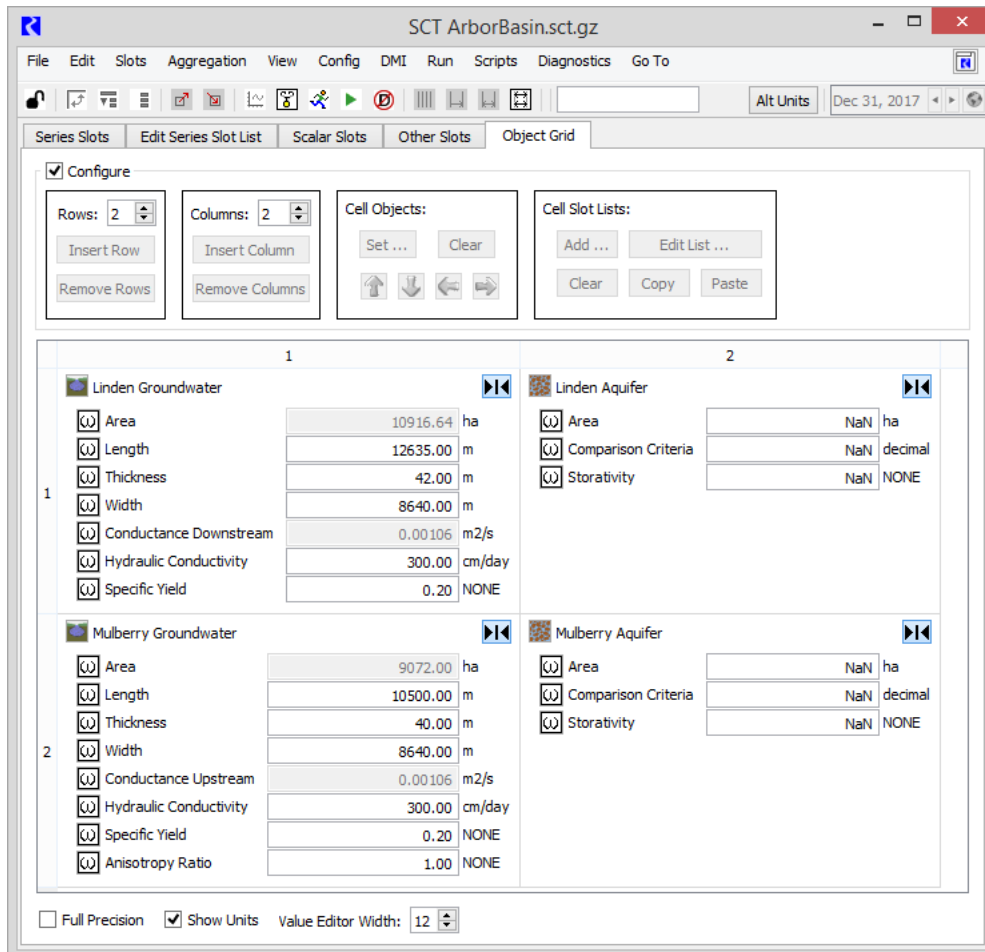
This tab displays a user-configured grid of objects and their included scalar and single value table slots. It provides a convenient location for you to organize and specify parameters on objects that interact with one another.

Each cell on the object grid represents an object, and each cell includes a user-specified list of scalar slots and single-cell table slots, called *cell slot lists*. You can add and arrange the objects and slots to fit your needs.

The object grid is particularly useful for a groundwater or aquifer network, in which the physical characteristics of the objects are related to one another. It is a useful tool when calibrating these types of systems, particularly when they become more complex.

Chapter 2 SCT Basics

Figure 2.7 Object Grid tab

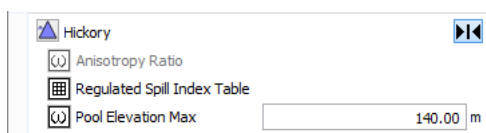


You can perform the following actions on this tab:

- Create a grid of objects with a specified number of columns and rows.
- Add, delete, and arrange objects in the grid.
- Open objects in the Object Viewer.
- Add, delete, and order slot lists for each object.
- Set slot values directly; alternatively, you can open slots in the Slot Viewer and edit values there.

The slots in a cell slot list are displayed in one of the following ways:

- Scalar slots and single-cell table slots are displayed with an active slot type button and a value, which is editable.



- Slots that do not exist on the object are not available (dimmed).

- Table slots with more than one cell are displayed with an active slot type button and no value.

Related topics

- [Scalar Data on the Object Grid](#), page 31
- [Organizing Scalar Data in the Object Grid](#), page 220

Time Series Data

This section includes details about displaying and manipulating values on series slots.

Topics

- [Timestep size](#), page 15
- [Time range](#), page 15
- [Timestep orientation](#), page 16
- [Editing Slot Values](#), page 19
- [Sheet dividers and slot dividers](#), page 19
- [Summary statistics](#), page 20
- [Aggregation](#), page 21
- [Display units](#), page 25
- [Flags and priorities](#), page 26
- [Custom slot labels](#), page 26
- [Notes on series slots](#), page 29

Timestep size

By default, the SCT uses the same timestep size as the model. Optionally, you can configure the SCT to use any timestep size supported in RiverWare. Each SCT can display only one timestep size for all series slot sheets; however, you can create and view multiple SCTs that show different timestep sizes. When an SCT is shown for a different timestep than the model, the timestep size is added to the title bar of the SCT.

Related topics

- [Configure timesteps](#), page 165

Time range

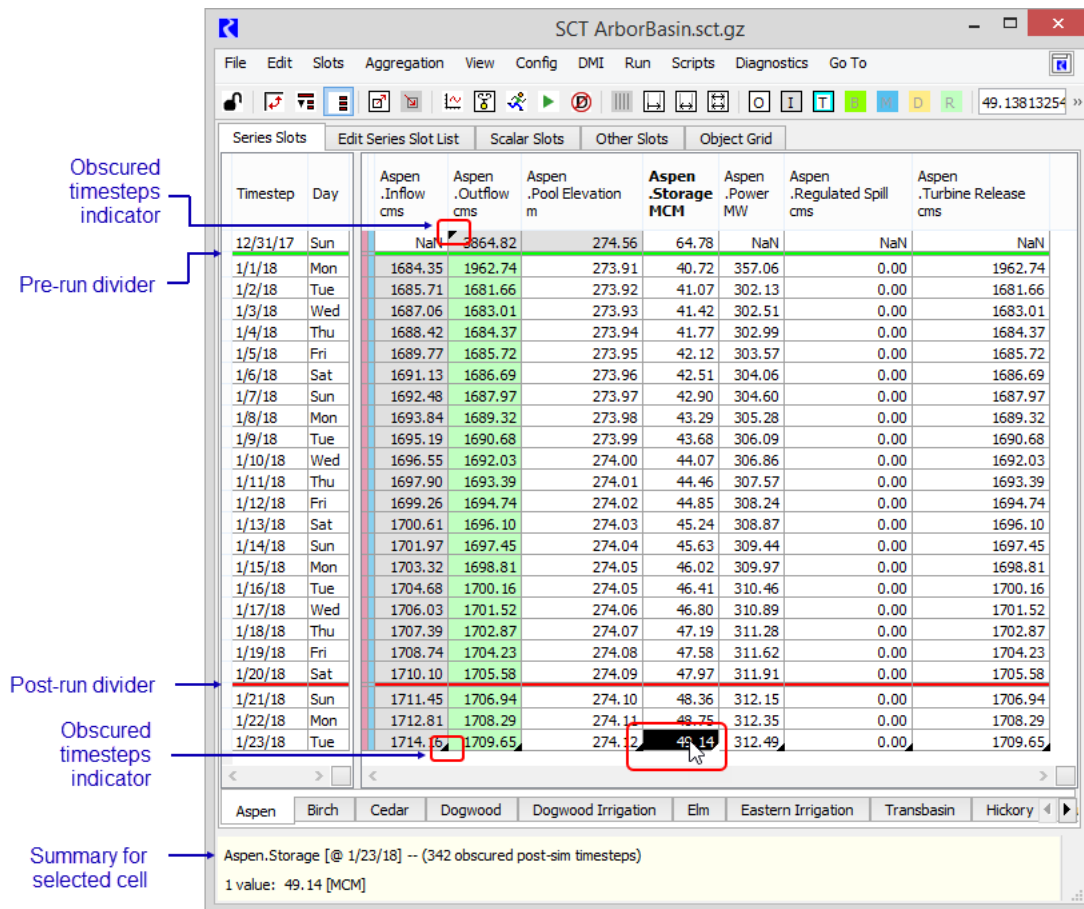
By default, the SCT uses the same time range as the model run control. Optionally, you can add pre-simulation and post-simulation timesteps to the SCT.

Chapter 2

SCT Basics

When data at pre-simulation or post simulation timesteps are obscured, a small corner triangle is displayed in the first or last cell in that slot. If you select the detail cell, the number of obscured timesteps is displayed in the SCT Status area. To display obscured timesteps for a slot, extend the time range of the SCT for pre or post-simulation timesteps to cover the full time range of the slot. SCT cells that correspond to a timestep that is out of the defined range for a slot display as shaded / inactive. [Figure 2.8](#) illustrates.

Figure 2.8 Time range indicators



Related topics

- [Configure timesteps](#), page 165

Timestep orientation

The Series Slots tab offers two timestep orientations, and you can easily switch between them.

- In vertical timestep orientation, timesteps run down the page, from top to bottom, and each slot is a separate column.
- In horizontal timestep orientation, timesteps run across the page, from left to right, and each slot is a separate row.

One orientation may be preferable to the other, depending on the tasks you are performing and the number of slots and timesteps included in the SCT. You can display either orientation in aggregated or non-aggregated view.

Vertical timestep orientation

In this orientation, timesteps are displayed vertically, running from top to bottom. Column headers are slot names, and row headers are one of the following:

- Timesteps
- Timestep aggregate summaries
- Custom summary rows

Timestep	Day	Aspen .Inflow cms	Aspen .Outflow cms	Aspen .Pool Elevation m	Aspen .Storage MCM	Aspen .Power MW	Aspen .Regulated Spill cms	Aspen .Turbine Release cms
12/27/17	Wed	NaN	NaN	NaN	NaN	NaN	NaN	NaN
12/28/17	Thu	NaN	NaN	NaN	NaN	NaN	NaN	NaN
12/29/17	Fri	NaN	3227.72	NaN	NaN	NaN	NaN	NaN
12/30/17	Sat	NaN	3253.74	NaN	NaN	NaN	NaN	NaN
12/31/17	Sun	NaN	3864.82	274.56	64.78	NaN	NaN	NaN
1/1/18	Mon	1684.35	1962.74	273.91	40.72	357.06	0.00	1962.74
1/2/18	Tue	1685.71	1681.66	273.92	41.07	302.13	0.00	1681.66
1/3/18	Wed	1687.06	1683.01	273.93	41.42	302.51	0.00	1683.01
1/4/18	Thu	1688.42	1684.37	273.94	41.77	302.99	0.00	1684.37
1/5/18	Fri	1689.77	1685.72	273.95	42.12	303.57	0.00	1685.72
1/6/18	Sat	1691.13	1686.69	273.96	42.51	304.06	0.00	1686.69
1/7/18	Sun	1692.48	1687.97	273.97	42.90	304.60	0.00	1687.97
1/8/18	Mon	1693.84	1689.32	273.98	43.29	305.28	0.00	1689.32
1/9/18	Tue	1695.19	1690.68	273.99	43.68	306.09	0.00	1690.68
1/10/18	Wed	1696.55	1692.03	274.00	44.07	306.86	0.00	1692.03
1/11/18	Thu	1697.90	1693.39	274.01	44.46	307.57	0.00	1693.39

This orientation provides a view similar to the Slot Viewer and Slot Editor. Long slot names can get obscured because of limited column width.

Each summary row corresponds to one timestep aggregation. The details rows represent the individual timesteps within each aggregation interval. Details for a single summary row correspond to one timestep aggregation across all slots.

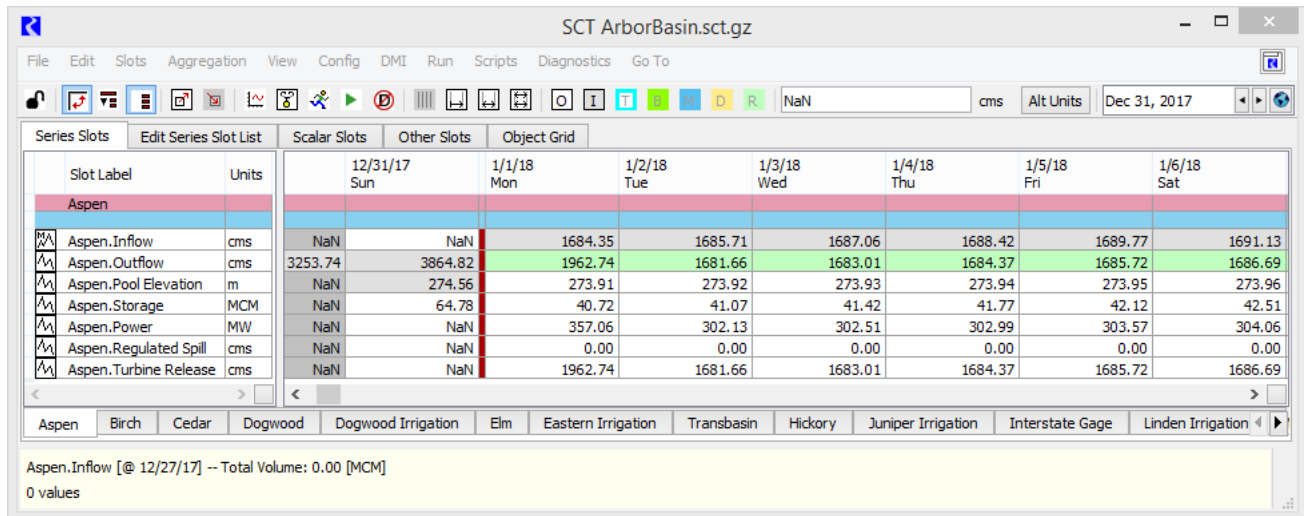
Horizontal timestep orientation

In this orientation, timesteps are displayed across the page, running from left to right. Row headers are slot names, and column headers are one of the following:

- Timesteps

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- Timestep aggregate summaries



Each summary row corresponds to a slot. Therefore, showing or hiding the details for a summary row displays or hides the details for that slot.

Related Topics

- [Change the timestep orientation](#), page 98
- [Customize a slot label](#), page 51

Synchronized Scrolling

When scrolling the SCT, other dialogs will also scroll to keep the same timesteps visible if Synchronized Scrolling is enabled.

Use the **View** and then **Synchronized Series Slot Scrolling** to toggle the setting on or off. Alternatively, use the tool bar buttons to toggle it as shown in the following figures.

Figure 2.9 Screenshot with Synchronized Scrolling enabled. Select the button to disable it

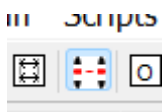
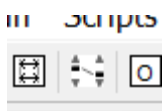


Figure 2.10 Screenshot with Synchronized Scrolling disabled. Select to disable it



Only the scroll bars on the Series Slot tab are affected by this setting, but the controls will work in either time orientation. This setting is a RiverWare (session) wide setting. Changing the setting in the SCT will affect the analogous options in the Slot Viewer and other times series dialogs, and vice versa.

Aggregated views are not affected by this setting; that is, aggregated views will never synchronize scrolling.

The toolbar button can be shown or hidden as described in [“Show or hide buttons on the SCT toolbar”](#) on page 181.

Related Topics

- [“Show or hide buttons on the SCT toolbar”](#) on page 181
- [“Synchronize Scrolling”](#) in *User Interface*

Editing Slot Values

The SCT is a window into the model that not only displays the model data, but allows you to edit slot values directly. Following are general guidelines for editing slot values through the SCT. See [“Manipulating Slot Values”](#) on page 67 for details.

- You do *not* need to unlock the SCT. The SCT lock mechanism only locks the SCT configuration, not the data values. See [“SCT Configuration”](#) on page 33 for details.
- All values are stored in the model itself; you must *save the model* to preserve the new values.
- Enter values manually using the following approaches.
 - Enter numbers directly in one or more selected cells. When multiple cells are selected, the entered value is applied to all of them.
 - Copy and paste values from one or more cells in the SCT to other cells in the SCT.
 - Import Paste data that was copied to the Windows system clipboard from external applications.
 - Use utilities to Adjust Values or Interpolate over a time range, as in other slot editing dialogs.
- Export Copy data out of RiverWare to the Windows system clipboard, making the data available for pasting to external applications.
- All timestep values are assigned a flag. Display this flag as a cell background color, a flag letter, or both.
- Clear values and flags to get rid of unwanted data.
- Set or change flags using menu options, toolbar buttons, or right-click context menus.
- You cannot edit certain types of slots, including expression slots and certain accounting system slots. Values for these slots are read only, which is indicated by crosshatching in a user-defined Read-only color.
- Undo and Redo of data edits using the Edit menu or Ctrl+Z and Ctrl+Shift+Z keyboard shortcuts.

Sheet dividers and slot dividers

You can use slot dividers and sheet dividers to organize the Series Slot display to meet your needs. Slot group dividers identify the start of a logical grouping of slots on a sheet. Sheet dividers identify the start of a new series slot sheet; each sheet appears as a tab at the bottom of the series slot display.

By default, slot dividers are blue, and sheet dividers are pink. You can optionally define a custom color for each.

In the vertical timestep orientation, dividers display as columns, and they are drawn as thick lines. In this orientation, you cannot enter text in the slot divider.

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SCT Basics

In the horizontal timestep orientation, dividers display as rows that are tall enough for a line of text. In this orientation, you can enter text in the Slot Label field of the slot divider.

Figure 2.11 Sheet and slot group dividers in horizontal timestep orientation

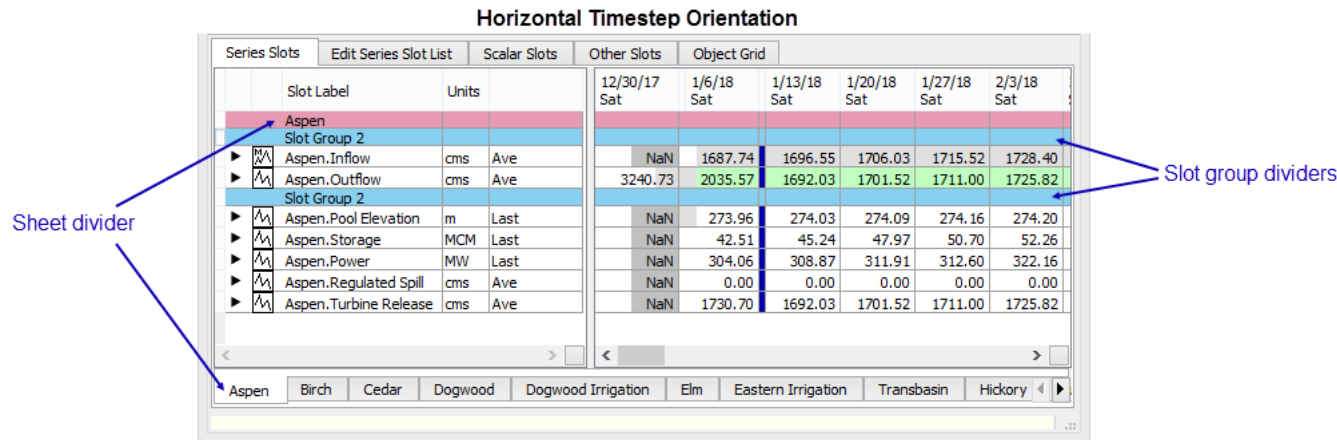
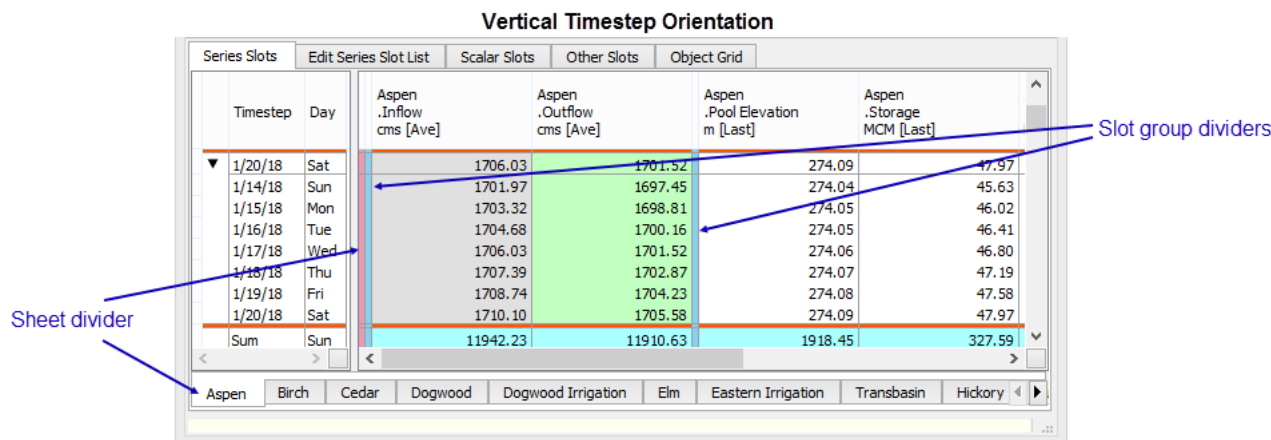


Figure 2.12 Sheet and slot dividers in vertical timestep orientation



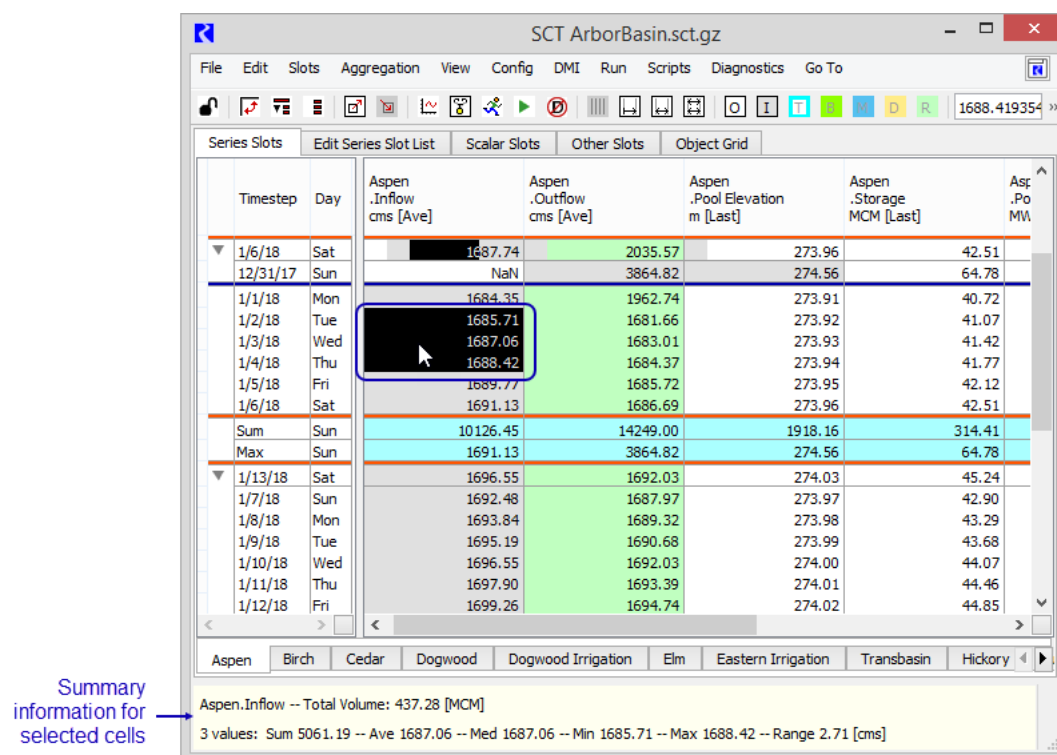
Related topics

- [Add slot dividers](#), page 54
- [Add sheet dividers](#), page 56
- [Customize divider and cell colors](#), page 190

Summary statistics

The Status Area at the bottom of the Series Slots tab displays summary information on selected cells. This includes the name or number of the slots selected and summary statistics. Statistics include Sum, Average, Median, Min,

Max, Range, and Difference. All, some, or none of these values may be shown depending on the number of cells selected and the units of the cells.



Aggregation

For display and some data export purposes, you can aggregate timesteps within each series slot according to user-selected summary functions. Timestep data is aggregated into an aggregation interval size larger than the timestep interval size. You can aggregate a fixed number of timesteps per summary (for example, aggregate three timesteps per summary), or you can aggregate timesteps into a regular interval (for example, aggregate one-hour timesteps into daily summaries). See “[Timestep size](#)” on page 15 for details.

Note: Synchronized timestep scrolling is not supported for Aggregated vies. See “[Synchronized Scrolling](#)” on page 18 for more information on this feature.

Aggregated and non-aggregated views

You can choose to display series data with aggregations enabled or disabled.

The No Aggregation view displays timestep detail only, and summaries are not available. [Figure 2.13](#) illustrates.

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Figure 2.13 No Aggregation view

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid		
Timestep	Day	Aspen .Inflow cms	Aspen .Outflow cms	Aspen .Pool Elevation m	Aspen .Storage MCM	Aspen .Power MW	Aspen .Regulate cms
1/1/18	Mon	1684.35	1962.74	273.91		40.72	357.06
1/2/18	Tue	1685.71	1681.66	273.92		41.07	302.13
1/3/18	Wed	1687.06	1683.01	273.93		41.42	302.51
1/4/18	Thu	1688.42	1684.37	273.94		41.77	302.99
1/5/18	Fri	1689.77	1685.72	273.95		42.12	303.57
1/6/18	Sat	1691.13	1686.69	273.96		42.51	304.06
1/7/18	Sun	1692.48	1687.97	273.97		42.90	304.60
1/8/18	Mon	1693.84	1689.32	273.98		43.29	305.28
1/9/18	Tue	1695.19	1690.68	273.99		43.68	306.09
1/10/18	Wed	1696.55	1692.03	274.00		44.07	306.86
1/11/18	Thu	1697.90	1693.39	274.01		44.46	307.57
1/12/18	Fri	1699.26	1694.74	274.02		44.85	308.24
1/13/18	Sat	1700.61	1696.10	274.03		45.24	308.87
1/14/18	Sun	1701.97	1697.45	274.04		45.63	309.44

Aspen Birch Cedar Dogwood Dogwood Irrigation Elm Eastern Irrigation Transbasin Hickory

The Hide All Details view displays summary values only, and details are not available. [Figure 2.14](#) illustrates.

Figure 2.14 Hide All Details view

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid		
Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	Aspen .Power MW	Aspen .Regulate cms
▶ 3/24/18	Sat	2112.25	2116.76	273.97		42.90	
▶ 3/31/18	Sat	2258.38	2262.55	273.90		40.37	
▶ 4/7/18	Sat	2446.73	2455.71	273.75		34.95	
▶ 4/14/18	Sat	2666.77	2675.71	273.60		29.54	
▶ 4/21/18	Sat	2886.80	2902.35	273.34		20.13	
▶ 4/28/18	Sat	3106.83	3122.79	273.07		10.48	
▶ 5/5/18	Sat	3286.94	3271.96	273.32		19.54	
▶ 5/12/18	Sat	3384.17	3357.42	273.77		35.72	
▶ 5/19/18	Sat	3479.57	3460.46	274.08		47.28	
▶ 5/26/18	Sat	3574.97	3559.70	274.32		56.52	
▶ 6/2/18	Sat	3645.60	3634.95	274.51		62.96	
▶ 6/9/18	Sat	3466.82	3463.68	274.56		64.86	

Aspen Birch Cedar Dogwood Dogwood Irrigation Elm Eastern Irrigation Transbasin Hickory

The Show All Details view displays all summary and detail values. [Figure 2.15](#) illustrates.

Figure 2.15 Show All Details view

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid				
Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	Aspen .Pool MW			
1/6/18	Sat	1687.74	2035.57	273.96		42.51			
12/31/17	Sun	NaN	3864.82	274.56		64.78			
1/1/18	Mon	1684.35	1962.74	273.91		40.72			
1/2/18	Tue	1685.71	1681.66	273.92		41.07			
1/3/18	Wed	1687.06	1683.01	273.93		41.42			
1/4/18	Thu	1688.42	1684.37	273.94		41.77			
1/5/18	Fri	1689.77	1685.72	273.95		42.12			
1/6/18	Sat	1691.13	1686.69	273.96		42.51			
Sum	Sun	10126.45	14249.00	1918.16		314.41			
Max	Sun	1691.13	3864.82	274.56		64.78			
Min	Sun	1684.35	1681.66	273.91		40.72			
DIFF	Sun	6.77	2183.16	0.65		24.05			
Pool Elevat	Sun	NaN	NaN	273.96		NaN			
1/13/18	Sat	1696.55	1692.03	274.03		45.24			
1/7/18	Sun	1692.48	1687.97	273.97		42.90			
1/8/18	Mon	1693.84	1689.32	273.98		43.29			
1/9/18	Tue	1695.19	1690.68	273.99		43.68			
1/10/18	Wed	1696.55	1692.03	274.00		44.07			
1/11/18	Thu	1697.90	1693.39	274.01		44.46			
1/12/18	Fri	1699.26	1694.74	274.02		44.85			
1/13/18	Sat	1700.61	1696.10	274.03		45.24			
Sum	Sun	11875.84	11844.24	1917.98		308.48			
Max	Sun	1700.61	1696.10	274.03		45.24			
Aspen		Birch	Cedar	Dogwood	Dogwood Irrigation	Elm	Eastern Irrigation	Transbasin	Hickory

The Show/Hide Details view displays summary and detail values in a dynamic expand/collapse format. [Figure 2.16](#) illustrates.

Figure 2.16 Show/Hide Details view

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid				
Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	Aspen .Pool MW			
▶ 1/6/18	Sat	1687.74	2035.57	273.96		42.51			
▼ 1/13/18	Sat	1696.55	1692.03	274.03		45.24			
1/20/18	Sat	1706.03	1701.52	274.09		47.97			
1/14/18	Sun	1701.97	1697.45	274.04		45.63			
1/15/18	Mon	1703.32	1698.81	274.05		46.02			
1/16/18	Tue	1704.68	1700.16	274.05		46.41			
1/17/18	Wed	1706.03	1701.52	274.06		46.80			
1/18/18	Thu	1707.39	1702.87	274.07		47.19			
1/19/18	Fri	1708.74	1704.23	274.08		47.58			
1/20/18	Sat	1710.10	1705.58	274.09		47.97			
Sum	Sun	11942.23	11910.63	1918.45		327.59			
Max	Sun	1710.10	1705.58	274.09		47.97			
Min	Sun	1701.97	1697.45	274.04		45.63			
DIFF	Sun	8.13	8.13	0.06		2.34			
Pool Elevat	Sun	NaN	NaN	274.09		NaN			
▼ 1/27/18	Sat	1715.52	1711.00	274.16		50.70			
Aspen		Birch	Cedar	Dogwood	Dogwood Irrigation	Elm	Eastern Irrigation	Transbasin	Hickory

To customize the display according to your needs, you can assign summary functions in any combination of the following ways:

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- Globally by unit type. For example all slots with the flow type could use average, and all slots with the area type could use maximum.
- Globally by slot name. For example, all Input slots could use average, and all Spill slots could use maximum.
- Individually by slot. For example, most Input slots could use average, but the Input slot in Aspen reservoir could use minimum and the Input slot in Birch reservoir could use maximum.

Related topics

- [Displaying Aggregated Values](#), page 123

Custom summary rows

For aggregated, vertical timestep view, you can specify any number of custom summary rows at the end of each time aggregation. Each custom summary row can be configured to display the value of a simple arithmetic function or the value of a referenced series or scalar slot. Optionally, you can assign a user-defined background color to each custom summary row.

Custom summary row values are automatically updated whenever any detail values are changed. Custom summary row values for simple functions are read only, but the read-only crosshatch is *not* displayed.

In the case of referenced slots, summary values are displayed only for aggregated slots with the same unit type as the referenced slot. Also, you can designate whether referenced slot values can be edited directly from the summary cell.

Figure 2.17 Sample custom summary rows definition

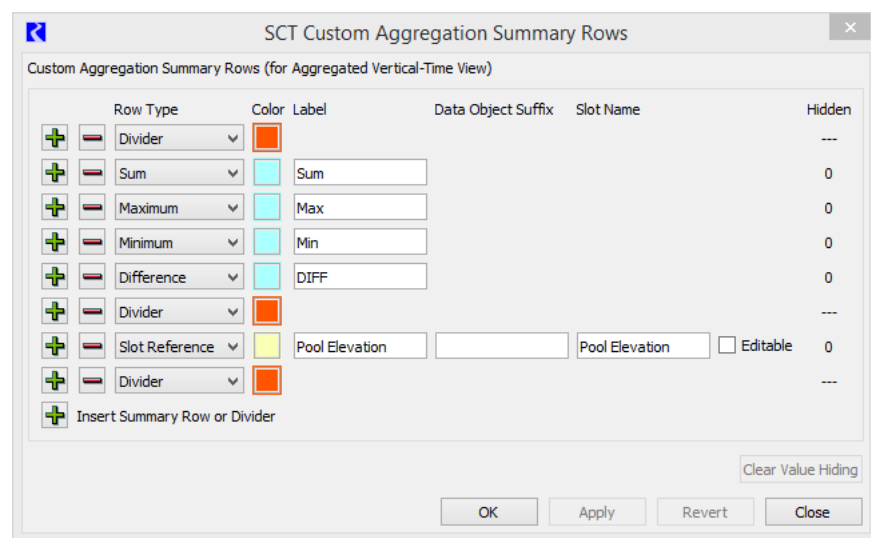


Figure 2.18 Sample custom summary rows display

The screenshot shows the SCT ArborBasin.sct.gz window. The table displays data for various time steps and summary rows. A red box highlights the summary rows: Sum, Max, Min, DIFF, and Pool Elevat.

Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	Aspen .Pool MCM
1/1/18	Mon	1684.35	1962.74	273.91		40.72
1/2/18	Tue	1685.71	1681.66	273.92		41.07
1/3/18	Wed	1687.06	1683.01	273.93		41.42
1/4/18	Thu	1688.42	1684.37	273.94		41.77
1/5/18	Fri	1689.77	1685.72	273.95		42.12
1/6/18	Sat	1691.13	1686.69	273.96		42.51
Sum	Sun	10126.45	14249.00	1918.16		314.41
Max	Sun	1691.13	3864.82	274.56		64.78
Min	Sun	1684.35	1681.66	273.91		40.72
DIFF	Sun	6.77	2183.16	0.65		24.05
Pool Elevat	Sun	NaN	NaN	273.96		NaN
1/13/18	Sat	1696.55	1692.03	274.03		45.24
1/7/18	Sun	1692.48	1687.97	273.97		42.90

Related topics

- [Displaying Aggregated Values](#), page 123
- [Create custom summary rows for all slots](#), page 137

Display units

Numeric time series slot values are stored internally as double-precision floating point numbers. Cell values are displayed in the SCT, as with the rest of the RiverWare user interface, in a user-defined display precision. Display precision specifies the number of fractional decimal digits to which the internal value is rounded. The SCT uses the display precision for each slot as defined in the active unit scheme; see “[Unit Schemes](#)” in *User Interface* for details.

When you select a single cell, the value is also displayed with model precision in the Value field in the SCT toolbar.

Note: Model precision is not necessarily the full precision of the internal value which is represented in internal units.

If the Display “NaN” setting is enabled in the SCT configuration, the SCT displays all undefined values as “NaN” (Not a Number). If this setting is disabled, undefined values are displayed as blanks.

If the global Show Commas in Numbers setting is enabled on the workspace, the SCT uses the comma as the thousands separator for all cell values. If this setting is disabled, the SCT does not display a thousands separator.

Related topics

- [Configure NaN display](#), page 167
- “[Show Commas in Numbers](#)” in *User Interface*

Flags and priorities

The SCT provides options for displaying slot value flags and rule priority numbers. Priority numbers are relevant when using rulebased simulation. See [“Flags on the Timestep I/O Menu” in *User Interface*](#) for details about flags on series slots. See [“Slot Priorities and Flags” in *Solution Approaches*](#) for details about slot value priorities.

Flags and cell colors

By default, each series slot cell is displayed with a background color that indicates the value flag, such as Input (I), Output (O), Rules (R), and so on. The background color provides a visual cue to help you interpret series slot data. Optionally, you can also choose to display the one-letter flag indicator.

Each value type flag has a default color. Optionally, you can define custom colors for any or all of the flags.

Optionally, you can configure whether flag colors are displayed when aggregated view is enabled.

Flag letters and priority numbers

Optionally display the flag letter for each slot value. This is useful if custom highlighting or threshold alerts have overridden the flag colors. In addition, if you are using rulebased simulation, you can optionally display the priority number for each slot value. See [“Display flag letters and priority numbers”](#) on page 103 for details.

Tooltips

When you hover over an area on the SCT, a context-sensitive tooltip is displayed.

The tooltip for a slot name or label displays the full slot name and any description defined on that slot.

The tooltip for a cell value indicates which rule or Data Management Interface (DMI) was responsible for the value, where applicable.

Related topics

- [Display flag letters and priority numbers](#), page 103
- [Customize flag colors](#), page 189
- [Set flags on series slot values](#), page 198

Custom slot labels

RiverWare slots are identified with a fully qualified name, which includes the name of the simulation object, account, or other entity the slot is on. SCT slots for which a match cannot be found on the workspace are indicated as disabled—no data is shown, and you cannot provide data for such slots.

Slot names

RiverWare uses slot names to refer to a particular slot. Generally, the slot name includes the simulation object or account that the slot is on. You cannot change a slot name from the SCT.

Slot labels

Slot labels are text strings that you can use to label a slot within a particular SCT. Slot labels are always displayed in the SCT as either row or column headers.

Each slot can have two separate slot labels, as follows:

- A single-line text string for use in the horizontal timestep orientation
- A multiple-line text string for use in the vertical timestep orientation

The slot label is also displayed in the SCT Status area when a single slot is selected.

Related topics

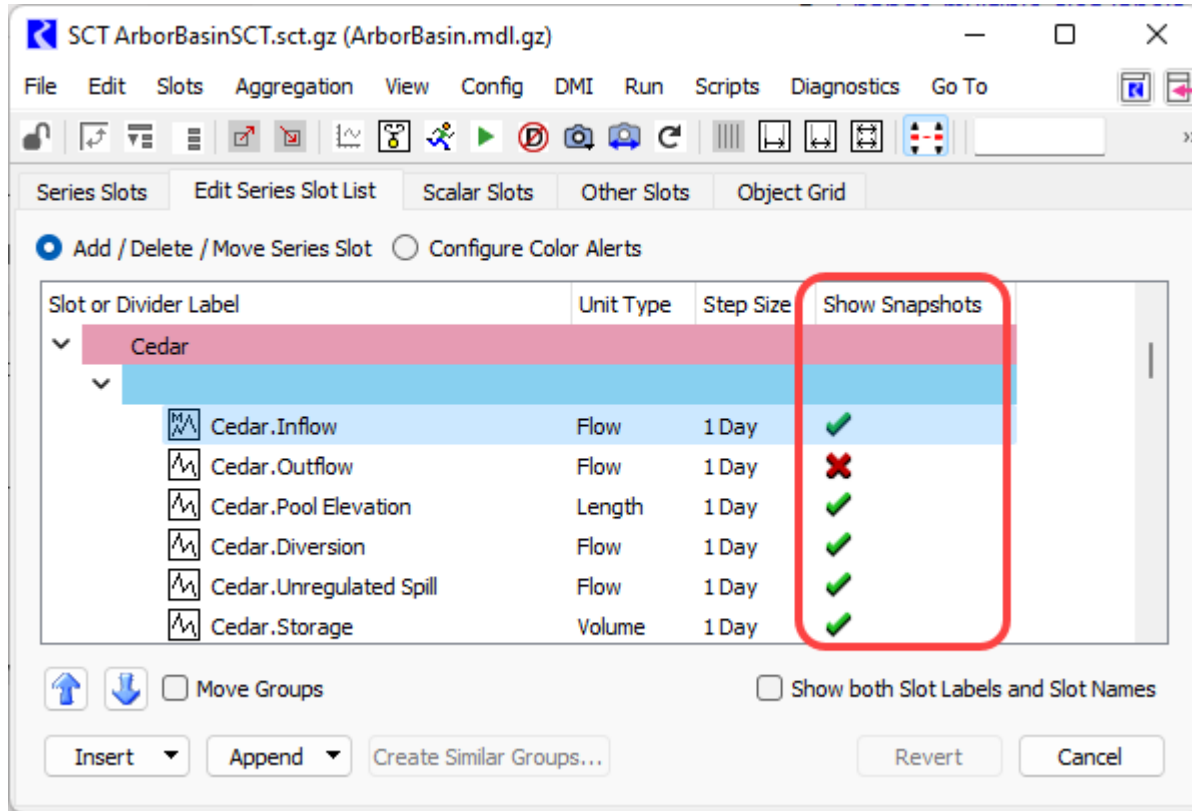
- [Customize a slot label](#), page 51
- [Change multiple slot labels](#), page 52
- [View and manage a slot description](#), page 212

Snapshot slots

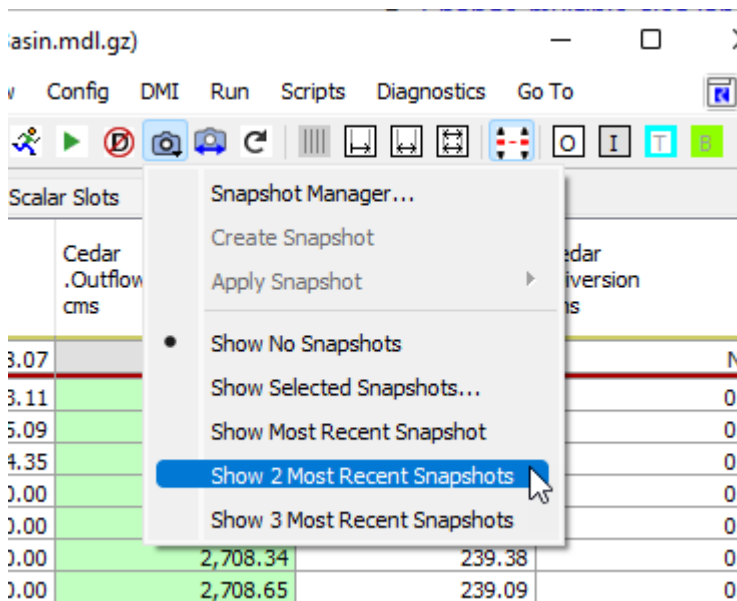
The SCT can be configured to show slots from snapshot objects together with their source slots. This can allow you to quickly see the values for a slot from previous runs together with the current values on the slot. See [“Snapshots” in *Output Utilities and Data Visualization*](#) for more information on snapshots.

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There are two settings to control the display of snapshot slots. The first is a per-slot setting on the Edit Series Slot List tab that controls whether snapshot slots will be shown for each slot.



The second is a selection from the snapshot button menu on the tool bar to Show Recent Snapshots.



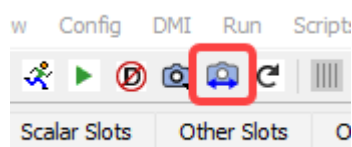
If the selection is to show one or more recent snapshots, and the slot has the green check mark for Show Snapshots on the Edit Series Slot List tab, and the slot has corresponding slots in the selected snapshots, then the snapshot slots will be shown to the right of the source slot if slots are columns or below the source slot if slots are rows.

Source slot **Snapshot slots**

Timestep	Day	Cedar .Pool Elevation m	Init Elev Very Low .Cedar_Pool Elevation m	Init Elev Low .Cedar_Pool Elevation m	Cedar .Diversion cms
1/1/20	Wed	46.73	240.18	228.00	232.00
1/2/20	Thu	95.86	240.41	228.97	232.57
1/3/20	Fri	54.39	240.51	229.72	232.99
1/4/20	Sat	01.82	240.26	229.84	232.98
1/5/20	Sun	07.59	239.97	229.88	232.91
1/6/20	Mon	07.99	239.68	229.92	232.84
1/7/20	Tue	08.34	239.38	229.96	232.77
1/8/20	Wed	08.65	239.09	230.00	232.70
1/9/20	Thu	08.92	238.80	230.04	232.63

Aspen Birch Cedar Dogwood Dogwood Irrigation Elm Eastern Irrigation Transbasin

In addition, the “Show All Snapshot Slots” button can be used to show snapshot slots for the selected snapshots for all slots in the SCT, regardless of the per slot setting on the Edit Series Slot List tab. Clicking on the button again will return to showing snapshot slots based on the per slot setting.



Snapshot slots are always read-only.

Related Topics

- [Displaying Snapshot Slots](#), page 145

Notes on series slots

The SCT displays Series Slot Notes on the Series Slots tab. See “[Notes on Series Slots](#)” in *User Interface* for more information on Series Slot Notes.

If a single SCT cell is selected, the note text associated with that cell is displayed in the SCT selection status bar, at the bottom of the SCT. As with the series data values displayed in the SCT, series slot notes are saved with the slots themselves, not with the SCT configuration.

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When the SCT is in the non-aggregated vertical timestep orientation, the text for all notes in each timestep row are displayed in a row header column, depending on whether the Show Column: Notes setting is enabled. See “[Vert Time tab](#)” on page 291 for details.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid				
Timestep	Day	Notes Column	Aspen .Inflow cms	Aspen .Outflow cms	Aspen .Regulated Spill cms	Aspen .Turbine Release cms	Aspen .Pool Elevation m	Aspen .Power MW	
5/22/20	Fri	Above Flood Stage	3,561.34	3,544.88	0.00	3,544.88	274.18		6
5/23/20	Sat	Above Flood Stage; Excess Spill	3,574.97	3,558.51	135.00	3,558.51	274.22		6
5/24/20	Sun	Above Flood Stage; Excess Spill	3,588.60	3,572.47	135.00	3,572.47	274.25		6
5/25/20	Mon	Above Flood Stage; Excess Spill	3,602.23	3,589.76	135.00	3,589.76	274.29		6
5/26/20	Tue	Above Flood Stage; High Flow	3,615.86	3,603.39	0.00	3,603.39	274.32		6
5/27/20	Wed	Above Flood Stage; High Flow	3,629.49			3,617.01	274.36		6
5/28/20	Thu	Above Flood Stage; High Flow	3,643.11				274.39		6
5/29/20	Fri	Above Flood Stage; High Flow	3,656.74				274.43		6
5/30/20	Sat	Above Flood Stage; High Flow	3,670.37				274.46		6
5/31/20	Sun	Above Flood Stage; High Flow	3,684.00				274.50		6
6/1/20	Mon	Above Flood Stage; High Flow	3,634.50				274.51		6
6/2/20	Tue	Above Flood Stage	3,600.96				274.51		6
6/3/20	Wed	Above Flood Stage	3,567.42				274.52		6
6/13/20	Sat	Above Flood Stage	3,232.07				274.59		5
6/14/20	Sun	Above Flood Stage	3,198.54				274.59		5
6/15/20	Mon	Above Flood Stage	3,165.00				274.60		5
6/16/20	Tue	Above Flood Stage	3,132.20				274.61		5
6/17/20	Wed	Above Flood Stage	3,099.40				274.61		5
6/18/20	Thu	Above Flood Stage	3,066.60						5
6/19/20	Fri	Above Flood Stage	3,033.80						5
6/20/20	Sat	Above Flood Stage	3,001.00						5
6/21/20	Sun		2,968.20						5
6/22/20	Mon		2,935.40						5
6/23/20	Tue		2,902.60						5
6/24/20	Wed		2,869.80						5
6/25/20	Thu		2,837.00						5
6/26/20	Fri		2,804.20	2,801.12	0.00	2,801.12	274.67		5
6/27/20	Sat		2,771.40	2,768.22	0.00	2,768.22	274.68		5

Set to Input
Set to Output
Clear Outputs
Copy Cells
Paste Cells
Set Values over Time Range...
Add Note
Paste Note
High Flow
Remove All Notes
Undo Set O Flag
Redo
Open Rule 32...

Copy Note
Edit Note...
Replace with New Note...
Remove Note

Aspen	Birch	Cedar	Dogwood	Dogwood Irrigation	Elm	Eastern Irrigation	Transbasin	Hickory	Juniper Irrigation	Interst
-------	-------	-------	---------	--------------------	-----	--------------------	------------	---------	--------------------	---------

Aspen.Outflow [@ 5/27/20] -- Total Volume: 312.51 [MCM]
1 value: 3,617.01 [cms] -- High Flow

The right-click context menu for a cell selection provides multiple options for adding a note to all slot timesteps in the selection as well as an option to remove all notes from the selected slot timesteps.

Series Slots	Edit Series Slot List	Scalar Slots	Other Slots	Object Grid				
Timestep	Day	Birch .Inflow cms	Birch .Outflow cms	Birch .Pool Elevation m	Birch .Storage MCM	Birch .Power MW	Birch .Regulated Spill cms	Birch .Turbine f cms
5/22/20	Fri	3,521.03	3,521.03	253.00	268.99	286.08	0.00	
5/23/20	Sat	3,521.03	3,521.03	253.00	268.99	286.83	0.00	
5/24/20	Sun	3,521.03	3,521.03	253.00	268.99	287.58	0.00	
5/25/20	Mon	3,521.03	3,521.03	253.00	268.99	288.32	0.00	
5/26/20	Tue	3,521.03	3,521.03	253.00	268.99	289.11	0.00	
5/27/20	Wed	3,521.03	3,521.03	253.00	268.99	289.98	0.00	
6/9/20	Tue	3,521.03	3,521.03	253.00	268.99	279.80	0.00	
6/10/20	Wed	3,521.03	3,521.03	253.00	268.99	277.54	0.00	
6/11/20	Thu	3,521.03	3,521.03	253.00	268.99	275.25	0.00	
6/12/20	Fri	3,521.03	3,521.03	253.00	268.99	273.05	0.00	
6/13/20	Sat	3,521.03	3,521.03	253.00	268.99			
6/14/20	Sun	3,521.03	3,521.03	253.00	268.99			
6/15/20	Mon	3,521.03	3,521.03	253.00	268.99			
6/16/20	Tue	3,521.03	3,521.03	253.00	268.99			
6/17/20	Wed	3,521.03	3,521.03	253.00	268.99			
6/18/20	Thu	3,521.03	3,521.03	253.00	268.99			
6/19/20	Fri	3,521.03	3,521.03	253.00	268.99			
6/20/20	Sat	3,521.03	3,521.03	253.00	268.99			
6/21/20	Sun	3,022.52	3,022.52	253.00	268.99			
6/22/20	Mon	2,989.72	3,300.00	252.24	242.19			
6/23/20	Tue	2,956.92	3,300.00	251.30	212.54			
6/24/20	Wed	2,924.12	3,237.15	250.50	185.50			
6/25/20	Thu	2,891.32	2,891.32	250.50	185.50			
6/26/20	Fri	2,858.52	2,858.52	250.50	185.50			
6/27/20	Sat	2,825.72	2,825.72	250.50	185.50			
6/28/20	Sun	2,792.92	2,792.92	250.50	185.50			

Aspen Birch Cedar Dogwood Dogwood Irrigation Elm Eastern Irrigation

Birch.Inflow -- Total Volume: 1,228.75 [MCM]
4 values: Sum 14,221.74 -- Ave 3,555.44 -- Med 3,555.14 -- Min 3,534.66 -- Max 3,576.79 -- Range 42.13 [cms]

Related topics

- [Display and edit series slot notes](#), page 109

Scalar Data on the Object Grid

Scalar slot data can be shown as a list of slots on the Scalar Slots tab or as part of a grid of objects on the Object Grid tab. This section describes the Object Grid and gives an overview of the features.

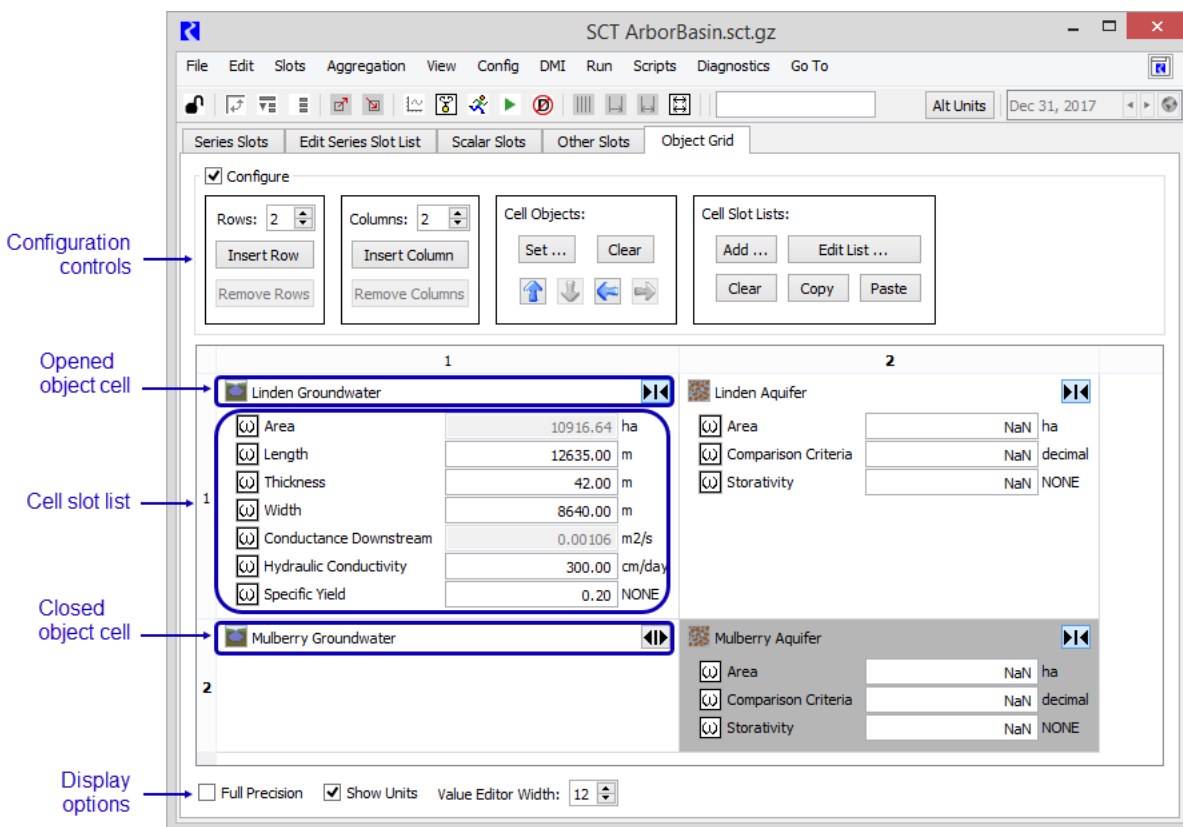
Chapter 2 SCT Basics

The Object Grid tab displays a user-configured grid of objects and their included scalar and single value table slots. It provides a convenient location to organize and specify parameters on objects that interact with one another. The object grid is particularly useful in calibration of a groundwater or aquifer network, in which the physical characteristics of the objects are related to one another.

Each cell on the object grid represents an object, and each cell includes a user-specified list of scalar slots and single-cell table slots, called *cell slot lists*.

Figure 2.19 shows the main areas of the Object Grid, which are described below.

Figure 2.19 Object Grid tab



Configuration controls

The primary configuration controls are optionally shown at the top of the object grid, based on whether the Configure check box is selected.

The Rows and Columns configuration specify the number of rows and columns in the Object Grid.

The Cell Objects buttons allow you to specify the object represented in each cell. The four arrow buttons shift the set of selected cells in the indicated direction.

The Cell Slot List buttons allow you to specify the slots displayed in the selected object cells.

Object cells

Object cells represent a single object and can be opened or closed. When opened, they show the full cell slot list. When closed, they collapse to just the object name and icon.

Each cell includes the following buttons:

- Object icon—opens the Open Object dialog.
- Slot icon (displayed when the cell is opened)—opens the Open Slot dialog.
- Open/Close—shows or hides the slots in the cell.

Cell slot list

The cell slot list lists the scalar and table slots and their values and units.

Display options

The controls at the bottom of the Object Grid are used to modify how data is displayed for the entire grid, as follows.

- Full Precision—switches between configured display attributes and full precision.
- Show Units—shows or hides the units.
- Value Editor Width—sets the width of slot value editors to ensure visibility of the specified number of characters. The supported range is 3 to 22; the default is 12.

Related topics

- [Object Grid tab](#), page 13
- [Organizing Scalar Data in the Object Grid](#), page 220

SCT Configuration

The SCT configuration includes the settings that can be saved and reloaded as part of an SCT. These settings allow you to customize the format, or *look and feel*, of each SCT, as well as the content displayed.

- SCT format settings include the following:
 - Axis orientation
 - Text and cell background colors and flag colors
 - Display fonts
 - Buttons and flags included in the SCT toolbar
 - Specifications for displaying missing data
 - Aggregation functions for multiple timesteps
- SCT content settings include the following:
 - Lists of included slots

Chapter 2

SCT Basics

- Slot labels
- Color Alerts for each slot
- Aggregation functions for each slot
- Object and slot ordering on the Object Grid tab

You can define and apply a *default* SCT configuration, which serves as the base configuration. The default configuration is automatically applied to all new SCTs and can be applied manually to any SCT. See [“Using the Default SCT Configurations”](#) on page 252 for details.

Related topics

- [Configuring SCT Settings](#), page 163
- [SCT Configuration Window](#), page 284
- [Using the Default SCT Configurations](#), page 252

Accessing Other Utilities From the SCT

The SCT provides direct access to commonly used RiverWare utilities.

Topics

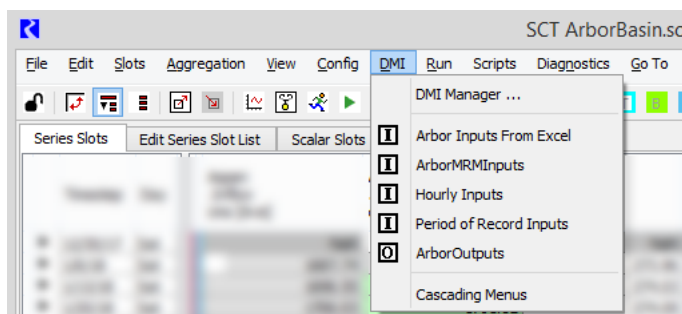
- [Data Management Interfaces \(DMIs\)](#), page 34
- [Scripts](#), page 35
- [Diagnostics](#), page 35

Data Management Interfaces (DMIs)

The SCT DMI menu includes all Input and Output DMIs and DMI Groups defined for the model. You can execute any DMI directly from the SCT.

Caution: Selecting a DMI or Group immediately invokes that DMI.

You can also open the DMI Manager to import or export data using the Data Management Interface.

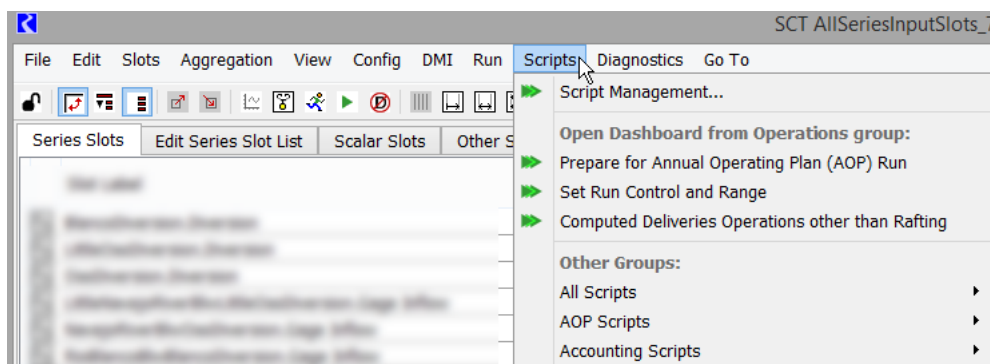


Related topics

- [SCT DMI menu](#), page 270

Scripts

RiverWare scripts can be an integral part of operating a model. The SCT Scripts menu includes all scripts defined for the model. You can open the Script Dashboard directly from the SCT and use it to configure settings and run a selected script. You can also open the Script Manager to edit scripts.



Related topics

- [SCT Scripts menu](#), page 271

Diagnostics

From the SCT, you can display Diagnostics Output in a standalone window or as an integrated pane within the SCT. Either display option is available from all SCT tabs.

The Diagnostics Output displays diagnostics messages for a simulation run. Error and warning messages are always shown, and informational messages may be displayed depending on user-defined settings. See [“Diagnostics” in *Debugging and Analysis*](#) for details.

Related topics

- [Show or hide simulation diagnostic messages](#), page 204
- [SCT Diagnostics menu](#), page 271

Chapter 2

SCT Basics

Chapter 3

Using the SCT

Topics

- [Getting Started With the SCT, page 37](#)
- [Adding Series Slots, page 46](#)
- [Organizing Series Slots, page 53](#)
- [Manipulating Slot Values, page 67](#)
- [Copying and Pasting Slot Values, page 81](#)
- [Exporting and Importing Slot Values, page 90](#)
- [Changing How Data is Displayed, page 97](#)
- [Navigating an SCT, page 119](#)
- [Displaying Aggregated Values, page 123](#)
- [Displaying Snapshot Slots, page 145](#)
- [Configuring Color Alerts, page 150](#)
- [Configuring SCT Settings, page 163](#)
- [Using an SCT in Operations, page 193](#)
- [Plotting and Printing, page 207](#)
- [Accessing Slot Dialogs From the SCT, page 209](#)
- [Organizing Scalar and Table Slots, page 213](#)
- [Organizing Scalar Data in the Object Grid, page 220](#)
- [Using the Default SCT Configurations, page 252](#)

Getting Started With the SCT

This section includes basic procedures to get started using an SCT.

Topics

- [Create a new SCT, page 38](#)
- [Load an existing SCT, page 39](#)
- [Create a new SCT from the Slot Viewer, page 40](#)
- [Create a new SCT from a series slot sheet, page 41](#)

Chapter 3

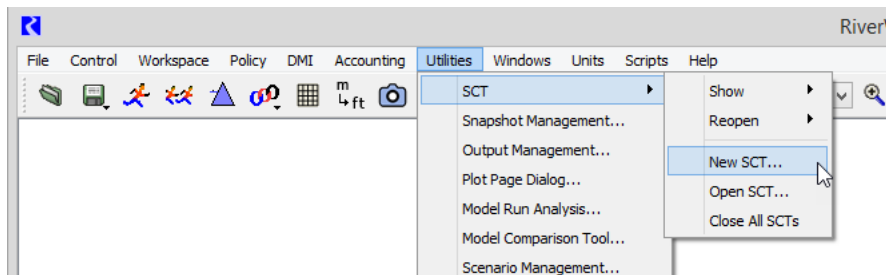
Using the SCT

- [Save an SCT](#), page 42
- [Close an SCT without saving](#), page 44
- [Process for setting up an SCT](#), page 45

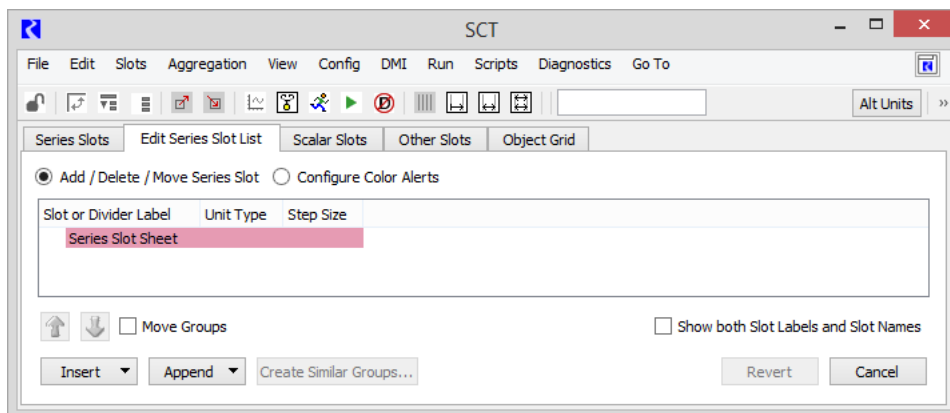
Create a new SCT

Use this procedure to create a new SCT. After performing this procedure, you can configure the SCT to view and edit any slot values in the model, group them, and arrange them in any order. You can also customize the appearance of the SCT.

1. From the workspace menu, select **Utilities**, then **SCT**, then **New SCT**.



A new SCT opens at the Edit Series Slot List tab.



2. Add slots and dividers, and configure the SCT to your needs. See the following sections for details:
 - [Add series slots to an SCT](#), page 46
 - [Add slot dividers](#), page 54
 - [Add sheet dividers](#), page 56
 - [Rearrange items on the Series Slots tab](#), page 60
 - [Save an SCT](#), page 42

Related topics

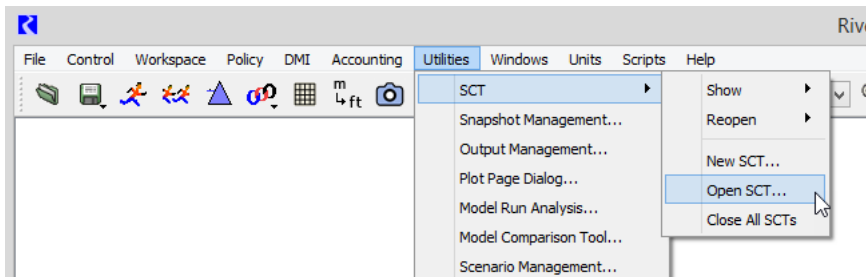
- [Create a new SCT from the Slot Viewer](#), page 40

- Create a new SCT from a series slot sheet, page 41

Load an existing SCT

Use this procedure to load an existing SCT.

1. Do either of the following:
 - On the workspace menu, select **Utilities**, then **SCT**, then **Open SCT**.



- On the workspace toolbar, select **SCT** .
2. Use the File Chooser to navigate to the folder where the SCT is saved. By default, the File Chooser goes to the last folder used.
 3. Select the SCT file, then select **Open**.

The SCT opens with the **Series Slots** tab active.

SCT ArborBasin.sct.gz

File Edit Slots Aggregation View Config DMI Run Scripts Diagnostics Go To

Series Slots Edit Series Slot List Scalar Slots Other Slots Object Grid

Timestep	Day	Aspen .Inflow cms	Aspen .Outflow cms	Aspen .Pool Elevation m	Aspen .Storage MCM	Aspen .Power MW	Aspen .Regulated Spill cms	Aspen .Turbine Release cms
12/27/17	Wed	NaN	NaN	NaN	NaN	NaN	NaN	NaN
12/28/17	Thu	NaN	NaN	NaN	NaN	NaN	NaN	NaN
12/29/17	Fri	NaN	3227.72	NaN	NaN	NaN	NaN	NaN
12/30/17	Sat	NaN	3253.74	NaN	NaN	NaN	NaN	NaN
12/31/17	Sun	NaN	3864.82	274.56	64.78	NaN	NaN	NaN
1/1/18	Mon	1684.35	1962.74	273.91	40.72	357.06	0.00	1962.74
1/2/18	Tue	1685.71	1681.66	273.92	41.07	302.13	0.00	1681.66
1/3/18	Wed	1687.06	1683.01	273.93	41.42	302.51	0.00	1683.01
1/4/18	Thu	1688.42	1684.37	273.94	41.77	302.99	0.00	1684.37
1/5/18	Fri	1689.77	1685.72	273.95	42.12	303.57	0.00	1685.72
1/6/18	Sat	1691.13	1686.69	273.96	42.51	304.06	0.00	1686.69
1/7/18	Sun	1692.48	1687.97	273.97	42.90	304.60	0.00	1687.97
1/8/18	Mon	1693.84	1689.32	273.98	43.29	305.28	0.00	1689.32
1/9/18	Tue	1695.19	1690.68	273.99	43.68	306.09	0.00	1690.68
1/10/18	Wed	1696.55	1692.03	274.00	44.07	306.86	0.00	1692.03
1/11/18	Thu	1697.90	1693.39	274.01	44.46	307.57	0.00	1693.39

Aspen Birch Cedar Dogwood Dogwood Irrigation Elm Eastern Irrigation Transbasin Hickory Juniper Irrigation Interstate Gage Lir

Aspen.Inflow [@ 12/27/17] -- Total Volume: 0.00 [MCM]
0 values

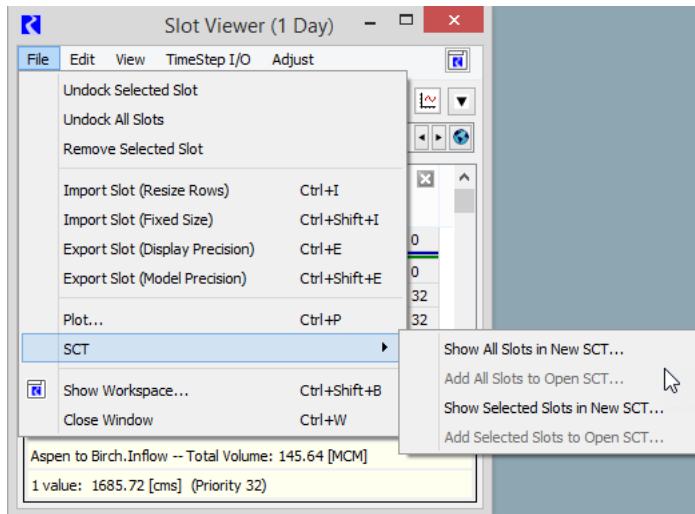
Related topics

- [Save an SCT](#), page 42

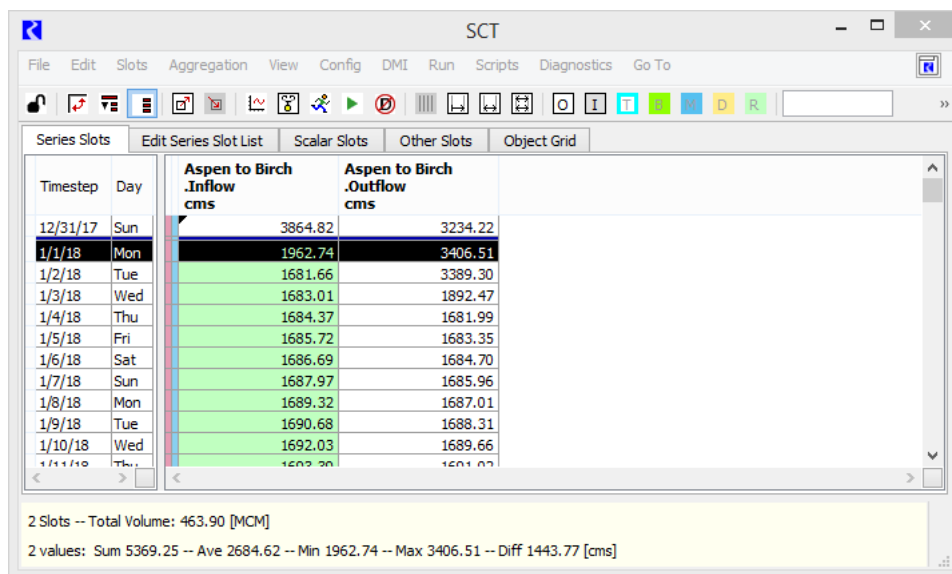
Create a new SCT from the Slot Viewer

Use this procedure to add one or more series slots from the Slot Viewer to a new SCT.

1. Open one or more series slots in the Slot Viewer; see [“Opening a Slot” in User Interface](#) for details.
2. Optionally, select the slots you want to include in the SCT. In the next step, you can choose whether to include all slots or only selected slots in the new SCT.
3. In the Slot Viewer menu, select **File**, then **SCT**, then select one of the following options:
 - **Show All Slots in New SCT**—adds all slots displayed in the Slot Viewer to a new SCT.
 - **Show Selected Slots in New SCT**—adds just the selected slots to a new SCT.



The new SCT opens, and the slots are located on the Series Slots tab.



- To add additional slots to this SCT, see [“Add series slots to an SCT”](#) on page 46 and [“Add series slots from the Slot Viewer”](#) on page 50.
- To save the SCT, see [“Save an SCT”](#) on page 42.

Related topics

- [“Slot Windows” in User Interface](#)
- [Add series slots from the Slot Viewer](#), page 50
- [Save an SCT](#), page 42
- [Create a new SCT](#), page 38
- [Create a new SCT from a series slot sheet](#), page 41

Create a new SCT from a series slot sheet

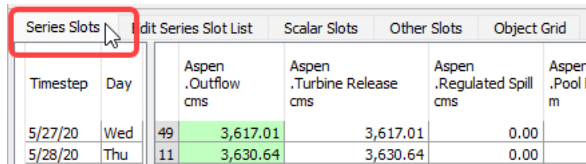
Use this procedure to copy a series slot sheet to a new SCT. You can perform this procedure to create SCTs from any number of sheets. This is useful for the following purposes:

- You can quickly build a new SCT using the selected sheet and its SCT configuration settings as a starting point. You can make changes to the copy and then save it as a new SCT file.
- You can easily view multiple sheets of an SCT side-by-side. When you are done viewing, you can close the copies without saving them.

Chapter 3

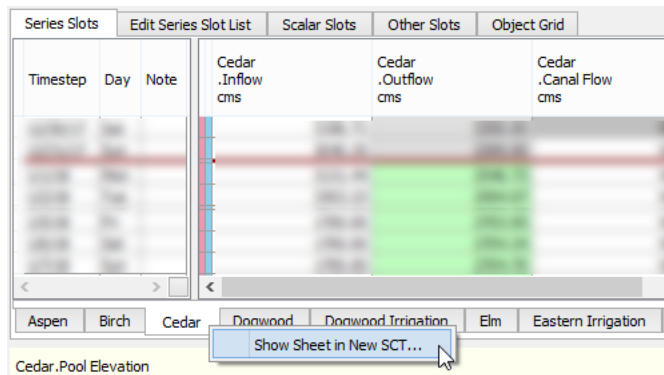
Using the SCT

1. Go to the **Series Slot** tab.



Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. In the bottom of the display, right-click the sheet tab you want to display in a separate SCT.
3. In the context menu, select **Show Sheet in New SCT**.



The sheet is displayed in a new SCT window.

Note: The sheet is not removed from the original SCT; it is copied to a new SCT window, and the original sheet remains unchanged in the original SCT.

4. To add additional slots to this SCT, see [“Add series slots to an SCT”](#) on page 46.
5. To save the SCT, see [“Save an SCT”](#) on page 42.

Related topics

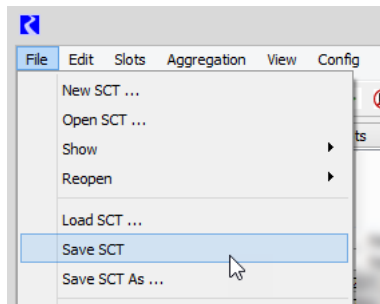
- [Add series slots to an SCT](#), page 46
- [Save an SCT](#), page 42
- [Create a new SCT](#), page 38
- [Create a new SCT from the Slot Viewer](#), page 40

Save an SCT

Use this procedure to save an SCT. You can save to the current file or save to a new name or location, which effectively duplicates the existing SCT.

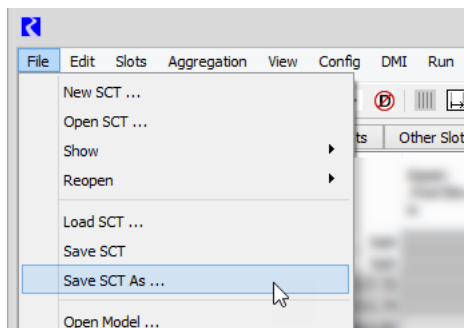
Note: The SCT file should be saved with a .sct or .sct.gz extension so RiverWare can recognize it as an SCT.

1. To save the SCT with the current file name and extension, on the SCT menu, select **File**, then **Save SCT**. Alternatively, press Ctrl+S.



If the file already exists, a confirmation dialog box appears. Select **Yes** to confirm the replacement.

2. To save the SCT to a new file name or location, on the SCT menu, select **File**, then **Save SCT As**. Alternatively, press Ctrl+Shift+S.

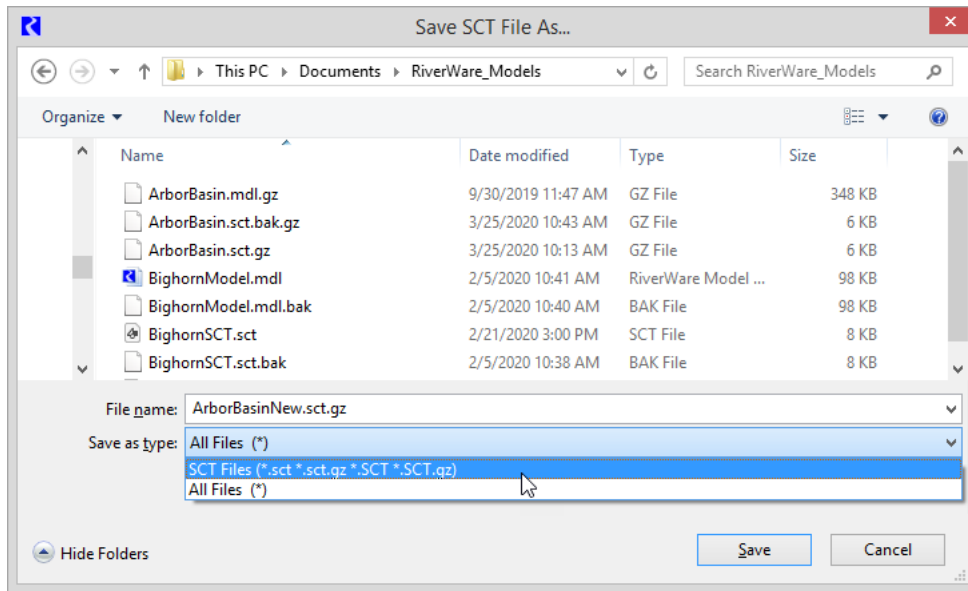


- a. Use the File Chooser to navigate to the folder where you want to save the SCT. By default, the File Chooser goes to the last folder used.
- b. In the **File name** field, enter a unique file name and an extension, as follows:
 - Enter the .sct extension to save the SCT as an uncompressed file. If you do not enter an extension, this one is provided automatically.

Chapter 3

Using the SCT

- Enter the .sct.gz extension to save the SCT as a compressed file.



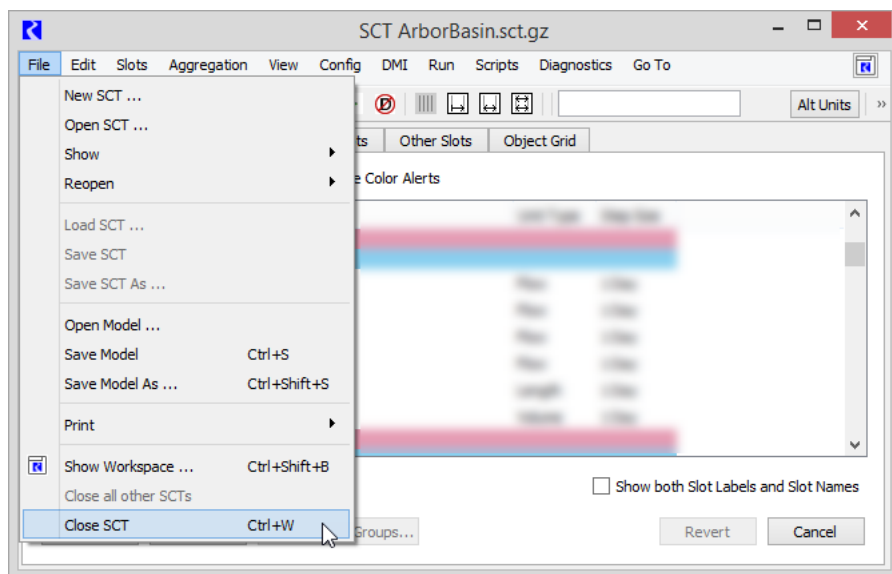
Related topics

- [Create a new SCT](#), page 38
- [Load an existing SCT](#), page 39
- [Close an SCT without saving](#), page 44

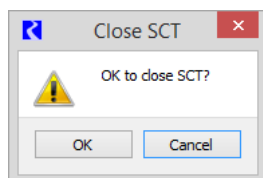
Close an SCT without saving

Use this procedure to close an SCT *without saving* changes you have made to its configuration or appearance. This allows you to cancel any changes you have made since the last time you saved the SCT.

1. From any SCT tab, select **File**, then **Close SCT**. Alternatively, press Ctrl+W.



The Close SCT confirmation dialog box appears.



2. Select **OK** to close without saving changes to the SCT. Select **Cancel** to cancel the operation and return to the SCT.

Related topics

- [Save an SCT](#), page 42

Process for setting up an SCT

Once you have created an SCT, you can add and organize slots and configure the display to meet your needs. The following process provides a summary of the tasks involved.

1. Add slots. See the following:
 - [Add series slots to an SCT](#), page 46
 - [Add series slots from the Slot Viewer](#), page 50
 - [Add scalar and table slots to an SCT](#), page 214
2. Organize series slots into logical groups that fit your needs. See the following:
 - [Add slot dividers](#), page 54
 - [Add sheet dividers](#), page 56

Chapter 3

Using the SCT

- [Cut or copy and paste items](#), page 62
- 3. Define aggregations for series slots. See the following:
 - [Define the aggregation interval for the SCT](#), page 125
 - [Define summary functions by unit type](#), page 129 and [Define summary functions by slot name](#), page 131
 - [Enable or disable aggregated view](#), page 127
- 4. Define the timestep orientation, and fine-tune the display. See the following:
 - [Change the timestep orientation](#), page 98
 - [Adjust column widths](#), page 99
 - [Customize flag colors](#), page 189
 - [Customize divider and cell colors](#), page 190
- 5. Define and configure Color Alerts. See the following:
 - [Define custom colors for highlights and color alerts](#), page 191
 - [Configure Color Alerts](#), page 150
- 6. Save the SCT; see [Save an SCT](#), page 42.

Adding Series Slots

This section includes procedures for adding series slots and labeling them in the SCT.

Note: All procedures in this section require the SCT configuration to be unlocked. See “[Lock or unlock the SCT Configuration](#)” on page 164 for details.

Note: See “[Organizing Scalar and Table Slots](#)” on page 213 and “[Organizing Scalar Data in the Object Grid](#)” on page 220 for procedures to add other types of slots to an SCT.

Topics

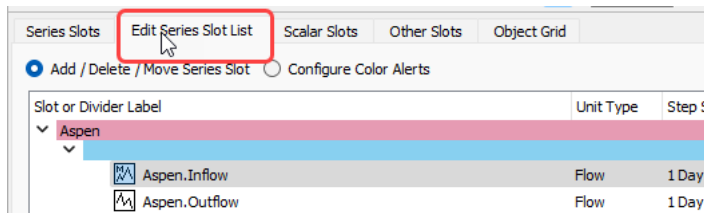
- [Add series slots to an SCT](#), page 46
- [Add series slots from the Slot Viewer](#), page 50
- [Customize a slot label](#), page 51
- [Change multiple slot labels](#), page 52

Add series slots to an SCT

Use this procedure to add one or more series slots to an existing SCT.

Note: While you build an SCT, be sure to save it frequently so you do not lose your changes.

1. Go to the **Edit Series Slot List** tab. Alternatively, press Alt+Shft+S.

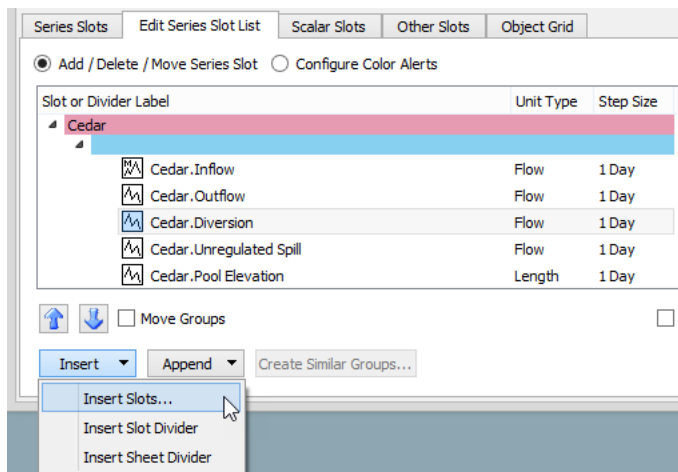


This tab provides a list of all series slots. It also provides a convenient place to add, delete, and organize the list of slots.

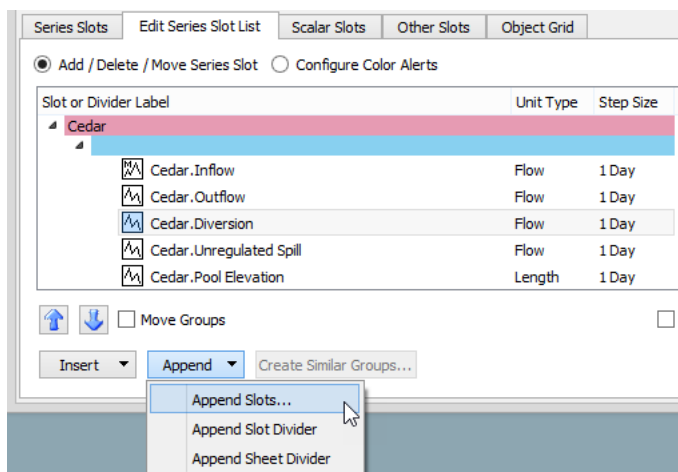
2. Select one of the following buttons at the bottom of the SCT window to add slots.

Note: If these buttons are not available, the SCT may be locked; you must unlock it before proceeding.

- Select **Insert**, then **Insert Slots**.



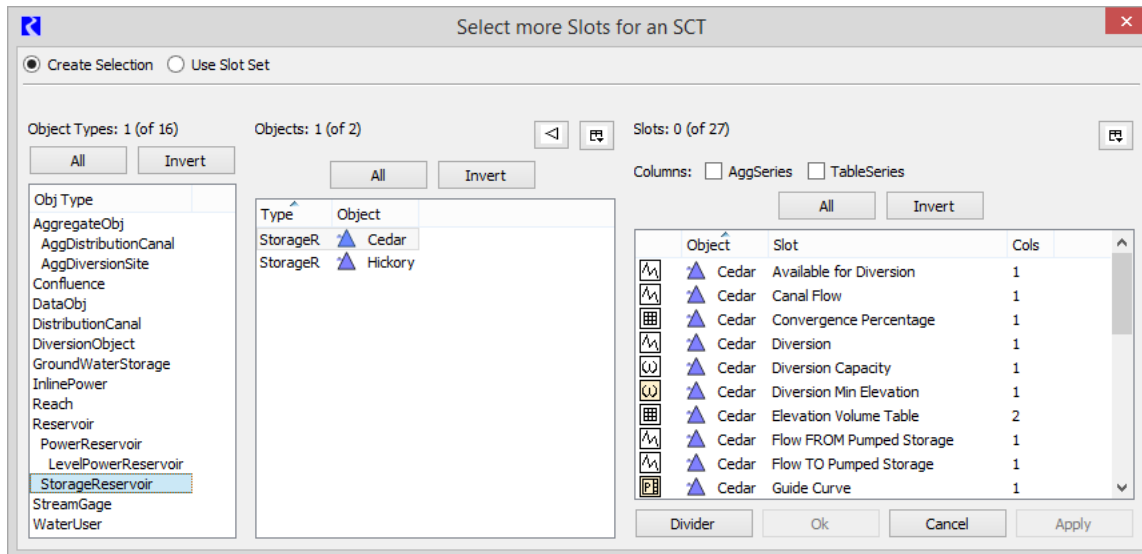
- Select **Append**, then **Append Slots**.



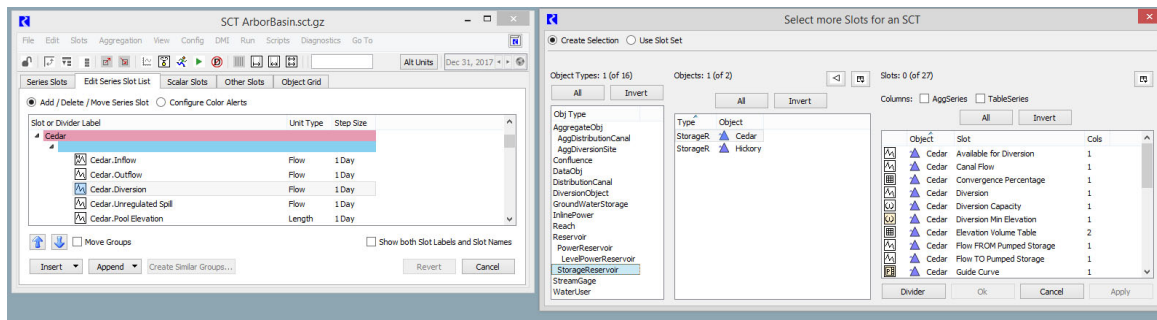
Chapter 3

Using the SCT

The Select more Slots for an SCT dialog box opens.



3. Optionally, position the dialog box so you can see it and the SCT together, side-by-side.



4. In the dialog box, select the slots you want to add to the SCT.

Note: You can only add slots that exist in the currently loaded RiverWare model.

Note: You can include the same slot multiple times within the same SCT. This is useful if you want to display more than one type of aggregation, such as Sum and Average, for the same slot.

- Select an object type.
- Select an object.
- Select one or more slots. Press Ctrl+Click or Shift+Click to select multiple slots in the slot list.

The slots are added to the **Edit Series Slot List** tab in the SCT at the position you have indicated.

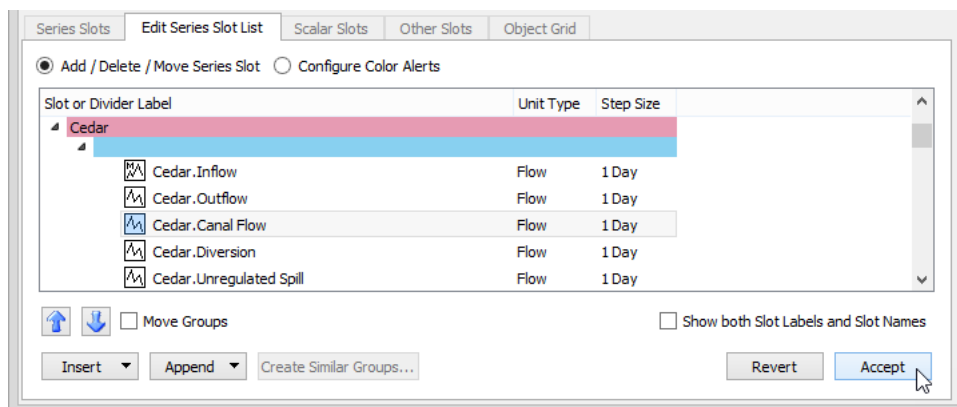
5. When you are done adding slots, select **Ok** or **Cancel** to close the Select more Slots for an SCT dialog box.

Note: You can add more slots, rearrange them, and make other modifications at a later time.

6. Review and verify the slots in the SCT.

7. In the SCT, select **Accept** to apply the new slots to the **Series Slot** tab.

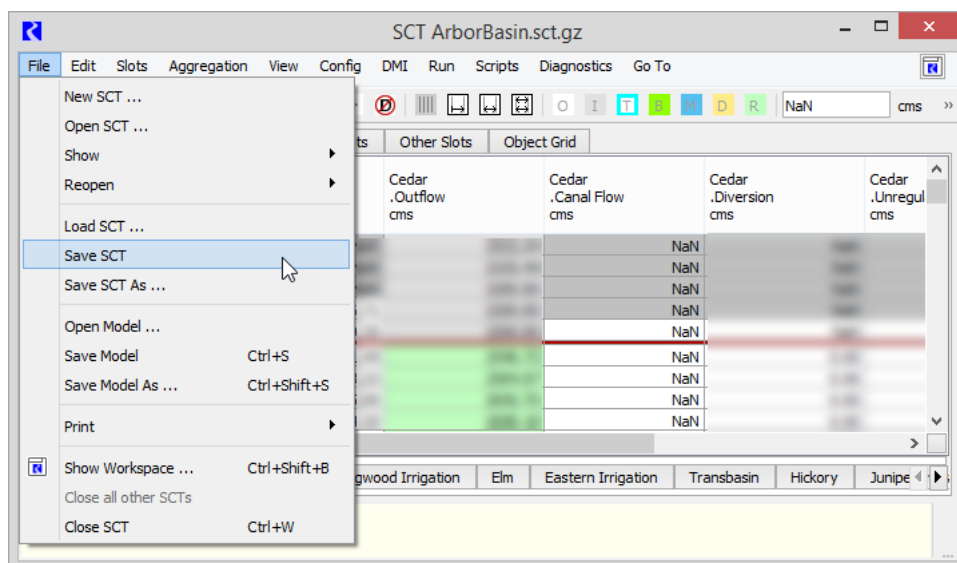
Note: The **Accept** and **Revert** buttons are enabled only when the Select Slots dialog box is closed.



The **Series Slot** tab becomes active. Slots are displayed as either rows or columns, depending on the selected timestep orientation, and current values from the model are displayed for each timestep.

8. Select **File**, then **Save SCT** or **Save SCT As** to save the new slots to the SCT. Alternatively, press Ctrl+S or Ctrl+Shift+S.

Caution: If you close the SCT without performing this step, your changes will not be saved to the SCT.



Related topics

- [Add scalar and table slots to an SCT](#), page 214
- [“Selecting Slots” in User Interface](#)

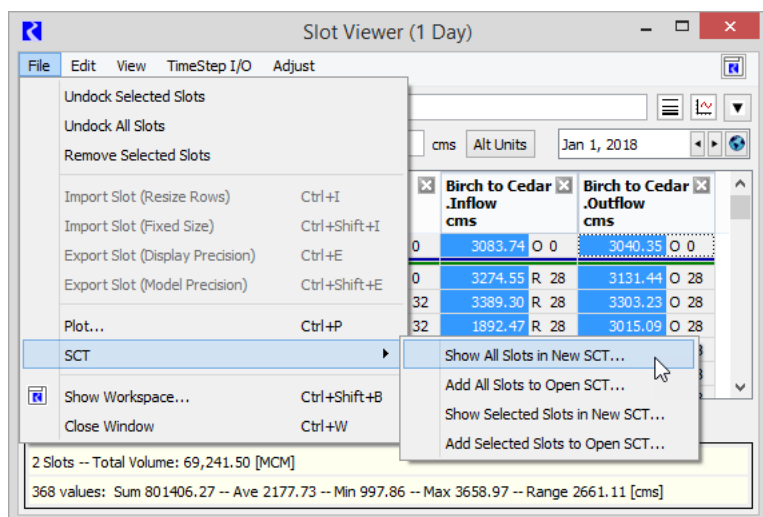
Alternate methods

- On the SCT menu, select **Slots**, then **Append Slots** or **Insert Slots**.

Add series slots from the Slot Viewer

Use this procedure to add one or more series slots from the Slot Viewer to an existing SCT.

1. Open the SCT you want to modify and close any other SCTs, if applicable.
Note: To perform this procedure, you can have only one SCT open at a time.
2. Open one or more series slots in the Slot Viewer; see [“Opening a Slot” in User Interface](#) for details.
3. Optionally, select the slots you want to add to the SCT. In the next step, you can choose whether to add all slots or just selected slots to the SCT.
4. In the Slot Viewer menu, select **File**, then **SCT**, then select one of the following options:
 - **Add All Slots to Open SCT**—adds all slots displayed in the Slot Viewer to the SCT.
 - **Add Selected Slots to Open SCT**—adds just the selected slots to the SCT.



The slots are added to the Series Slots tab in the SCT.

Timestep	Day	Aspen to Birch .Inflow cms	Aspen to Birch .Outflow cms	Birch to Cedar .Inflow cms	Birch to Cedar .Outflow cms
12/31/17	Sun	3864.82	3234.22	3083.74	3040.35
1/1/18	Mon	1962.74	3406.51	3274.55	3131.44
1/2/18	Tue	1681.66	3389.30	3389.30	3303.23
1/3/18	Wed	1683.01	1892.47	1892.47	3015.09
1/4/18	Thu	1684.37	1681.99	1700.00	1844.35
1/5/18	Fri	1685.72	1683.39	1700.00	1700.00
1/6/18	Sat	1686.69	1684.70	1700.00	1700.00
1/7/18	Sun	1687.97	1685.96	1700.00	1700.00
1/8/18	Mon	1689.32	1687.01	1700.00	1700.00
1/9/18	Tue	1690.68	1688.31	1700.00	1700.00
1/10/18	Wed	1692.03	1689.66	1700.00	1700.00
1/11/18	Thu	1693.38	1691.01	1700.00	1700.00

4 Slots -- Total Volume: 1,017.38 [MCM]
4 values: Sum 11775.23 -- Ave 2943.81 -- Med 3202.99 -- Min 1962.74 -- Max 3406.51 -- Range 1443.77 [cms]

- Repeat [Step 2.](#) through [Step 4.](#) to add additional slots to this SCT.
- Save the SCT to save your changes.

Related topics

- [Save an SCT](#), page 42
- [Create a new SCT from the Slot Viewer](#), page 40

Customize a slot label

The default slot label displayed in the SCT is the full name of the object and slot, in the following format:
`object.slot`

Use this procedure to change the slot label to make it more meaningful or compact. Optionally, you can specify different labels for a slot in the horizontal and vertical timestep orientations.

- Go to the **Series Slot** tab.

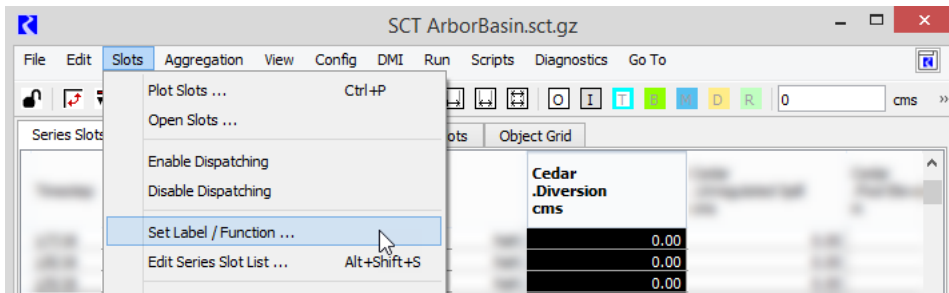
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

- Select the slot label you want to change.

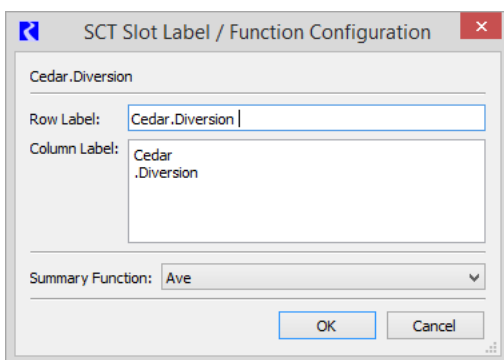
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Using the SCT

3. On the SCT menu, select **Slots**, then **Set Label / Function**.



The SCT Slot Label / Function Configuration dialog box opens.



4. Specify the label you want to assign to the slot in each orientation, as follows:
 - **Row Label**—enter the label to be displayed in the horizontal timestep orientation.
 - **Column Label**—enter the label to be displayed in the vertical timestep orientation.
5. Select **OK** to apply the changes.

The dialog box closes, and the label is changed in the SCT.

Related topics

- [Custom slot labels](#), page 26
- [Change multiple slot labels](#), page 52

Alternate methods

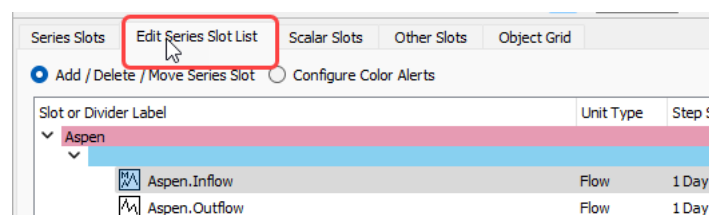
- In the horizontal timesteps view, double-click the slot row label to edit it.
- Go to the **Edit Series Slot List** tab and double-click the slot label to edit it.

Change multiple slot labels

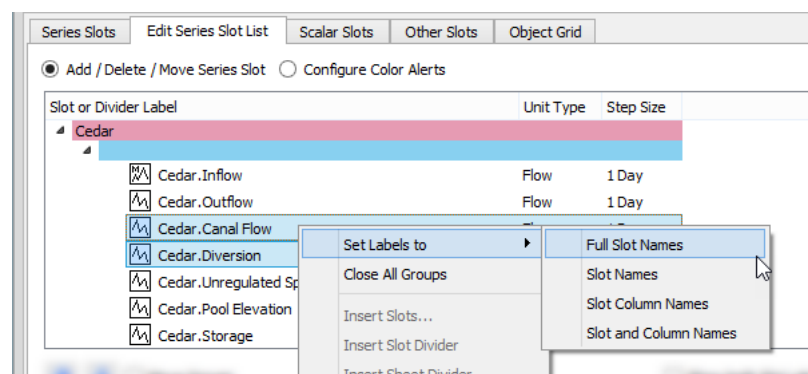
Use this procedure to set the labels for many slots at once.

Note: This operation changes both the row and column labels.

1. Go to the **Edit Series Slot List** tab. Alternatively, press Alt+Shft+S.



2. Select the slots you want to change, then right-click.
3. In the shortcut menu, select **Set Labels to**, then select one of the following options:
 - **Full Slot Names**—change the label to the default Object.Slot name; for example, “Mead.Inflow”.
 - **Slot Names**—change the label to just the slot name. For example, “Inflow”.
 - **Slot Column Names**—for AggSeries and other multi-column slots, use the column label; for example, “Rice”.
 - **Slot and Column Names**—for AggSeries and other multi-column slots, use the slot and column label; for example, “Evapotranspiration Rate by Crop.Rice”.



The labels are updated according to your selection.

4. Select **Accept** to accept the updates.

The **Series Slot** tab opens with the updated slot labels.

Related topics

- [Custom slot labels](#), page 26
- [Customize a slot label](#), page 51

Organizing Series Slots

These procedures allow you to create slot groups and organize the order of series slots in the SCT. In general, you perform these updates on the Edit Series Slot List tab and then view the results on the Series Slots tab.

Note: All procedures in this section require the SCT configuration to be unlocked. See [“Lock or unlock the SCT Configuration”](#) on page 164 for details.

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Topics

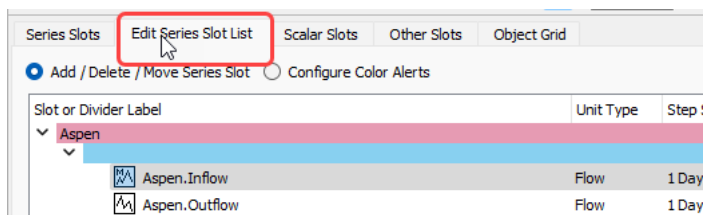
- [Add slot dividers](#), page 54
- [Add sheet dividers](#), page 56
- [Replicate a slot group to other objects](#), page 58
- [Rearrange items on the Series Slots tab](#), page 60
- [Cut or copy and paste items](#), page 62
- [Remove items](#), page 65

Add slot dividers

Use this procedure to add slot dividers to the series slot display. Slot Dividers are graphical elements that are used to separate slots on each sheet. The group of slots within the divider can be manipulated together such as moving or creating similar groups.

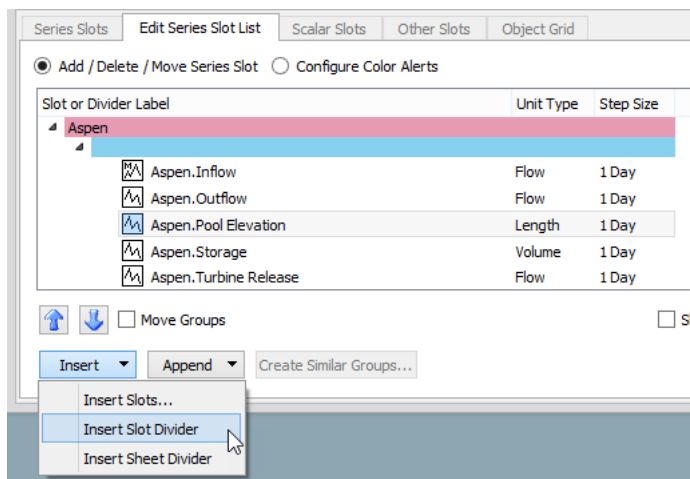
In vertical timestep orientation, the slot divider is a thin column, and in horizontal timestep orientation, the divider is a row tall enough to display a text label. By default, the slot divider color is blue. You can optionally customize this with the Slot Divider Background color; see [“Customize divider and cell colors”](#) on page 190 for details.

1. Go to the **Edit Series Slot List** tab. Alternatively, press Alt+Shft+S.

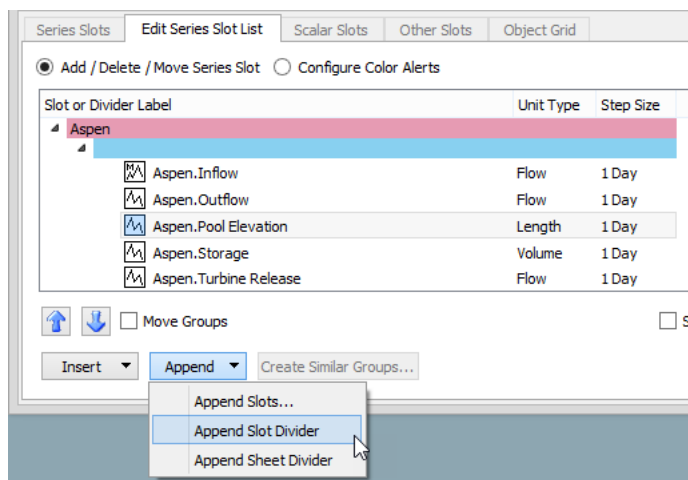


2. Do one of the following actions to add a slot divider.

- Select the slot where you want to start a slot group. Then select **Insert**, then **Insert Slot Divider** to insert the divider *directly above* the selected item.

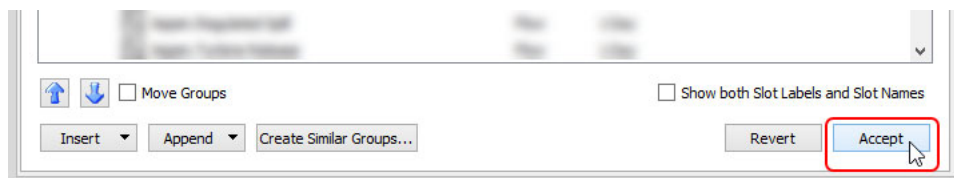


- Select **Append**, then **Append Slot Divider** to insert the divider *at the end of the slot list*.



A new slot group divider appears in the position you have selected.

3. Optionally, double-click the divider and enter a slot name to identify the divider. This label will be displayed on the divider row in horizontal timestep orientation.
4. When you are done making changes, select **Accept** to apply the changes to the Series Slot tab.



The Series Slot tab opens, and the new divider is displayed.

Timestep	Day	Note	Aspen .Inflow cms	Aspen .Outflow cms	Aspen .Pool Elevation m
12/30/17	Sat		NaN	3253.74	NaN
12/31/17	Sun		NaN	3864.72	274.56
1/1/18	Mon		1684.35	1962.74	273.91
1/2/18	Tue		1685.71	1681.66	273.92
1/3/18	Wed		1687.06	1683.01	273.93
1/4/18	Thu		1688.42	1684.77	273.94
1/5/18	Fri		1689.77	1685.72	273.95
1/6/18	Sat		1691.13	1686.49	273.96
1/7/18	Sun		1692.48	1687.77	273.97

Related topics

- [Sheet dividers and slot dividers](#), page 19
- [Add sheet dividers](#), page 56
- [Replicate a slot group to other objects](#), page 58

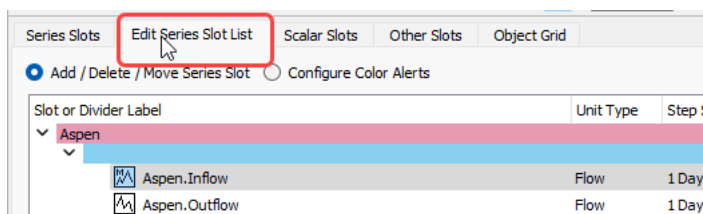
- [Customize divider and cell colors](#), page 190

Add sheet dividers

Use this procedure to add sheet dividers to the series slot display. These dividers allow you organize the display into multiple sheets, which is helpful when the SCT includes too many slots to scroll through easily. You can access a sheet by selecting its tab at the bottom of the Series Slot tab.

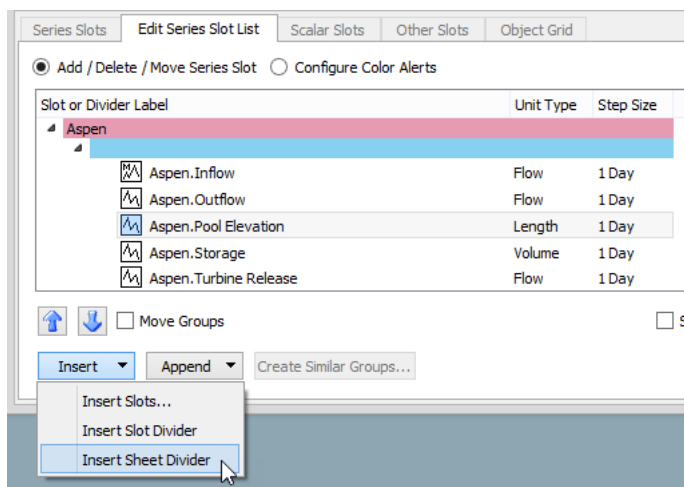
In vertical timestep orientation, the sheet divider is a thin column, and in horizontal timestep orientation, the divider is a row tall enough to display a text label. By default, the sheet divider color is pink. You can optionally customize this with the Series Sheet Divider Background color; see [“Customize divider and cell colors”](#) on page 190 for details.

1. Go to the **Edit Series Slot List** tab. Alternatively, press Alt+Shft+S.

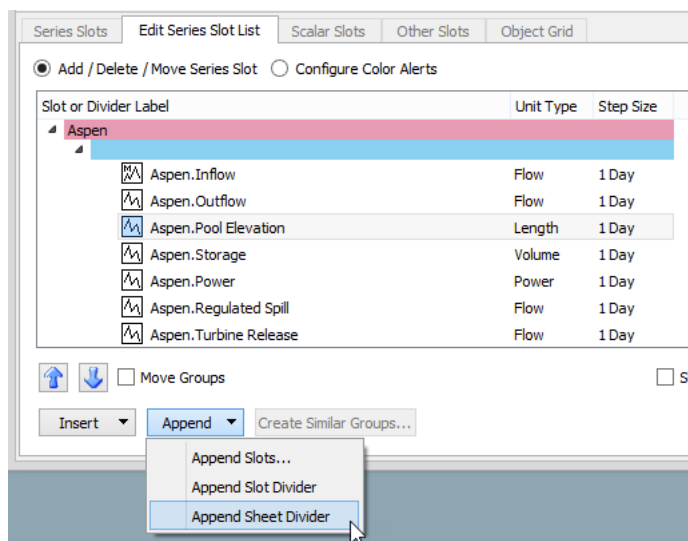


2. Do one of the following to add a sheet divider.

- Select the slot or slot divider where you want to start a new sheet. Then select **Insert**, then **Insert Sheet Divider** to insert the divider *directly above* the selected item.

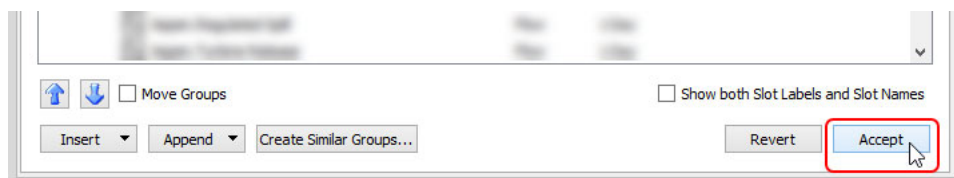


- Select **Append**, then **Append Sheet Divider** to insert the divider *at the end of the slot list*.

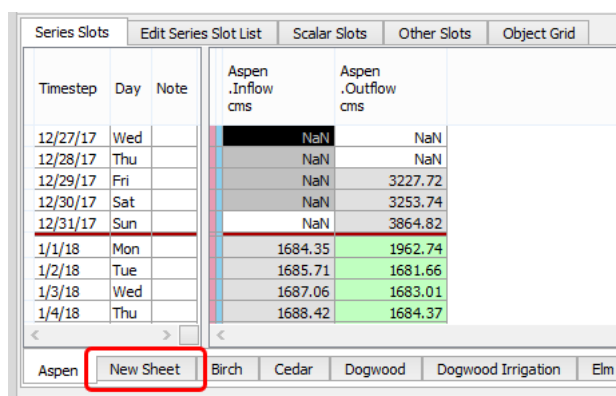


A new sheet divider appears in the position you have selected. It has the default name “Series Slot Sheet”.

3. Optionally, double-click the divider and enter a label to identify the sheet. This label will be displayed on the sheet tab on the Series Slot tab and on the sheet divider row in horizontal timestep orientation.
4. When you are done making changes, select **Accept** to apply the changes to the Series Slots tab.



The **Series Slot** tab opens, and new sheet appears as a tab at the bottom of the page.



5. Select the tab to display the new sheet.

Related topics

- [Sheet dividers and slot dividers](#), page 19
- [Add slot dividers](#), page 54

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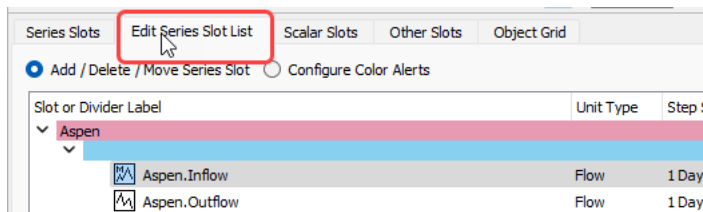
- [Replicate a slot group to other objects](#), page 58
- [Customize divider and cell colors](#), page 190

Replicate a slot group to other objects

Use this procedure to add slots to an SCT by applying the slot group structure for one object to multiple objects that are assigned the same, or mostly the same, slots. The new slot group includes the same slots as the original group if these slots are included in the new object.

Note: You can replicate the structure of a slot group only if it contains slots from a single object or account; slot groups containing slots from multiple objects are not eligible.

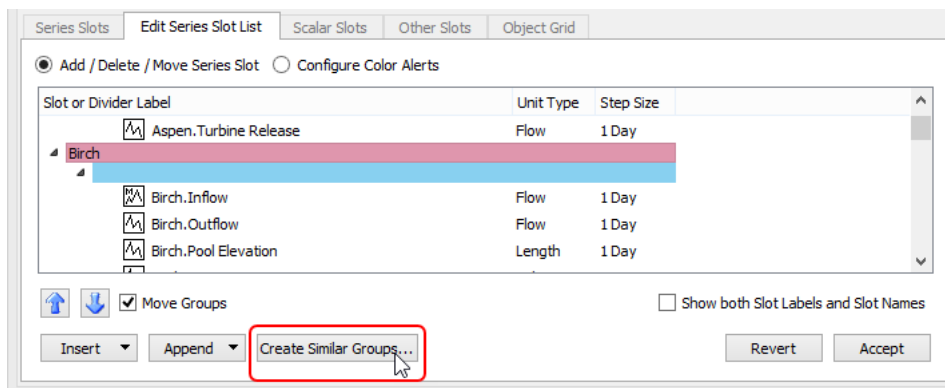
1. Go to the **Edit Series Slot List** tab. Alternatively, press Alt+Shift+S.



2. Select the divider for the slot group you want to replicate; you can select one only. Typically, you would select a slot group for an object that has the same, or mostly the same, slots as the objects you want to add to the SCT.

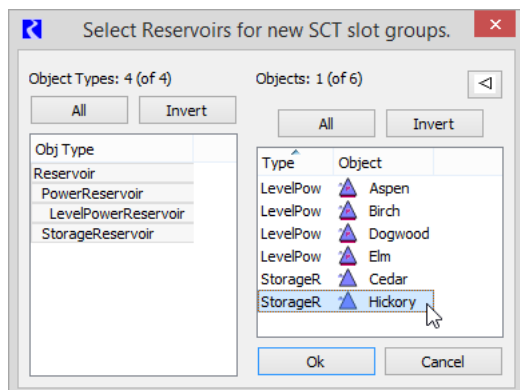
The **Create Similar Groups** button is activated.

3. Select **Create Similar Groups**.



The Select object for new SCT slot groups dialog box opens, displaying a list of all objects in the model that have the same object type as the object to which the selected slot group belongs.

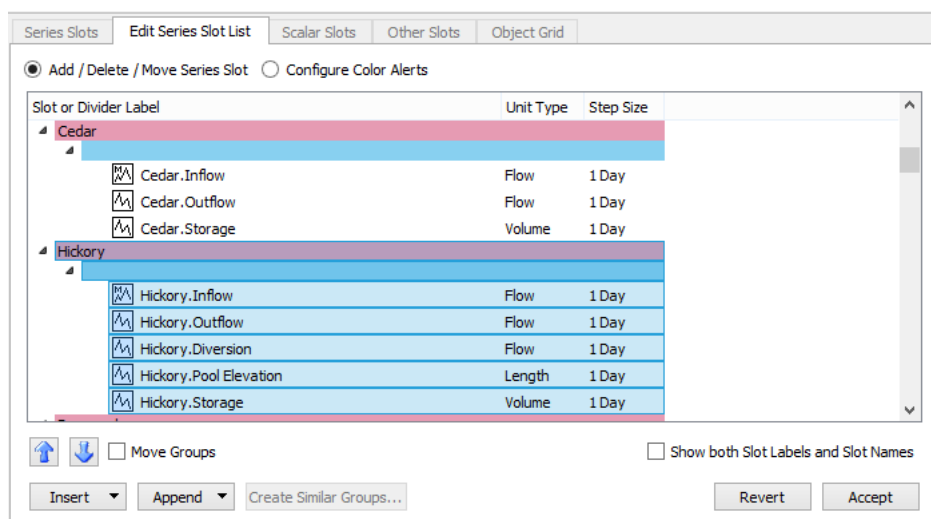
4. Select the objects you want to add to the SCT. You can select any number of objects; one slot group will be created for each object you select.



5. Select **Ok** to confirm the selection; the dialog box closes.

The new slot groups are added to the slot list, immediately after the selected one. There is one new slot group for each new object you have selected, and each group contains all slots from the original group that are also assigned to the new object. If a slot in the original group is not assigned to the new object, it is not included in the new slot group for the new object.

Slots in the new group are listed in the same order as the original group. If the object name is included in the original slot group's divider, the new object's name is also included in the new slot group divider.



6. Review the new groups and, optionally, make any changes you want at this time.
7. When the slot list is arranged as you want, select **Accept** to accept the changes.

The Series Slots tab becomes active, and the new slots are included in the positions you have defined. Slot values are from the model.

Related topics

- [Add slot dividers](#), page 54

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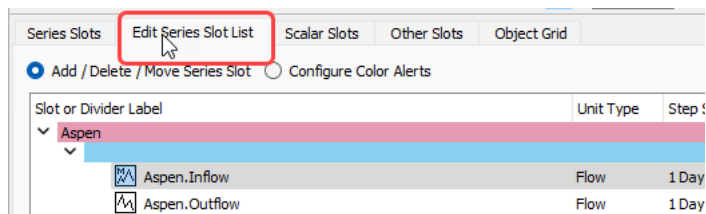
- [Add sheet dividers](#), page 56
- [Rearrange items on the Series Slots tab](#), page 60

Rearrange items on the Series Slots tab

Use this procedure to create the slot order and groupings you want on the Series Slots tab by rearranging any of the following items:

- Individual slots
- Individual slot group dividers or sheet dividers
- Entire slot groups or sheets with their slots

1. Go to the **Edit Series Slot List** tab.

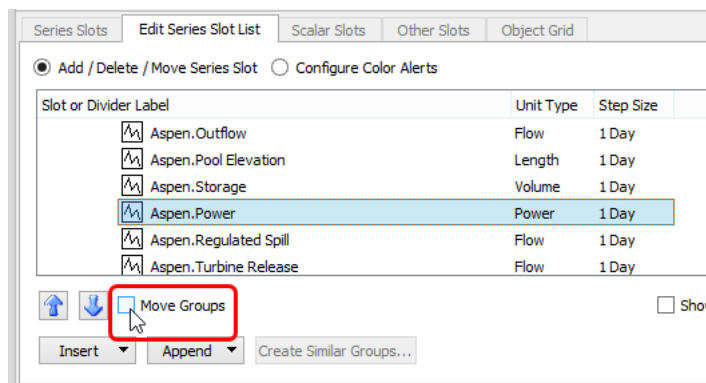


2. Use the following steps to rearrange slots and dividers by performing any of the following actions:

- Rearrange slots within their current groups.
 - Move slots to different groups or sheets.
 - Move the locations of slot group dividers or sheet dividers.
- a. Select the items to move. You can select any combination of contiguous or non-contiguous slots, slot group dividers, and sheet dividers.

Note: If you select multiple items, they will be moved in unison.

b. Clear the **Move Groups** check box.



c. Select the **Up** or **Down** arrow one or more times.

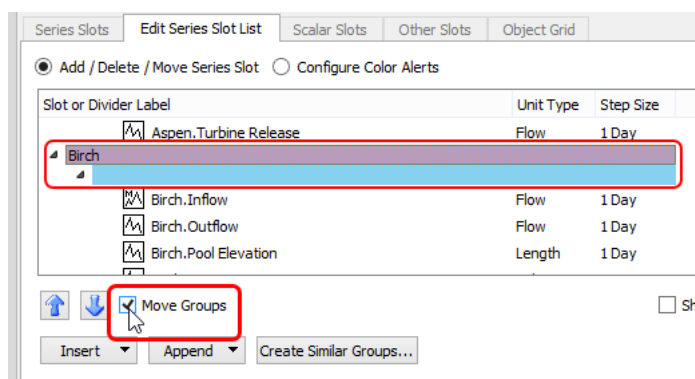
Note: If a selected item is already at the top or bottom of the list, the corresponding arrow is not available.

The selected slots and dividers are moved one step at a time in the direction you select. You can select the **Up** and **Down** arrows multiple times to move the items into the positions you want within the slot list.

3. Use the following steps to rearrange whole slot groups and sheets. The order of individual slots within each group is preserved while the group or sheet is moved as a unit.
 - a. Select the dividers of the items to move. You can select any combination of contiguous or non-contiguous slot group dividers and sheet dividers.

Note: If you select multiple items, they will be moved in unison.

- b. Select the **Move Groups** check box.



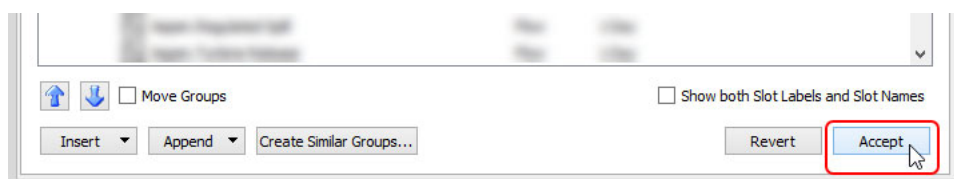
- c. Select the **Up** or **Down** arrow one or more times.

Note: When the Move Groups check box is selected, the arrows are available only if at least one slot group or sheet divider is selected. You can select individual slots, but they will not be moved apart from their group or sheet.

Note: If a selected item is already at the top or bottom of the list, the corresponding arrow is not available.

The selected groups and sheets are moved one step at a time in the direction you select. You can select the **Up** and **Down** arrows multiple times to move the items into the positions you want within the slot list.

4. Optionally, repeat [Step 2.](#) and [Step 3.](#) to rearrange more items.
5. When you have arranged the slots and dividers into the groupings and order you want, select **Accept** to apply the changes to the Series Slots tab.



The Series Slots tab opens, and the slots are displayed in the groupings and sheets you have configured.

Related topics

- [Add slot dividers](#), page 54
- [Add sheet dividers](#), page 56

- [Cut or copy and paste items](#), page 62

Cut or copy and paste items

Use this procedure to cut and paste or copy and paste any of the following items on the Series Slots tab:

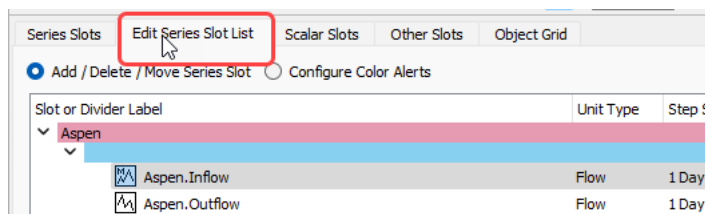
- Individual slots
- Individual slot dividers or sheet dividers
- Entire slot groups or sheets with their slots

The cut or copied items are placed on the RiverWare clipboard, allowing you to paste them one or more times to other locations in this SCT or another SCT for the same model.

Note: Slots on the RiverWare clipboard can also be pasted to other locations in RiverWare, such as the slot list for certain output devices; see [“Output Manager” in Output Utilities and Data Visualization](#) for details.

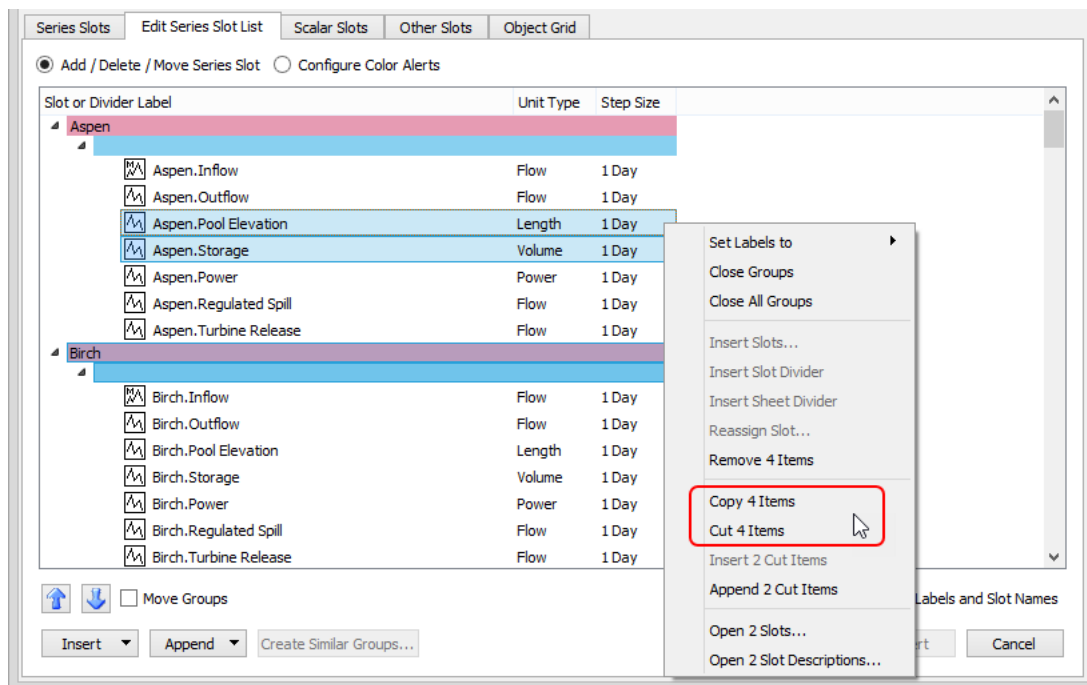
Note: Cutting a slot from the SCT does not remove it from the RiverWare model, nor does it affect the slot data.

1. Go to the **Edit Series Slot List** tab. Alternatively, press Alt+Shft+S.



2. Use the following steps to cut or copy *individual slots and dividers only*.
 - a. Select the items to remove. You can select any combination of contiguous or non-contiguous slots, slot group dividers, and sheet dividers.
 - b. Right-click and select one of the following options on the shortcut menu. The options depend on the items you have selected.
 - **Copy *n* Slots**—available if you have selected slots only and no dividers. Copies the indicated number of slots.
 - **Cut *n* Slots**—available if you have selected slots only and no dividers. Cuts the indicated number of slots.
 - **Copy *n* Dividers**—available if you have selected slot group and sheet dividers only and no slots. Copies just the indicated number of dividers.
 - **Cut *n* Dividers**—available if you have selected slot group and sheet dividers only and no slots. Cuts just the indicated number of dividers; the slots under the dividers are retained in their current positions.
 - **Copy *n* Items**—available if you have selected both slots and dividers. Copies the indicated number of slots and dividers.

- **Cut n Items**—available if you have selected both slots and dividers. Cuts the indicated number of slots and dividers. All unselected slots are retained in their current positions.



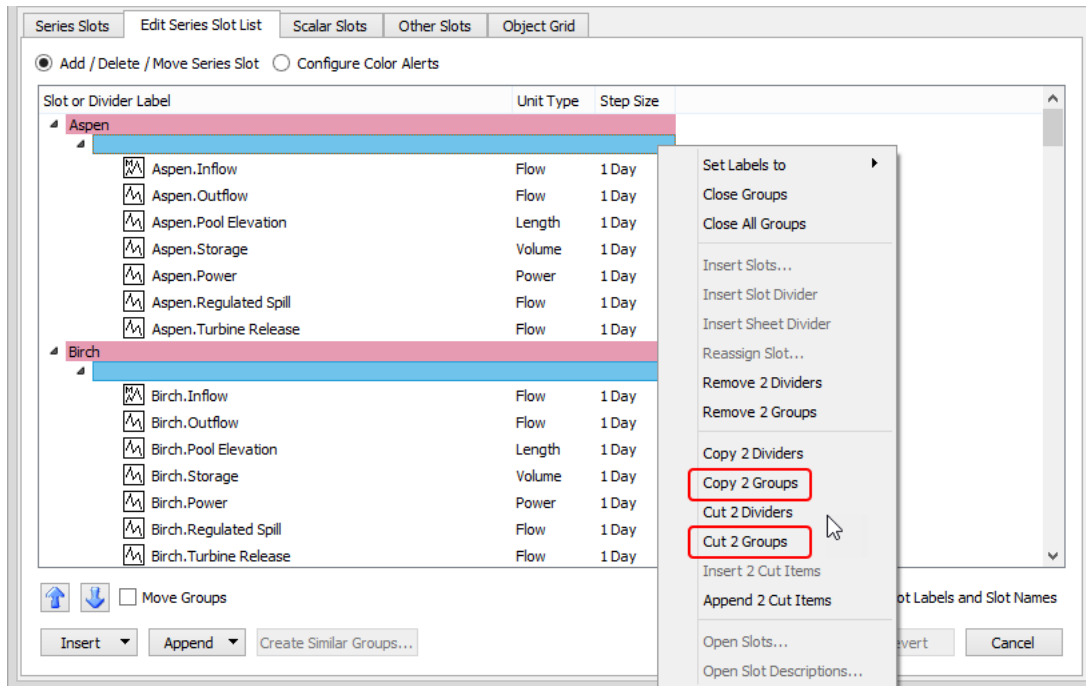
The items are cut or copied and placed on the RiverWare clipboard.

- Use the following steps to cut or copy *entire slot groups and sheets*.
 - Select the dividers of the items to cut or copy. Do *not* select any slots. You can select any combination of contiguous or non-contiguous slot group dividers and sheet dividers.
 - Right-click and select one of the following options on the shortcut menu.
 - **Copy n Groups**—available if you have selected slot group and sheet dividers only and no slots. Copies the indicated number of slot groups and sheets with their slots.

Chapter 3

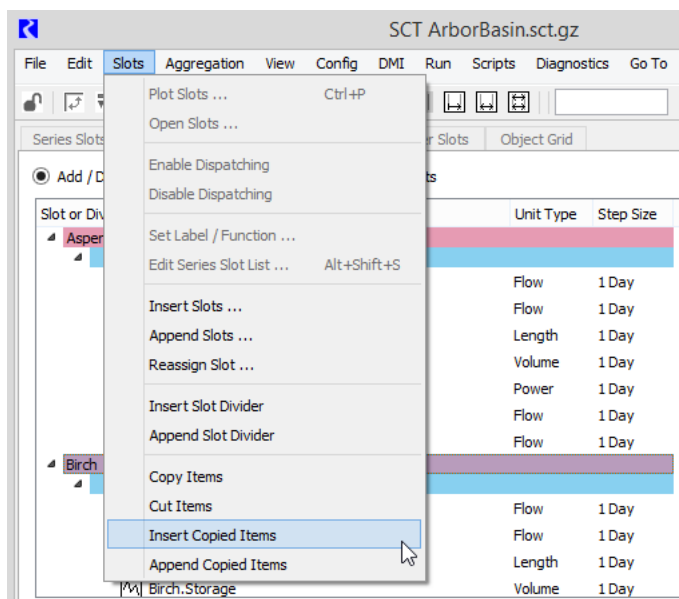
Using the SCT

- **Cut n Groups**—available if you have selected slot group and sheet dividers only and no slots. Cuts the indicated number of slot groups and sheets with their slots.

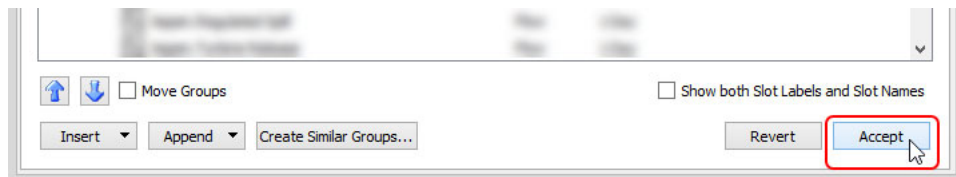


The groups or sheets are cut or copied and placed on the RiverWare clipboard.

4. Select the location where you want to paste the items. This can be in any SCT for this model.
5. On the SCT menu, select **Slots**, then one of the following selections:
 - **Insert Copied Items**—the items are pasted *directly above* the selected item.
 - **Append Copied Items**—the items are pasted *at the end of the slot list*.



6. Optionally, repeat [Step 4.](#) and [Step 5.](#) to paste the items multiple times to the SCT.
7. Review your changes in the updated slot list and select **Accept** to apply them to the Series Slots tab.



The Series Slots tab opens with your updates applied.

Related topics

- [Add slot dividers](#), page 54
- [Add sheet dividers](#), page 56
- [Rearrange items on the Series Slots tab](#), page 60
- [Remove items](#), page 65

Remove items

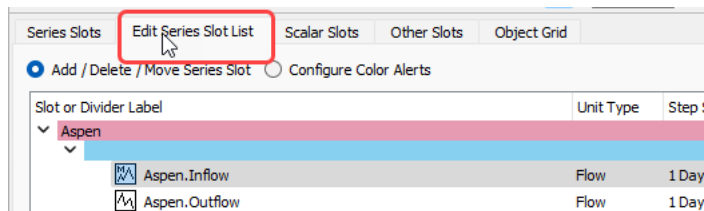
Use this procedure to remove any of the following items from the Series Slots tab:

- Individual slots
- Individual slot group dividers or sheet dividers
- Entire slot groups or sheets with their slots

Note: Items removed using this procedure are *not* placed on the RiverWare clipboard and therefore cannot be recovered or pasted to another location.

Note: Removing a slot from the SCT does not remove it from the RiverWare model, nor does it affect the slot data.

1. Go to the **Edit Series Slot List** tab. Alternatively, press Alt+Shft+S.

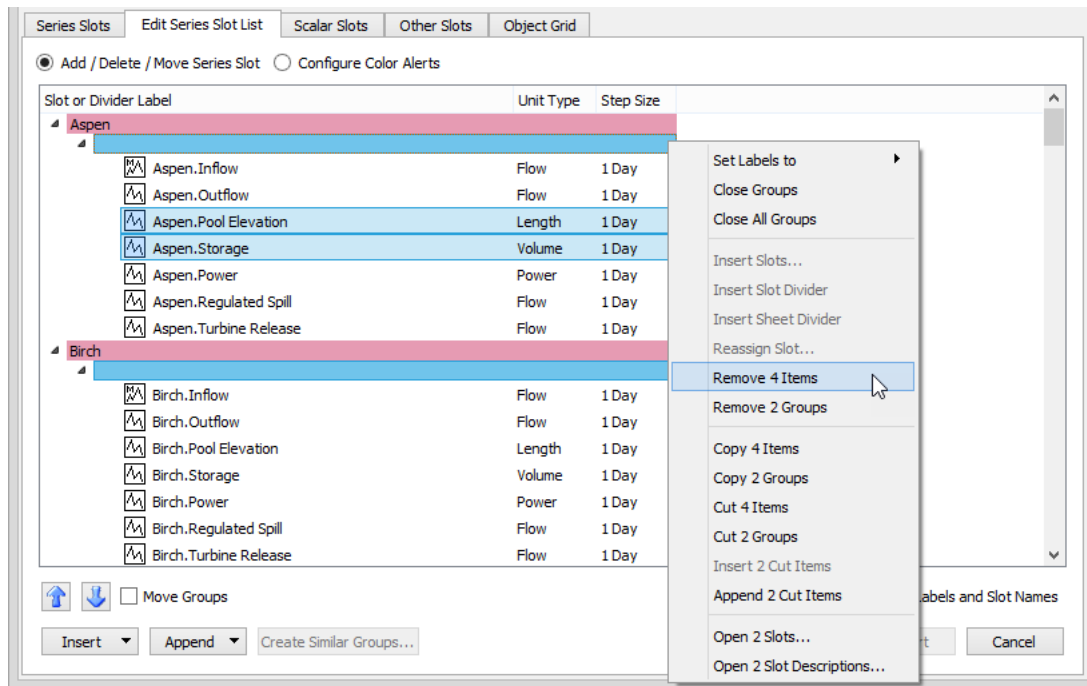


2. Use the following steps to remove individual slots and dividers only.
 - a. Select the items to remove. You can select any combination of contiguous or non-contiguous slots, slot group dividers, and sheet dividers.
 - b. Right-click and select one of the following options on the shortcut menu. The options depend on the items you have selected.
 - **Remove *n* Slots**—available if you have selected slots only and no dividers. Removes the indicated number of slots.

Chapter 3

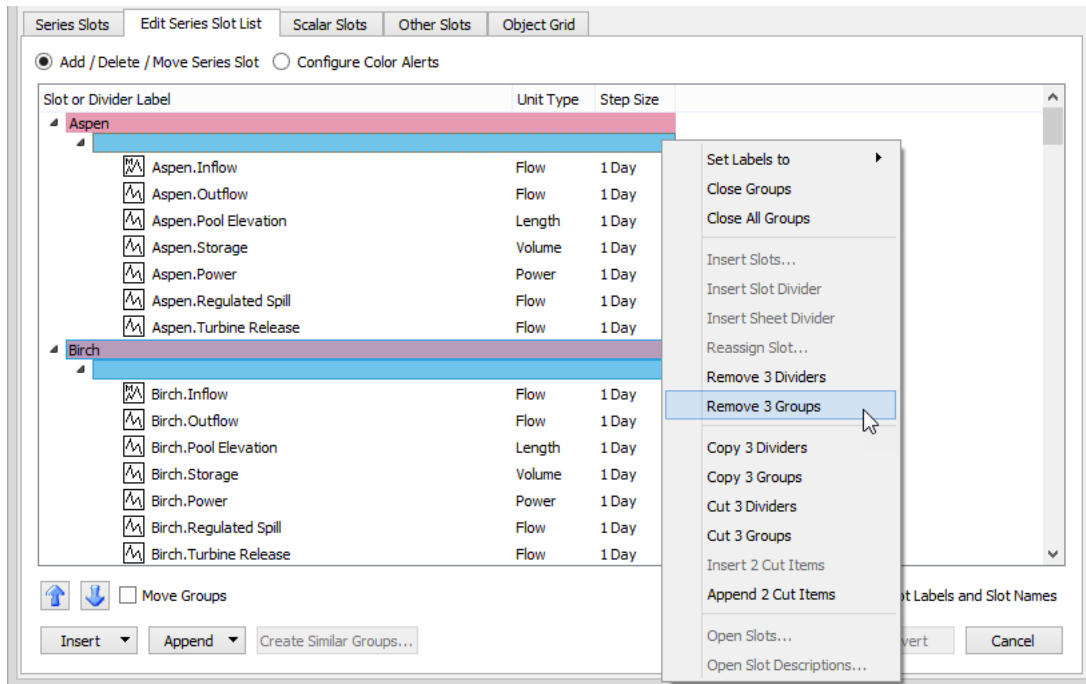
Using the SCT

- **Remove n Dividers**—available if you have selected slot group and sheet dividers only and no slots. Removes just the indicated number of dividers; the slots under the dividers are retained in their current positions.
- **Remove n Items**—available if you have selected both slots and dividers. Removes the indicated number of slots and dividers only. All unselected slots are retained in their current positions.



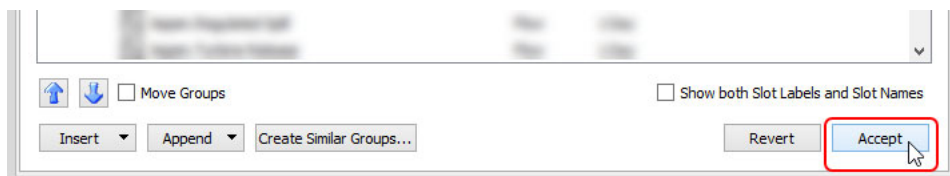
- Use the following steps to remove entire slot groups and sheets.
 - Select the dividers of the items to move. Do *not* select any slots. You can select any combination of contiguous or non-contiguous slot group dividers and sheet dividers.
 - Right-click and select the following option on the shortcut menu.

- **Remove n Groups**—available if you have selected slot group and sheet dividers only and no slots. Removes the indicated number of slot groups and sheets with their slots.



The items are removed from the SCT.

4. Optionally, repeat [Step 2.](#) and [Step 3.](#) to remove more items.
5. Review the changes and select **Accept** to apply them to the Series Slots tab.



The Series Slots tab opens with your updates applied.

Related topics

- [Cut or copy and paste items](#), page 62

Manipulating Slot Values

This section describes how to use the SCT to set values in the model.

Topics

- [Guidelines for manipulating slot values](#), page 68
- [Edit slot values manually](#), page 68

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- [Change slot value flags](#), page 70
- [Clear slot values](#), page 72
- [Interpolate slot values](#), page 75
- [Set slot values over a time range](#), page 77
- [Adjust a range of slot values](#), page 79

Guidelines for manipulating slot values

Following are important considerations when manipulating slot values in the SCT.

- You do *not* need to unlock the SCT to manipulate slot values.
- The SCT is just a *window* into the model, and all values are stored in the model itself. You must save the model to preserve the new values.
- The new value is not applied until you press Return, Enter or click elsewhere on the desktop. To cancel an edit, press Escape.
- When you enter a numeric value, the cell type is set to Input (I) regardless of the original type. The exception to this is Target (T) type cells, which retain that type after the new value is applied.
- If you enter “n” or “NaN” (in any combination of upper- and lower-case), the cell value is cleared and the cell type is set to Output (O).
- In aggregated view, editing the value of a summary cell applies the edits to all timesteps in the aggregation interval.
- Undo and redo of slot value editing is available from the Edit menu and Ctrl+Z for undo and Ctrl+Shift+Z for redo.
- You cannot edit certain types of slots, including expression slots and certain accounting system slots. These slots are read only, which is indicated by crosshatching in a user-defined Read-only color.
- If you want expression slots to evaluate after you enter a new value, the Auto-Evaluate Expression Slots on Edits option must be enabled. See [“Configure cell value editing options”](#) on page 169 for details.

Related topics

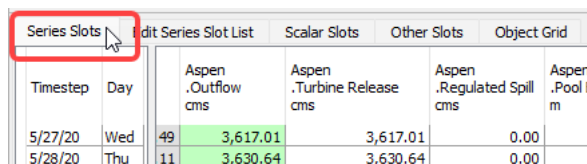
- [Data in the SCT](#), page 5

Edit slot values manually

Use this procedure to manually change the value of one or more cells in a slot. The new value is applied to all selected cells and the cell type is changed to “Input”.

Note: Before performing this procedure, see [“Manipulating Slot Values”](#) on page 67 for important details on updating values.

1. Go to the **Series Slot** tab.



Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

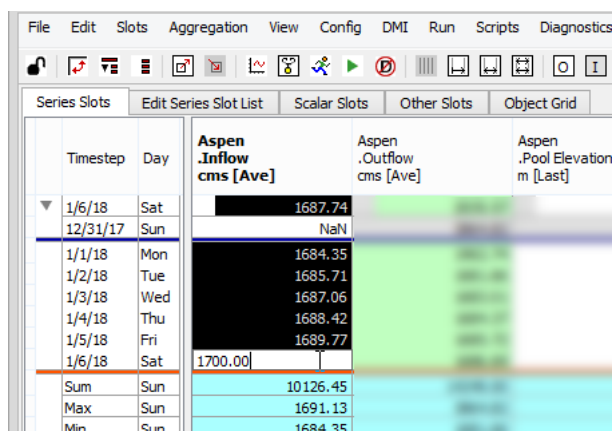
2. Locate the slot you want to modify, and select one or more cells in the slot.

Caution: The new value will be applied to all selected cells regardless of the original unit type (Flow or Area, for example). It is up to you to ensure that the new value will be appropriate for all selected cells.

Note: In aggregated view, if you select a summary cell, the new value will be applied to all timesteps in the aggregation interval

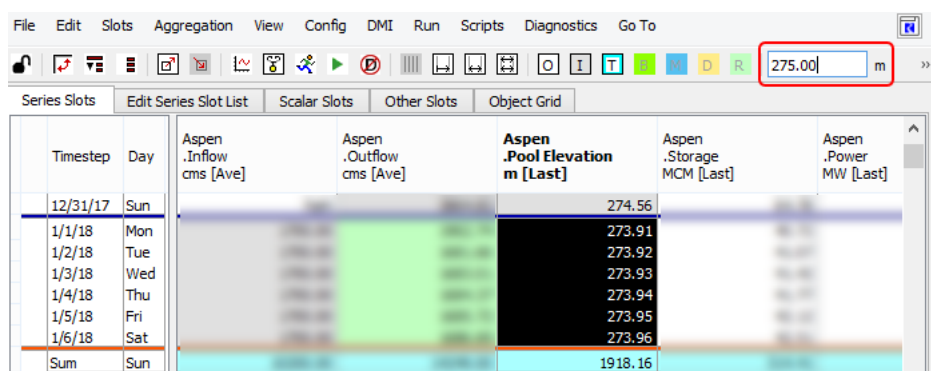
3. Enter the new value as follows.

- Enter the new value in the active cell (the last cell selected) and press Enter.



Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	
1/6/18	Sat	1687.74			
12/31/17	Sun	NaN			
1/1/18	Mon	1684.35			
1/2/18	Tue	1685.71			
1/3/18	Wed	1687.06			
1/4/18	Thu	1688.42			
1/5/18	Fri	1689.77			
1/6/18	Sat	1700.00			
Sum	Sun	10126.45			
Max	Sun	1691.13			
Min	Sun	1684.35			

- Enter the new value in the **Value** field in the SCT toolbar and press Enter. This field displays the value with full precision.



Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]
12/31/17	Sun			274.56	
1/1/18	Mon			273.91	
1/2/18	Tue			273.92	
1/3/18	Wed			273.93	
1/4/18	Thu			273.94	
1/5/18	Fri			273.95	
1/6/18	Sat			273.96	
Sum	Sun			1918.16	

The new value is applied to all selected cells, and, with the exception of Target type cells, the cell type is set to Input. Target type cells retain their original type.

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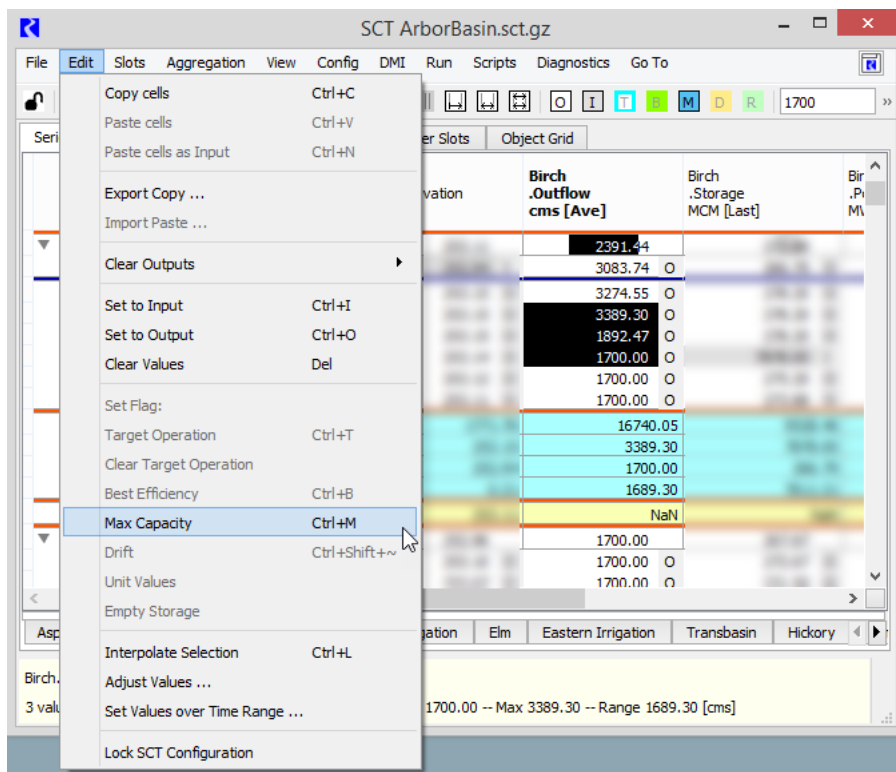
Related topics

- [Manipulating Slot Values](#), page 67
- [Clear slot values](#), page 72

Change slot value flags

Use this procedure to change the status of a series slot cell value by changing the flag.

1. Optionally, enable the display of flag letters for all slot values; see [“Display flag letters and priority numbers”](#) on page 103 for details.
2. Select one or more cells with editable values.
Note: In aggregated view, if you select a summary cell, the update will be applied to all timesteps in the aggregation interval
3. In the SCT menu, select **Edit**, then an available flag option; see [“Available flags by slot type”](#) on page 71 for details about the available flags.



The flag is updated for the selected cells. The cells are shaded in the flag color defined on the flag legend, except for Target Operations. Target Operations are indicated by a frame around all timestep cells in the target range. The Target time is also shaded as an Input value.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Birch .Inflow cms [Ave]	Birch .Pool Elevation m [Last]	Birch .Outflow cms [Ave]	Birch .Storage MCM [Last]
1/1/18	Mon			3274.55 O	
1/2/18	Tue			3389.30 M	
1/3/18	Wed			1892.47 M	
1/4/18	Thu			1700.00 M	
1/5/18	Fri			1700.00 O	
1/6/18	Sat			1700.00 O	
Sum	Sun			16740.05	
Max	Sun			3389.30	
Min	Sun			1700.00	
DIFF	Sun			1689.30	
Pool Elevat	Sun			NaN	
1/13/18	Sat			1700.00	
1/7/18	Sun			1700.00 O	
1/8/18	Mon			1700.00 O	
1/9/18	Tue			1700.00 O	
1/10/18	Wed			1700.00 O	

Available flags by slot type

- General flags available to all series slots
 - I—Input
 - O—Output
 - Z—DMI Input
- Maximum outflow flag available to Outflow slots
 - M—Max Capacity
- Target flags available to Storage and Pool Elevation slots
 - TB—Target Begin
 - T—Target
- Drift flag available to Regulated Spill and Bypass slots
 - D—Drift
- Surcharge release flag available to the Surcharge Release slot
 - S—Surcharge Release
- Regulation discharge flag available to the Regulation Discharge slot
 - G—Regulation Discharge
- Rule flag set by RiverWare
 - R—Set by Rule

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- Computed input set during iterative multiple run management (MRM)
 - i—MRM Input
- Power calculation flags available to Energy slots
 - B—Best Efficiency
 - M—Max Capacity
- Unit values flag available to Turbine Release and Energy slots on power reservoirs with the Unit Power method selected
 - U—Unit Values

Related topics

- [Display flag letters and priority numbers](#), page 103
- [“Flags on the Timestep I/O Menu” in *User Interface*](#)

Clear slot values

Use these procedures to clear editable slot values. The cell values are set to NaN and the cell value flag is set to Output (O). Generally, editable slot values are ones that would be computed during a model run, including values with the following flags:

- O—Output
- B—Best Efficiency
- D—Drift
- M—Max Capacity
- U—Unit Values

Clear selected values

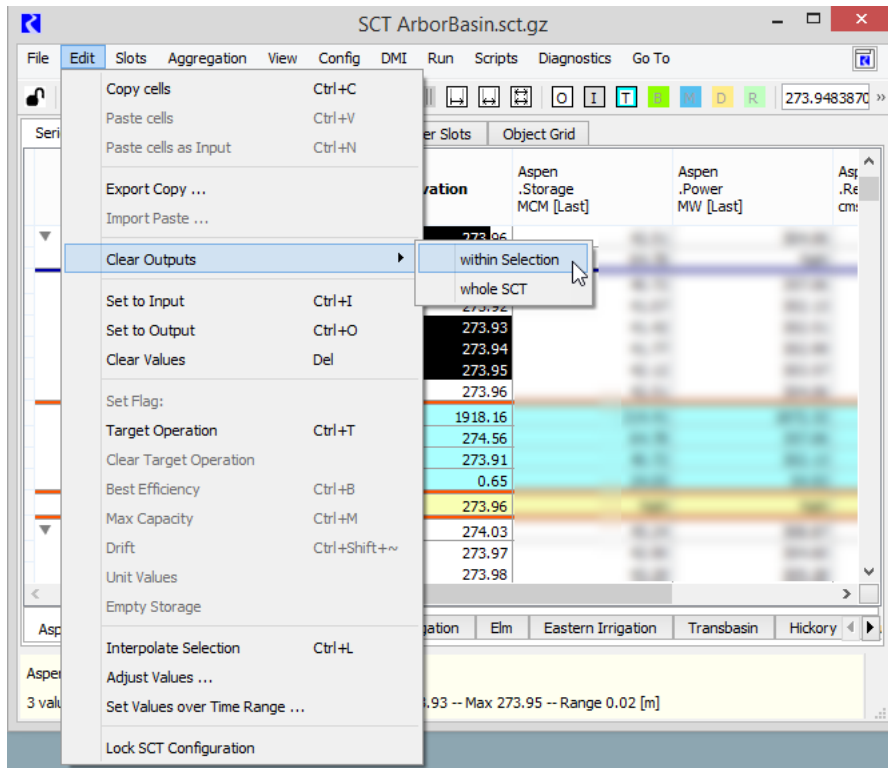
Use this procedure to clear selected values only.

1. Select one or more cells with editable values.

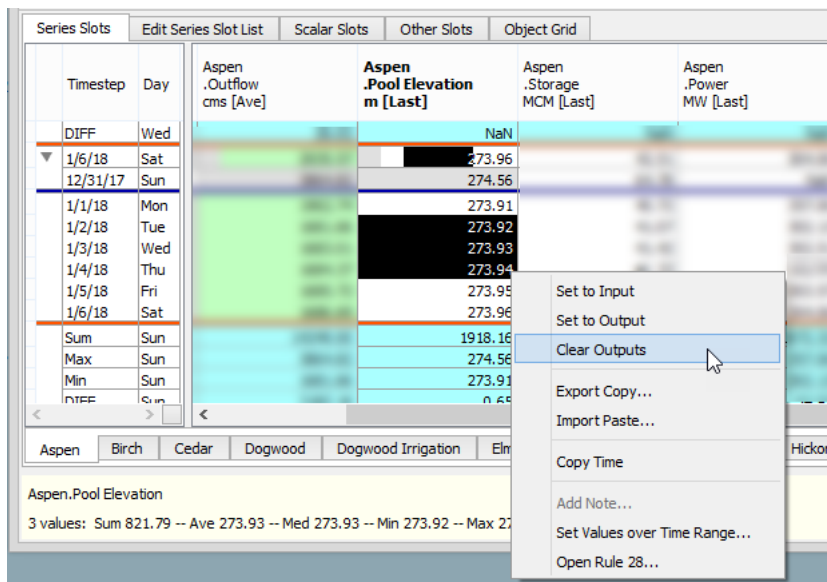
Note: In aggregated view, if you select a summary cell, the update will be applied to all timesteps in the aggregation interval

2. Do one of the following to clear the selected values.
 - Press Delete.

- Select **Edit**, then **Clear Outputs**, then **within Selection**.



- Right-click and select **Clear Outputs** in the shortcut menu.



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The cell values are set to NaN and the cell value flags are set to Output (O).

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	
DIFF	Wed		NaN		
1/6/18	Sat		273.96		
12/31/17	Sun		274.56		
1/1/18	Mon		273.91		
1/2/18	Tue		NaN		
1/3/18	Wed		NaN		
1/4/18	Thu		NaN		
1/5/18	Fri		273.95		
1/6/18	Sat		273.96		
Sum	Sun		1096.37		

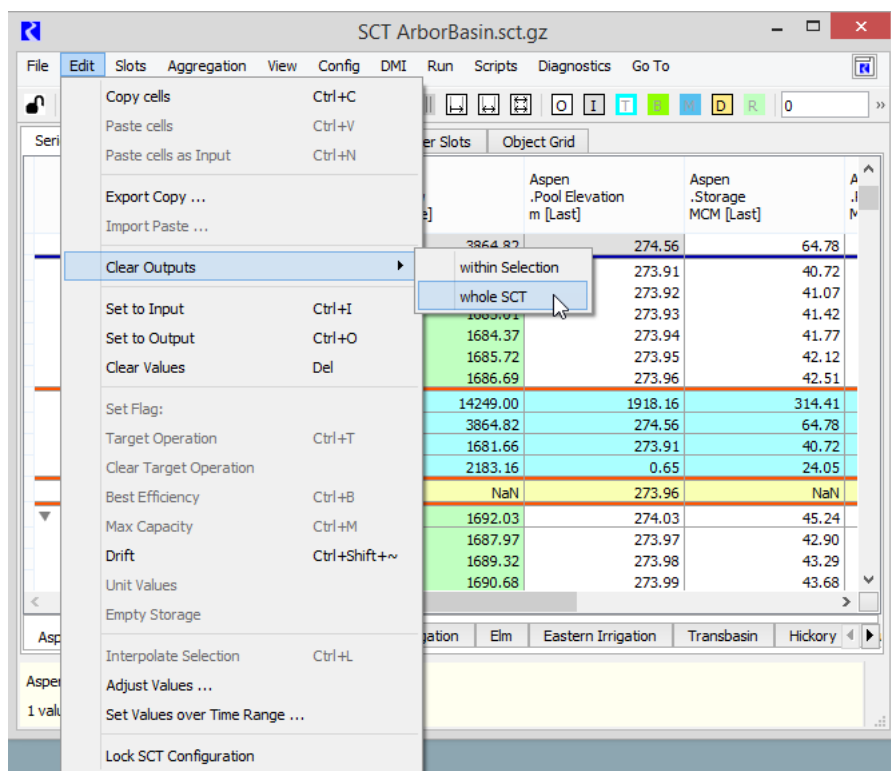
Related topics

- [Manipulating Slot Values](#), page 67
- [Edit slot values manually](#), page 68
- [Clear all values](#), page 74
- [“Flags on the Timestep I/O Menu” in *User Interface*](#)

Clear all values

Use this procedure to clear all editable values in the SCT.

1. On the SCT menu, select **Edit**, then **Clear Outputs**, then **Whole SCT**.



All editable values are cleared. The cell values are set to NaN and the cell value flags are set to Output (O).

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid				
Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	Aspen .Power MW [Last]			
DIFF	Wed	NaN	26.02	NaN	NaN	NaN			
▼ 1/6/18	Sat	1687.74	3864.82	NaN	NaN	NaN			
12/31/17	Sun	NaN	3864.82	274.56	NaN	NaN			
1/1/18	Mon	1684.35	NaN	NaN	NaN	NaN			
1/2/18	Tue	1685.71	NaN	NaN	NaN	NaN			
1/3/18	Wed	1687.06	NaN	NaN	NaN	NaN			
1/4/18	Thu	1688.42	NaN	NaN	NaN	NaN			
1/5/18	Fri	1689.77	NaN	NaN	NaN	NaN			
1/6/18	Sat	1691.13	NaN	NaN	NaN	NaN			
Sum	Sun	10126.45	3864.82	274.56	NaN	NaN			
Max	Sun	1691.13	3864.82	274.56	NaN	NaN			
Min	Sun	1684.35	3864.82	274.56	NaN	NaN			
DIFF	Sun	6.77	0.00	0.00	NaN	NaN			
< >									
Aspen	Birch	Cedar	Dogwood	Dogwood Irrigation	Elm	Eastern Irrigation	Transbasin	Hickory	Juniper Irrigation

Related topics

- [Manipulating Slot Values](#), page 67
- [Edit slot values manually](#), page 68
- [Clear selected values](#), page 72
- “Flags on the Timestep I/O Menu” in *User Interface*

Interpolate slot values

Use this procedure to interpolate linearly between two input values. You can choose to overwrite all inputs between the end points or just fill in the missing values between them.

Note: All interpolation is done in internal units and then converted to display units. For 1 Month timesteps, this takes into account the differing lengths of time among months. Since the standard internal units for per-time values are never per month (for example, flow values are cms), using internal units avoids the potentially undesirable effect different amounts of time represented by different months would have on the interpolation function.

1. Go to the **Series Slot** tab and locate the slot you want to update.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

2. Select the range of cells you want to update. The first and last cells define the starting and ending interpolation values.

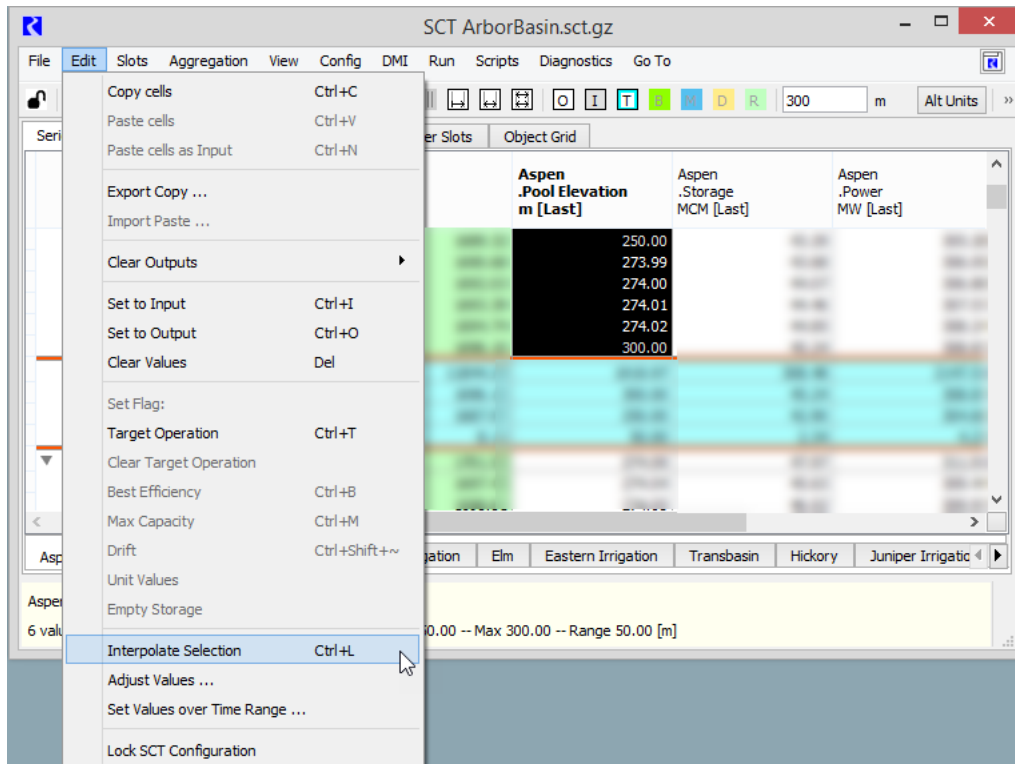
For interpolation to be available, the selection must meet all the following criteria:

- The cell range selection is rectangular and includes at least three contiguous timesteps.
- The first and last slot values are defined (not NaN).

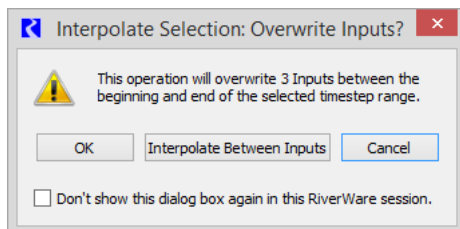
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3. On the SCT menu, select **Edit**, then **Interpolate Selection**; alternatively, press Ctrl+L.



The Overwrite Inputs confirmation dialog appear.



4. Proceed as follows:

- Select **Cancel** to cancel the operation.
- Select **OK** to overwrite all inputs between the two end values.
- Select **Interpolate Between Inputs** to preserve your sparse inputs. This is useful for filling gaps in series data.

The values in the selected range of cells are updated.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]
1/8/18	Mon			250.00	
1/9/18	Tue			260.00	
1/10/18	Wed			270.00	
1/11/18	Thu			280.00	
1/12/18	Fri			290.00	
1/13/18	Sat			300.00	
Sum	Sun			1900.00	

Related topics

- [Manipulating Slot Values](#), page 67
- [Set slot values over a time range](#), page 77

Set slot values over a time range

Use this procedure to perform either of the following updates on a range of cells:

- Set the cells to a specified value.
- Interpolate the cell values linearly to a specified value.

Note: This procedure is useful for updating cells over an extended time period, which might be difficult to select on the SCT worksheet because of the size.

1. Go to the **Series Slot** tab and locate the slot you want to update.

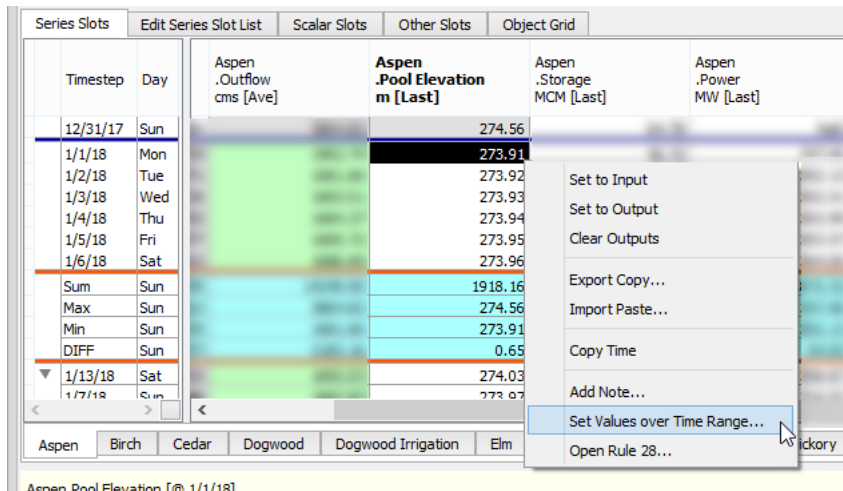
Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool l m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. Select the first cell in the range and right-click.

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- On the shortcut menu, select **Set Values over Time Range**.



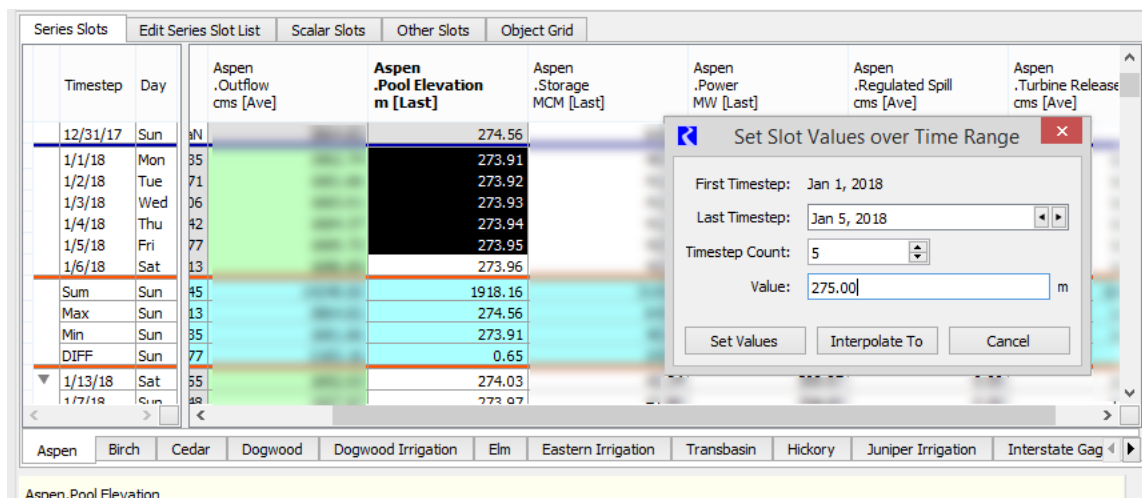
The Set Slot Values over Time Range dialog box appears.

- Complete the dialog box fields as follows.

- Define the time range by specifying either the **Last Timestep** or the **Timestep Count**.

As you change the time range, the corresponding cells are selected on the SCT, and the alternate entry field on the dialog box is updated to match.

- In the **Value** field, enter the value you want to set.



- Specify the operation you want to perform, as follows:

- Select **Set Values** to set all selected cells to the specified value.
- Select **Interpolate To** to set the final timestep to the specified value and interpolate linearly all values between the first and last cells.

Note: This button is enabled only if the value of the first cell is *not* NaN.

- Select **Cancel** to cancel the operation.

The cell values are updated and their types are set to Input (I).

Related topics

- [Manipulating Slot Values](#), page 67
- [Interpolate slot values](#), page 75

Adjust a range of slot values

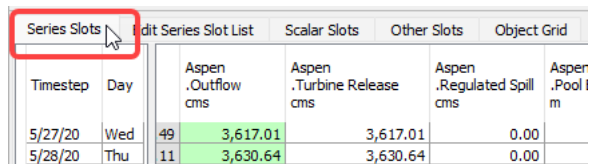
Use this procedure to perform either of the following updates on a range of cells:

- Add or subtract a constant offset to each of the values. This option is available only when all cells in the range have the same scale and unit.
- Scale all the values by a specified percentage.

This update does not affect NaN values nor read-only values, which are indicated with crosshatching in a user-defined Read-only color.

Note: Since slots can appear in an SCT more than once, this update is careful to adjust each selected slot/timestep value only once. Optionally, you can view a summary of the slots updated.

1. Go to the **Series Slot** tab and locate the slot you want to update.



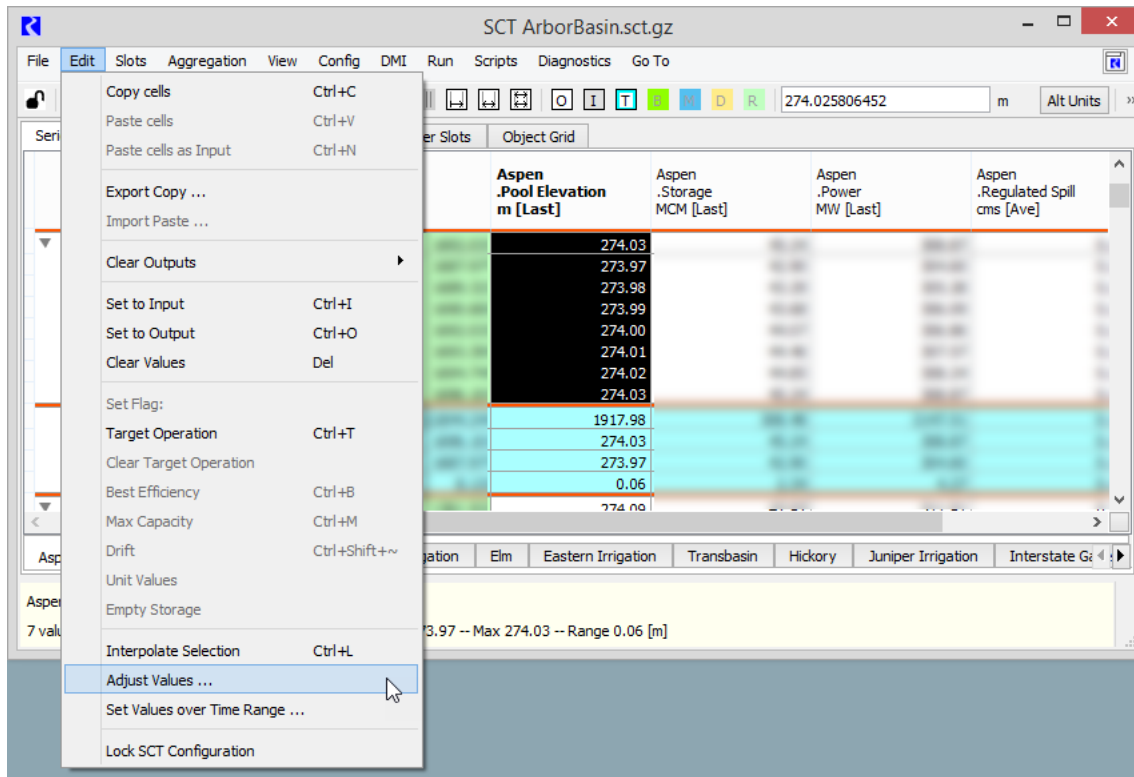
Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. Select the range of cells you want to update.

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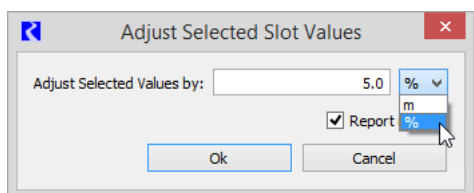
3. On the SCT menu, select **Edit**, then **Adjust Values**.



The Adjust Selected Slot Values dialog box opens.

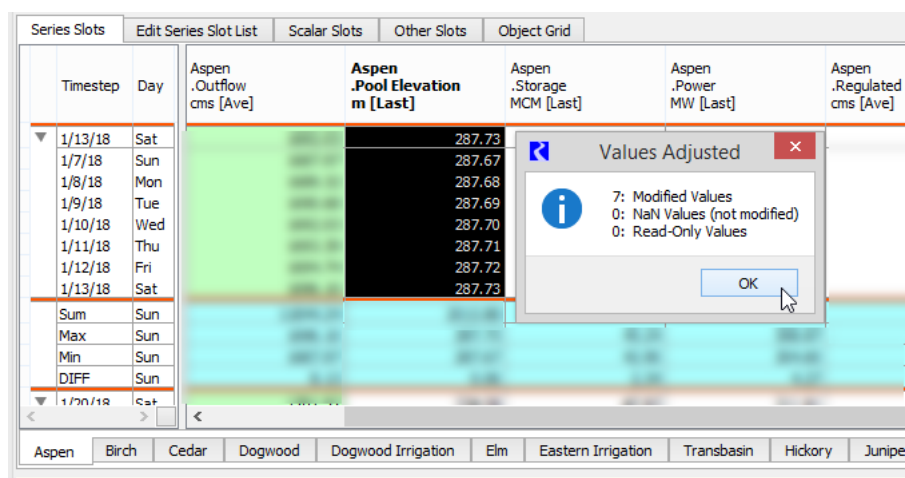
4. Complete the dialog box as follows.

- In the **Adjust Selected Values by** field, enter the constant or percentage adjustment you want to make.
- In the menu, select the type of update you want to make:
 - Select **%** to scale by the specified percentage.
 - Select the unit type to add or subtract the specified constant.
- Optionally, select the **Report Results** check box to view a summary of the slots changed.



5. Select **Ok** to perform the update.

The cells are updated in the SCT. If you selected Report Results, a dialog box displays summary information.



Related topics

- [Manipulating Slot Values](#), page 67

Copying and Pasting Slot Values

This section includes procedures for copying and pasting values across slots. Following are important considerations.

Topics

- [Copy and paste guidelines](#), page 81
- [Copy and paste one or more slot values](#), page 84
- [Copy and paste multiple values to the same timesteps](#), page 87
- [Copy and paste all values for a timestep](#), page 88

Copy and paste guidelines

- Copy and paste slot values even when the SCT configuration is locked.
- As with other updates, pasted values are updated in the model itself, and you must save the model to preserve the new values.
- Copy values from only one slot at a time; similarly, you can paste to only one slot at a time. To copy from or paste to multiple slots, you must perform multiple operations.
- Paste values to a different slot only if the destination slot has the same unit type as the source slot.
- Similarly, paste values to slots in a different SCT, provided the unit types are the same.
- You cannot paste to read-only slots, such as expression slots and certain accounting system slots. Read-only slots are indicated by crosshatching in a user-defined Read-only color.

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- In a default paste, non-Input values are pasted to the destination cells as NaNs, as follows.
 - Best Efficiency (B), Drift (D), Max Capacity (M), and Unit Values (U) values are pasted to the destination cells as NaNs while retaining the source type flag.
 - Other non-Input values are pasted as NaNs with an Output (O) flag.
- To paste all values as Input (I), you must explicitly specify the Paste Cells as Input option (or press Ctrl+N).
- The final timestep value of a Target Operation is copied to the destination with the Target (T) flag, and the Begin Target is pasted if an explicit Begin Target timestep was defined—but only if the entire Target Operation was included in the copied selection.
- Pasted values are assigned to the destination cells using standard units, which are then converted to display units. This means that if the source and target slots have different display units (cubic feet per second vs cubic meters per second, for example), the pasted values will appear to be different than the source values; the underlying quantities represented by the display values will be the same, however.
- Copy and paste operations use the RiverWare internal clipboard and therefore can be done within RiverWare only. To copy and paste values to and from external applications, see [“Exporting and Importing Slot Values”](#) on page 90.
- If you copy non-contiguous cells, the gaps are honored by the paste operation. The values are pasted in the same non-contiguous pattern, starting with the first target cell selected, and destination cells that correspond to gaps in the pattern are left unchanged. For example, if you copy every other cell for a total of three cells, the values are pasted to every other cell starting with the first target cell selected.
- In aggregated view, copying a summary cell copies the values of all timesteps in the aggregation interval.
- The ways values are pasted vary based on the number of values you are copying, the target selection, and whether aggregation is enabled in the SCT.
 - [Table 3.1](#) provides details about copying and pasting when aggregation is *not* enabled.
 - [Table 3.2](#) provides details about copying and pasting when aggregation is enabled.

Table 3.1 Copying and pasting values when aggregation is *not* enabled

Copy source	Paste target	Paste result
One detail cell	One detail cell	Pastes the copied value to the selected timestep.
	Multiple detail cells	Pastes the copied value to all selected timesteps.
	Entire selected slot	Pastes the copied value to all timesteps in the slot.
Multiple detail cells	One or more detail cells	Pastes all copied values in sequence, starting with the first selected timestep. If the copied cells are non-contiguous, the pattern is honored when pasted.
	Entire selected slot	Pastes the values to the same timesteps from which they were copied.

Table 3.1 Copying and pasting values when aggregation is *not* enabled (Continued)

Copy source	Paste target	Paste result
Entire slot	One detail cell	Pastes all copied values in sequence, starting with the selected timestep.
	Multiple detail cells	Not available
	Entire slot	Pastes all values to the same timesteps from which they were copied.

Table 3.2 Copying and pasting values when aggregation is enabled

Copy source	Paste target	Paste result
One detail cell	One detail cell	Pastes the copied value to the selected timestep.
	Multiple detail cells; cells can be in different aggregation intervals.	Pastes the copied value to all selected timesteps.
	One or more summary cells	Pastes the copied value to all timesteps in all selected aggregation intervals.
	Entire slot	Pastes the copied value to all timesteps in the selected slot.
Multiple detail cells	One or more detail cells	Pastes all copied values in sequence, starting with the first selected timestep. If the copied cells are non-contiguous, the pattern is honored when pasted.
	One or more summary cells	Pastes all copied values in sequence, starting with the first timestep in the first aggregation interval. If the copied cells are non-contiguous, the pattern is honored when pasted.
	Entire selected slot	Pastes the values to the same timesteps from which they were copied.
One summary cell	One summary cell	Pastes all values in the copied aggregation interval to the selected interval.
	Multiple summary cells	Pastes all values in the copied aggregation interval to the first selected interval only.
	One or more detail cells	Pastes all values in the copied aggregation interval in sequence, starting with the first selected timestep.
	Entire slot	Pastes all values to the same timesteps from which they were copied.
Multiple summary cells	One or more summary cells	Pastes all values in the copied aggregation intervals in sequence, starting with the first selected interval. If the copied cells are non-contiguous, the pattern is honored when pasted.
	Entire slot	Pastes all values to the same timesteps from which they were copied.

Table 3.2 Copying and pasting values when aggregation is enabled (Continued)

Copy source	Paste target	Paste result
Entire slot	One detail cell	Pastes all values in sequence, starting with the selected timestep.
	Multiple detail cells	Not available
	One or more summary cells	Not available
	Entire slot	Pastes all values to the same timesteps from which they were copied.

Related topics

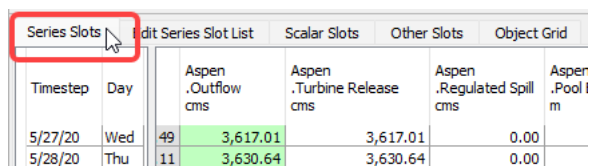
- [Data in the SCT](#), page 5
- [Manipulating Slot Values](#), page 67
- [Copy and paste one or more slot values](#), page 84
- [Copy and paste multiple values to the same timesteps](#), page 87
- [Copy and paste all values for a timestep](#), page 88

Copy and paste one or more slot values

Use this procedure to copy one or more timestep values from a slot and paste them to timesteps in the same slot, a different slot, or a different SCT.

Note: Before performing this procedure, see “[Copying and Pasting Slot Values](#)” on page 81 for important details about these operations.

1. Go to the **Series Slot** tab.

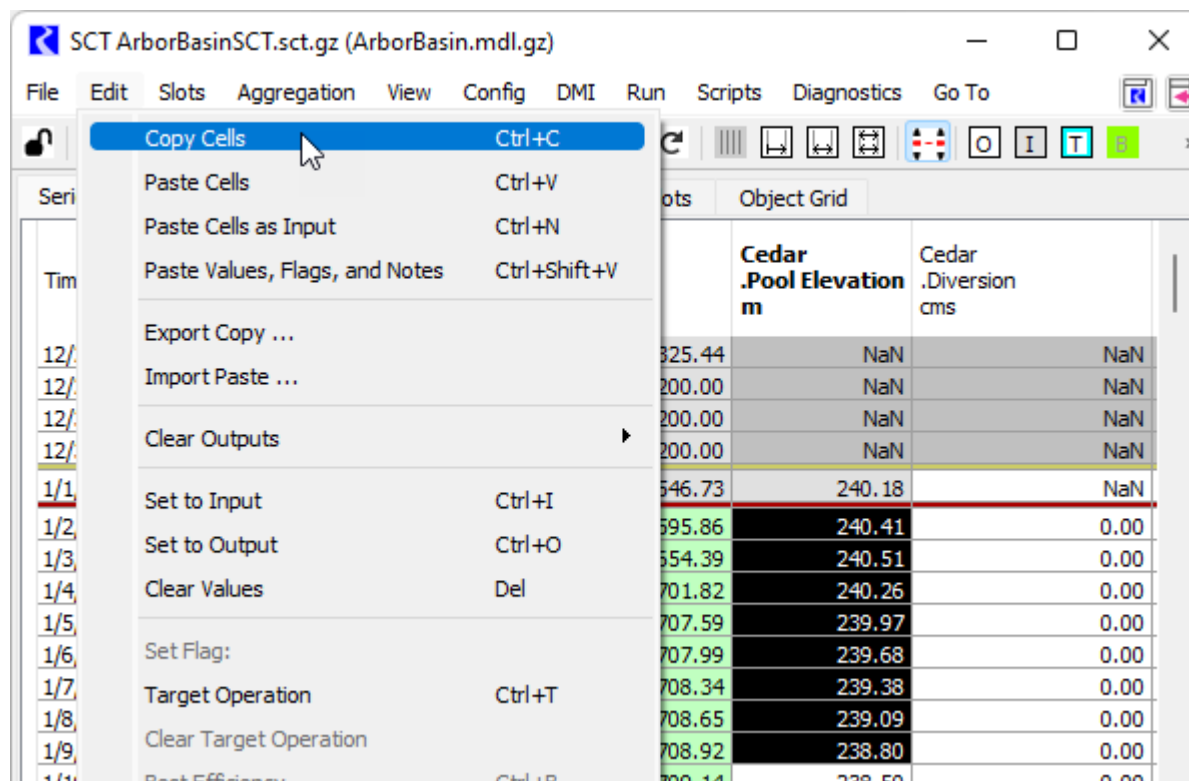


Series Slots					
Edit Series Slot List					
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool l m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

2. Select the cells to copy. All cells must be in the same slot. You can select one cell, multiple contiguous cells, or multiple non-contiguous cells.

Note: In aggregated view, if you select a summary cell, the values for all timesteps in the aggregation interval will be copied. If you select a detail cell, just those values will be copied.

- On the SCT menu, select **Edit**, then **Copy Cells**; alternatively, press Ctrl+C.



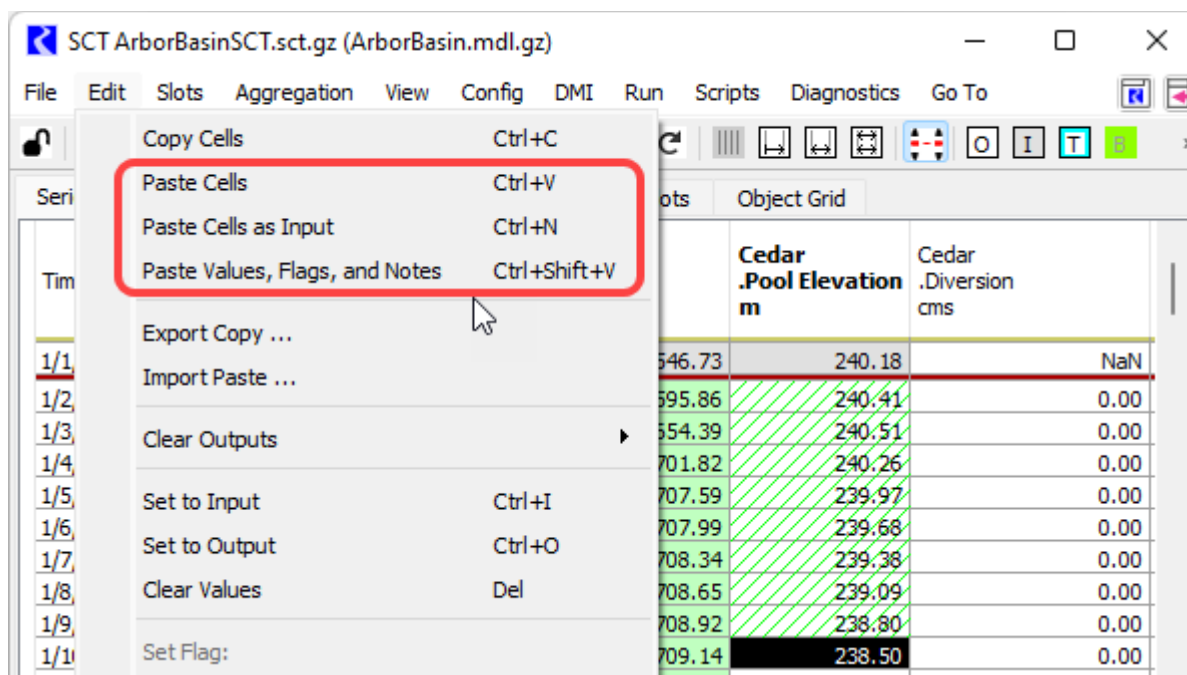
The timestep values are copied to the RiverWare clipboard, and the cells are displayed with crosshatching in a user-defined Copy color.

Timestep	Day	Cedar .Inflow cms	Cedar .Outflow cms	Cedar .Pool Elevation m	Cedar .Diversion cms
1/1/20	Wed	3,158.07	2,546.73	240.18	NaN
1/2/20	Thu	3,383.11	2,595.86	240.41	0.00
1/3/20	Fri	3,015.09	2,654.39	240.51	0.00
1/4/20	Sat	1,844.35	2,701.82	240.26	0.00
1/5/20	Sun	1,700.00	2,707.59	239.97	0.00
1/6/20	Mon	1,700.00	2,707.99	239.68	0.00
1/7/20	Tue	1,700.00	2,708.34	239.38	0.00
1/8/20	Wed	1,700.00	2,708.65	239.09	0.00
1/9/20	Thu	1,700.00	2,708.92	238.80	0.00
1/10/20	Fri	1,700.00	2,709.14	238.50	0.00
1/11/20	Sat	1,700.00	2,709.30	238.21	0.00

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Using the SCT

4. Select one or more paste targets. Targets can be in the same slot as the source values, a different slot with the same unit type, or a different SCT. See [Table 3.1](#) and [Table 3.2](#) for details on how the copied values will be pasted to the targets.
5. Paste the values, as follows:
 - To perform a default paste, select **Edit**, then **Paste Cells**; alternatively, press Ctrl+V. Input values are pasted as Input (I), and non-Input values are pasted as NaNs, as follows:
 - Best Efficiency (B), Drift (D), Max Capacity (M), and Unit Values (U) values are pasted to the destination cells as NaNs while retaining the source type flag.
 - Other non-Input values are pasted as NaNs with an Output (O) flag.
 - To paste all cell values as Input (I), select **Edit**, then **Paste Cells as Input**; alternatively, press Ctrl+N.
 - To paste all values with the original flag from the copied value and any Series Slot Notes, select **Edit**, then **Paste Values, Flags, and Notes**; alternatively, press Ctrl+Shift+V.



The values are pasted to the appropriate timesteps.

6. Optionally, you can repeat [Step 4](#). and [Step 5](#). to paste the same values to additional cells in the same or different slots.

Related topics

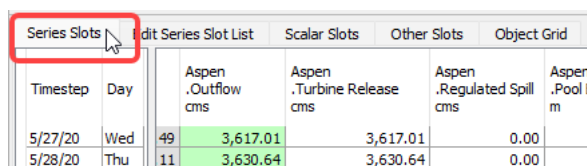
- [Manipulating Slot Values](#), page 67
- [Copying and Pasting Slot Values](#), page 81
- [Copy and paste multiple values to the same timesteps](#), page 87
- [Copy and paste all values for a timestep](#), page 88

Copy and paste multiple values to the same timesteps

Use this procedure to copy multiple timeseries values from one slot and paste them to the same timesteps on a different slot. The target slot can be in the same or a different SCT, and it must have the same unit type as the source slot.

Note: Before performing this procedure, see “[Copying and Pasting Slot Values](#)” on page 81 for important details about these operations.

1. Go to the **Series Slot** tab.

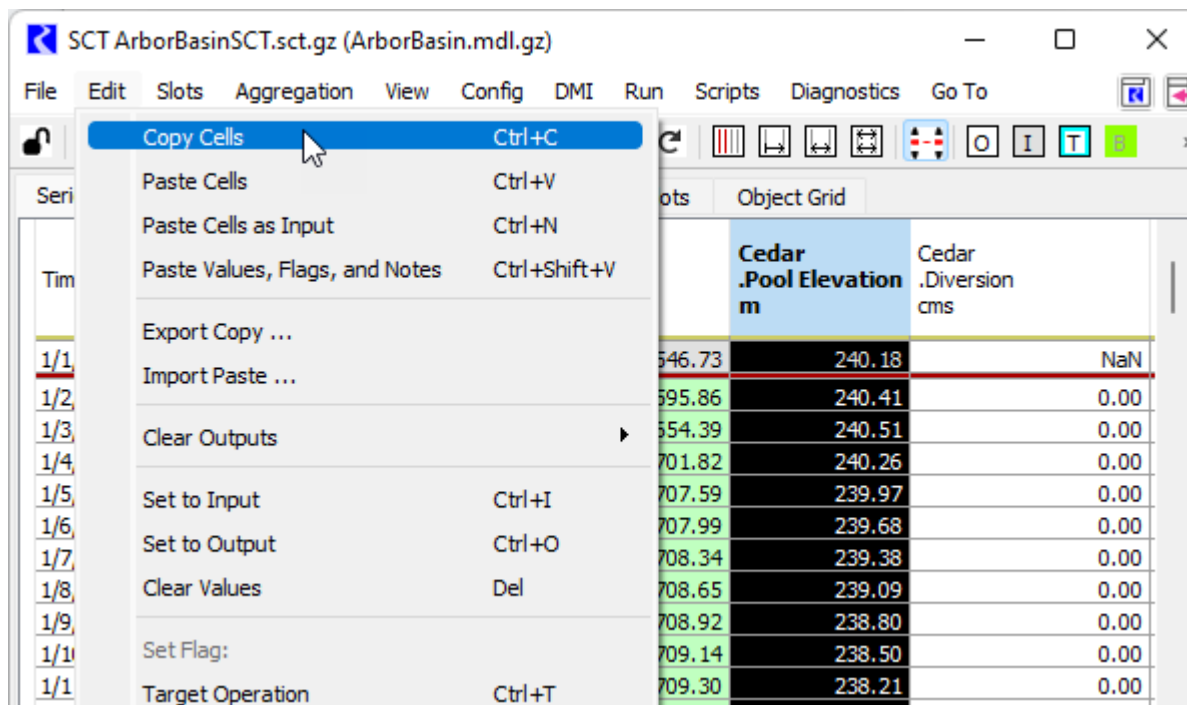


Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool Elevation m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

2. Select multiple contiguous or non-contiguous cells to copy. To select all timesteps in the slot, select the slot header.

Note: In aggregated view, if you select a summary cell, the values for all timesteps in the aggregation interval will be copied. If you select a detail cell, just those values will be copied.

3. On the SCT menu, select **Edit**, then **Copy Cells**; alternatively, press Ctrl+C.



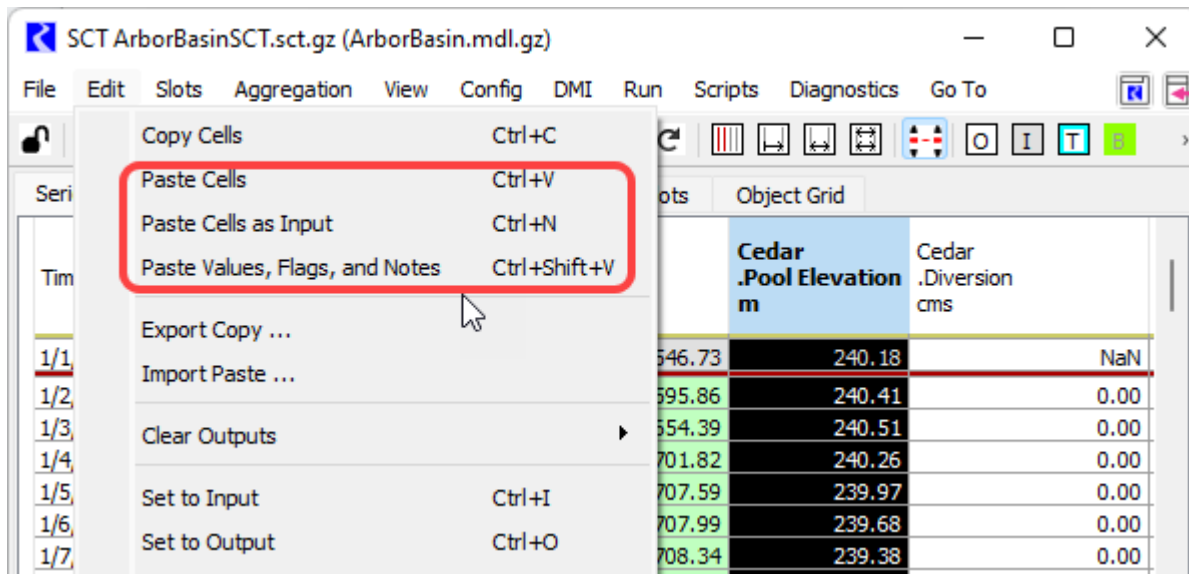
The selected cell values are copied to the RiverWare clipboard, and the cells are displayed with crosshatching in a user-defined Copy color.

4. Select the slot header of the target slot. The target slot must have the same unit type as the source slot.
5. Paste the values as follows.

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Using the SCT

- To perform a default paste, select **Edit**, then **Paste Cells**; alternatively, press Ctrl+V. Input values are pasted as Input (I), and non-Input values are pasted as NaNs, as follows:
 - Best Efficiency (B), Drift (D), Max Capacity (M), and Unit Values (U) values are pasted to the destination cells as NaNs while retaining the source type flag.
 - Other non-Input values are pasted as NaNs with an Output (O) flag.
- To paste all cell values as Input (I), select **Edit**, then **Paste Cells as Input**; alternatively, press Ctrl+N.
- To paste all values with the original flag from the copied value and any Series Slot Notes, select **Edit**, then **Paste Values, Flags, and Notes**; alternatively, press Ctrl+Shift+V.



The values are pasted to the same timesteps from which they were copied.

6. Optionally, you can repeat [Step 4](#). and [Step 5](#). to paste the same slot values to additional slots.

Related topics

- [Manipulating Slot Values](#), page 67
- [Copying and Pasting Slot Values](#), page 81
- [Copy and paste one or more slot values](#), page 84
- [Copy and paste all values for a timestep](#), page 88

Copy and paste all values for a timestep

Use this procedure to copy the entire set of values for a timestep (that is, the values for all slots at that timestep) and paste it to other timesteps within the same SCT. You can copy and paste multiple timesteps if they are contiguous.

Note: Before performing this procedure, see “[Copying and Pasting Slot Values](#)” on page 81 for important details about these operations.

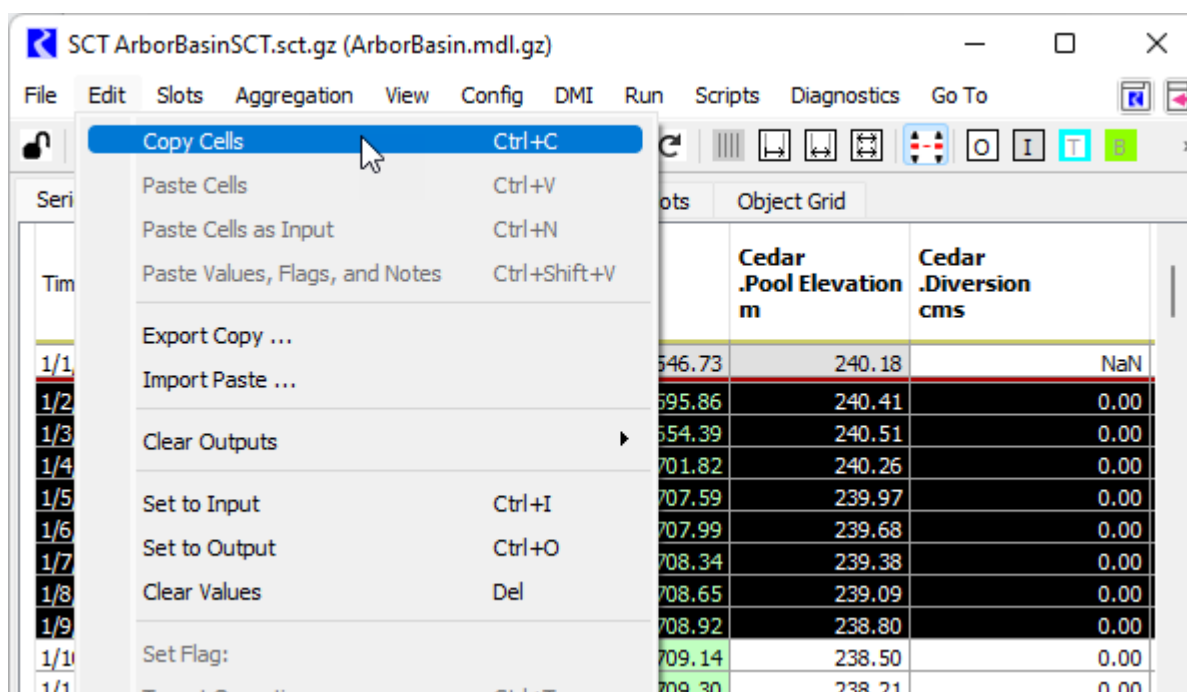
1. Go to the **Series Slot** tab.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. Select one or more contiguous timesteps to copy.

Note: In aggregated view, if you select a summary cell, the values for all timesteps in the aggregation interval will be copied. If you select a detail cell, just those values will be copied.

3. On the SCT menu, select **Edit**, then **Copy Cells**; alternatively, press Ctrl+C.



The cell values are copied to the RiverWare clipboard, and the cells are displayed with crosshatching in a user-defined Copy color.

4. Specify the starting target by selecting a timestep header. The timestep values will be pasted to this and subsequent timesteps.

Note: The target must be within the same SCT as the source; you cannot paste values at a timestep across all slots to a different SCT.

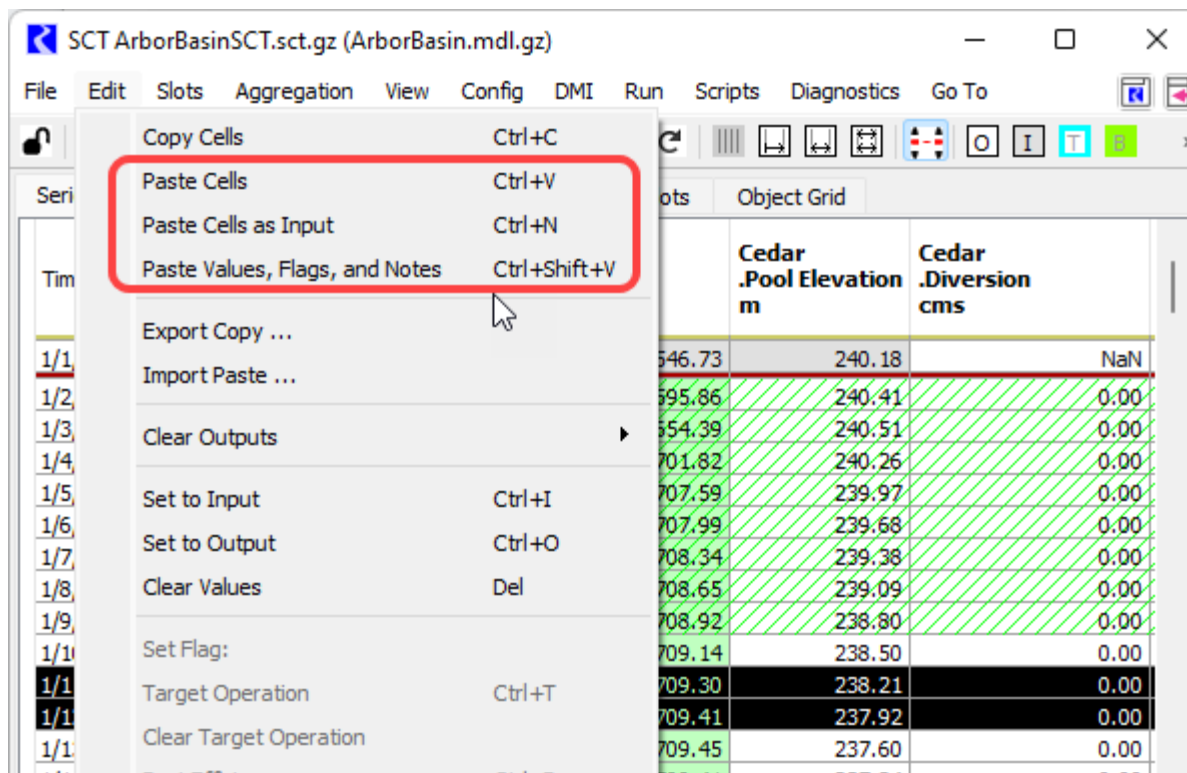
5. Paste the values, as follows:

- To perform a default paste, select **Edit**, then **Paste Cells**; alternatively, press Ctrl+V. Input values are pasted as Input (I), and non-Input values are pasted as NaNs, as follows:
 - Best Efficiency (B), Drift (D), Max Capacity (M), and Unit Values (U) values are pasted to the destination cells as NaNs while retaining the source type flag.
 - Other non-Input values are pasted as NaNs with an Output (O) flag.

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- To paste all cell values as Input (I), select **Edit**, then **Paste Cells as Input**; alternatively, press Ctrl+Shift+N.
- To paste all values with the original flag from the copied value and any Series Slot Notes, select **Edit**, then **Paste Values, Flags, and Notes**; alternatively, press Ctrl+Shift+V.



The values are pasted to the selected timesteps.

6. Optionally, you can repeat [Step 4](#). and [Step 5](#). to paste the same time slice values to additional timesteps.

Related topics

- [Manipulating Slot Values](#), page 67
- [Copying and Pasting Slot Values](#), page 81
- [Copy and paste one or more slot values](#), page 84
- [Copy and paste multiple values to the same timesteps](#), page 87

Exporting and Importing Slot Values

This section includes procedures for importing and exporting slot values to and from RiverWare and external applications, such as Microsoft Excel.

Topics

- [Export values to an external application](#), page 91

- [Import values from an external application](#), page 93

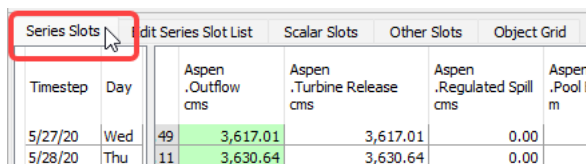
Export values to an external application

Use this procedure to copy values from the SCT to the Windows clipboard so the values can be pasted into an external application. You can copy any number of values, up to the entire SCT, at one time.

Note: Values on the RiverWare clipboard can also be pasted to other locations in RiverWare where an Import Paste is supported. See “[Slots](#)” in *User Interface* for details.

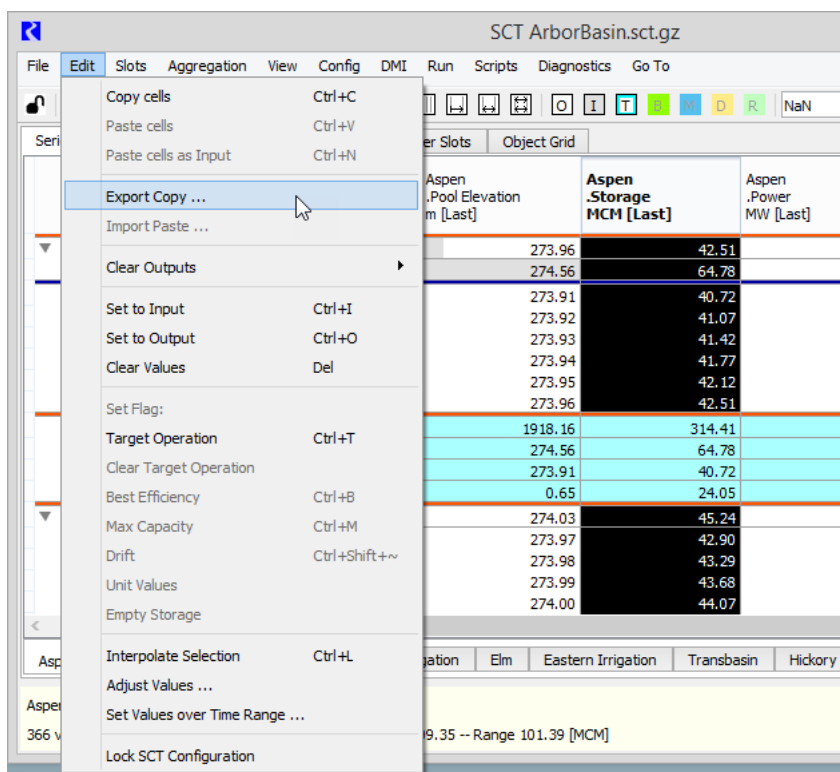
Note: You can also export series slot values from the Slot Viewer; see “[Open a slot dialog](#)” on page 209 for details.

1. Go to the **Series Slot** tab.



Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen Outflow cms	Aspen Turbine Release cms	Aspen Regulated Spill cms	Aspen Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. Select the cells to copy. You can select any combination of cells, including one cell, multiple non-contiguous cells, a range of cells, one or more entire slots, or an entire sheet.
3. On the SCT menu, select **Edit**, then **Export Copy**.

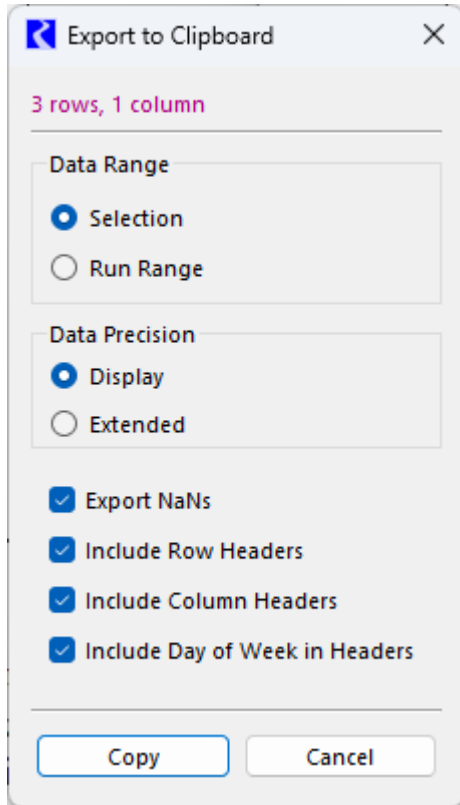


The Export to Clipboard window appears.

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Using the SCT

4. In the **Export to Clipboard** window, shown below, specify the copy options. See “[Export Copy](#)” in *User Interface* for a full description of the options in the Export Copy window.



Note: Headers are copied as currently displayed in the SCT. Depending on the timestep orientation and the SCT display options, the headers may include units and days of the week.

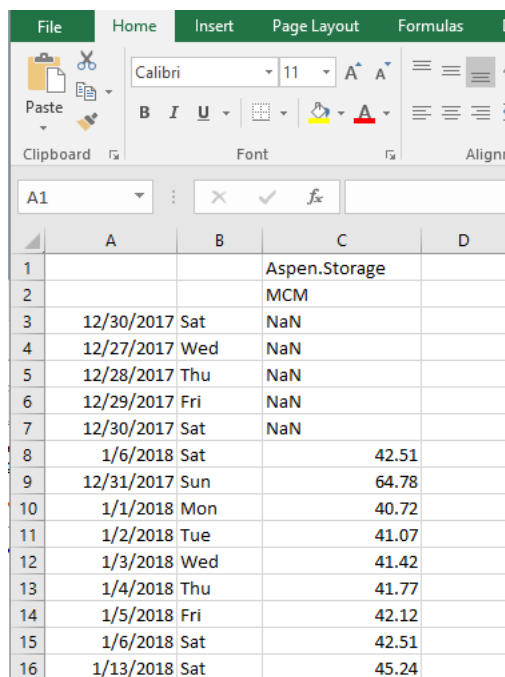
5. Select **Copy**.

The values are copied to the Windows clipboard. The values are copied as follows, depending on whether aggregated view is currently enabled.

- Hide Detail—summary values only
- Show/Hide Details—summary values only, even if details are shown
- Show All Detail—details only, no summary
- No Aggregation—details only, no summary

Note: Windows clipboard data format is ASCII-encoded decimal numeric values, or “NaN” for undefined values. Values within a row are separated by tab characters. Rows are separated by new line characters.

6. Paste the values into a Windows application, including Excel, Notepad, Word, and RiverWare itself.



	A	B	C	D
1			Aspen.Storage	
2			MCM	
3	12/30/2017	Sat	NaN	
4	12/27/2017	Wed	NaN	
5	12/28/2017	Thu	NaN	
6	12/29/2017	Fri	NaN	
7	12/30/2017	Sat	NaN	
8	1/6/2018	Sat		42.51
9	12/31/2017	Sun		64.78
10	1/1/2018	Mon		40.72
11	1/2/2018	Tue		41.07
12	1/3/2018	Wed		41.42
13	1/4/2018	Thu		41.77
14	1/5/2018	Fri		42.12
15	1/6/2018	Sat		42.51
16	1/13/2018	Sat		45.24

Related topics

- [“Export Copy” in User Interface](#)
- [Data in the SCT](#), page 5
- [Import values from an external application](#), page 93

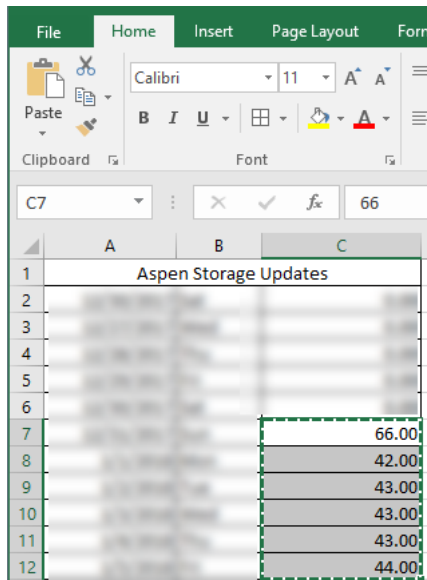
Import values from an external application

Use the following procedure to paste data from the Windows clipboard into the SCT.

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Using the SCT

1. Copy data in an external application, such as Excel, Word, or Notepad.



The data is copied to the Windows clipboard.

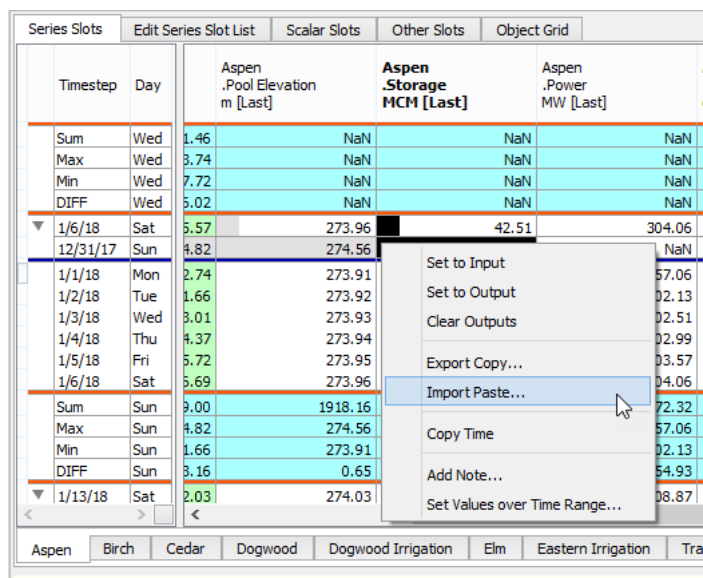
2. Open the SCT and go to the **Series Slot** tab.

Series Slots		Edit Series Slot List		Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m	
5/27/20	Wed	49 3,617.01	3,617.01	0.00		
5/28/20	Thu	11 3,630.64	3,630.64	0.00		

3. Select the location in the SCT where you want to paste the values.

Note: If you select a summary cell, the values will be pasted into the detail cells for that aggregation interval.

4. On the SCT menu, select **Edit**, then **Import Paste**.



The Import from Clipboard dialog box opens showing the values that are on the Windows clipboard.

The options on this dialog box vary, depending on the number of values you have copied and the timestep orientation of the SCT. Proceed as follows:

- If the SCT is in vertical timestep orientation, go to [Step 5](#).
- If the SCT is in horizontal timestep orientation, go to [Step 6](#).

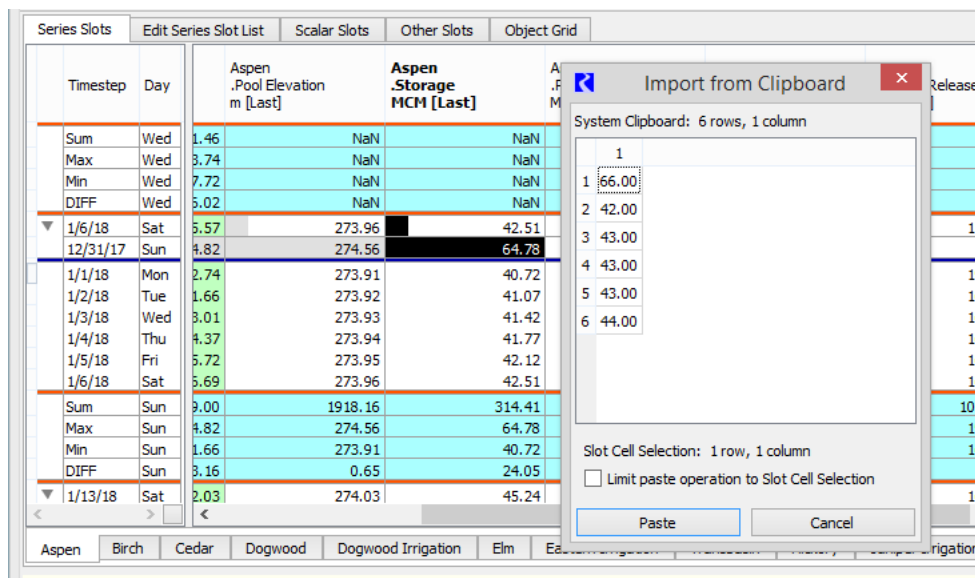
5. In vertical timestep orientation, complete the dialog box as follows.

- **Limit paste operation to Slot Cell Selection**—this selection applies only if the number of values on the Windows clipboard exceeds the number of cells you have selected in the SCT; for example, there are seven values on the Windows clipboard, but you have selected only two SCT cells.
 - Select the check box to paste the copied values only to the selected cells in the SCT. Using the example above, only the first two values will be pasted.

Chapter 3

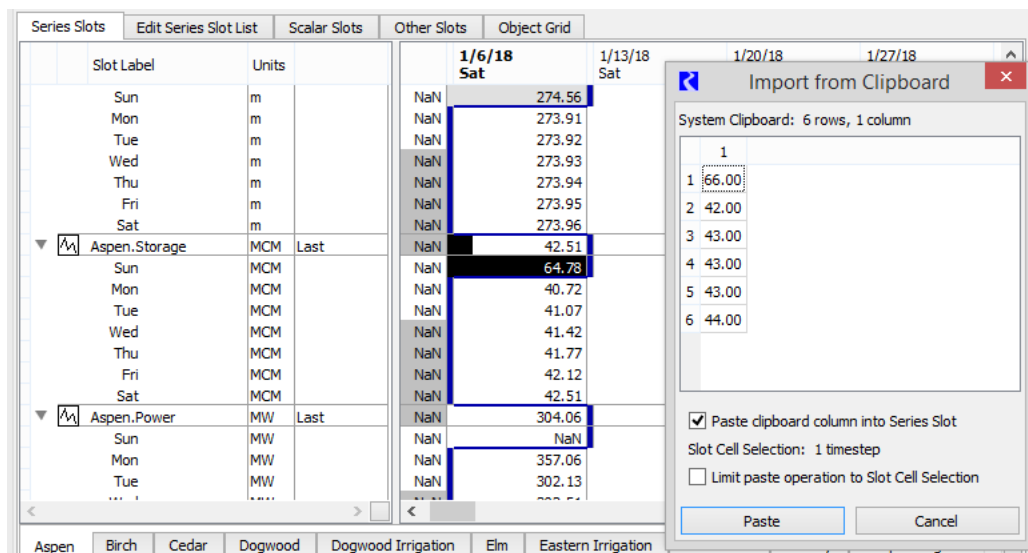
Using the SCT

- Clear the check box to paste all copied values on the Windows clipboard to the corresponding number of cells, starting with the selected cells.



- In horizontal timestep orientation, complete the dialog box as follows.
 - **Paste clipboard row into Series Slot**—this option appears when you have copied *a single value or a single row* of data.
 - Select the check box to paste the row horizontally, as data for multiple timesteps within a single slot.
 - Clear the check box to paste the data vertically, as data for a single timestep across multiple slots.
 - **Paste clipboard column into Series Slot**—this option appears when you have copied a *single column* of data.
 - Select the check box to paste the row horizontally, as data for multiple timesteps within a single slot.
 - Clear the check box to paste the row vertically, as data for a single timestep across multiple slots.
 - **Limit paste operation to Slot Cell Selection**—this selection applies only if the number of values on the Windows clipboard exceeds the number of cells you have selected in the SCT; for example, there are seven values on the Windows clipboard, but you have selected only two SCT cells.
 - Select the check box to paste the copied values only to the selected cells in the SCT. Using the example above, only the first two values will be pasted.

- Clear the check box to paste all copied values on the Windows clipboard to the corresponding number of cells, starting with the selected cells. Using the example above, all seven values will be pasted.



7. Select **Paste** to complete the operation.

The SCT is updated according to your selections.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	Aspen .Power MW [Last]	
Sum	Wed			NaN	
Max	Wed			NaN	
Min	Wed			NaN	
DIFF	Wed			NaN	
1/6/18	Sat		42.51		
12/31/17	Sun		66.00		
1/1/18	Mon		42.00		
1/2/18	Tue		43.00		
1/3/18	Wed		43.00		
1/4/18	Thu		43.00		
1/5/18	Fri		44.00		
1/6/18	Sat		42.51		
Sum	Sun		323.51		
Max	Sun		66.00		

Related topics

- [Data in the SCT](#), page 5
- [Import values from an external application](#), page 93

Changing How Data is Displayed

The procedures in this section allow you to modify a variety of display characteristics of the SCT.

Chapter 3

Using the SCT

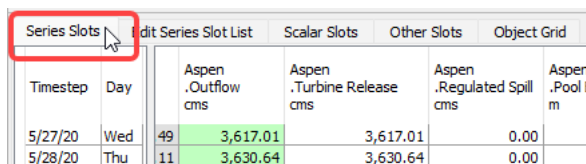
Topics

- [Change the timestep orientation](#), page 98
- [Adjust column widths](#), page 99
- [Display summary information and statistics](#), page 102
- [Display flag letters and priority numbers](#), page 103
- [Manage display units](#), page 104
- [Display flow and volume values in alternate units](#), page 107
- [Display and edit series slot notes](#), page 109
- [Apply custom colors to timestep values](#), page 117

Change the timestep orientation

Use this procedure to switch between horizontal and vertical timestep orientation on the Series Slots tab. Depending on the tasks you are performing and the number of slots and timesteps in the SCT, you may find one orientation to be more useful than the other, and you can easily switch between them at any time.

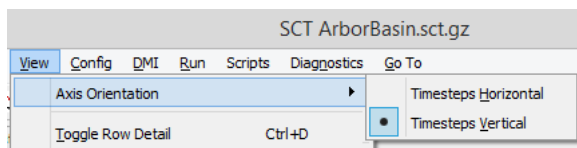
1. Go to the **Series Slots** tab.



Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

2. Change the orientation, as follows:

- To display horizontal timesteps (running horizontally across the page):
 - On the SCT menu, select **View**, then **Axis Orientation**, then **Timesteps Horizontal**.
- To display vertical timesteps (running vertically, from top to bottom of the page):
 - On the SCT menu, select **View**, then **Axis Orientation**, then **Timesteps Vertical**.



- Use the toolbar Swap Axis button.



Related topics

- [Timestep orientation](#), page 16

Adjust column widths

Use any of the following procedures to resize all columns on the current tab; other tabs are not affected.

On the Series Slots tab, this procedure applies to both horizontal and vertical timestep orientation, and only the current sheet is affected.

Many, but not all, column width adjustments are preserved when you save the SCT.

Note: You can also resize individual columns on each tab by dragging the dividers between the column headers.

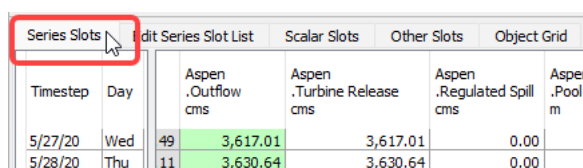
Depending on the setting in the SCT configuration, column widths may be automatically resized at various times. See [“Automatically Resize Column Widths to Fit Data”](#) on page 287 for information.

Resize all data columns to the same width

Use this procedure to resize all columns to the width of a selected column.

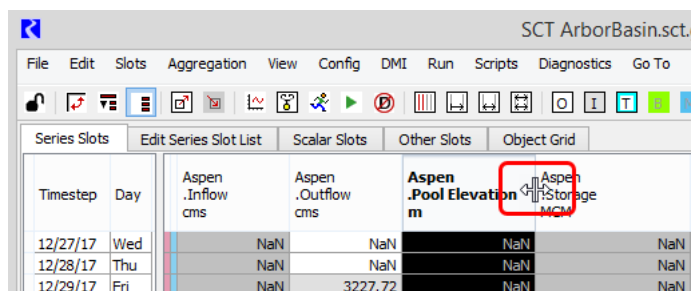
Note: This procedure applies to the Series Slots tab only.

1. Go to the **Series Slots** tab.



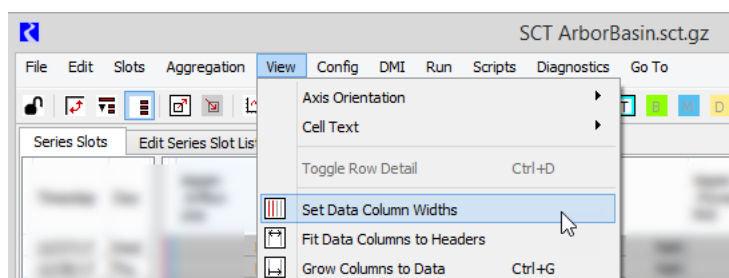
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool Elevation m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

2. Select a column and then drag its column header divider so the column is the width you want.



Timestep	Day	Aspen .Inflow cms	Aspen .Outflow cms	Aspen .Pool Elevation m	Aspen .Storage MCM
12/27/17	Wed	NaN	NaN	NaN	NaN
12/28/17	Thu	NaN	NaN	NaN	NaN
12/29/17	Fri	NaN	3227.72	NaN	NaN

3. In the SCT menu, select **View**, then **Set Data Column Widths**. Alternatively, select the  button on the toolbar.



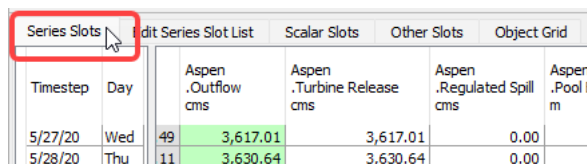
All columns are resized to the same width as the selected column.

Resize all columns to fit their contents

Use this procedure to resize all columns to fit their data, column headers, or both.

Note: This procedure applies to the Series Slots, Edit Series Slot List, Scalar Slots, and Other Slots tabs.

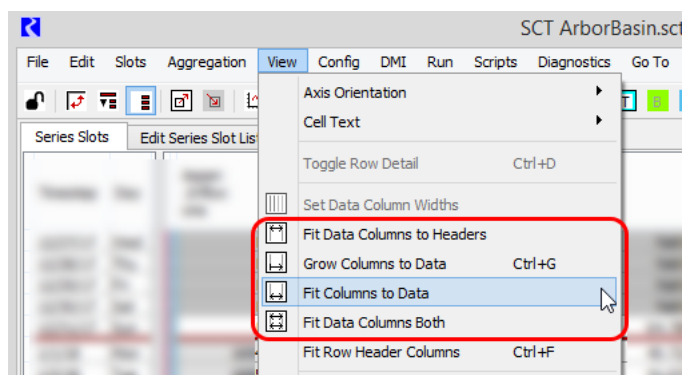
1. Go to the SCT tab you want to adjust.



Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

2. Select **View**, then one of the following options. Alternatively, select the indicated button on the toolbar.

- **Fit Data Columns to Headers**—resizes each column to fit the text in its column header. Columns are grown or shrunk as needed.
- **Grow Columns to Data**  —resizes each column to be at least as wide as its widest data value. Columns are grown only; columns wider than their widest data value are not changed. Keyboard shortcut is Ctrl+G.
- **Fit Columns to Data**  —resizes each column to the same width as its widest data value. Columns are grown or shrunk as needed.
- **Fit Data Columns Both**  —resizes each column to fit its widest header or data value. Columns are grown or shrunk as needed.



The columns are resized as necessary.

Resize all row header columns

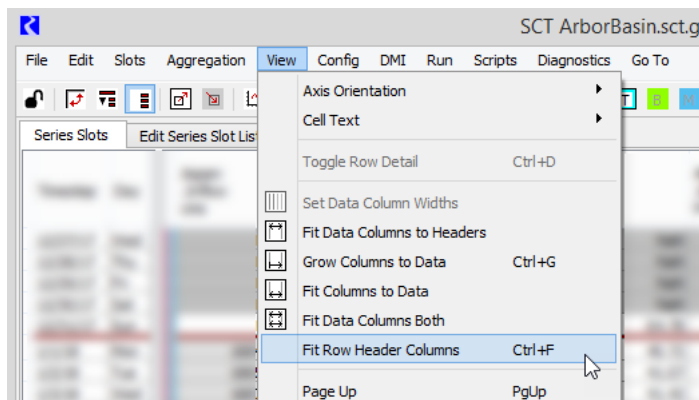
Use this procedure to resize the columns in the row header table to fit the column headers and cell content. The splitter between the row header table and the SCT data table is also moved to fit the row header table exactly.

Note: This procedure applies to the Series Slots tab only.

1. Go to the **Series Slots** tab.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. In the SCT menu, select **View**, then **Fit Row Header Column**.



Each row header column is resized to fit its widest header or cell content. Columns are grown or shrunk as needed. The splitter between the row header table and the SCT data table is moved to fit the row header table exactly.

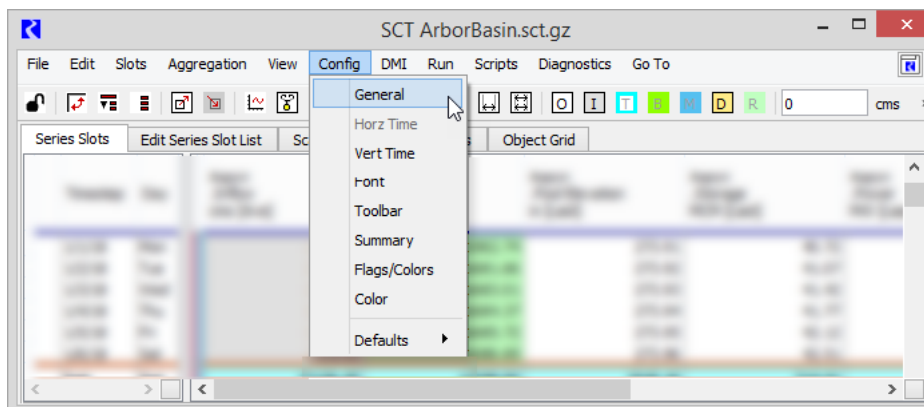
Related topics

- [SCT Layout](#), page 6

Configure Automatic Width Resizing

Use this procedure to configure whether to automatically resize column widths to fit the data.

1. On the SCT menu, select **Config**, then **General**.

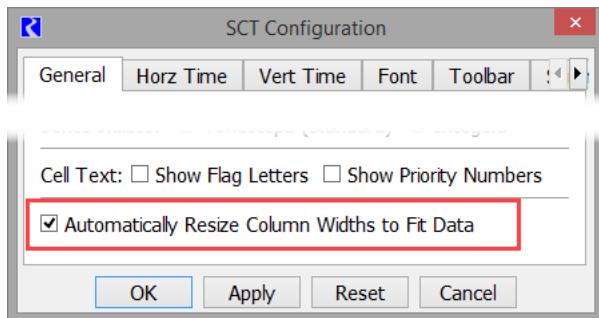


The SCT Configuration window opens at the General tab.

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Using the SCT

2. Select or clear the **Automatically Resize Column Widths to Fit Data** check box, as follows:
 - Select to fit the column widths to the data during certain operations, like stopping a run. In most cases, the columns will only grow, but some operations cause the column width to shrink.
 - Clear if you want the specified column widths to be preserved.



3. Select **Apply** or **OK** to apply the change to the SCT.
The SCT display is immediately updated.

Related topics

- [Adjust column widths](#), page 99
- “[SCT View menu](#)” on page 267 for resizing options.
- [Automatically Resize Column Widths to Fit Data](#), page 287

Display summary information and statistics

Use this procedure to display basic statistics or summary information for selected cells. The following information is displayed:

- Name or number of the selected slots
- Units and scale
- Number of values in the selection
- Sum, average, median, min, max, and range of the selected values

Note: If you select multiple slots, the SCT computes summary information only if the units are compatible.

1. In the SCT, go to the **Series Slots** tab.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. Select one or more cells. You can select one cell, multiple contiguous cells, or multiple non-contiguous cells.

The cells are highlighted and the summary information for these cells is displayed in the Status Area at the bottom of the SCT.

1/5/18	Fri		1685.72			
1/6/18	Sat		1686.69			
1/7/18	Sun		1687.97			
1/8/18	Mon		1689.32			
1/9/18	Tue		1690.68			
1/10/18	Wed		1692.03			
1/11/18	Thu		1693.39			
1/12/18	Fri		1694.74			

Aspen Birch Cedar Dogwood Dogwood Irrigation Elm Eastern Irrigation Transbasin

Aspen.Outflow -- Total Volume: 438.22 [MCM]
3 values: Sum 5072.04 -- Ave 1690.68 -- Med 1690.68 -- Min 1689.32 -- Max 1692.03 -- Range 2.71 [cms]

Related topics

- [SCT Layout](#), page 6

Display flag letters and priority numbers

Use this procedure to display slot value flags and rule priority numbers for all cells on the Series Slots tab.

1. Go to the **Series Slots** tab.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. On the SCT menu, select **Config**, then **General**

SCT Configuration

General Horz Time Vert Time Font Toolbar

General Settings

☐ Always Synchronize Range with Slots

Pre-Run Timesteps: 5

Post-Run Timesteps: 0

Run Control Timestep: 1 Day

☐ Timestep Size Override 5 Minute

☒ Show Run Status in SCT

☒ Display "NaN"

☒ Show Grid

☒ Crosshatch Read Only Cells

☐ Double Click Data Cell Toggles Detail Rows

☒ Full Precision Editing

☐ Auto-Evaluate Expression Slots on Edits

Series indices: ☒ Timesteps (standard) ☐ Integers

Cell Text: ☐ Show Flag Letters ☐ Show Priority Numbers

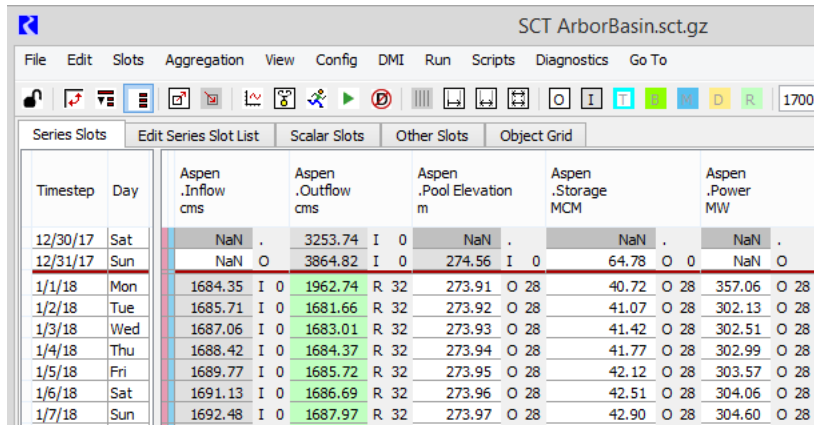
OK Apply Reset Cancel

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Using the SCT

3. In the **Cell Text** area, select one or both of the following selections:
- **Show Flag Letters**: show the flag letter(s) for the cell value (I, O, R, for example). This is useful when threshold alert colors override the flag colors.
 - **Show Priority Numbers**: show the rule priority for the cell value. This is useful for rulebased simulation models.

The display is updated for all series slot cells on the SCT. For example, if both items are shown:



The screenshot shows the SCT ArborBasin.sct.gz interface. The table displays data for Aspen Inflow, Outflow, Pool Elevation, Storage MCM, and Power MW over time. The table has columns for Timestep, Day, and five data series. Each data series has a value, a flag letter (I, O, R), and a priority number (32, 28, or 0). The table is filtered to show data from 12/30/17 to 1/7/18.

Timestep	Day	Aspen .Inflow cms	Aspen .Outflow cms	Aspen .Pool Elevation m	Aspen .Storage MCM	Aspen .Power MW
12/30/17	Sat	NaN	3253.74	NaN	NaN	NaN
12/31/17	Sun	NaN	3864.82	274.56	64.78	NaN
1/1/18	Mon	1684.35	1962.74	273.91	40.72	357.06
1/2/18	Tue	1685.71	1681.66	273.92	41.07	302.13
1/3/18	Wed	1687.06	1683.01	273.93	41.42	302.51
1/4/18	Thu	1688.42	1684.37	273.94	41.77	302.99
1/5/18	Fri	1689.77	1685.72	273.95	42.12	303.57
1/6/18	Sat	1691.13	1686.69	273.96	42.51	304.06
1/7/18	Sun	1692.48	1687.97	273.97	42.90	304.60

Related topics

- [Flags and cell colors](#), page 26

Manage display units

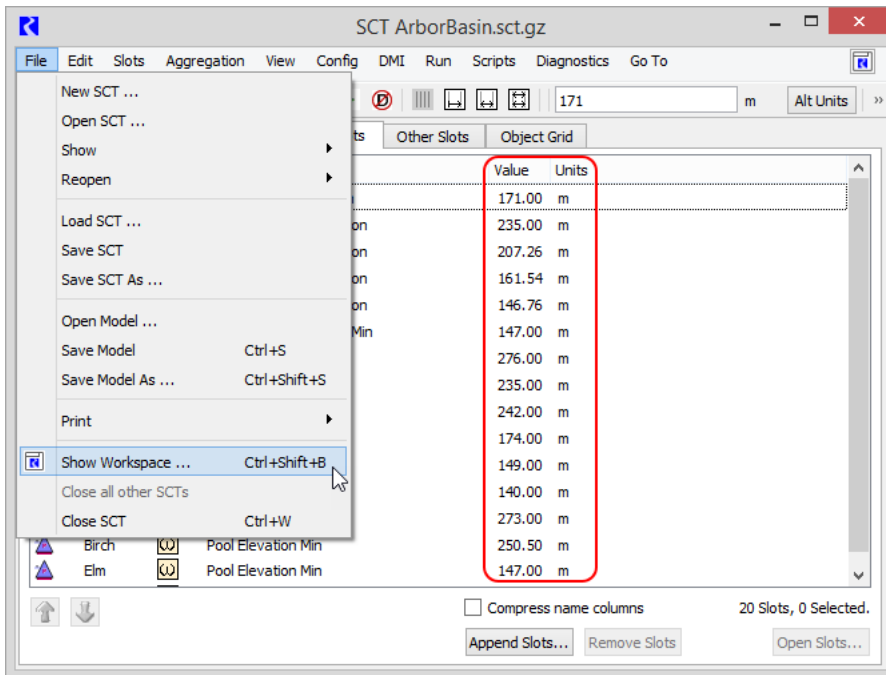
The slot values displayed in the SCT use the units defined by the unit scheme for each slot. Unit schemes define the display units, scale, format, and precision for each unit type. Exceptions for individual slots can also be configured. You cannot change the slot display units directly through the SCT, but you can use any of the following procedures to modify the display.

- [Modify display units by switching the unit scheme](#), page 104
- [Modify display units by editing the slot configuration](#), page 106
- [Display flow and volume values in alternate units](#), page 107

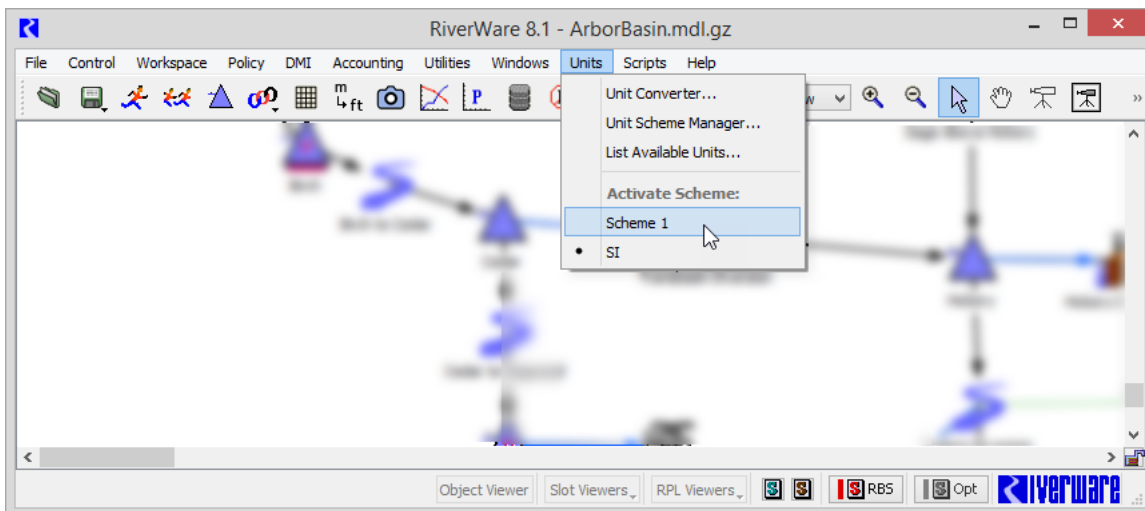
Modify display units by switching the unit scheme

Use this procedure to change the units for displayed values by activating an existing alternate unit scheme. Before using this procedure, at least one alternate scheme must be defined; see [“Units” in User Interface](#) for details.

1. In the SCT, select **File**, then **Show Workspace**.



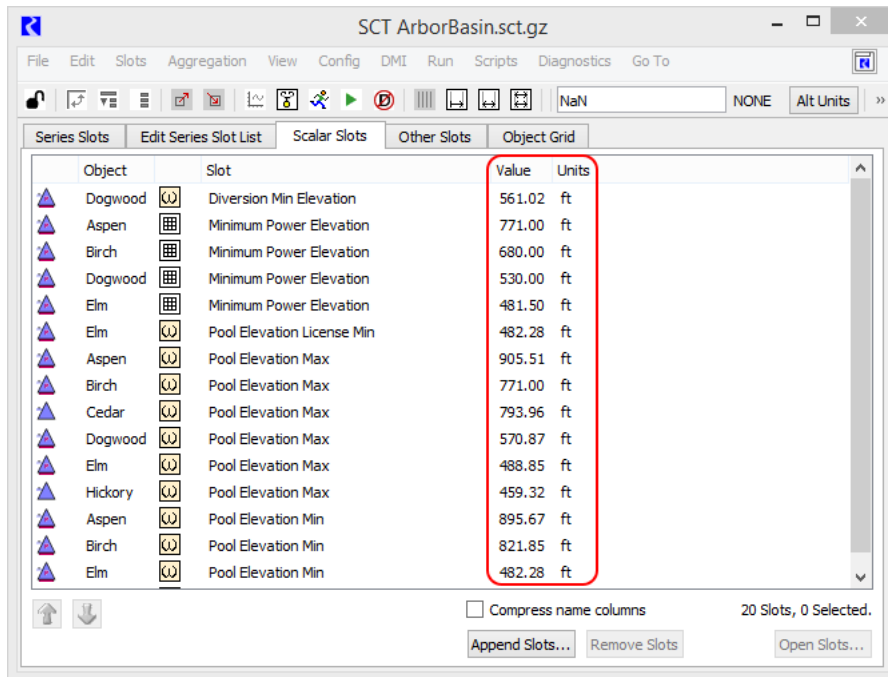
2. In the Workspace menu, select **Units**, then in the Activate Scheme section, select the scheme to use.



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All slot values are updated to the new unit scheme.



Object	Slot	Value	Units
Dogwood	Diversion Min Elevation	561.02	ft
Aspen	Minimum Power Elevation	771.00	ft
Birch	Minimum Power Elevation	680.00	ft
Dogwood	Minimum Power Elevation	530.00	ft
Elm	Minimum Power Elevation	481.50	ft
Elm	Pool Elevation License Min	482.28	ft
Aspen	Pool Elevation Max	905.51	ft
Birch	Pool Elevation Max	771.00	ft
Cedar	Pool Elevation Max	793.96	ft
Dogwood	Pool Elevation Max	570.87	ft
Elm	Pool Elevation Max	488.85	ft
Hickory	Pool Elevation Max	459.32	ft
Aspen	Pool Elevation Min	895.67	ft
Birch	Pool Elevation Min	821.85	ft
Elm	Pool Elevation Min	482.28	ft

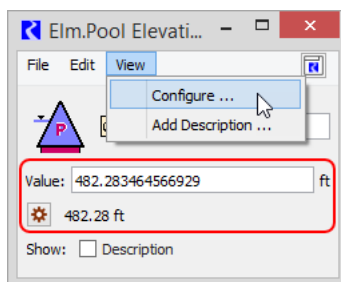
Related topics

- [“Units” in User Interface](#)

Modify display units by editing the slot configuration

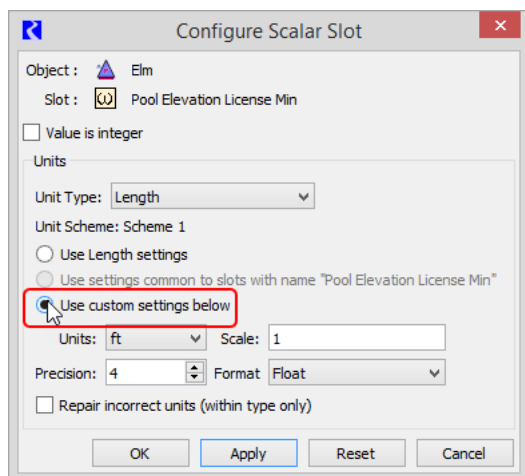
Use this procedure to change how the values for a selected slot are displayed by editing the slot configuration to define a new unit scheme exception for the slot.

1. Open a slot in the SCT; see [Open series slots](#), page 209 or [Open other slots](#), page 211 for details.
2. In the Slot Viewer or Slot dialog box, select **View**, then **Configure**.

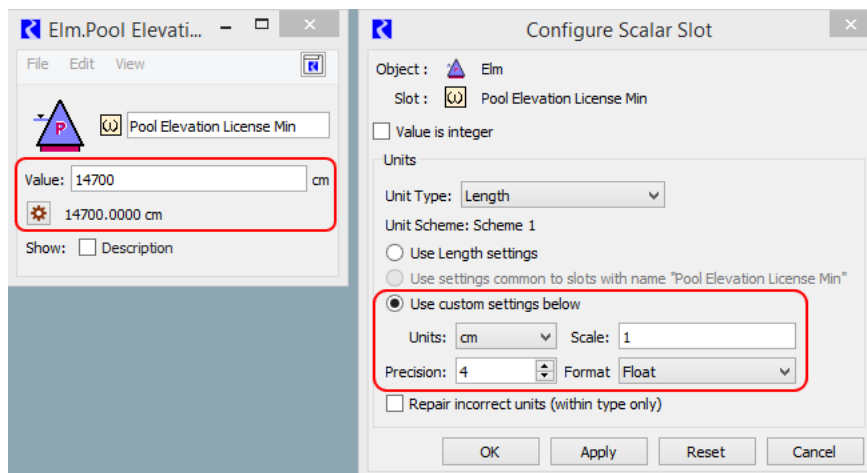


The Configure Slot dialog box opens.

3. Select the **Use custom settings below** option.



4. Modify the settings for the four unit scheme attributes—Units, Scale, Precision, and Format—then select **OK**.
The values for the selected slot are updated using the new units.



Related topics

- [“Unit Schemes” in User Interface](#)
- [Open series slots](#), page 209
- [Open other slots](#), page 211

Display flow and volume values in alternate units

Use this procedure to specify how values for series slots that use Flow and Volume units should be displayed.

The units for both the default and alternate units are defined by the unit scheme. The unit type applies unless an exception has been created for the slot. Exceptions can be created for both Flow and Volume unit types for a particular slot.

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1. Go to the **Series Slots** tab.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. On the SCT toolbar, select **Alt Units**, then select one of the following options:

- **Standard**—displays Flow slots as flow and Volume slots as volume, with no conversion.
- **Flow**—displays Flow and Volume slots as flow, as appropriate. Some Volume slots can be shown as flow.
- **Volume**—displays Flow and Volume slots as volume. All Flow slots on the SCT can be displayed as volume over the timestep using the default unit scheme definition for volume; see “[Slots](#)” in *User Interface* for more information.

The screenshot shows the SCT ArborBasin.sct.gz application window. The 'Series Slots' tab is selected. The 'Alt Units' dropdown menu is open, showing three options: 'Standard - Display flows and volumes without conversion' (selected), 'Flow - Display volumes as flows (as appropriate)', and 'Volume - Display Flows as Volumes'. The main data table displays various flow and volume slots for different timesteps and days.

Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	Aspen .Power MW [Last]
1/6/18	Sat	1687.74	2035.57	273.96	42.51	304.06
12/31/17	Sun	NaN	3864.82	274.56	64.78	NaN
1/1/18	Mon	1684.35	1962.74	273.91	40.72	357.06
1/2/18	Tue	1685.71	1681.66	273.92	41.07	302.13
1/3/18	Wed	1687.06	1683.01	273.93	41.42	302.51
1/4/18	Thu	1688.42	1684.37	273.94	41.77	302.99
1/5/18	Fri	1689.77	1685.72	273.95	42.12	303.57
1/6/18	Sat	1691.13	1686.69	273.96	42.51	304.06
Sum	Sun	10126.45	14249.00	1918.16	314.41	1872.32
Max	Sun	1691.13	3864.82	274.56	64.78	357.06
Pool Elevat	Sun	NaN	NaN	273.96	NaN	NaN
1/8/18	Mon	1693.84	1689.37	273.98	43.29	305.28

The display is updated as appropriate.

Note: Slots shown in alternate units are marked with an asterisk next to the unit in the slot header, if included in the display.

Timestep	Day	Aspen .Inflow MCM* [Last]	Aspen .Outflow MCM* [Last]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	Aspen .Power MW [Last]	Aspen .Regulated Spill MCM* [Last]
1/6/18	Sat	146.11	145.73	273.96	42.51	304.06	
12/31/17	Sun	NaN	333.92	274.56	64.78	357.06	NaN
1/1/18	Mon	145.53	169.58	273.91	40.72	302.13	
1/2/18	Tue	145.65	145.30	273.92	41.07	302.51	
1/3/18	Wed	145.76	145.41	273.93	41.42	302.99	
1/4/18	Thu	145.88	145.53	273.94	41.77	303.57	
1/5/18	Fri	146.00	145.65	273.95	42.12	304.06	
1/6/18	Sat	146.11	145.73	273.96	42.51	304.06	
Sum	Sun	874.93	1231.11	1918.16	314.41	1872.32	
Max	Sun	146.11	333.92	274.56	64.78	357.06	
Pool Elevat	Sun	NaN	NaN	273.96	NaN	NaN	NaN
1/8/18	Mon	146.35	145.96	273.98	43.29	305.28	

Related topics

- “Unit Schemes” in *User Interface*

Display and edit series slot notes

Use this procedure to display and edit series slots notes. Notes are short text strings associated with a series slot at a timestep.

Note: This procedure is a summary of actions you can perform on notes from the SCT. See “Notes on Series Slots” in *User Interface* for details.

1. Go to the **Series Slots** tab.

Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool Elevation m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

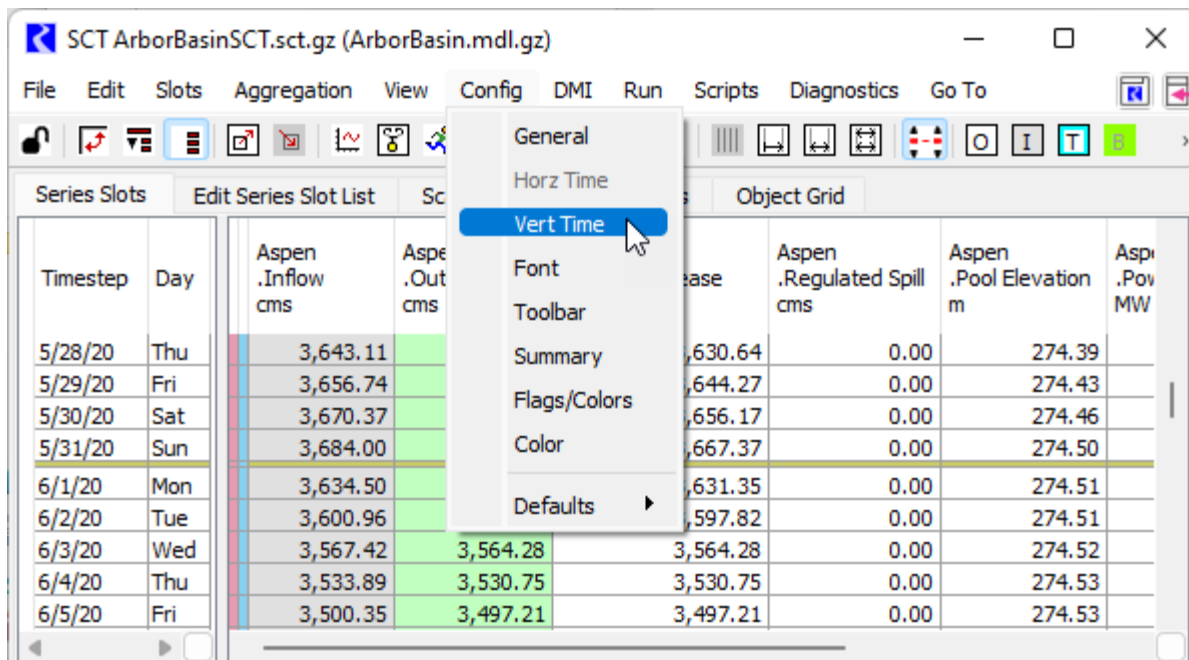
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Each note is indicated by a Note icon in the corresponding cell.

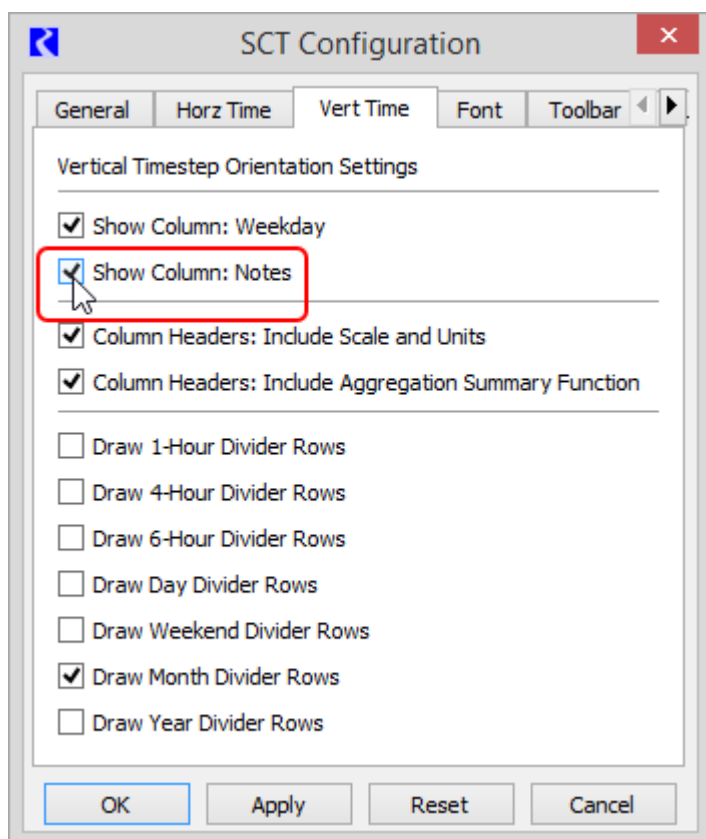
Series Slots		Edit Series Slot List	Scalar Slots		Other Slots	Object Grid
Timestep	Day	Aspen .Inflow cms	Aspen .Out cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool m
5/28/20	Thu	11	3,630.64	3,630.64	0.00	
5/29/20	Fri	74	3,644.27	3,644.27	0.00	
5/30/20	Sat	37	3,656.17	3,656.17	0.00	
5/31/20	Sun	00	3,667.37	3,667.37	0.00	
6/1/20	Mon	50	3,631.35	3,631.35	0.00	
6/2/20	Tue	96	3,597.82	3,597.82	0.00	
6/3/20	Wed	42	3,564.28	3,564.28	0.00	

- Hover over a note icon to see the note text in a tooltip.
 - Click on the cell with the note to see the note text in the status bar at the bottom of the SCT.
- Optionally, in vertical timestep orientation, you can add the Note column, which displays the text of all notes at each timestep. Proceed as follows:
 - Switch to vertical timestep orientation, if necessary.
 - In the SCT menu, select **Config**, then **Vert Time**.



The SCT Configuration window opens at the Vert Time tab.

- c. Select the **Show Column: Notes** checkbox, then select **Apply** or **OK**.



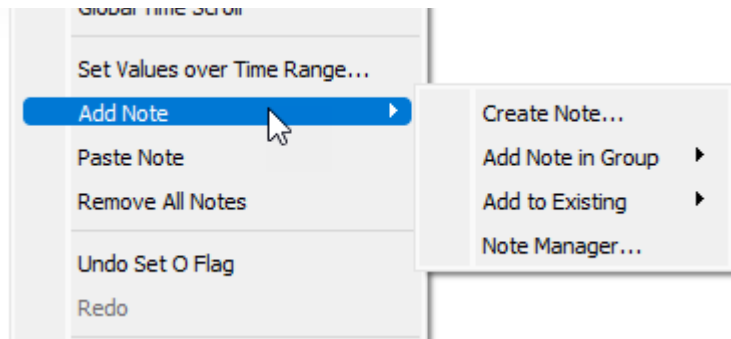
The Note column is added to the row header. The text of all notes is displayed in this column.

Timestep	Day	Note	Aspen .Inflow cms	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms
5/28/20	Thu		3,643.11	3,630.64	3,630.64	0.00
5/29/20	Fri		3,656.74	3,644.27	3,644.27	0.00
5/30/20	Sat	Max Capacity	3,670.37	3,656.17	3,656.17	0.00
5/31/20	Sun	Max Capacity	3,684.00	3,667.37	3,667.37	0.00
6/1/20	Mon		3,634.50	3,631.35	3,631.35	0.00
6/2/20	Tue		3,600.96	3,597.82	3,597.82	0.00
6/3/20	Wed		3,567.42	3,564.28	3,564.28	0.00
6/4/20	Thu		3,533.89	3,530.75	3,530.75	0.00
6/5/20	Fri		3,500.35	3,497.21	3,497.21	0.00

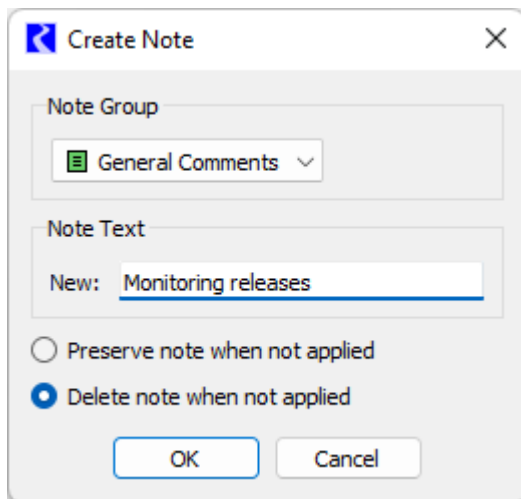
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3. To add a new note, proceed as follows:
 - a. Select any cells with values you want to annotate.
 - b. Right-click and move the cursor over **Add Note** in the shortcut menu. Four options will appear.

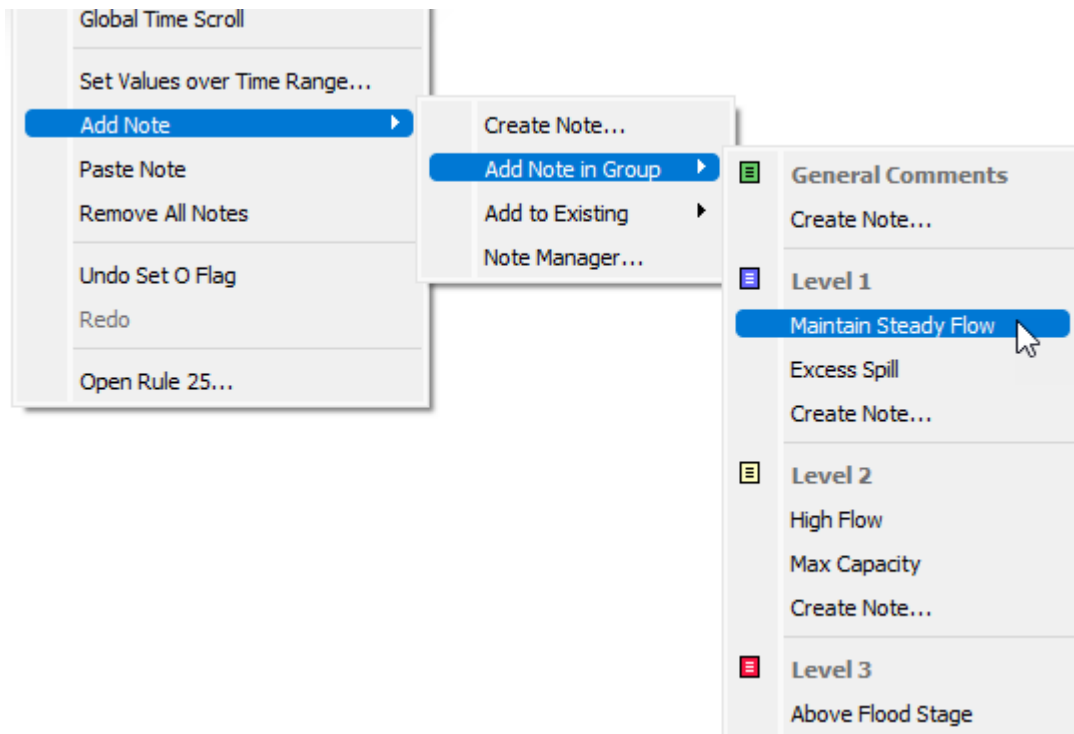


- **Create Note**—opens the Create Note dialog; you can type the desired note text, and select the Note Group to which you want to assign the note. Note Groups are configured in the Note Manager. See [“Notes on Series Slots” in User Interface](#) for details. If no Note Groups have been configured, a Default Note Group will be available. When the Create Note dialog is opened from the SCT, the option to **Delete note when not applied** will be selected by default. If you want the note to remain in the Note Manager even if it is no longer applied for any note instances, so that it can be selected for later use, select the option for **Preserve note when not applied**. When you select **OK**, the note is added on the selected timesteps, and the note gets added to the Note Manager.



- **Add Note in Group**—allows you to select from notes already configured in the Note Manager (see [“Notes on Series Slots” in User Interface](#) for details), organized by Note Group. You can also choose to

create a new note within the specified group. When you select a note, it is immediately applied on the selected timesteps.

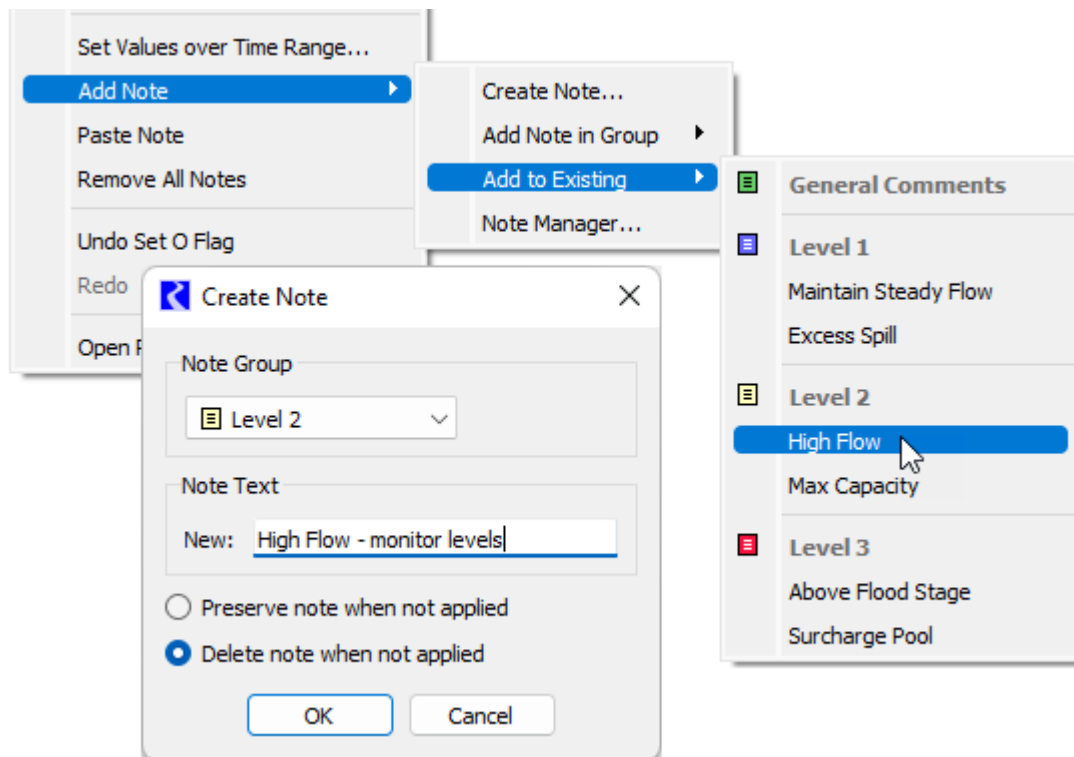


- **Add to Existing**—allows you to select from notes already configured in the Note Manager (see “[Notes on Series Slots](#)” in *User Interface* for details), but instead of applying the selected note directly, it opens the Create Note dialog to allow you to add to or modify the selected note. When the Create Note dialog is opened from the SCT, the option to **Delete note when not applied** will be selected by default. If you want the note to remain in the Note Manager even if it is no longer applied for any note instances, so that it can be selected for later use, select the option for **Preserve note when not applied**. When you select

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OK, the note is added on the selected timesteps, and the note gets added to the Note Manager as a new note.



- **Note Manager**—opens the Note Manager window. From there you can add or edit notes and apply them to specified slots. See “Notes on Series Slots” in *User Interface* for details.

The Note Manager window is divided into three main sections: Note Groups, Notes, and Note Instances.

Note Groups: This section shows a list of note groups: General Comments, Level 1, Level 2, and Level 3. Level 1 is currently selected. Buttons for 'Create Group', 'Edit Selected Group', 'Delete Selected', 'Import', 'Export Selected', and 'Select/Deselect All' are available.

Notes: This section shows a list of notes: 'Excess Spill' and 'Maintain Steady Flow', both associated with 'Level 1' and set to 'Preserve note when not applied'. Buttons for 'Create Note', 'Edit Selected Note', 'Delete Selected Notes', and 'Select/Deselect All' are available.

Note Instances: This section displays a table of note instances. The table has columns for Slot, Timestep, Note, Group, Slot Value, and Timestamp. The 'Maintain Steady Flow' note is applied to 'Birch.Outflow' at 24:00 on Jan 4 through Jan 9, 2020, with a slot value of 1,700.00 cms. A checkbox 'Delete note instances outside the slot time series range' is checked. At the bottom, there is a field 'Add Entry Using Context:' with a value of 'Aspen.Outflow / 24:00 Jun 5 2020'.

Slot	Timestep	Note	Group	Slot Value	Timestamp
Birch.Outflow	24:00 Jan 4 2020	Maintain Steady Flow	Level 1	1,700.00 cms	11:48:04 Oct 27 2023
Birch.Outflow	24:00 Jan 5 2020	Maintain Steady Flow	Level 1	1,700.00 cms	11:48:04 Oct 27 2023
Birch.Outflow	24:00 Jan 6 2020	Maintain Steady Flow	Level 1	1,700.00 cms	11:48:04 Oct 27 2023
Birch.Outflow	24:00 Jan 7 2020	Maintain Steady Flow	Level 1	1,700.00 cms	11:48:04 Oct 27 2023
Birch.Outflow	24:00 Jan 8 2020	Maintain Steady Flow	Level 1	1,700.00 cms	11:48:04 Oct 27 2023
Birch.Outflow	24:00 Jan 9 2020	Maintain Steady Flow	Level 1	1,700.00 cms	11:48:04 Oct 27 2023

When a note is added a Note icon appears in the cell.

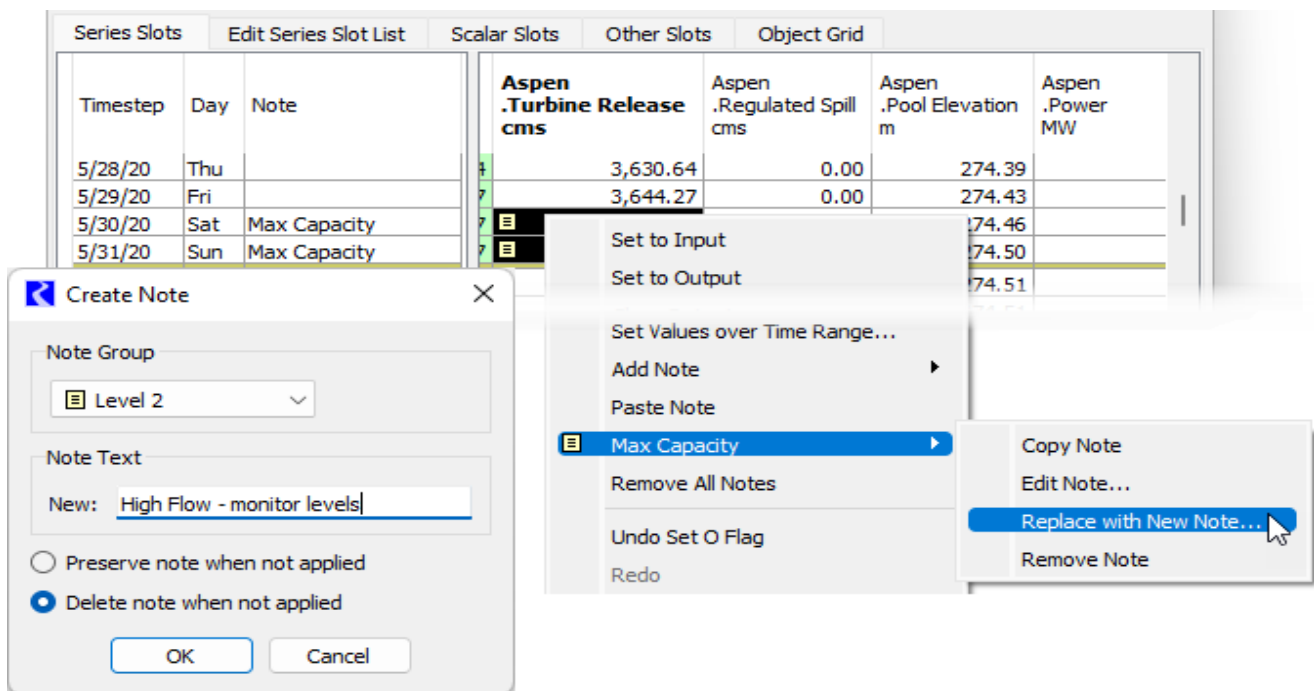
- To edit an existing note, proceed as follows:
 - Select the cells where the Note icon appears.

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Note: For a multiple selection, all of the selected cells must have the same notes applied for the note editing options to be available.

- b. Right-click a cell in the selection, select the note in the shortcut menu, and then select one of the following options:
 - **Copy Note**—copies the note to the RiverWare clipboard; you can paste the note multiple times to other slots.
 - **Edit Note**—opens the Note Manager window, allowing you to edit the note.
 - **Replace with New Note**—opens the Create Note dialog to allow you to add to or modify the selected note. When you select **OK**, the modified note replaces the note on the selected timesteps, and the note gets added to the Note Manager as a new note.



- **Remove Note**—immediately removes the note from the slot on the selected timesteps. No confirmation message is displayed.
5. To remove all notes from a selection, proceed as follows:
 - a. Select the cells from which you want to remove all notes.
 - b. Right-click a cell in the selection. Then in the shortcut menu, select **Remove All Notes**.
 - c. A confirmation dialog will appear asking if you want to remove all notes. Select **Yes**. The notes will be removed.

Related topics

- [“Notes on Series Slots” in User Interface](#)

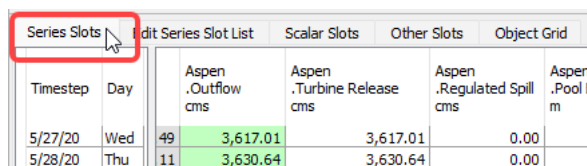
Apply custom colors to timestep values

Use this procedure to manually highlight selected cells with custom colors. You can save these highlights as part of the SCT.

Note: Before using this procedure, you must create and label custom colors to use; see “[Define custom colors for highlights and color alerts](#)” on page 191 for instructions.

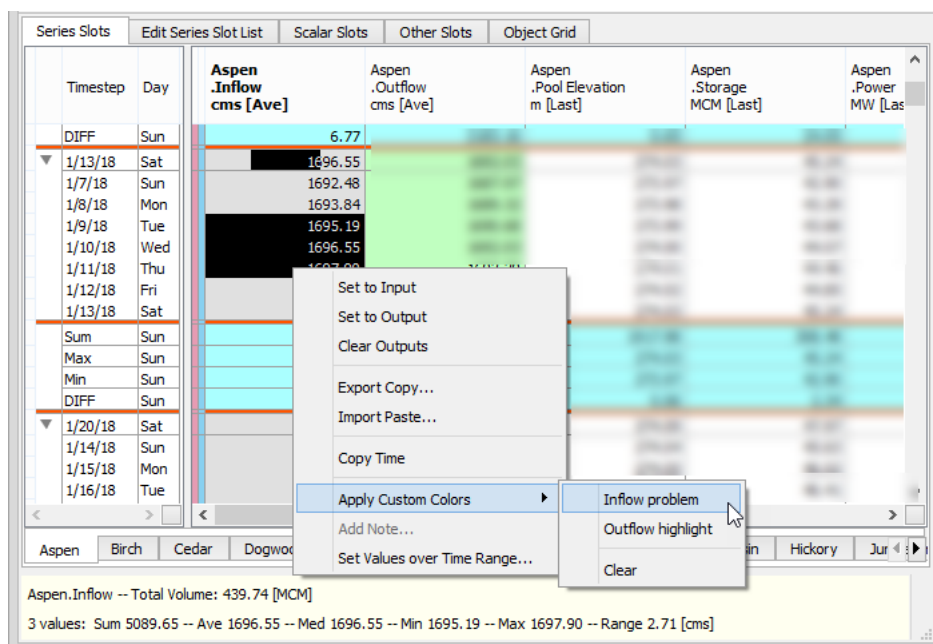
Note: Custom SCT colors can be applied manually to any set of selected cells, but Color Alerts have precedence over manually assigned colors; see “[Configuring Color Alerts](#)” on page 150 for details.

1. In the SCT, go to the **Series Slots** tab.



Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. Select the cells you want to highlight.
3. Right-click, and in the shortcut menu, select **Apply Custom Colors**, then select the desired user-defined color.



The colors are applied to the text and background as specified in the definition.

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Note: When SCT cells are selected, the foreground and background colors of the numeric portion of the cells are reversed, so initially, the colors appear reversed. Move the cursor to another area of the screen to view the selected colors.

Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	Aspen .Power MW [Last]
DIFF	Sun	6.77				
1/13/18	Sat	1696.55				
1/7/18	Sun	1692.48				
1/8/18	Mon	1693.84				
1/9/18	Tue	1695.19				
1/10/18	Wed	1696.55				
1/11/18	Thu	1697.90				
1/12/18	Fri	1699.26				
1/13/18	Sat	1700.61				
Sum	Sun	11875.84				
Max	Sun	1700.61				
Min	Sun	1692.48				
DIFF	Sun	8.13				

Note: The cell / custom color associations are stored in the SCT, not in the RiverWare model. So the SCT must be saved to retain these color associations.

Note: When cells are highlighted with custom colors, the custom text and background colors are used instead of the standard value flag colors. You can hover over a cell to display a tooltip showing the flag information and custom color label or threshold alert interval name.

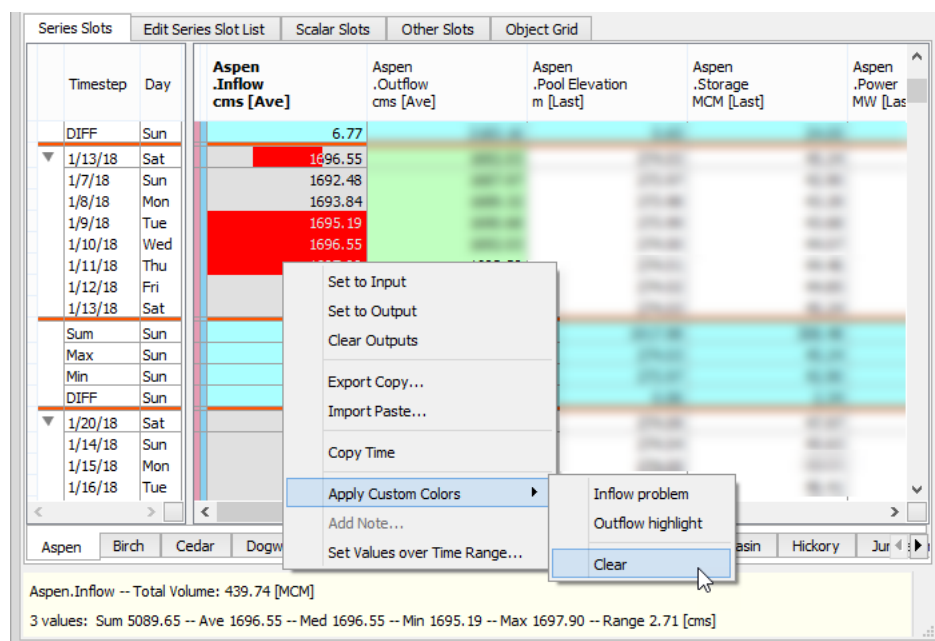
Navajo Reservoir .Inflow cfs	Navajo Reservoir .Outflow cfs	Navajo Reservoir .Storage acre-ft	Navajo Reservoir .Pool Elevation ft
589.19	R 12 450.00	O 6 967,840.93	O 6 6,025.89
633.56	R 12 400.00	O 6 967,524.73	O 6 6,025.85
662.70	R 12 400.00	O 6 967,256.09	O 6 6,025.83
659.34	R 12 400.00	O 6 966,989.38	O 6 6,025.80
643.78	R 12 400.00	O 6 966,701.21	O 6 6,025.77
655.93	R 12 400.00	O 6 966,503.85	O 6 6,025.75
651.13	R 12 400.00	O 6 966,268.69	O 6 6,025.72
640.12	R 12 400.00	O 6 966,079.30	O 6 6,025.70
668.64	R 12 400.00	O 6 965,930.10	O 6 6,025.69
624.23	R 12 400.00	O 6 965,640.21	O 6 6,025.66
611.99	R 12 400.00	O 6 965,254.51	O 6 6,025.62
624.58	R 12 400.00	O 6 964,885.59	O 6 6,025.60
627.88	R 12 400.00	O 6 964,625.83	O 6 6,025.57
648.63	R 12 400.00	O 6 964,485.59	O 6 6,025.54
647.33	R 12 450.00	O 6 964,121.37	O 6 6,025.50
554.98	R 12 450.00	O 6 963,806.19	O 6 6,025.47

4. If you want to save the highlights, you must save the SCT.

5. To clear the custom colors, select the cells again.

Note: To clear all custom cell colors for one or more slots on the SCT, select the slot name header to select all the cells for that slot.

6. Right-click, and in the shortcut menu, select **Apply Custom Colors**, then **Clear**.



The selected cells are returned to the colors associated with the series flag.

Related topics

- [Configure Color Alerts](#), page 150

Navigating an SCT

The procedures in this section provide instructions for locating specific slots and moving through the SCT data tables. See “[SCT Layout](#)” on page 6 and “[SCT Tab Details](#)” on page 278 for details about the SCT tabs.

Tip: See also “[Synchronized Scrolling](#)” on page 18 for information keeping series displays synchronized when scrolling.

Topics

- [Moving the cursor through timestep values](#), page 119
- [Go to a series slot sheet](#), page 120
- [Go to a slot group divider](#), page 121
- [Search for a series slot by name](#), page 122

Moving the cursor through timestep values

You can use the following methods to move through timestep cells on the Series Slot tab.

Chapter 3

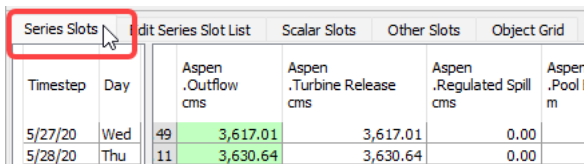
Using the SCT

- Press Enter to select the next cell in the timeseries. This applies to both vertical and horizontal timestep orientations.
- Press Tab to move to the next slot at the same timestep. This applies to both vertical and horizontal timestep orientations.
- Use the keyboard arrow keys to move from cell to cell in the indicated direction, as you would in a spreadsheet.
- Enter a new date in the date time spinner in the tool bar to go to that date.

Go to a series slot sheet

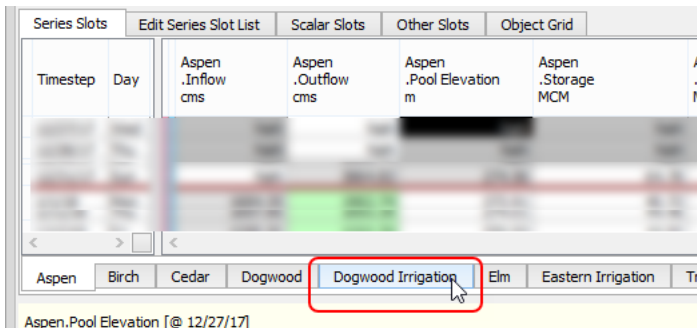
Use this procedure to go directly to a series slot sheet or the All Slots sheet. The All Slots sheet displays all slots in the SCT.

1. Go to the **Series Slots** tab.



Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool Elevation m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

2. Use one of the following methods to go to a sheet tab.
 - Select the slot sheet name from the sheet tabs below the data table.



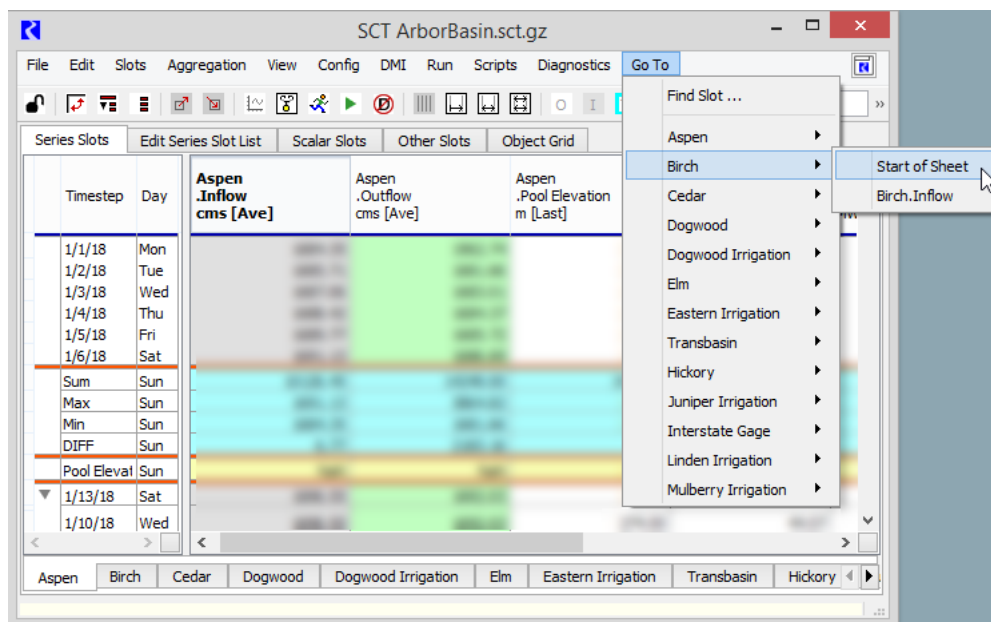
Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Inflow cms	Aspen .Outflow cms	Aspen .Pool Elevation m	Aspen .Storage MCM
[Blurred data rows]					

Aspen Birch Cedar Dogwood **Dogwood Irrigation** Elm Eastern Irrigation Tri

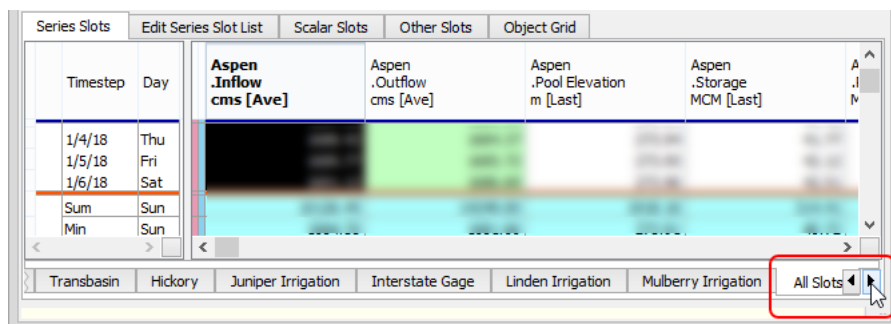
Aspen.Pool Elevation [@ 12/27/17]

- In the SCT menu, select **Go To**, then the sheet name, then **Start of Sheet**, if applicable.

Note: The **Start of Sheet** option is available if there is at least one slot group divider on the sheet.



- To go to the All Slots sheet, select the right arrow in the sheet tabs area to scroll to the end of the sheet list. Then select the **All Slots** tab.



The selected sheet is displayed.

Related topics

- [Add sheet dividers](#), page 56

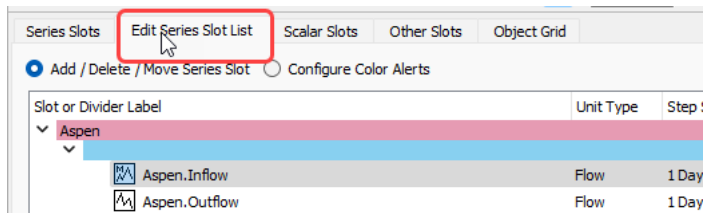
Go to a slot group divider

Use this procedure to go directly to a slot divider on a sheet. This is useful if the SCT has many sheets and slot groups.

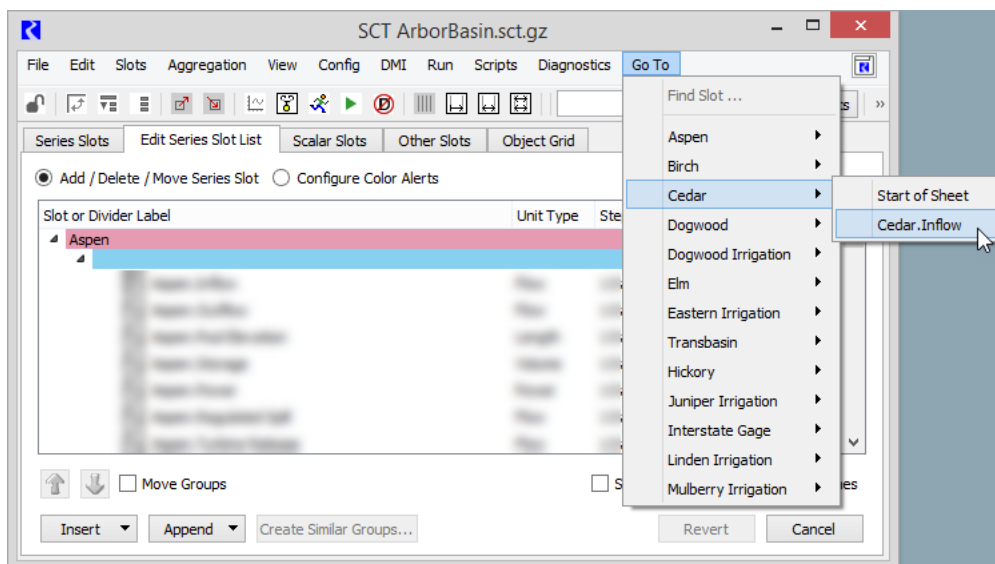
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1. Go to the **Series Slots** tab or the **Edit Series Slot List** tab.



2. In the SCT menu, select **Go To**, then the sheet name, then one of the following options, as applicable:
 - **Slot divider name**—available for named slot dividers. Goes to the selected divider.
 - **Slot name**—available for unnamed slot group dividers. Goes to the first slot in the group.



The divider is brought to the focus of the display.

Related topics

- [Add slot dividers](#), page 54

Search for a series slot by name

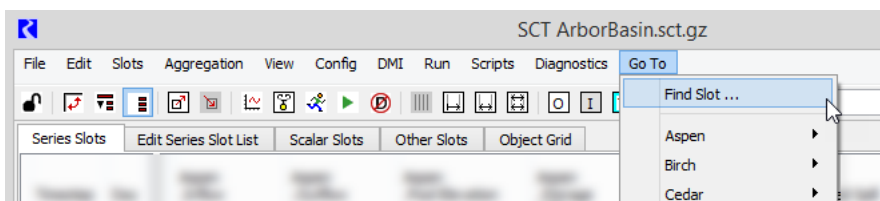
Use this procedure to go directly to a series slot on the Series Slots tab. This is useful if the SCT has many slots.

1. Go to the **Series Slots** tab.

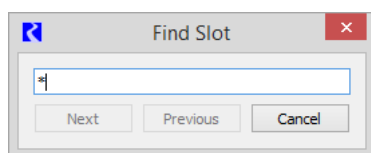
The screenshot shows the 'Series Slots' tab in the SCT interface. The 'Aspen' slot is expanded, showing two sub-slots: 'Aspen.Outflow' and 'Aspen.Turbine Release'. The 'Unit Type' for both is 'cms' and the 'Step' is '1 Day'.

Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

2. On the SCT menu, select **Go To**, then **Find Slot**.



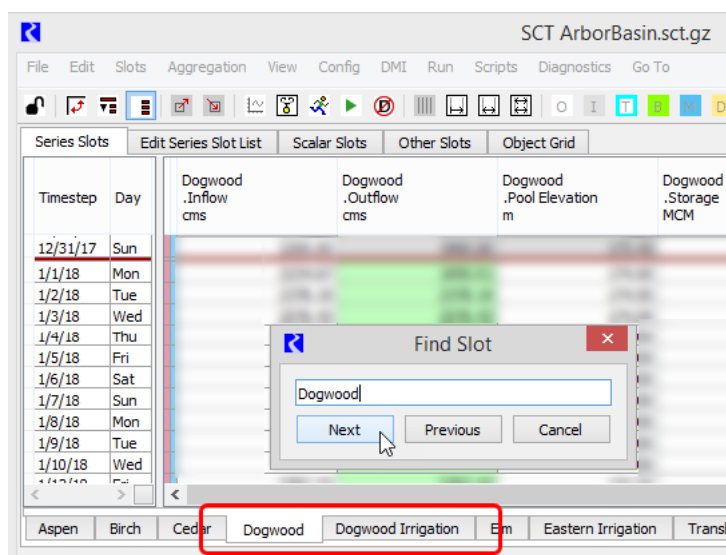
The Find Slot dialog box opens with the **Next** and **Previous** buttons deactivated.



3. In the text field, delete the asterisk (*) and enter the object or slot name you want to locate.

If there is at least one matching item, the **Next** and **Previous** buttons are made available. You can continue entering text to refine the search as much as you want.

4. Select **Next** and **Previous** to step through the matching slots, one at a time.



Related topics

- [Add series slots to an SCT](#), page 46
- [Customize a slot label](#), page 51

Displaying Aggregated Values

The procedures in this section are used to aggregate series slot values up to a larger timestep size. Displaying aggregations can help you to analyze data in different ways, especially in models with small timesteps.

Slot summary functions are stored with the SCT configuration when you save the SCT.

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Note: The procedures in this section apply to series slots only.

Note: Unless noted otherwise, you must unlock the SCT configuration to perform the procedures in this section. See [“Lock or unlock the SCT Configuration”](#) on page 164 for details.

Topics

- [Supported summary functions](#), page 124
- [Process for defining aggregations](#), page 125
- [Define the aggregation interval for the SCT](#), page 125
- [Enable or disable aggregated view](#), page 127
- [Define summary functions by unit type](#), page 129
- [Define summary functions by slot name](#), page 131
- [Define a slot-specific summary function](#), page 134
- [Restore the default summary functions to all slots](#), page 136
- [Create custom summary rows for all slots](#), page 137
- [Open a referenced slot in the Slot Viewer](#), page 140
- [Hide a custom summary row value](#), page 142
- [Restore display of hidden custom summary row values](#), page 143

Supported summary functions

The SCT supports the following functions for timestep aggregations.

Note: For the arithmetic functions, undefined values (NaNs) are ignored.

- 1st—displays the first value in the aggregation interval.
- Ave—displays the average of all timesteps in the aggregation interval.
- Divider—displays a thin visual divider, which can have a user-defined background color.
- Last—displays the last value in the aggregation interval.
- Max—displays the maximum value in the aggregation interval.
- Med—displays the median value in the aggregation interval. Since computing the median requires the timestep values to be sorted, this can slow the speed of SCT refreshes. If there is an even number of non-NaN values, the median is defined as the mean of the middle two values.
- Min—displays the minimum value in the aggregation interval.
- Difference—displays the difference of the values in the previous two intervals. Available for custom summary rows only.
- None—does not perform an aggregation; the summary cell for each aggregation interval is blank.
- Nth (particular n; for example, 5th)—displays the nth value in the aggregation interval.

- Slot Reference—available for custom summary rows only. Displays the last value in a referenced slot. A value is displayed only if the referenced slot uses the same units as the current slot.
- Sum—displays the sum of all values in the aggregation interval.

Related topics

- [Aggregation](#), page 21

Process for defining aggregations

The general process for defining aggregations is as follows:

1. Unlock the SCT configuration; see “[Lock or unlock the SCT Configuration](#)” on page 164 for details.
2. [Define the aggregation interval for the SCT](#), page 125
3. [Enable or disable aggregated view](#), page 127
4. Perform any combination of the following to define summary functions for most slots.
 - [Define summary functions by unit type](#), page 129
 - [Define summary functions by slot name](#), page 131
5. [Define a slot-specific summary function](#), page 134 to refine individual slots to use different summary functions.
6. Optionally, [Create custom summary rows for all slots](#), page 137 to display additional types of aggregations.

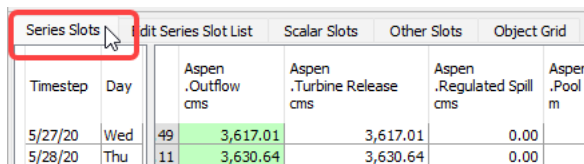
Define the aggregation interval for the SCT

Use this procedure to define the aggregation interval used in the SCT.

You can choose to aggregate a fixed number of timesteps per summary (for example, three timesteps), or you can aggregate values into some regular interval (for example, weekly, monthly, and so on). You can also specify the first timestep in each aggregation period.

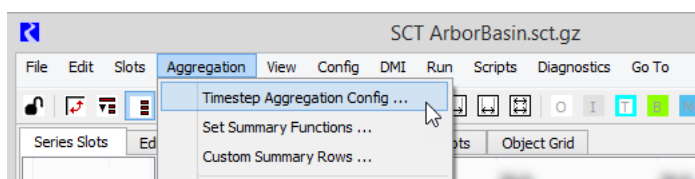
Across the SCT, values for the specified interval period are aggregated according to the summary function defined for each slot.

1. Go to the **Series Slots** tab.



Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

2. On the SCT menu, select **Aggregation**, then **Timestep Aggregation Config**.



Chapter 3

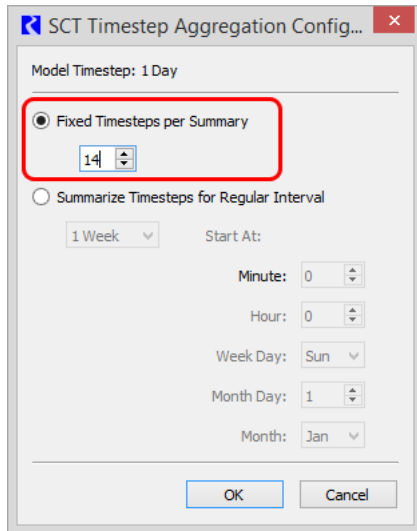
Using the SCT

The SCT Timestep Aggregation Configuration dialog opens.

3. Proceed as follows, depending on the aggregation interval you want to use.

– To aggregate values by a fixed number of timesteps, use the following steps:

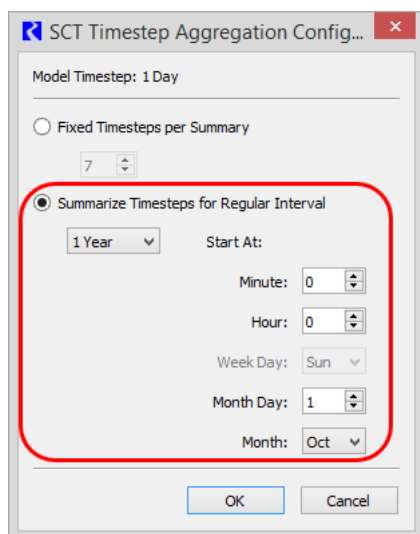
1. Select the **Fixed Timesteps per Summary** option.
2. In the entry field, enter the number of timesteps in the aggregation interval.



– To aggregate values by a regular time period (weekly, monthly, quarterly, or annually), use the following steps:

1. Select the **Summarize Timesteps for Regular Interval** option.
2. In the menu, select the interval to use. Options are 1Week, 1Month, 3Months, 1Year.
3. In the **Start At** fields, specify the start of the first interval. The available fields are determined by the interval option you select in the menu. For example, if you select the 3Months interval, you can specify the time (minute and hour), day of the month, and month when the first interval begins; subsequent intervals will start three, six, nine months, and so on, from then. If you select the 1Week interval, you can

specify the time (minute and hour) and day of the week when the first interval begins; subsequent intervals will start every week at the same time and day.



4. Select **OK** to apply the updates.

The dialog box closes, and the new summary interval is applied across the SCT. To display the aggregations, you must enable aggregated view.

Related topics

- [Enable or disable aggregated view](#), page 127

Enable or disable aggregated view

Use this procedure to specify whether to display series slot data in aggregated or non-aggregated view.

- In *non-aggregated* view, each data cell displays the value of exactly one timestep, and no values are aggregated.
- In *aggregated* view, timestep values are summarized by the aggregation interval defined for the model, and the summary values are displayed. You can also select whether to display detailed values for each timestep in the aggregation interval. This view is useful for analyzing model data in different ways, especially in models with small timesteps.

Note: The aggregation function can vary by slot.

Note: You do *not* need to unlock the SCT configuration to perform this procedure.

1. Go to the **Series Slots** tab.

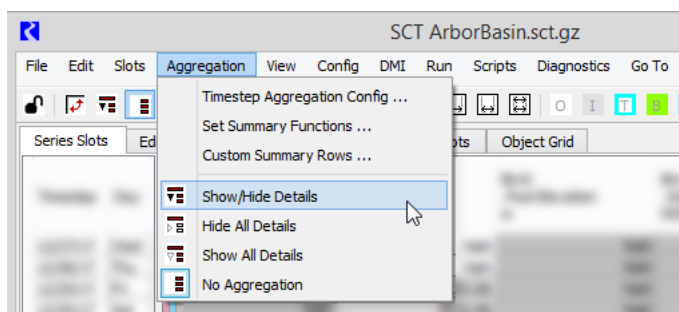
Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. On the SCT menu, select **Aggregation**, then one of the following options:

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- **Show/Hide Details**—displays summary and detail values in a dynamic expand/collapse format. Summary values for each aggregation interval are displayed by default, and you can selectively show and hide timestep detail for each interval using the following methods:
 - Select the arrow icon next to the summary row header.
 - Select one or more summary row headers and press Ctrl+D.
 - Double-click a cell. This applies *only* if the Double Click Data Cell Toggles Detail Rows option is enabled in the SCT general configuration. See “[General tab](#)” on page 285.
- **Hide All Details**—displays summary values only; the display is locked, so you cannot expand aggregation intervals to display timestep details.
- **Show All Details**—displays summary and detail values for each aggregation interval; the display is locked, so you cannot collapse aggregation intervals to hide timestep details.
- **No Aggregation**—displays timestep detail only; values are not aggregated.



The display is updated according to your selection.

Note: The date label for each summary cell is the last day in the aggregation period. For example, if you have defined monthly aggregations starting on the first day of the month, each interval ends with the last day of the month, and the summary cell is labeled with that date: Jan 31, Feb 28 or 29, Mar 31, April 30, for example. If the first or last aggregation period is not complete, the last date in the aggregation is displayed.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid					
Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	Aspen .Power MW [Last]	Aspen .Regulated Spill cms [Ave]			
▶ 12/30/17	Sat	NaN	3240.73	NaN	NaN	NaN	NaN			
▼ 1/6/18	Sat	1687.74	2035.57	273.96	42.51	304.06				
12/31/17	Sun	NaN	3864.82	274.56	64.78	NaN				
1/1/18	Mon	1684.35	1962.74	273.91	40.72	357.06				
1/2/18	Tue	1685.71	1681.66	273.92	41.07	302.13				
1/3/18	Wed	1687.06	1683.01	273.93	41.42	302.51				
1/4/18	Thu	1688.42	1684.37	273.94	41.77	302.99				
1/5/18	Fri	1689.77	1685.72	273.95	42.12	303.57				
1/6/18	Sat	1691.13	1686.69	273.96	42.51	304.06				
▶ 1/13/18	Sat	1696.55	1692.03	274.03	45.24	308.87				
▶ 1/20/18	Sat	1706.03	1701.52	274.09	47.97	311.91				
▶ 1/27/18	Sat	1715.52	1711.00	274.16	50.70	312.60				
▶ 2/3/18	Sat	1728.40	1725.82	274.20	52.26	322.16				
▶ 2/10/18	Sat	1762.25	1762.25	274.20	52.26	343.34				
▶ 2/17/18	Sat	1762.25	1762.25	274.20	52.26	343.34				
Aspen	Birch	Cedar	Dogwood	Dogwood Irrigation	Elm	Eastern Irrigation	Transbasin	Hickory	Juniper Irrigation	Interstate Gage

Related topics

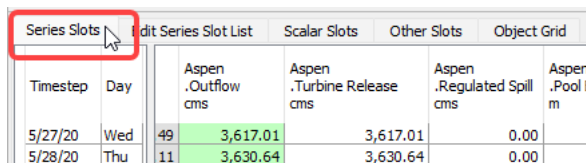
- [Define the aggregation interval for the SCT](#), page 125
- [General tab](#), page 285

Define summary functions by unit type

Use this procedure to define the summary function used to aggregate slots with a specified unit type. For example, you could specify to show the Minimum value for slots that use the Flow unit, and the Average value for slots that use Volume and slots that use Length. You can create as many definitions as you want. The resulting values are displayed in the summary cells for each aggregation interval.

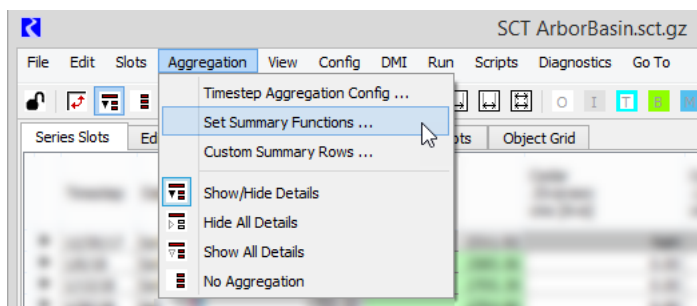
This procedure is useful when you want most slots to use the same function; you can update many slots at once, and then refine the assignments by updating individual slots to use a different function, as needed.

1. Go to the **Series Slot** tab.



Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day		Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

2. On the SCT menu, select **Aggregation**, then **Set Summary Functions**.



The Set SCT Aggregation Summary Functions dialog box opens, displaying the current summary function definitions for the SCT. Each function definition row specifies the function used to aggregate slots with the specified unit type or slot name. You can define any number of function definition rows, in any combination of definitions by unit type or by slot name.

Note: To define summary functions by slot name, see [“Define summary functions by slot name”](#) on page 131.

3. Specify the function definitions, as follows:

- a. Optionally add a row by selecting **Plus**  or delete a row by selecting **Minus** .

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- b. For each row, select the **Unit Type** option. In the menu, select the unit type to which you want to assign a summary function.

Set SCT Aggregation Summary Functions

Specify summary functions for SCT slot items based on Unit Type or Slot Name. The first row matching a particular slot is used for that slot.

Unit Type	Slot Name*	Summary Function	Slots
Flow		Ave	138
Flow		Last	6
Volume		Last	6
Length		Last	0
Area		Last	8

*Slot: AreaPerTime slot's name containing the specified slot name text.

Restore RiverWare Default Cancel Changes Apply to SCT Close

- c. In the **Summary Function** menu for that row, select the summary function you want to assign to that unit type. All slots with the specified unit type will be aggregated using this function.

Set SCT Aggregation Summary Functions

Specify summary functions for SCT slot items based on Unit Type or Slot Name. The first row matching a particular slot is used for that slot.

Unit Type	Slot Name*	Summary Function	Slots
Flow		Sum	138
Volume		1st	6
Length		Last	6
Time		Sum	0
		Ave	8

*Slot name matches are based on the slot's name containing the specified slot name.

Restore RiverWare Default Cancel Changes Apply to SCT Close

The number of slots that match the specified criteria is displayed in the Slots column.

Note: If a slot matches multiple definition criteria, only the first definition listed in the dialog box will be used. For example, if a row specifies Flow should use Average, and a later row specifies Flow should use Minimum, all slots with the Flow type will use the first definition listed, which is Average.

4. When you are finished specifying function definitions, select **Apply to SCT**.

The new functions are applied to all slots, across all objects, that meet the defined criteria. The slot headers are updated with the new function. If aggregated view is enabled in the SCT, the summary cells for the slots are updated to display the new values.

Timestep	Day	Cedar .Inflow cms [Sum]	Cedar .Outflow cms [Sum]	Cedar .Diversion cms [Sum]	Cedar .Unregulated Spill cms [Sum]	Cedar .Pool Elevation m [Last]	Cedar .Storage MCM [Last]
12/30/17	Sat	3196.71	9247.68	NaN	NaN	NaN	
1/6/18	Sat	17734.46	18098.92	0.00	0.00	239.65	4506
1/13/18	Sat	11900.00	18937.72	0.00	0.00	237.57	3896
1/20/18	Sat	11900.00	18934.27	0.00	0.00	235.06	3290
1/27/18	Sat	11900.00	18880.46	0.00	0.00	232.58	2681
2/3/18	Sat	12125.00	15214.96	0.00	0.00	231.39	2420
2/10/18	Sat	12600.00	10729.28	0.00	0.00	232.14	2581
2/17/18	Sat	12600.00	10783.30	0.00	0.00	232.79	2735
2/24/18	Sat	12600.00	10688.26	0.00	0.00	233.47	2904

- Optionally, select **Close** to close the dialog box.
- Save the SCT to save the updates.

Related topics

- [Supported summary functions](#), page 124
- [Define summary functions by slot name](#), page 131
- [Define a slot-specific summary function](#), page 134
- [Enable or disable aggregated view](#), page 127

Define summary functions by slot name

Use this procedure to define the summary function used to aggregate slots whose names match a specified string. For example, you could specify to show the Maximum value for all slots whose names include the string “Irrigation.Diversion”. The change would apply to all slots, across all objects, with names that match this string, including “Irrigation.Diversion”, “Irrigation.Diversion Requested”, “Irrigation.Diversion Shortage”, and so on.

You can create as many definitions as you want. The resulting values are displayed in the summary cells for each aggregation interval.

This procedure is useful when you want most slots to use the same function; you can update many slots at once, and then refine the assignments by updating individual slots to use a different function, as needed.

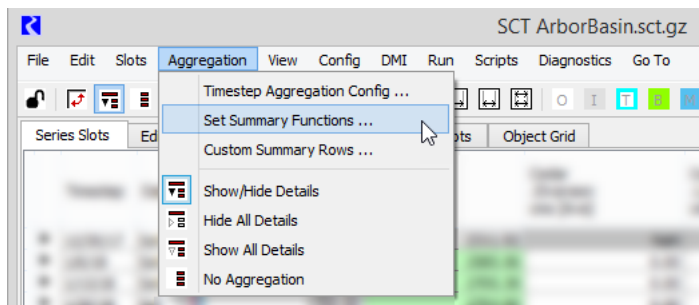
- Go to the **Series Slot** tab.

Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool Elevation m
49	5/27/20 Wed	3,617.01	3,617.01	0.00	
11	5/28/20 Thu	3,630.64	3,630.64	0.00	

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Using the SCT

- On the SCT menu, select **Aggregation**, then **Set Summary Functions**.

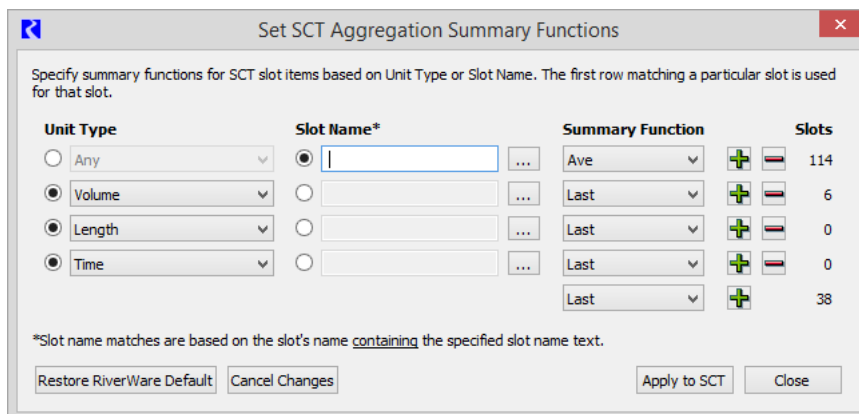


The Set SCT Aggregation Summary Functions dialog box opens, displaying the current summary function definitions for the SCT. Each function definition row specifies the function used to aggregate slots with the specified unit type or slot name. You can define any number of function definition rows, in any combination of definitions by unit type or by slot name.

Note: To define summary functions by unit type, see “[Define summary functions by unit type](#)” on page 129.

- Complete the dialog box as follows.

- Optionally add a row by selecting **Plus**  or delete a row by selecting **Minus** .
- For each row, select the **Slot Name*** option.



- Enter all or part of a slot name directly into the **Slot Name*** entry field. Alternatively, select the slot name from the Slot Selector, as follows:

- Select the **Chooser**  button.

The Slot Selector dialog box opens, listing all the objects and slots included in the SCT.

- In the Slot Selector dialog box, select the slot name you want to specify for the summary function.
- Select **Ok** to close the dialog box.

The selected slot name is entered into the **Slot Name*** field.

- d. In the **Summary Function** menu for that row, select the summary function you want to assign. All slots with names matching the string specified in the **Slot Name*** field will be aggregated using this function.

The number of slots that match the specified criteria is displayed in the Slots column.

Note: If a slot matches multiple definition criteria, only the first definition listed in the dialog box will be used. For example, if a row specifies that slots matching “Pool Elevation” should use Average, and a later row specifies the same slots should use Minimum, all slots matching “Pool Elevation” will use the first definition listed, which is Average.



Set SCT Aggregation Summary Functions

Specify summary functions for SCT slot items based on Unit Type or Slot Name. The first row matching a particular slot is used for that slot.

Unit Type	Slot Name*	Summary Function	Slots
☐ Any	☒ Diversion	Ave	40
☒ Volume	☐	Last	6
☒ Length	☐	Last	6
☒ Time	☐	Last	0
		Last	106

*Slot name matches are based on the slot's name containing the specified slot name text.

4. When you are finished defining the definitions, select **Apply to SCT**.

The new functions are applied to all slots, across all objects, that meet the defined criteria. The slot headers are updated with the new function. If aggregated view is enabled in the SCT, the summary cells for the slots are updated to display the new values.

Series Slots		Edit Series Slot List		Scalar Slots	Other Slots	Object Grid					
Timestep	Day	Cedar .Inflow cms [Last]	Cedar .Outflow cms [Last]	Cedar .Diversion cms [Ave]	Cedar .Unregulated Spill cms [Last]	Cedar .Pool Elevation m [Last]	Cedar .Storage MCM [Last]				
▶ 12/30/17	Sat	3196.71	2200.00	NaN	NaN	NaN					
▶ 1/6/18	Sat	1700.00	2704.34	0.00	0.00	239.65	450.00				
▶ 1/13/18	Sat	1700.00	2705.81	0.00	0.00	237.57	389.00				
▶ 1/20/18	Sat	1700.00	2703.41	0.00	0.00	235.06	329.00				
▶ 1/27/18	Sat	1700.00	2689.48	0.00	0.00	232.58	268.00				
▶ 2/3/18	Sat	1800.00	1524.42	0.00	0.00	231.39	242.00				
▶ 2/10/18	Sat	1800.00	1537.92	0.00	0.00	232.14	258.00				
▶ 2/17/18	Sat	1800.00	1540.01	0.00	0.00	232.79	273.00				
▶ 2/24/18	Sat	1800.00	1507.95	0.00	0.00	233.47	290.00				
Aspen		Birch	Cedar	Dogwood	Dogwood Irrigation	Elm	Eastern Irrigation	Transbasin	Hickory	Juniper Irrigation	Interstate Gate

5. Optionally, select **Close** to close the dialog box.
6. Save the SCT to save the updates.

Related topics

- [Supported summary functions](#), page 124
- [Define summary functions by unit type](#), page 129
- [Define a slot-specific summary function](#), page 134
- [Enable or disable aggregated view](#), page 127

Define a slot-specific summary function

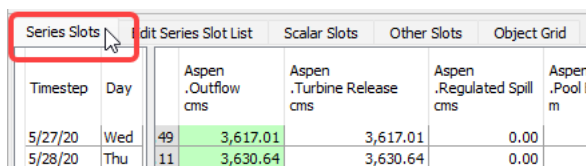
Use this procedure to define the summary function for an individual slot. For example, you can choose to show the Average for one slot, the Minimum value for another slot, and the Last value for another.

Note: You may want to include the same slot multiple times in the SCT and define each instance to be aggregated using a different function. This allows you to view the average, minimum value, and last value for the same slot, for example.

The summary function is displayed in each slot header, in brackets; for example, [Ave], [Min], or [Last].

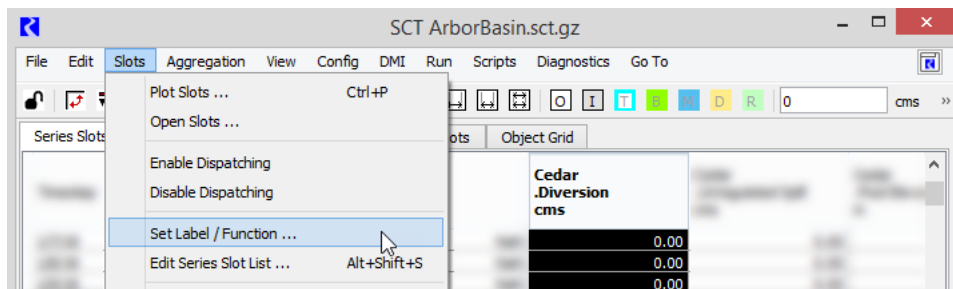
Slot summary functions are stored with the SCT configuration; therefore, you must save the SCT to save the updates.

1. Go to the **Series Slot** tab.

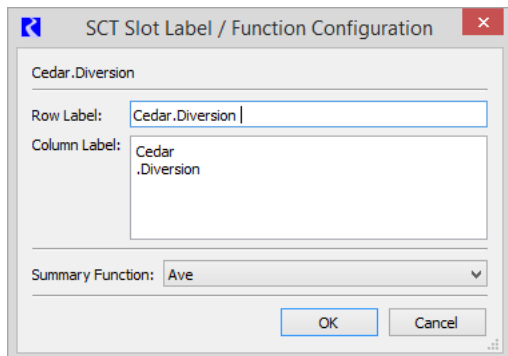


Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

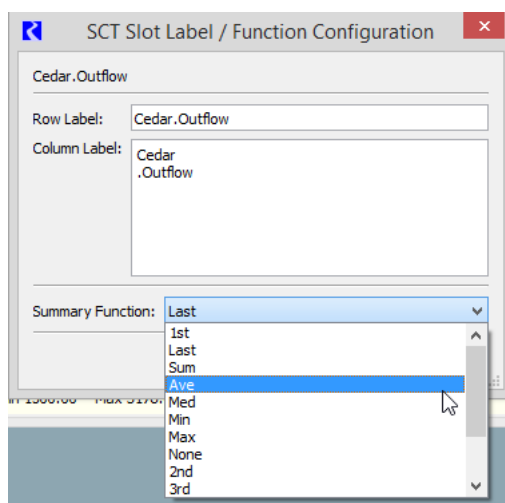
2. Select the header of the slot you want to update.
3. On the SCT menu, select **Slots**, then **Set Label / Function**.



The SCT Slot Label / Function Configuration dialog box opens.



- On the **Summary Function** menu, select the function to assign.



- Select **OK** to apply the change.

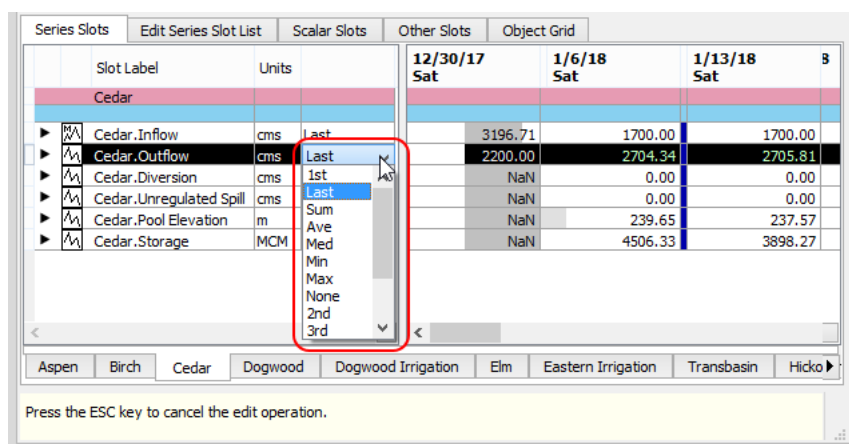
The dialog box closes. The new function is applied to the selected slot, and the function is included in the slot header. If aggregated view is enabled, the summary cells for the slot are updated to display the new values.

- Save the SCT to save the update.

Alternate method

In aggregated horizontal timestep view, if the Show Column: Aggregation Summary Function setting is enabled, you can use the following alternate method to select the summary function; see [“Configure horizontal timestep display”](#) on page 173 for details on enabling this column.

- Double-click the slot summary function to activate the summary function menu.
- On the menu, select the function to assign.



The new function is applied to the selected slot, and the function is included in the slot header. The summary cells for the slot are updated to display the new values.

Related topics

- [Supported summary functions](#), page 124
- [Define summary functions by unit type](#), page 129
- [Define summary functions by slot name](#), page 131
- [Enable or disable aggregated view](#), page 127

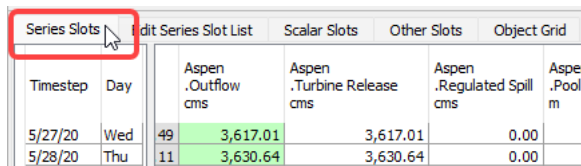
Restore the default summary functions to all slots

Use this procedure to restore all summary functions to the RiverWare defaults, which are based on unit type, as follows:

- Flow = Average
- Volume = Last
- Length = Last

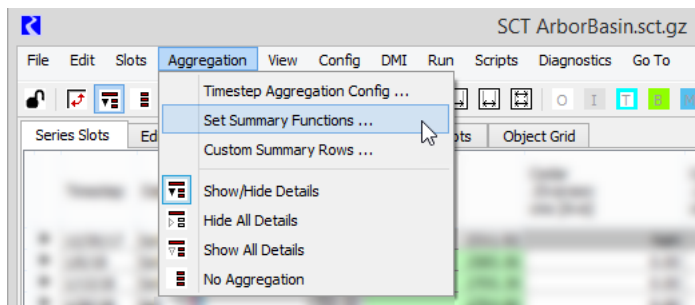
Caution: Using this procedure, all user-defined summary function definitions are removed from the SCT. Also, any slot-specific summary functions are overridden and returned to the default.

1. Go to the **Series Slot** tab.



Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. On the SCT menu, select **Aggregation**, then **Set Summary Functions**.



The Set SCT Aggregation Summary Functions dialog box opens, displaying the current summary function definitions for the SCT. Each function definition row specifies the function used to aggregate slots with the specified unit type or slot name.

3. Select **Restore RiverWare Default**.

Note: This button is available only if the SCT includes user-defined functions.

Specify summary functions for SCT slot items based on Unit Type or Slot Name. The first row matching a particular slot is used for that slot.

Unit Type	Slot Name*	Summary Function	Slots
☐ Any	☐ Diversion	Ave	40
☒ Volume	☐	Last	6
☒ Length	☐	Last	6
☒ Time	☐	Last	0
		Last	106

*Slot name matches are based on the slot's name containing the specified slot name text.

Restore RiverWare Default Cancel Changes Apply to SCT Close

The dialog box is updated with the default function definitions. All user-defined definition rows are deleted.

Specify summary functions for SCT slot items based on Unit Type or Slot Name. The first row matching a particular slot is used for that slot.

Unit Type	Slot Name*	Summary Function	Slots
☒ Flow	☐	Ave	138
☒ Volume	☐	Last	6
☒ Length	☐	Last	6
		Last	8

*Slot name matches are based on the slot's name containing the specified slot name text.

Restore RiverWare Default Cancel Changes Apply to SCT Close

4. Review the changes, and proceed as follows:

- Select **Apply to SCT** to apply the changes to the SCT, restoring the RiverWare defaults and removing any user-defined functions, including slot-specific functions.
- Select **Cancel Changes** to cancel the changes and restore the user-defined functions to the dialog box. Select **Close** to close the dialog box without making any changes to the SCT.

Related topics

- [Supported summary functions](#), page 124
- [Define summary functions by unit type](#), page 129
- [Define summary functions by slot name](#), page 131
- [Enable or disable aggregated view](#), page 127

Create custom summary rows for all slots

Use this procedure to create one or more custom summary rows for all slots in the SCT. You can assign a different summary function to each row, and you can optionally assign a user-defined color for each row background. You can create as many custom summary rows as you want.

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Using the SCT

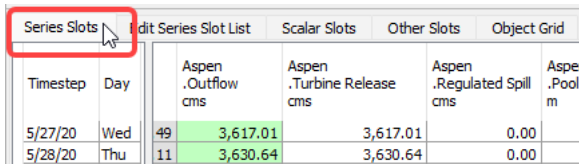
You can configure each custom summary row to display the value of a simple arithmetic function or the value of a referenced series or scalar slot for each simulation object column. For referenced slots, the primary reference simulation object is the object containing the SCT column's associated series slot.

The custom rows are added to the end of each aggregation interval, and they are automatically updated whenever any detail values are changed. If you change the aggregation interval, the custom summary rows remain on the SCT and the values are recomputed according to the new interval.

Note: Custom summary rows are displayed in vertical timestep orientation only. You must also enable aggregated view using either Show/Hide Details or Show All Details. The summary rows are not visible when timestep detail is hidden.

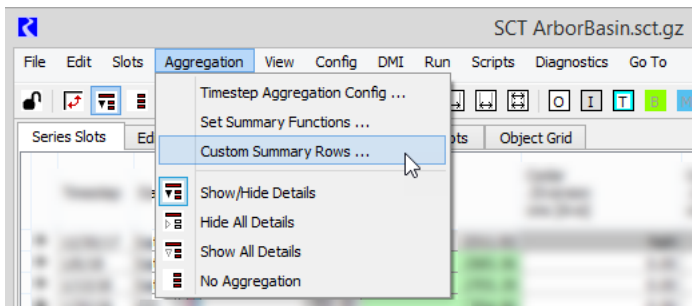
Note: For the arithmetic functions, undefined values (NaNs) are ignored.

1. Go to the **Series Slot** tab.

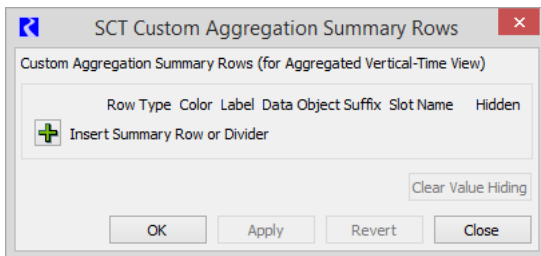


Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

2. On the SCT menu, select **Aggregation**, then **Custom Summary Rows**.



The SCT Custom Aggregation Summary Rows dialog box opens.



3. Optionally add a row by selecting **Plus**  or delete a row by selecting **Minus** .
4. In the **Row Type** menu, select one of the following options:
 - To add a thin divider row, for visual separation, select **Divider**. You can add as many divider rows as you want.
 - To add a function, select one of the numeric functions (Sum, Average, Maximum, Minimum, or Difference).

- To add a row that displays the last value for a specified slot, select **Slot Reference**, then complete the dialog box fields, as follows:

Note: A value is displayed only if the referenced slot uses the same units as the displayed slot.

- **Data Object Suffix**—enter an object name suffix to specify a Data Object associated with the referenced object. This entry is optional and can be left blank.
- **Slot Name**—enter the exact name of the slot you want to reference. If you entered a Data Object Suffix, this is the exact name of a slot on that object or associated data object.
- **Editable**—select the check box if you want the cell values in the summary row to be editable; clear the check box if you want the values to be display only.

5. Select the **Color** button to assign a user-defined color to the summary row.

The Select Color dialog box opens.

6. Use one of the color selector tools in the dialog box to define the color to apply, then select **OK**.

The Select Color dialog box closes, and the **Color** button is updated with the selected color.

7. In the **Label** field, enter a row label, which will appear in the row header.

Row Type	Color	Label	Data Object Suffix	Slot Name	Hidden
Divider	[Orange]				---
Sum	[Cyan]	Sum			0
Maximum	[Cyan]	Max			0
Minimum	[Cyan]	Min			0
Difference	[Cyan]	DIFF			0
Divider	[Orange]				---
Slot Reference	[Yellow]	Pool Elevation		Pool Elevation	☒ Editable 0
Divider	[Orange]				---
Insert Summary Row or Divider					

Clear Value Hiding

OK Apply Revert Close

8. In the SCT Configuration window, select **OK** or **Apply** to apply the changes to the SCT.

The summary rows are added to the end of each aggregation interval on the SCT. This update applies to all series slot sheets.

Note: To view the rows, you must enable aggregated view with details (either Show/Hide Details or Show All Details). The summary rows are not visible when timestep detail is hidden.

Note: Custom summary row values for simple functions are read only, but the Read-only crosshatching is *not* displayed.

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Note: For slot reference rows, a value is displayed only for slots whose unit matches the unit of the referenced slot. If a slot's unit does not match the referenced slot, the cell is blank.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid				
Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	Aspen .Power MW [Last]			
	12/31/17 Sun	NaN	3864.82	274.56	64.78	NaN			
	1/1/18 Mon	1684.35	1962.74	273.91	40.72	357.06			
	1/2/18 Tue	1685.71	1681.66	273.92	41.07	302.13			
	1/3/18 Wed	1687.06	1683.01	273.93	41.42	302.51			
	1/4/18 Thu	1688.42	1684.37	273.94	41.77	302.99			
	1/5/18 Fri	1689.77	1685.72	273.95	42.12	303.57			
	1/6/18 Sat	1691.13	1686.69	273.96	42.51	304.06			
	Sum Sun	10126.45	14249.00	1918.16	314.41	1872.32			
	Max Sun	1691.13	3864.82	274.56	64.78	357.06			
	Min Sun	1684.35	1681.66	273.91	40.72	302.13			
	DIFF Sun	6.77	2183.16	0.65	24.05	54.93			
	Pool Elevat Sun	NaN	NaN	273.96	NaN	NaN			
▼	1/13/18 Sat	1696.55	1692.03	274.03	45.24	308.87			
	1/7/18 Sun	1692.48	1687.97	273.97	42.90	304.60			
	1/8/18 Mon	1693.84	1689.32	273.98	43.29	305.28			
	1/9/18 Tue	1695.19	1690.68	273.99	43.68	306.09			
< > < >									
Aspen	Birch	Cedar	Dogwood	Dogwood Irrigation	Elm	Eastern Irrigation	Transbasin	Hickory	Juniper Irrigation

9. Save the SCT to save the updates with the SCT.

Related topics

- [Custom summary rows](#), page 24
- [Supported summary functions](#), page 124
- [Enable or disable aggregated view](#), page 127

Open a referenced slot in the Slot Viewer

Note: You do not need to unlock the SCT configuration to perform this procedure.

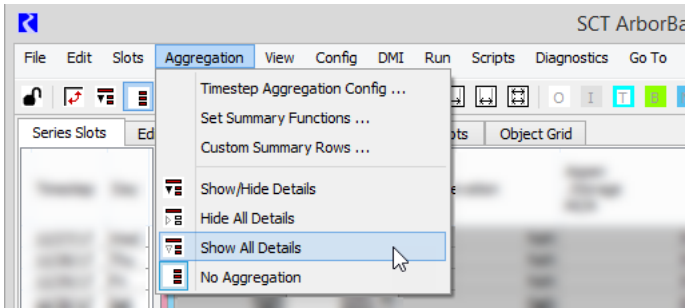
Note: This procedure applies only to custom summary rows that use the Slot Reference function.

Use this procedure to open a slot referenced in a custom summary row in the Slot Viewer. You can view and optionally edit all detail for the referenced slot.

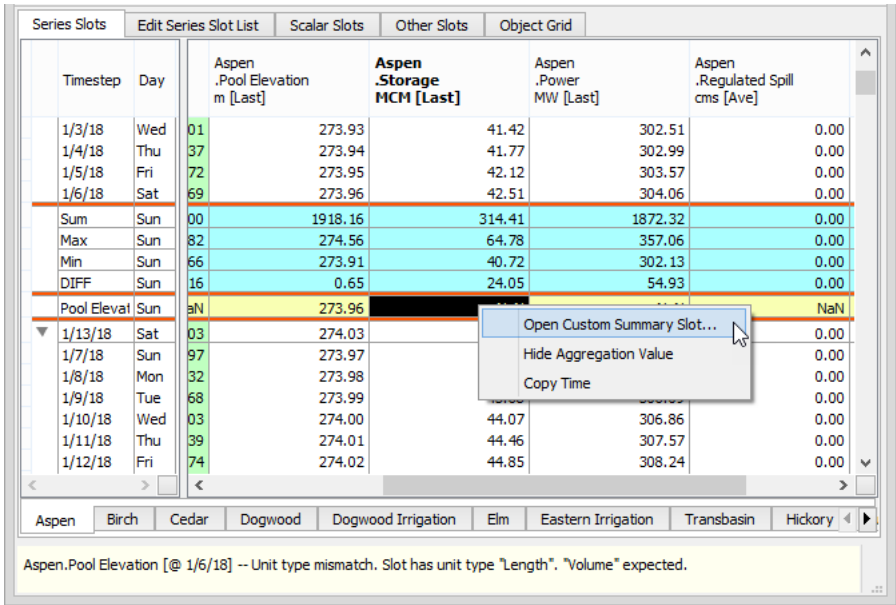
1. Go to the **Series Slot** tab.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. Enable aggregated view with details (either Show/Hide Details or Show All Details).



3. Locate a value in a referenced slot row. Right-click the cell and select **Open Custom Summary Slot** in the shortcut menu.



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The referenced slot is opened in the Slot Viewer, at the last timestep in the selected interval.

Timestep	Day	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	Aspen .Power MW [Last]	Aspen .Regulated Spill cms [Ave]
1/3/18	Wed	273.93	41.42		
1/4/18	Thu	273.94	41.77		
1/5/18	Fri	273.95	42.12		
1/6/18	Sat	273.96	42.51		
Sum	Sun	1918.16	314.41		
Max	Sun	274.56	64.78		
Min	Sun	273.91	40.72		
DIFF	Sun	0.65	24.05		
Pool Elevat	Sun	273.96	NaN		
1/13/18	Sat	274.03	45.24		
1/7/18	Sun	273.97	42.90		
1/8/18	Mon	273.98	43.29		
1/9/18	Tue	273.99	43.68		
1/10/18	Wed	274.00	44.07		
1/11/18	Thu	274.01	44.46		
1/12/18	Fri	274.02	44.85		

Aspen.Pool Elevation m [Last]

Value: 273.958064516 m Jan 6, 2018

Aspen .Pool Elevation m

Date	Value	Priority
01-05-2018 Fri	273.95	28
01-06-2018 Sat	273.96	28
01-07-2018 Sun	273.97	28
01-08-2018 Mon	273.98	28
01-09-2018 Tue	273.99	28
01-10-2018 Wed	274.00	28
01-11-2018 Thu	274.01	28

Show: ☐ Description

Aspen.Pool Elevation [@ 24:00 January 6, 2018]

1 value: 273.96 [m] (Priority 28)

- You can use the Slot Viewer to display, edit, or plot the referenced slot. If you edit any values, the SCT is updated accordingly.

Related topics

- [Enable or disable aggregated view](#), page 127
- [Create custom summary rows for all slots](#), page 137
- [Enable or disable aggregated view](#), page 127

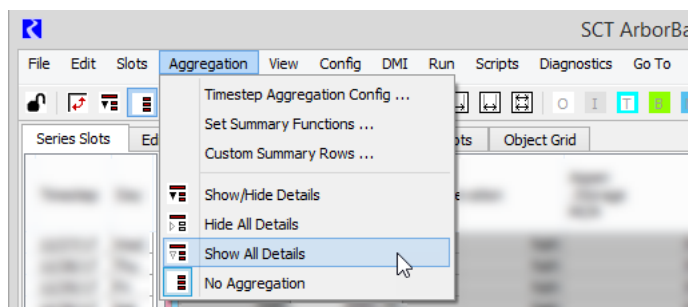
Hide a custom summary row value

Use this procedure to hide the value of a custom summary row cell. This can be useful when the aggregation does not make sense for a particular slot.

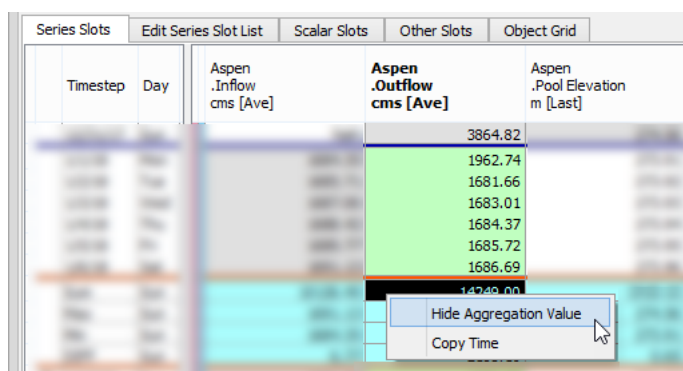
- Go to the **Series Slot** tab.

Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

2. Enable aggregated view with details (either Show/Hide Details or Show All Details).



3. Locate the cell you want to hide. Right-click the cell and select **Hide Aggregation Value** in the shortcut menu.



The cell value is hidden, making the cell display blank.

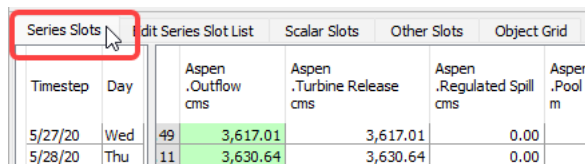
Related topics

- [Create custom summary rows for all slots](#), page 137
- [Restore display of hidden custom summary row values](#), page 143
- [Enable or disable aggregated view](#), page 127

Restore display of hidden custom summary row values

Use this procedure to restore the display of custom summary row values that have been hidden manually using the “[Hide a custom summary row value](#)” on page 142) procedure. You can restore display of selected cells or of all cells at once.

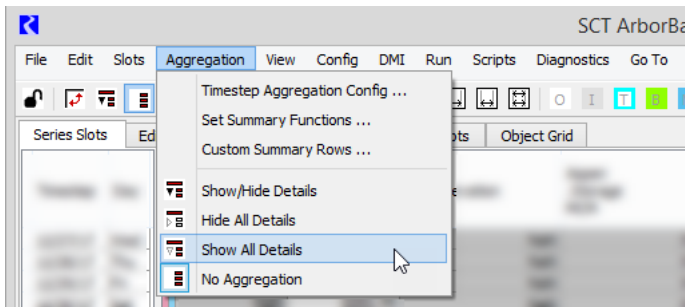
1. Go to the **Series Slot** tab.



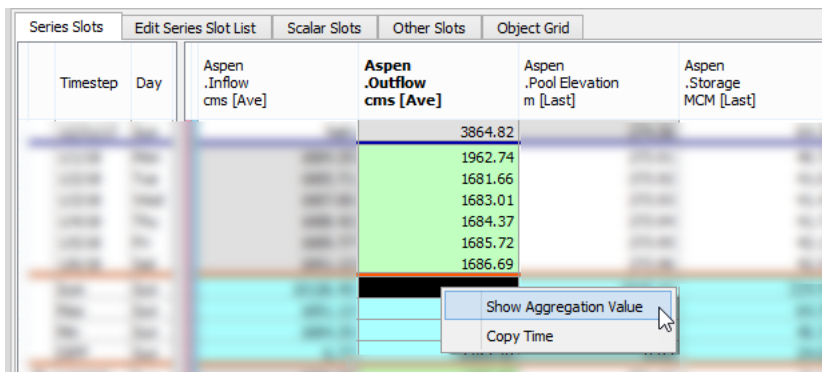
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2. Enable aggregated view with details (either Show/Hide Details or Show All Details).

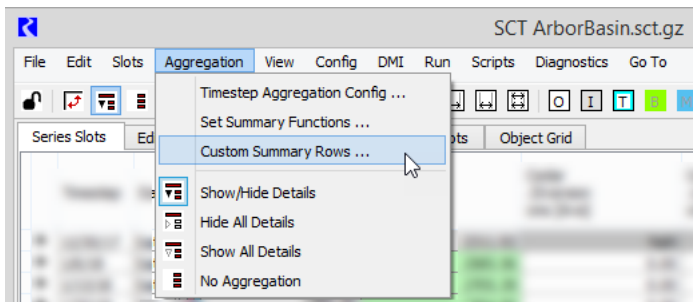


3. To restore selected hidden cells, right-click the blank cell and select **Show Aggregation Value** in the shortcut menu.



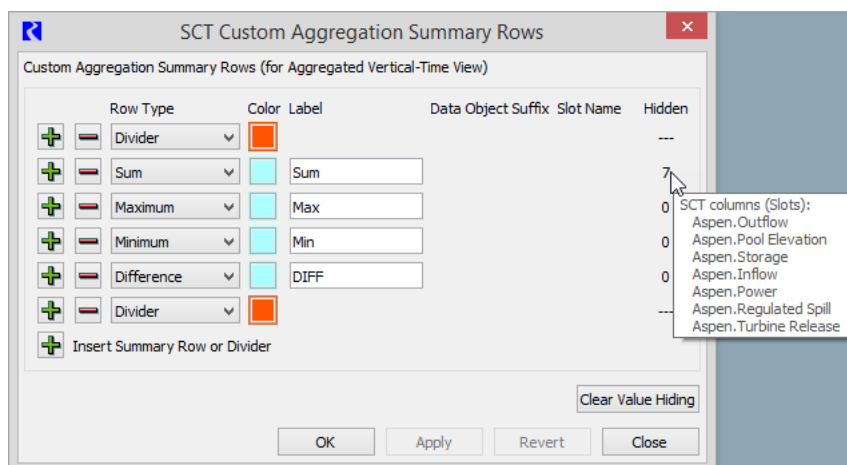
The summary value is restored in the cell.

4. To display all hidden cells at once, use the following steps:
 - a. On the SCT menu, select **Aggregation**, then **Custom Summary Rows**.

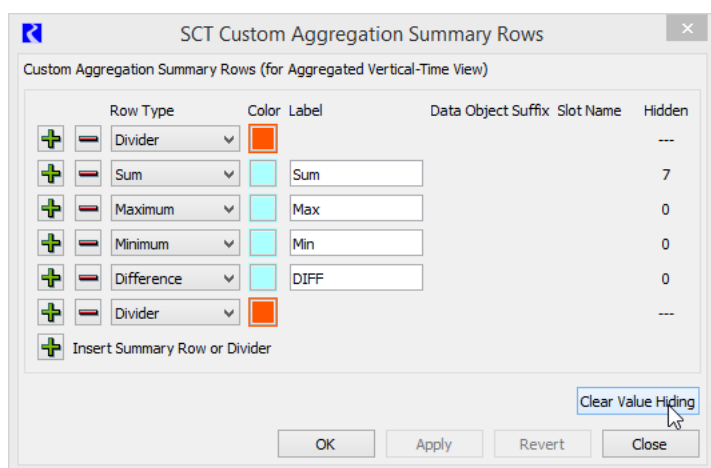


The SCT Custom Aggregation Summary Rows dialog box opens.

Note: The Hidden field displays the number of custom summary cells that are hidden. You can hover over the number to display a tooltip listing the names of the slots with hidden cells.



b. Select **Clear Value Hiding**, then select **OK** or **Apply** to apply the change to the SCT.



All hidden summary values are restored on all SCT sheets.

Related topics

- [Create custom summary rows for all slots](#), page 137
- [Hide a custom summary row value](#), page 142
- [Enable or disable aggregated view](#), page 127

Displaying Snapshot Slots

Snapshots preserve the values from a selected set of slots on a “snapshot object” so that they can later be compared with the values from later runs. See “[Snapshots](#)” in *Output Utilities and Data Visualization* for more information on snapshots. The SCT allows you to view slots from selected snapshots together with their source slots.

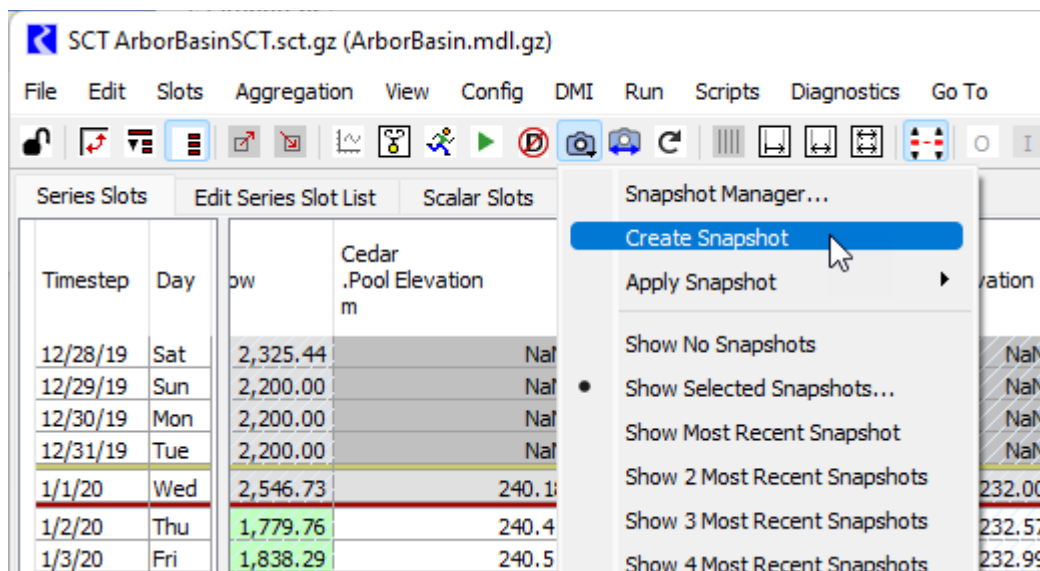
Topics

- [Create a snapshot from the SCT](#), page 146
- [Configure which snapshot slots to show](#), page 146
- [Show recent snapshots](#), page 148

Create a snapshot from the SCT

Use the following procedure to create a snapshot from the SCT.

1. Select the snapshot button on the SCT toolbar.
2. From the resulting menu, select **Create Snapshot**.



This will create a snapshot object using the Snapshot Template (slot selection and name) currently configured in the Snapshot Manager. See “[Snapshots](#)” in *Output Utilities and Data Visualization* for more information.

Alternatively, select **Snapshot Manager** from the menu, which will open the Snapshot Manager, where you can modify the Snapshot Template and then create a snapshot.

Related topics

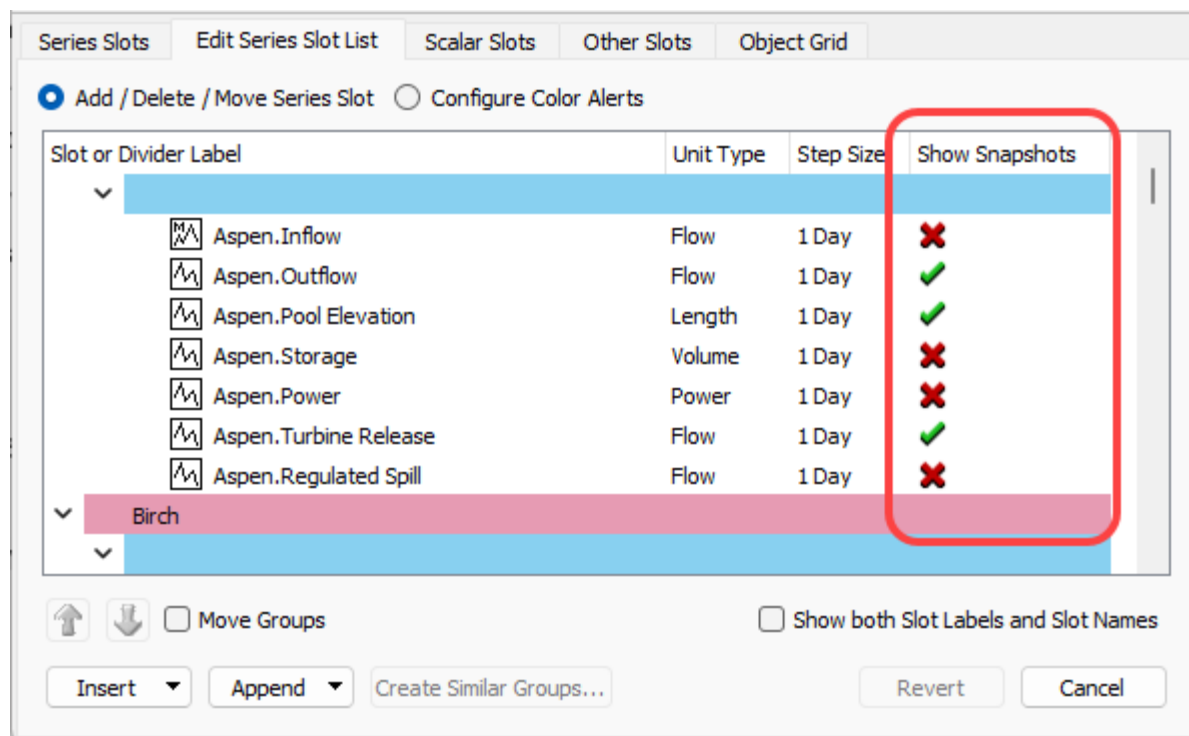
- [Configure which snapshot slots to show](#), page 146
- [Show recent snapshots](#), page 148

Configure which snapshot slots to show

To configure which slots in the SCT will have their snapshot slots shown, use the following procedure.

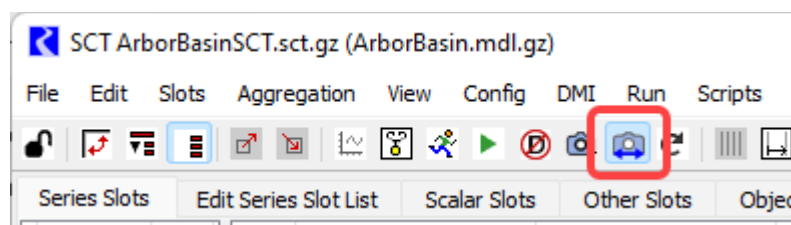
1. Select the **Edit Series Slot List** tab.

2. In the Show Snapshots column, click on the green check by a slot to change it to a red X so that snapshot slots are not shown for that slot. Click on a red X to change it back to a green check to show snapshot slots for the slot. By default, all slots have a green check so that their snapshot slots are shown.



Note: In order for snapshot slots to show for a slot in the SCT, 1) that slot must have a green check in the Show Snapshots column on the Edit Series Slot List tab, 2) a selection must be made to show recent snapshots from the snapshot button on the toolbar (see [Show recent snapshots](#), page 148), and 3) that slot must be included in one of the snapshots that is shown.

The Show All Snapshot Slots button on the SCT toolbar can override the per slot setting of whether to show snapshot slots.



When the Show All Snapshot Slots button is selected, all source slots in the SCT that are included in selected recent snapshots will have their corresponding snapshot slots displayed regardless of the per slot setting on the Edit Series Slot List tab. Click the button again to return to displaying snapshot slots based on the per slot settings.

Note: Snapshot slots can also be added to the SCT in the normal manner, by selecting Insert/Append Slots and then in the Slot Selector selecting the desired snapshot slots. However, when adding snapshot slots using the Slot Selector, you are selecting a specific slot on a specific snapshot object. If that snapshot object is deleted and replaced by a more recent snapshot, the slot will show up in the

Chapter 3

Using the SCT

SCT as unrecognized. Using the Show Recent Snapshots approach makes it easy to show snapshot slots even when the specific snapshot objects change.

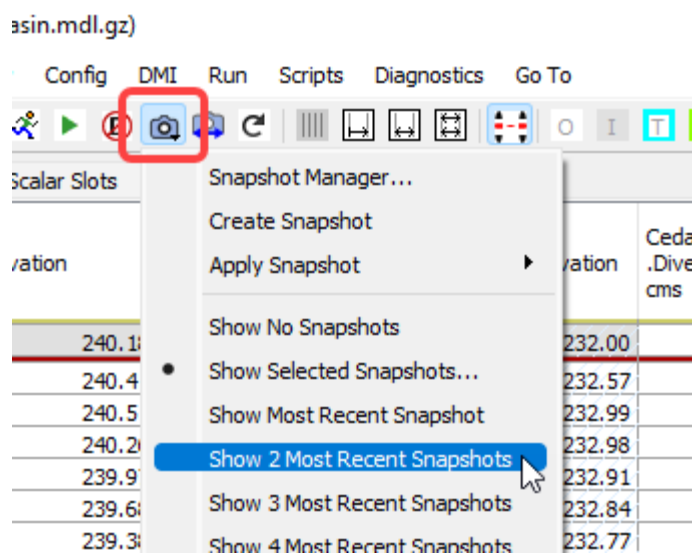
Related topics

- [Create a snapshot from the SCT](#), page 146
- [Show recent snapshots](#), page 148

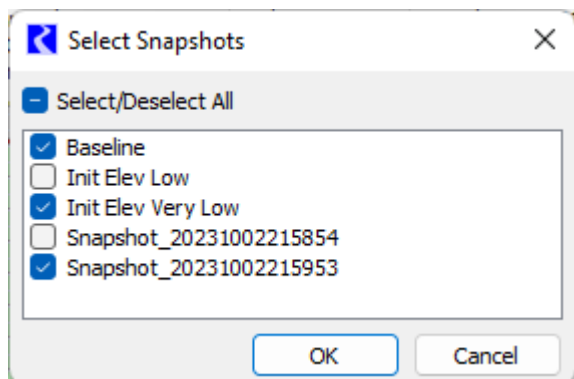
Show recent snapshots

To show slots from recent snapshots in the SCT together with their source slots, use the following procedure.

1. On the SCT toolbar, select the snapshot button.
2. In the resulting menu select on of the following options.
 - a. **Show *N* Most Recent Snapshots**—this option will show slots from the most recent snapshots, where most recent is based on the timestamp from when the snapshot was created.



- b. **Show Selected Snapshots**—this option opens the Select Snapshots dialog. Select the checkboxes for the snapshots you want to display. Then select **OK**.



After making a selection, any slots in the selected snapshots that have a source slot in the SCT will be shown in the SCT if either 1) the per slot setting on the Edit Series Slot List tab is set to show snapshot slots for that slot or 2) the override to show all snapshot slots is selected (see [Configure which snapshot slots to show](#), page 146). The snapshot slots will be shown to the right of the source slot if slots are columns and below the source slot if slots are rows. The snapshot slots are always read-only.

		Source slot		Snapshot slots		
Timestep	Day	Cedar .Pool Elevation m	Init Elev Very Low .Cedar_Pool Elevation m	Init Elev Low .Cedar_Pool Elevation m	Cedar .Diversion cms	
1/1/20	Wed	46.73 240.18	228.00	232.00		↑
1/2/20	Thu	95.86 240.41	228.97	232.57		0
1/3/20	Fri	54.39 240.51	229.72	232.99		0
1/4/20	Sat	01.82 240.26	229.84	232.98		0
1/5/20	Sun	07.59 239.97	229.88	232.91		0
1/6/20	Mon	07.99 239.68	229.92	232.84		0
1/7/20	Tue	08.34 239.38	229.96	232.77		0
1/8/20	Wed	08.65 239.09	230.00	232.70		0
1/9/20	Thu	08.92 238.80	230.04	232.63		0

Aspen Birch Cedar Dogwood Dogwood Irrigation Elm Eastern Irrigation Transbasin

Note: The Show Recent Snapshots options are global settings in the model and also control how snapshot slots are displayed in Plots. (See [“Show Recent Snapshots”](#) in *Output Utilities and Data Visualization* for more information.)

Related topics

- [Create a snapshot from the SCT](#), page 146
- [Configure which snapshot slots to show](#), page 146

Configuring Color Alerts

This section includes procedures for configuring alerts to show when values have either exceeded a defined threshold or meet a specified condition. A color alert is an indicator that operators might need to investigate further.

There are two approaches for configuring color alerts. The first is to use thresholds to define intervals. For example, you can define Normal, High, and Extra High intervals for a slot and assign text and cell color changes that provide a visual indicator that the slot value has exceeded the threshold you have defined. The second approach is to use SCT condition functions. You can define customized RPL functions, and if a function evaluates to true, the text and cell color are set based on the color associated with that condition.

The general process for configuring threshold alerts is as follows:

1. [Define custom colors for highlights and color alerts](#), page 191
2. [Configure Color Alerts](#), page 150

Topics

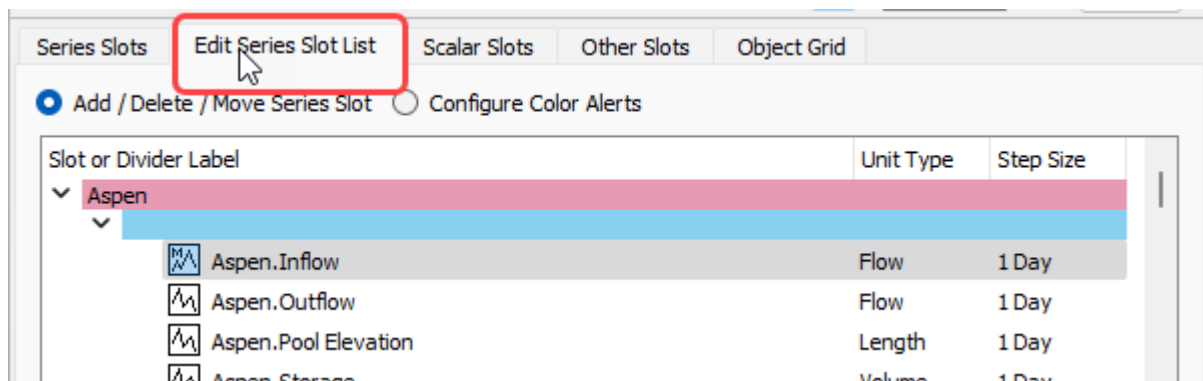
- [Configure Color Alerts](#), page 150
- [Copy Color Alert configurations to other slots](#), page 161

Configure Color Alerts

Use this procedure to define the conditions for which you want to trigger Color Alerts. You can define separate Color Alert configurations for any number of slots.

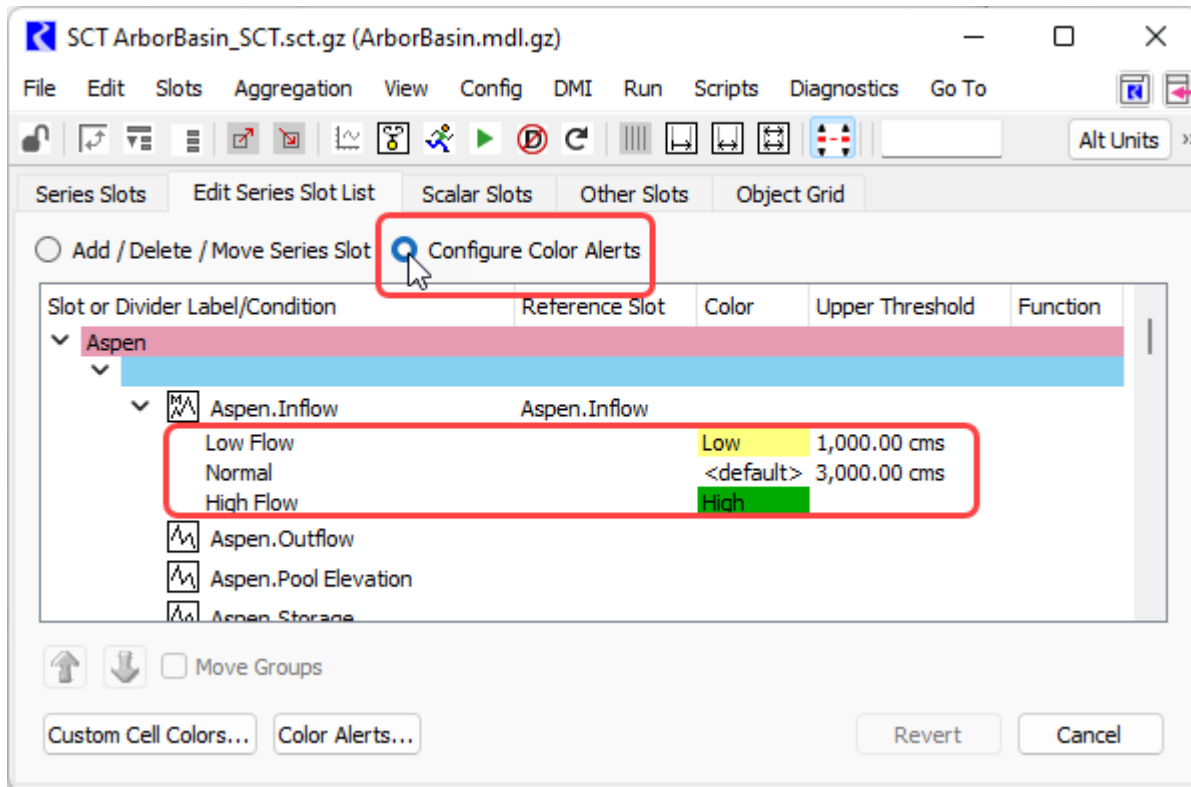
Note: Before performing this procedure, you should define the custom colors you want to use for the alerts. See [“Define custom colors for highlights and color alerts”](#) on page 191 for details.

1. Go to the **Edit Series Slot List** tab. Alternatively, press Alt+Shift+S.

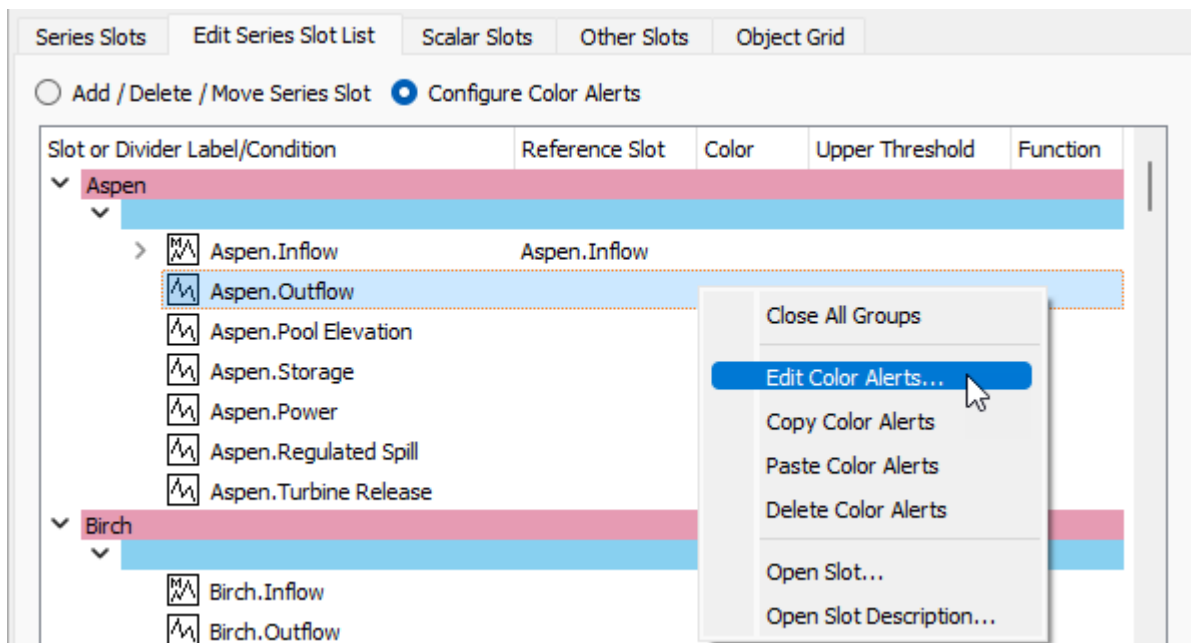


2. Select the **Configure Color Alerts** option.

The display changes to show color alert specifications that have already been defined.



3. Right-click a slot in the list and select **Edit Color Alerts** in the shortcut menu.

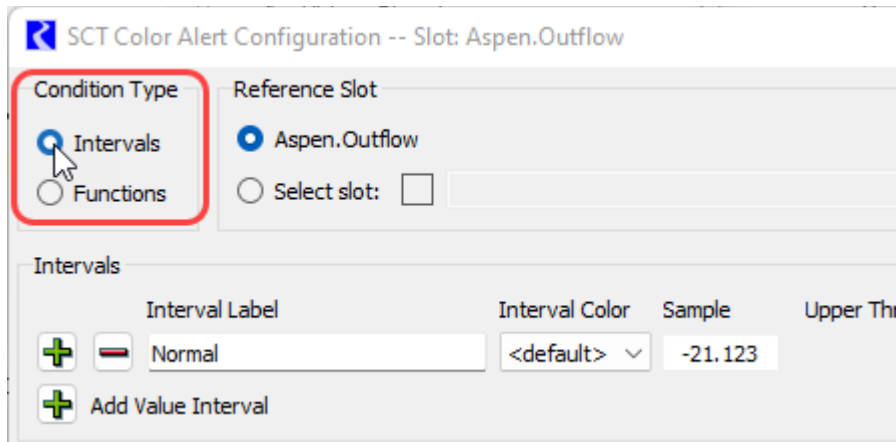


The SCT Color Alert Configuration dialog appears for that slot.

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4. Select one of the two Condition Type options:
 - **Intervals** (see “[Define Intervals with Thresholds](#)” on page 152)
 - **Functions** (see “[Define Condition Functions](#)” on page 155)



SCT Color Alert Configuration -- Slot: Aspen.Outflow

Condition Type

- ☒ Intervals
- ☐ Functions

Reference Slot

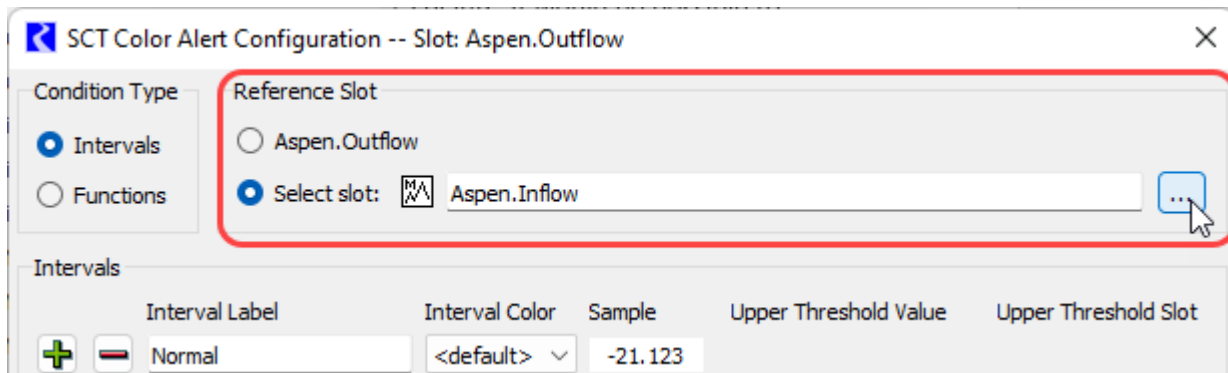
- ☒ Aspen.Outflow
- ☐ Select slot:

Intervals

	Interval Label	Interval Color	Sample	Upper Th
	Normal	<default>	-21.123	
	Add Value Interval			

5. Select the Reference Slot.

The Reference Slot is the slot on which the formatting is based. The default selection is the slot to which the Color Alerts are being applied. Alternatively, you can choose the option for **Select Slot** and then use the chooser button to open the Slot Selector. You would use this option if you want the formatting of a slot to be based on the value in another slot.



SCT Color Alert Configuration -- Slot: Aspen.Outflow

Condition Type

- ☒ Intervals
- ☐ Functions

Reference Slot

- ☐ Aspen.Outflow
- ☒ Select slot:

Intervals

	Interval Label	Interval Color	Sample	Upper Threshold Value	Upper Threshold Slot
	Normal	<default>	-21.123		

6. Define the Color Alert Conditions.

The display of the Conditions panel and the steps to define the conditions depend on which Condition Type is selected.

Define Intervals with Thresholds

Use this procedure to define the threshold values at which you want to trigger Color Alerts when you have selected **Intervals** as the Condition Type (see “[Select one of the two Condition Type options:](#)” on page 152). You can configure alerts to trigger with one of the following types of events:

- When the Reference Slot value exceeds a defined fixed value
- When the Reference Slot value exceeds the value of another slot

By default, there is a Normal interval, which uses the default cell flag color. You can define any number of alert intervals and their associated colors.

Note: Each interval is inclusive of its Upper Threshold Value.

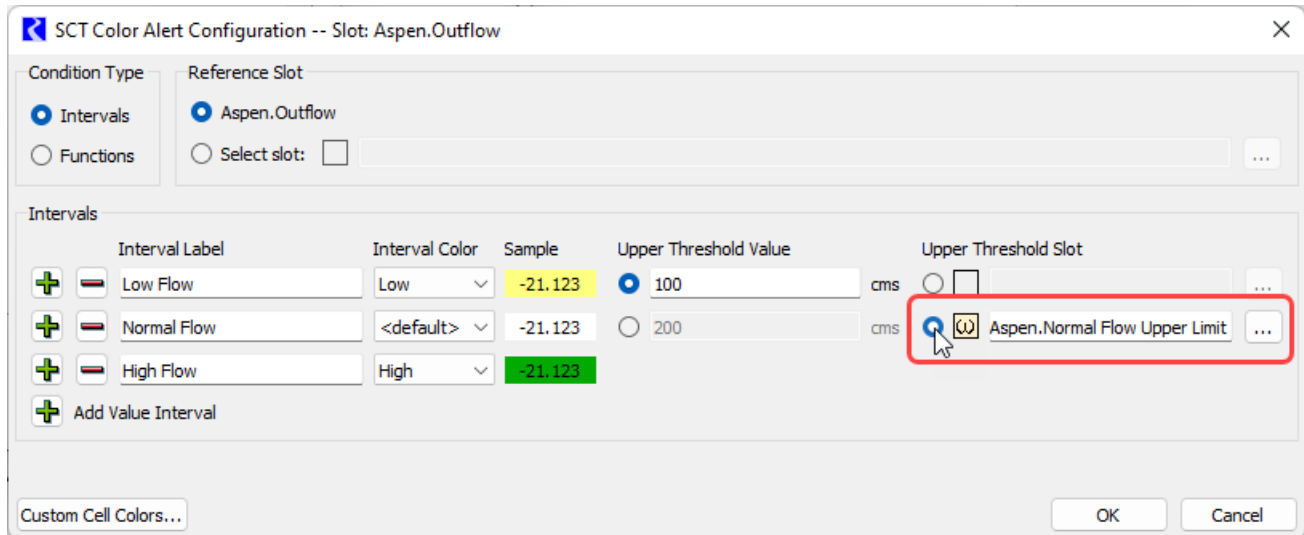
Interval Label	Interval Color	Sample	Upper Threshold Value	Upper Threshold Slot
Low Flow	Low	-21.123	100 cms	
Normal	<default>	-21.123	200 NONE	
High Flow	High	-21.123		

- Add an interval by selecting **Plus** .
- In the **Interval Label** field, enter an interval name.
The Interval Label will be displayed as a tool tip when hovering over any value in that interval.
- Proceed as follows, depending on how you want to define the upper threshold for the interval.
 - Go to [Step d.](#) to define the upper threshold as a fixed value.
 - Go to [Step e.](#) to obtain the upper threshold value from a slot.
- To define the upper threshold as a fixed value, proceed as follows.
 - Select the **Upper Threshold Value** option.
 - In the **Upper Threshold Value** field, enter the upper value.
- To obtain the upper threshold value from another slot, proceed as follows:
 - Select the **Upper Threshold Slot** option.

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2. Enter the exact slot name directly into the **Upper Threshold Slot** entry field. Alternatively, select the **Chooser**  button to use the Slot Selector to select the slot whose value you want to use to define the upper threshold for the interval.



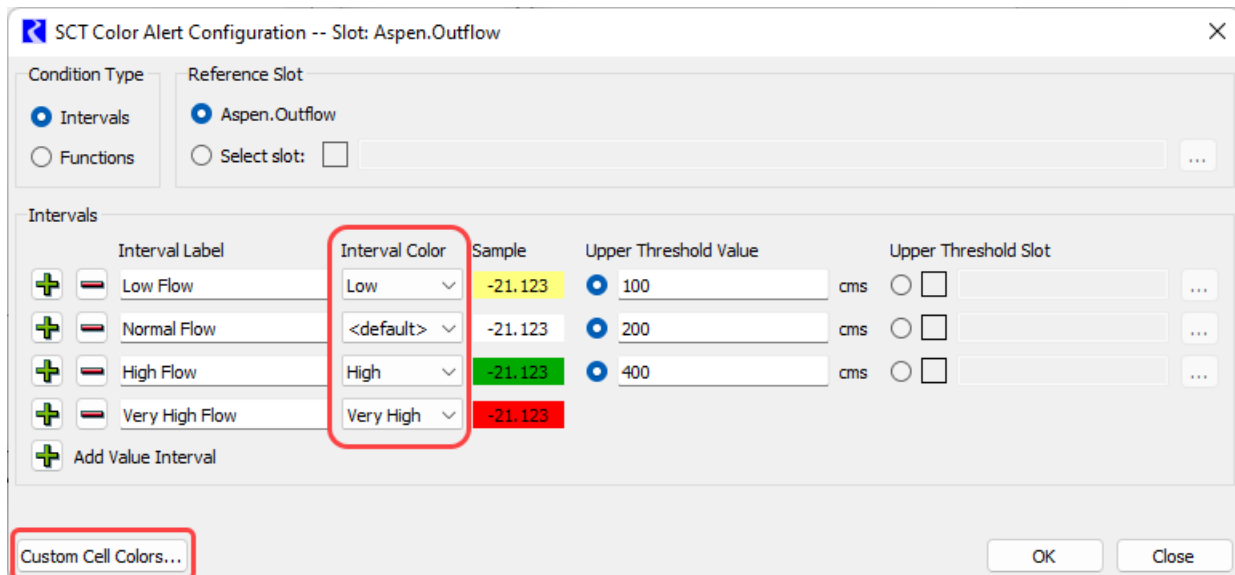
The dialog box is titled "SCT Color Alert Configuration -- Slot: Aspen.Outflow". It has two tabs: "Condition Type" and "Reference Slot". Under "Condition Type", "Intervals" is selected. Under "Reference Slot", "Aspen.Outflow" is selected. The "Intervals" section contains a table with columns: Interval Label, Interval Color, Sample, Upper Threshold Value, and Upper Threshold Slot. There are three intervals: Low Flow, Normal Flow, and High Flow. The "Upper Threshold Slot" column has a "Chooser" button (a square with a circle and a plus sign) next to each interval. The "Normal Flow" interval's "Upper Threshold Slot" is highlighted with a red box, and a mouse cursor is clicking the "Chooser" button. Below the table is a "Custom Cell Colors..." button. At the bottom are "OK" and "Cancel" buttons.

Interval Label	Interval Color	Sample	Upper Threshold Value	Upper Threshold Slot
Low Flow	Low	-21.123	100	cms
Normal Flow	<default>	-21.123	200	cms
High Flow	High	-21.123		

- f. In the **Interval Color** menu, select the color for the interval. The menu displays all custom colors defined for the SCT. The <default> selection indicates to use the cell flag color.

Note: A custom color entry may apply to cell text, the cell background, or both, depending on how the color entry has been defined.

Note: If there are no colors listed, or you want to define additional colors, select **Custom Cell Colors** to open the Custom SCT Cell Colors dialog box. See [“Define custom colors for highlights and color alerts”](#) on page 191 for details.



The dialog box is titled "SCT Color Alert Configuration -- Slot: Aspen.Outflow". It has two tabs: "Condition Type" and "Reference Slot". Under "Condition Type", "Intervals" is selected. Under "Reference Slot", "Aspen.Outflow" is selected. The "Intervals" section contains a table with columns: Interval Label, Interval Color, Sample, Upper Threshold Value, and Upper Threshold Slot. There are four intervals: Low Flow, Normal Flow, High Flow, and Very High Flow. The "Interval Color" column has a dropdown menu for each interval. The "High Flow" interval's "Interval Color" dropdown is highlighted with a red box, and a mouse cursor is clicking the dropdown arrow. The "Very High Flow" interval's "Sample" value is highlighted with a red box. Below the table is a "Custom Cell Colors..." button. At the bottom are "OK" and "Close" buttons.

Interval Label	Interval Color	Sample	Upper Threshold Value	Upper Threshold Slot
Low Flow	Low	-21.123	100	cms
Normal Flow	<default>	-21.123	200	cms
High Flow	High	-21.123	400	cms
Very High Flow	Very High	-21.123		

- g. Optionally, repeat [Step a.](#) through [Step f.](#) to define additional intervals. You can delete an interval by

selecting **Minus** 

Define Condition Functions

Use this procedure when you have selected **Functions** as the Condition Type to trigger Color Alerts with user-defined RPL functions (see [“Select one of the two Condition Type options:”](#) on page 152).

Note: When condition functions are used for Color Alerts, evaluation of the functions can take a noticeable amount of time, so the Color Alerts displayed in the SCT are not updated every time a value changes. Color Alerts are updated at the following times.

- When the SCT opens
 - At the completion of a run
 - When the SCT is manually refreshed, either by selecting **Refresh** from the **View** menu of the SCT, or by selecting the **Refresh** button  on the SCT tool bar.
- a. Create one or more RPL user-defined functions to use as the condition functions.

See [“Writing a User-defined Function” in *RiverWare Policy Language \(RPL\)*](#) for information on how to create a user-defined function. Functions used as SCT condition functions must meet the following criteria.

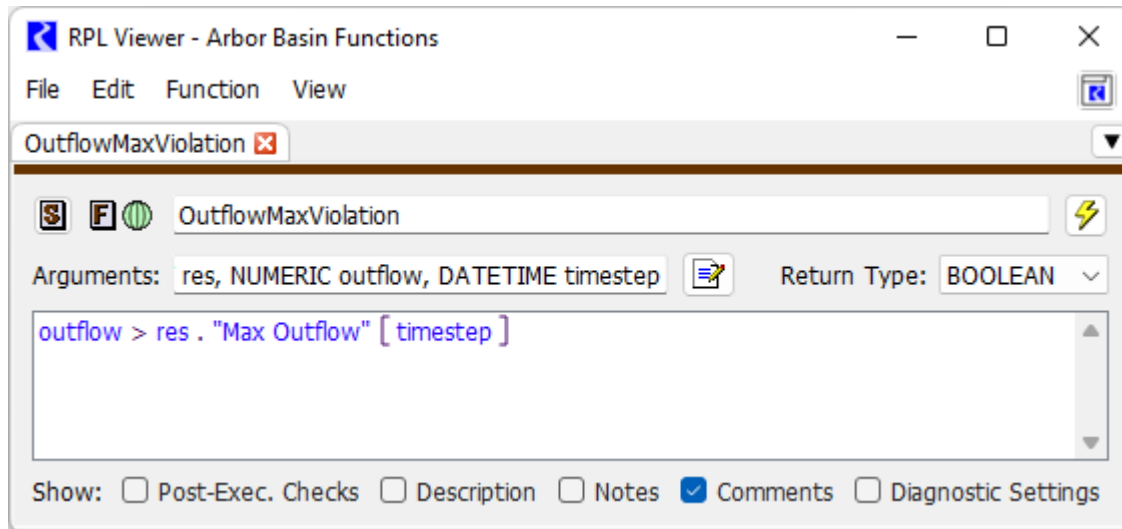
- Must be in a global functions set open in the model. See [“Global RPL Functions” in *RiverWare Policy Language \(RPL\)*](#).
- Must have a return type of Boolean (true/false).
- Can have no more than four arguments, one of each of the following types: OBJECT, SLOT, NUMERIC, TIMESTEP.

When a condition function uses arguments, the arguments will be filled at the time of function evaluation at each timestep based on the Reference Slot (see [“Select the Reference Slot.”](#) on page 152), where the NUMERIC is the Reference Slot value at that timestep.

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The following is an example of a valid SCT condition function.



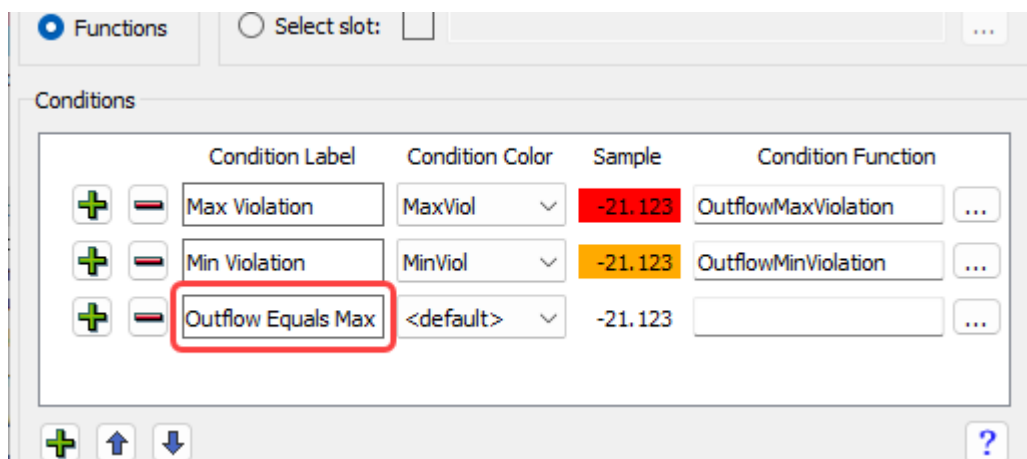
Assume that the Reference Slot is Aspen.Outflow, and at 12:00 March 16, 2024, Aspen.Outflow has a value of 2,453 cfs. On this timestep, the function will evaluate as
2,453 “cfs” > Aspen.Max Outflow [@"12:00 March 16, 2024"]

If the Aspen.Max Outflow is 2,000 cfs on that timestep, the function will evaluate to TRUE, and the Color Alert will be applied based on this condition.

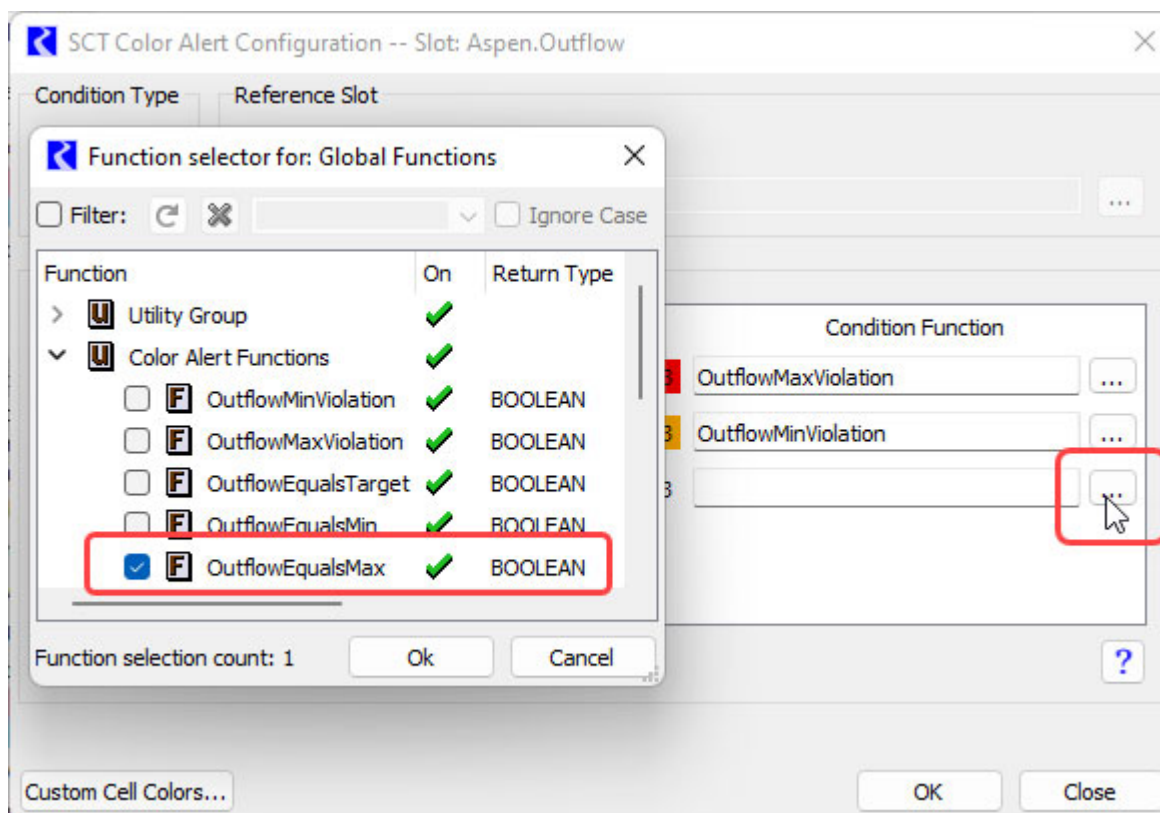
Tip: Most SCT condition functions are expected to be relatively simply and efficient, such as the example shown above. However, RPL is a powerful tool capable of expressing complex logic. Condition functions are evaluated for each timestep, so care should be taken regarding the use of overly complex logic to prevent negative impacts on SCT performance.

- b. In the SCT Color Alert Configuration, add a condition by selecting **Plus**  .
Selecting the **Plus** button at the bottom left of the dialog will append a new condition to the bottom of the condition list. Selecting the **Plus** button to the left of an existing condition will add a new condition above the selected condition.
- c. In the **Condition Label** field, enter a name for the condition.

The label will be displayed as a tool tip when hovering over a value for which the condition applies.



- d. In the **Condition Function** column select the **Chooser** button. Then use the Function Selector to select which function to use for the condition.



- e. In the **Condition Color** menu, select the color for the condition. The menu displays all custom colors defined for the SCT. The <default> selection indicates to use the cell flag color.

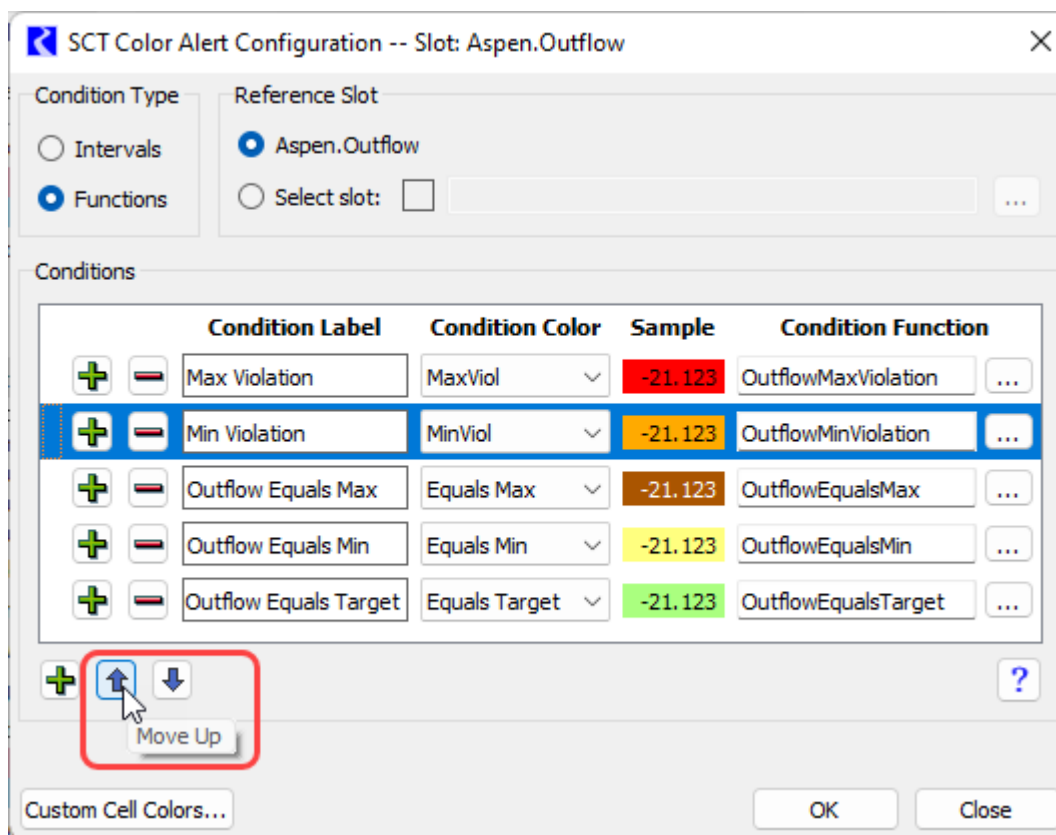
Note: A custom color entry may apply to cell text, the cell background, or both, depending on how the color entry has been defined.

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Note: If there are no colors listed, or you want to define additional colors, select **Custom Cell Colors** to open the Custom SCT Cell Colors dialog box. See “[Define custom colors for highlights and color alerts](#)” on page 191 for details.

- f. Optionally repeat [Step a.](#) through [Step e.](#) to define additional conditions. You can delete a condition by selecting **Minus** 
- g. To re-order the conditions, click on a condition row to select it; then use the up and down arrows at the bottom left to move the condition.



Click and drag over multiple condition rows or use Ctrl + click to select multiple condition rows to move together.

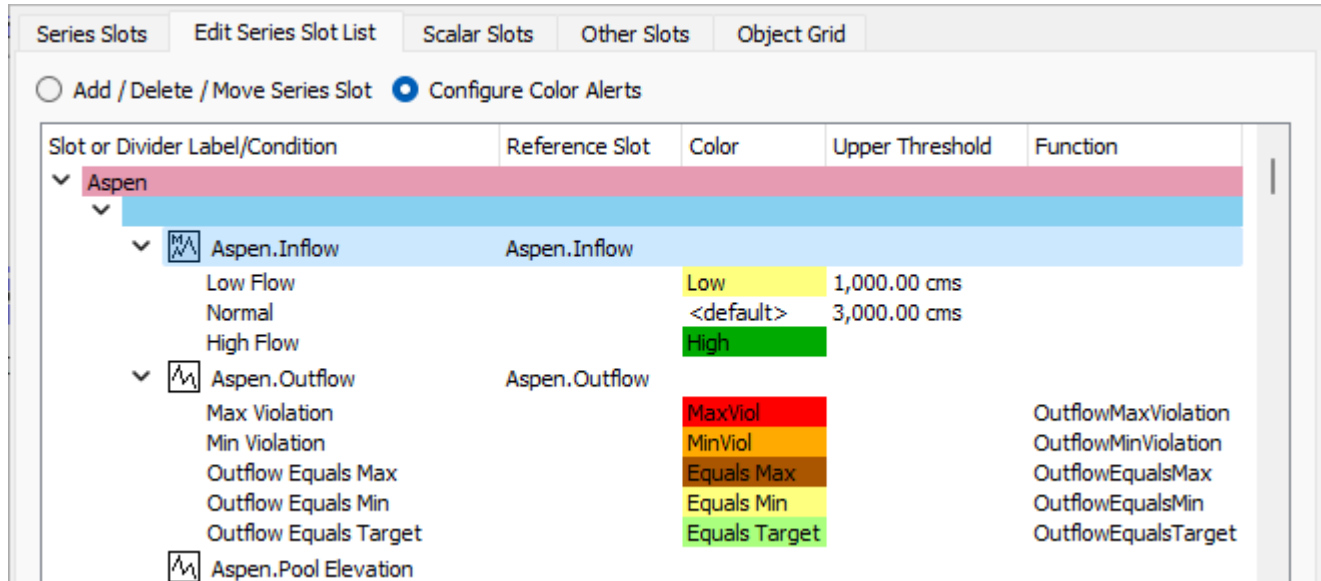
Note: Conditions are evaluated in order from the top of the list to the bottom, and the first condition that evaluates to TRUE is used to format the cell. In other words, if multiple conditions are true, the highest true condition in the list controls the formatting.

Note: If all conditions evaluate to FALSE, the cell is colored using the default flag color.

Note: In the context of SCT condition functions, if a function encounters a NaN during evaluation and thus cannot fully evaluate, it is treated as if it evaluates to FALSE.

7. When you are finished defining Color Alerts, select **OK** to apply the changes.

The SCT Color Alert Threshold Configuration dialog box closes, and the **Edit Series Slot List** tab is updated with the Color Alert specifications you have defined.



- To view the effect, go to the **Series Slots** tab.

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Any values that have exceeded the defined thresholds or satisfy one of the condition functions are displayed using the colors you have defined. Hover over a slot value to see a tool tip displaying the alert that has been applied to that value.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Inflow cms	Aspen .Outflow cms	Aspen .Pool Elevation m	Aspen .Storage MCM
9/14/20	Mon	1,030.53	1,030.53	273.00	7.9
9/15/20	Tue	986.00	986.00	273.00	7.9
9/16/20	Wed	1,033.43	1,033.43	273.00	7.9
9/17/20	Thu	1,080.87	1,080.87	273.00	7.9
9/18/20	Fri	1,128.30	1,128.30	273.00	7.9
9/19/20	Sat	1,175.73	1,175.73	273.00	7.9
9/20/20	Sun	1,223.17	1,223.17	273.00	7.9
9/21/20	Mon	1,270.60	1,270.60	273.00	7.9
9/22/20	Tue	1,318.03	1,318.03	273.00	7.9
9/23/20	Wed	1,365.47	1,365.47	273.00	7.9
9/24/20	Thu	1,412.90	1,412.90	273.00	7.9
9/25/20	Fri	1,460.33	1,460.33	273.00	7.9
9/26/20	Sat	1,507.77	1,500.00	273.02	8.6
9/27/20	Sun	1,555.20	1,500.00	273.15	13.4
9/28/20	Mon	1,602.64	1,500.00	273.40	22.2
9/29/20	Tue	1,650.07	1,500.00	273.76	35.2
9/30/20	Wed	1,697.50	1,638.11	273.90	40.3

Note: Color Alerts override manually applied cell highlights. See [“Apply custom colors to timestep values”](#) on page 117 for details.

Note: Color Alerts override the cell flag color. It can be helpful to add the flag letters to the SCT display as an additional indicator; see [“Display flag letters and priority numbers”](#) on page 103 for details.

- To manually update Color Alerts that use condition functions, select **Refresh** from the **View** menu of the SCT, or select the **Refresh** button  on the SCT tool bar.

Note: SCT condition functions for Color Alerts automatically evaluate at the following times.

- When the SCT is opened
- When a run completes

Note: Cells with Color Alert conditions that cannot be fully evaluated are drawn with a dotted pattern and have tool tips indicating the error. This could occur for interval thresholds if the Upper Threshold Slot is not defined, does not exist, or contains NaN. For condition functions, this could occur if a function does not exist, is not valid, or encounters an error on evaluation. Errors in condition functions will also be posted as red error messages in the Diagnostic Output Window.

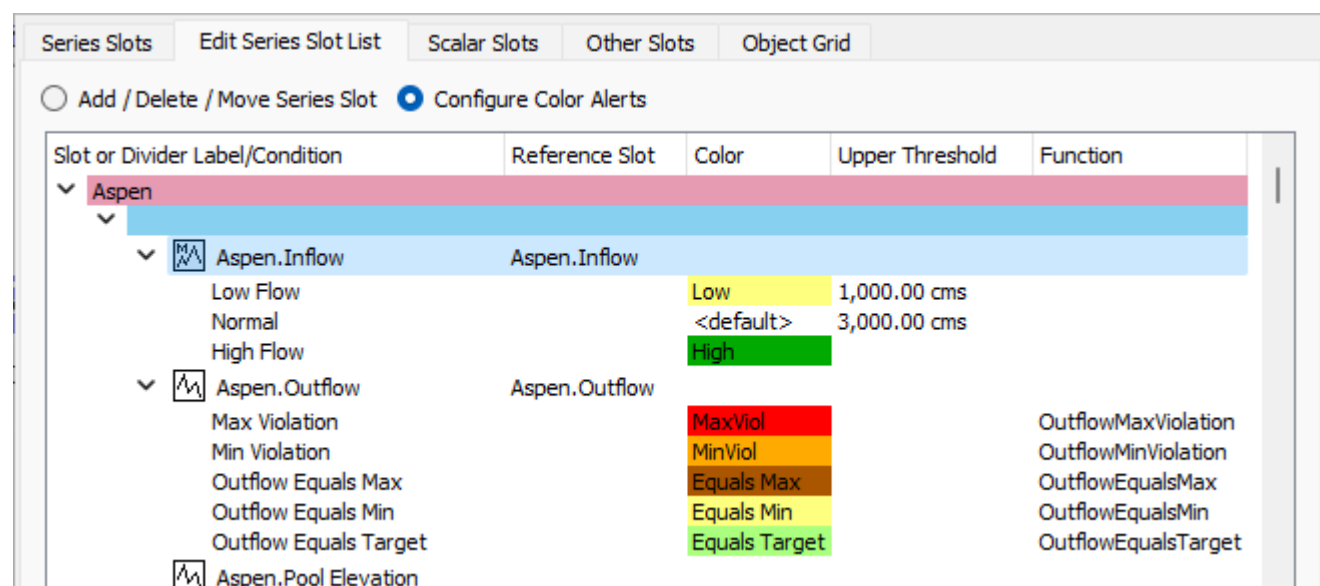
Related topics

- [Define custom colors for highlights and color alerts](#), page 191
- [Copy Color Alert configurations to other slots](#), page 161

Copy Color Alert configurations to other slots

Use this procedure to copy and paste Color Alert configurations from one slot to others. This procedure simplifies the process of defining Color Alerts for multiple slots; you can copy Color Alert configurations from one slot and paste them to one or more others, and then edit the copied Color Alert configurations to fine-tune the settings.

1. Go to the **Edit Series Slot List** tab. Alternatively, press Alt+Shift+S.
2. Select the **Configure Color Alerts** option.

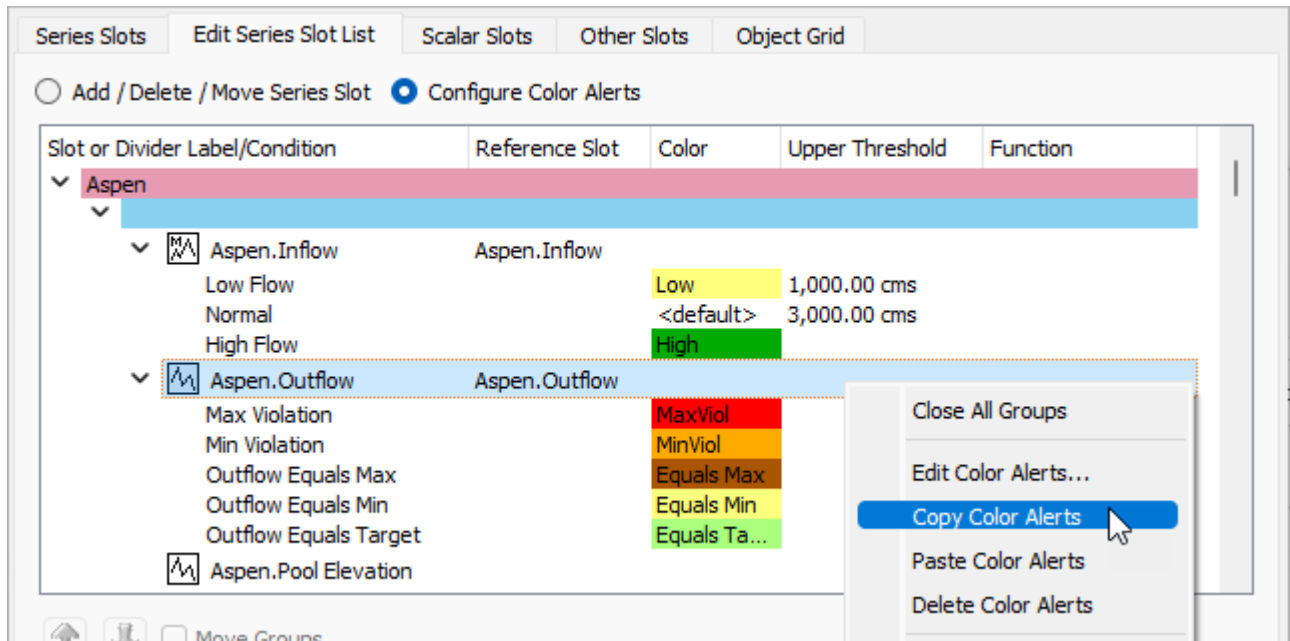


3. Select the slot whose Color Alert configuration you want to copy.

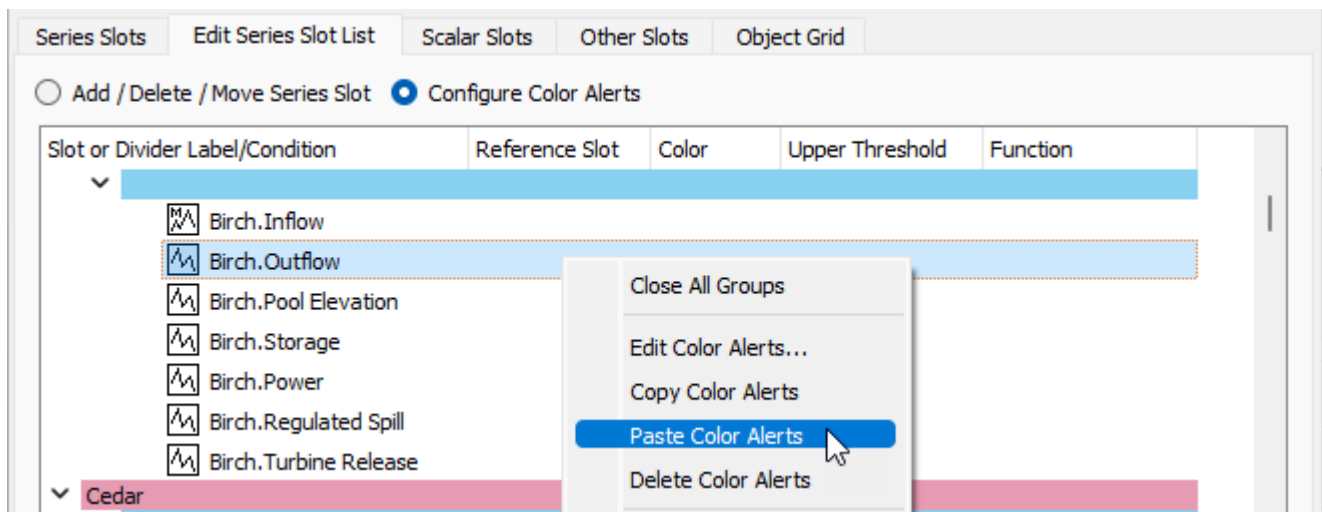
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- Right-click and select **Copy Color Alerts** in the shortcut menu.



- Select a slot to which you want to paste the Color Alert configuration.
- Right-click and select **Paste Color Alerts** in the shortcut menu.



The Reference Slot selection from the Color Alert configuration is applied to selected slot as follows (see [Select the Reference Slot.](#), page 152).

- If the default option to use the slot being formatted as the Reference Slot is selected in the copied configuration, then in the pasted configuration, the slot being pasted to will automatically become the Reference Slot.
- If the option to use an alternative Reference Slot is selected in the copied configuration, then that same alternative slot will be the Reference Slot in the pasted configuration.

The Condition Type and Conditions (interval thresholds or condition functions) are applied to the selected slot exactly as defined in the copied configuration (see [Define the Color Alert Conditions.](#), page 152).

7. Optionally, you can repeat [Step 5.](#) and [Step 6.](#) to paste the same Color Alert configuration to additional slots.
8. After pasting the Color Alert configuration, you can then edit individual settings to define conditions and colors that are unique to this slot, if you want.

Related topics

- [Configure Color Alerts](#), page 150

Configuring SCT Settings

This section includes tasks for customizing the appearance of the SCT, including timesteps used, flag and background colors, column widths, and whether to view NaNs. These settings are saved with the SCT.

Note: All procedures in this section require the SCT configuration to be unlocked. See [“Lock or unlock the SCT Configuration”](#) on page 164 for details.

Note: While you configure an SCT, be sure to save it frequently so you do not lose your changes.

Topics

- [Lock or unlock the SCT Configuration](#), page 164
- [Configure timesteps](#), page 165
- [Enable or disable the run status display](#), page 166
- [Configure NaN display](#), page 167
- [Configure worksheet display options](#), page 168
- [Configure cell value editing options](#), page 169
- [Specify the type of slot indexing for the SCT](#), page 171
- [Configure horizontal timestep display](#), page 173
- [Configure vertical timestep display](#), page 177
- [Configure and select the SCT font](#), page 179
- [Show or hide buttons on the SCT toolbar](#), page 181
- [Configure display of summary cells](#), page 184
- [Show or hide flag buttons on the SCT toolbar](#), page 187
- [Customize flag colors](#), page 189
- [Customize divider and cell colors](#), page 190
- [Define custom colors for highlights and color alerts](#), page 191

Lock or unlock the SCT Configuration

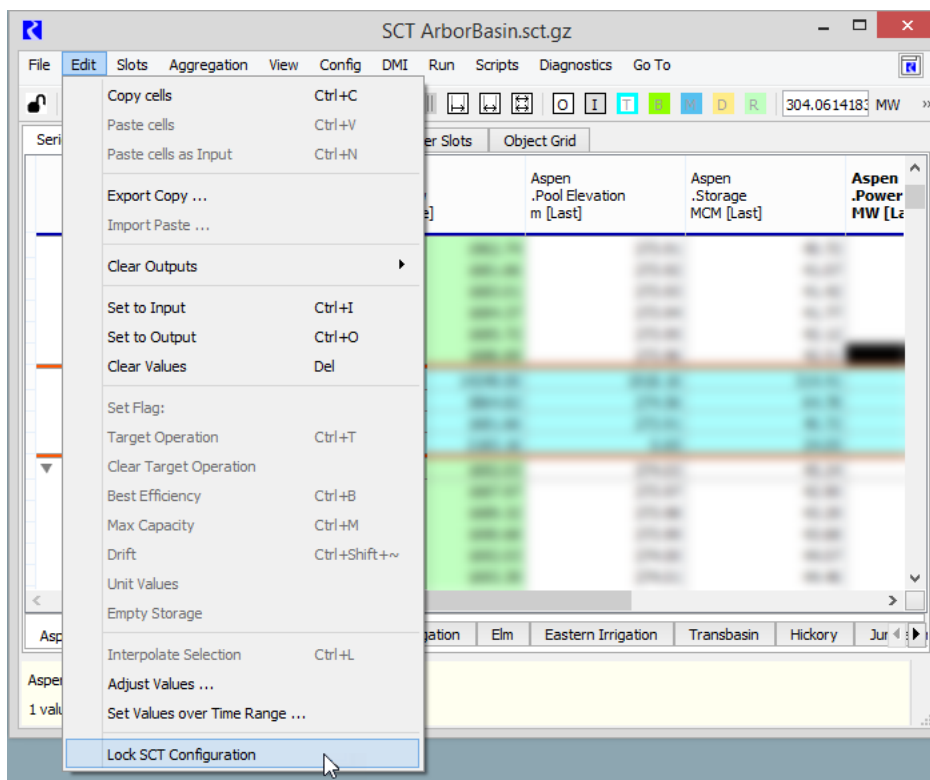
When you load an existing SCT, the configuration is locked by default, and you must unlock it to make changes. After you make your changes, it is good practice to lock the configuration so slots are not added, deleted, or moved inadvertently.

Note: You can still use the SCT to change data values when the configuration is locked.

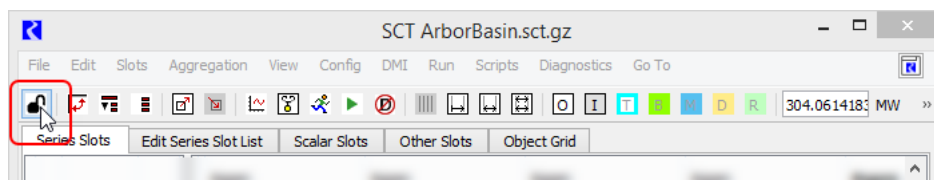
Locking the SCT has the following effects:

- The Edit Series Slot List tab is made unavailable.
- All options in the SCT Configuration window are made unavailable.
- You cannot add, move, or delete slots from the SCT.
- You cannot add, move, or remove slot or sheet dividers.
- You cannot change slot labels or horizontal slot divider labels.
- You cannot modify slot summary functions, which are used for aggregations.

1. To lock or unlock the SCT, do either of the following.
 - On the SCT menu, select **Edit**, then select or clear **Lock SCT Configuration**.



- On the SCT toolbar, select or clear **Lock SCT Config**.



Related topics

- [Configuring SCT Settings](#), page 163
- [SCT Configuration](#), page 33

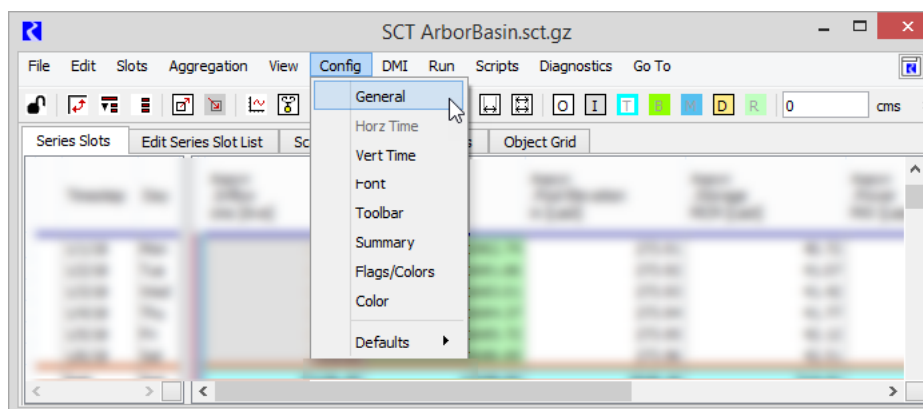
Configure timesteps

Use this procedure to configure how the SCT aligns with the timestep defined for the model. You can choose to do any of the following:

- Always synchronize the time range with the slots included in the SCT.
- Display additional timesteps before the beginning or after the end of the run.
- Override the model timestep size.

By default, the SCT displays the initialization timestep and all timesteps of the model. However, sometimes the simulation needs values for more than one timestep before the start time; for example, Reaches with multi-timestep lags need several previous values of Inflow to solve for the Outflow at the first timestep. Similarly, post-simulation timesteps may be affected by lags in a Reach.

1. On the SCT menu, select **Config**, then **General**.



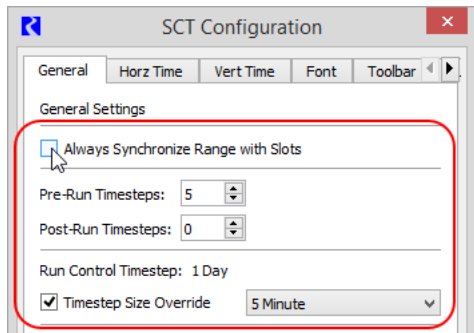
The SCT Configuration window opens at the General tab.

2. Select or clear the **Always Synchronize Range with Slots** check box, as follows:
 - Select if you want the SCT to always use the exact same timesteps as the included slots.
 - Clear if you want the SCT timesteps to be different from the included slots, then complete the following settings to specify the differences. See [“General tab”](#) on page 285 for details on these settings.

Chapter 3

Using the SCT

- Pre-Run Timesteps
- Post-Run Timesteps
- Run Control Timestep
- Timestep Size Override



3. Select **Apply** or **OK** to apply the changes to the SCT.

The SCT display is immediately updated according to your specifications.

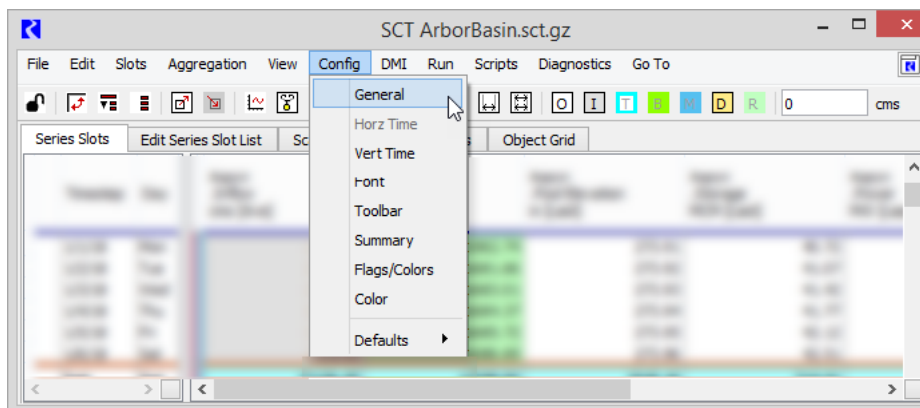
Related topics

- [Timestep size](#), page 15

Enable or disable the run status display

Use this procedure to configure whether the SCT displays model simulation run status information during and upon completion of a simulation run.

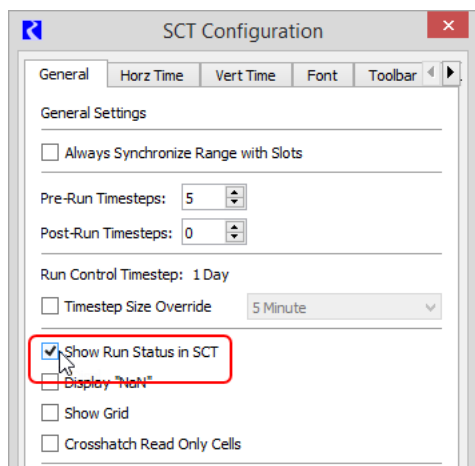
1. On the SCT menu, select **Config**, then **General**.



The SCT Configuration window opens at the General tab.

2. Select or clear the **Show Run Status in SCT** check box, as follows. See [“General tab”](#) on page 285 for details on this setting.
- Select to display model simulation run status.

- Clear to hide model simulation run status.



3. Select **Apply** or **OK** to apply the changes to the SCT.

The SCT is updated according to your specifications, and the settings will apply the next time you run a model simulation.

Related topics

- [Series Slots tab](#), page 7

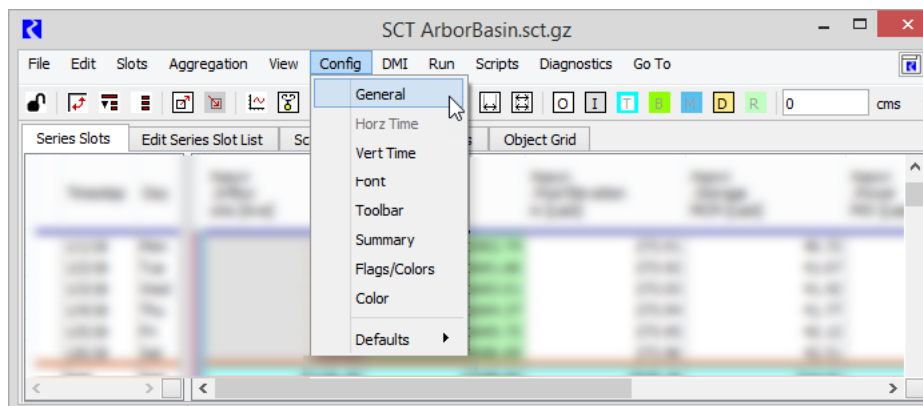
Configure NaN display

Use this procedure to specify whether blank cells in the SCT should be displayed as NaNs.

Blank cells in the SCT indicate memory that has not yet been allocated. NaNs are invalid values in slots. Until a value is set on a series slot used as Output in RiverWare, the memory to hold that value is not allocated.

When you save and reload a model, all series slots with no values are condensed into one NaN at the initial timestep. This is identical to the appearance of a new series slot in the Open Slot dialog box. NaNs displayed as blanks appear identical to unallocated values in the SCT.

1. On the SCT menu, select **Config**, then **General**.

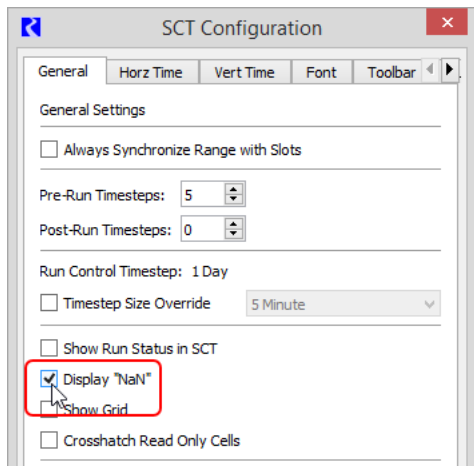


Chapter 3

Using the SCT

The SCT Configuration window opens at the General tab.

2. Select or clear the **Display “NaN”** check box as follows. See [“General tab”](#) on page 285 for details on this setting.
 - Select if you want all blank cells to display as “NaN”.
 - Clear if you want undefined numeric values to be shown as blanks (empty cells).



3. Select **Apply** or **OK** to apply the change to the SCT.

The SCT display is immediately updated.

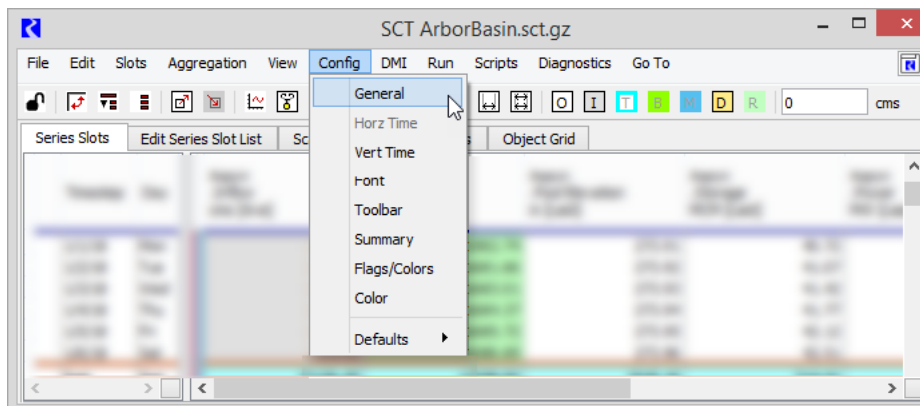
Related topics

- [Display units](#), page 25

Configure worksheet display options

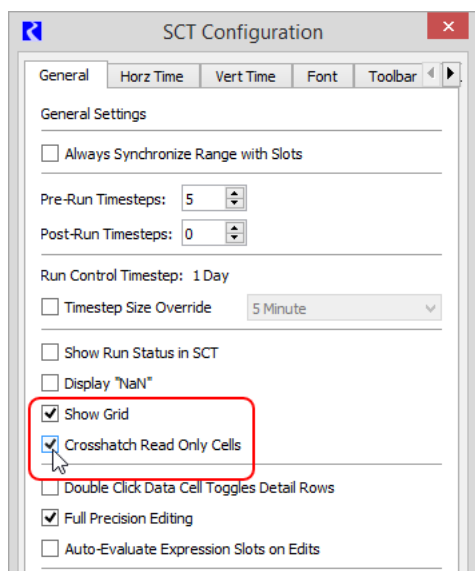
Use this procedure to specify display options for worksheets on the Series Slots tab. See [“General tab”](#) on page 285 for details on these options.

1. On the SCT menu, select **Config**, then **General**.



The SCT Configuration window opens at the General tab.

2. Select or clear the **Show Grid** check box, as follows:
 - Select if you want a thin border drawn around each cell.
 - Clear if you want no borders around cells.
3. Select or clear the **Crosshatch Read Only Cells** check box, as follows. Read-only cells contain non-editable values. These include cells for expression slots and accounting system slots.
 - Select if you want all read-only cells to have a crosshatched background in a user-defined Read Only color.
 - Clear if you want read-only cells to be displayed using their default background.



4. Select **Apply** or **OK** to apply the change to the SCT.

The SCT display is immediately updated.

Related topics

- [Apply custom colors to timestep values](#), page 117

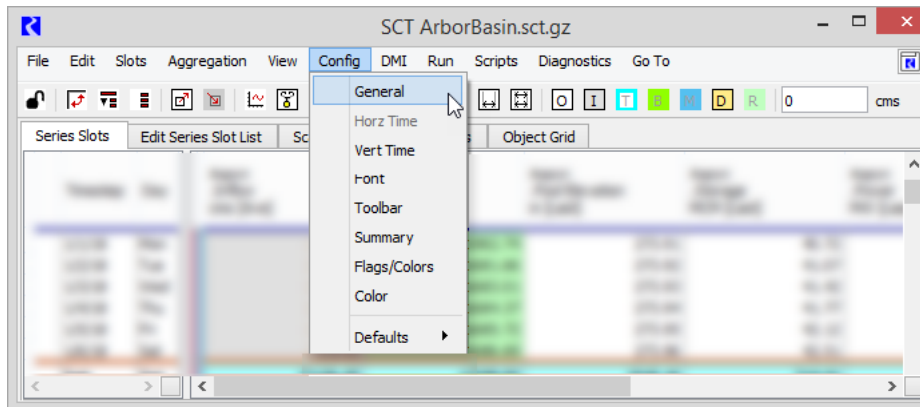
Configure cell value editing options

Use this procedure to configure methods for editing and updating timestep values on the Series Slots tab. See [“General tab”](#) on page 285 for details on these options.

Chapter 3

Using the SCT

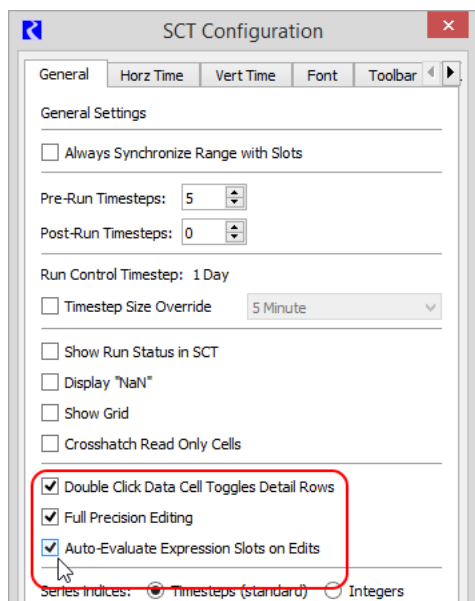
1. On the SCT menu, select **Config**, then **General**.



The SCT Configuration window opens at the General tab.

2. Select or clear the **Double Click Data Cell Toggles Detail Rows** check box, as follows:
 - Select to make all cells read only; double-clicking a cell does not open it for editing.
Note: For aggregated view with Show/Hide Details *only*, double-clicking a cell causes the aggregation interval display to switch between summary and detail view.
 - Clear to make all cells, except read-only cells, editable by double-clicking.
3. Select or clear the **Full Precision Editing** check box, as follows:
 - Select if you want cell values to be editable to full precision.
 - Clear if you want cell values to be editable to display precision only.
4. Select or clear the **Auto-Evaluate Expression Slots on Edits** check box, as follows:
 - Select if you want to fully evaluate across all timesteps whenever a single timestep value is edited.
Note: This may cause refresh delays in the SCT if there are many expression slots or if the slots do not evaluate quickly.

- Clear if you do not want expression slots to fully evaluate across all timesteps when a single value is edited.



5. Select **Apply** or **OK** to apply the change to the SCT.

The SCT display is immediately updated.

Related topics

- [Data in the SCT](#), page 5

Specify the type of slot indexing for the SCT

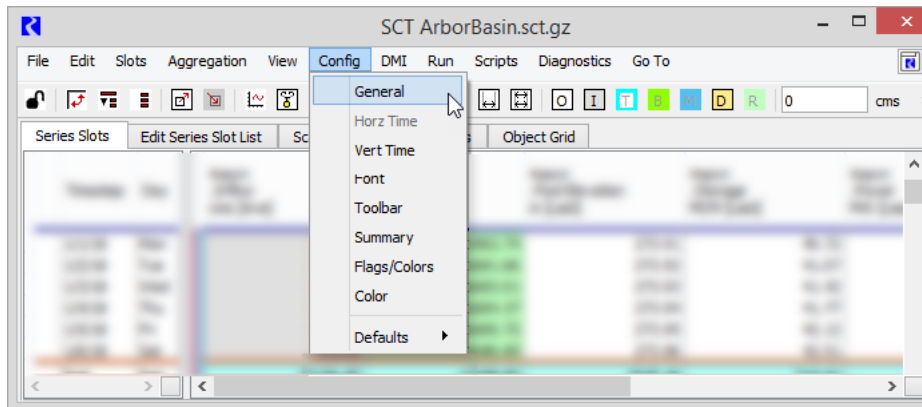
Use this procedure to specify the type of slots that can be included in the SCT. By default, the SCT displays timeseries slots, but you can configure an SCT to display integer-indexed slots instead. Each slot type has unique properties; therefore, an SCT can display one type or the other, not both. See [“Integer-indexed SCTs”](#) on page 302 for details.

Note: If you want to use an SCT for integer-indexed slots, you must use this procedure before adding any slots.

Chapter 3

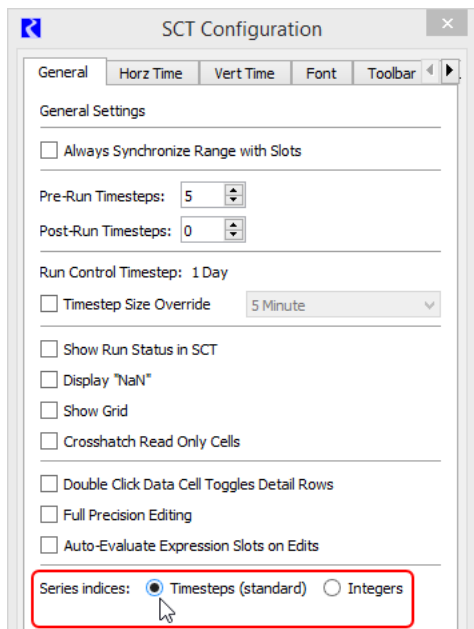
Using the SCT

1. On the SCT menu, select **Config**, then **General**.



The SCT Configuration window opens at the General tab.

2. Select one of the following Series Indices options. See [“General tab”](#) on page 285 for details.
 - Select **Timesteps (standard)** if you want the SCT to display and manage timeseries data only. This is the default option.
 - Select **Integers** if you want the SCT to display and manage integer-indexed slots.



3. Select **Apply** or **OK** to apply the change to the SCT.

The SCT display is immediately updated.

Note: If the SCT already included slots, any slots of the other type are retained, but their values are hidden.

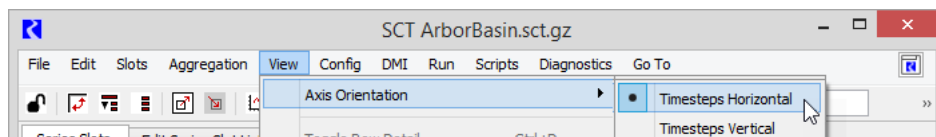
Related topics

- [Integer-indexed SCTs](#), page 302

Configure horizontal timestep display

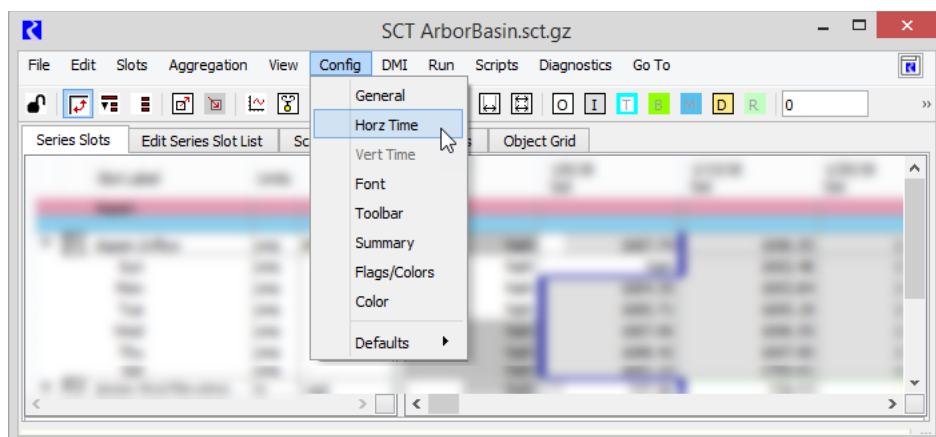
Use this procedure to configure the horizontal timestep orientation on the Series Slots tab. See [“Horz Time tab”](#) on page 288 for details on these settings.

1. Go to the **Series Slots** tab.
2. In the SCT menu, select **View**, then **Axis Orientation**, then **Timesteps Horizontal**.



The SCT is displayed in horizontal timestep orientation.

3. On the SCT menu, select **Config**, then **Horz Time**.



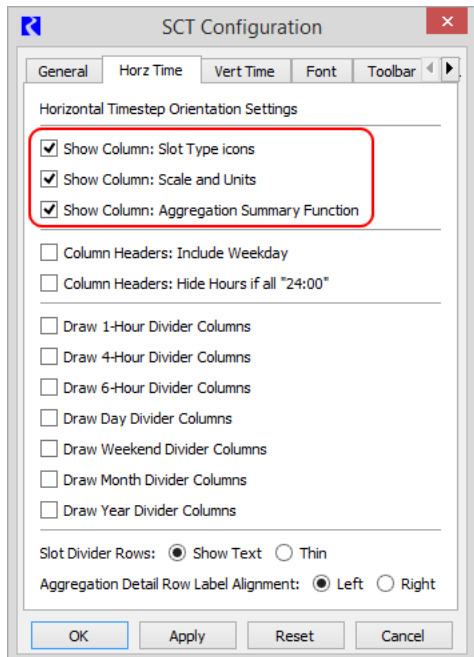
The SCT Configuration window opens at the Horizontal Timestep Orientation Settings tab.

4. Specify the columns you want to include in the slot row header. Select a check box to include the column, or clear the check box to hide it.
 - **Slot Type icons**—identifies the slot type of each slot.
 - **Scale and Units**—identifies the unit type for each slot.

Chapter 3

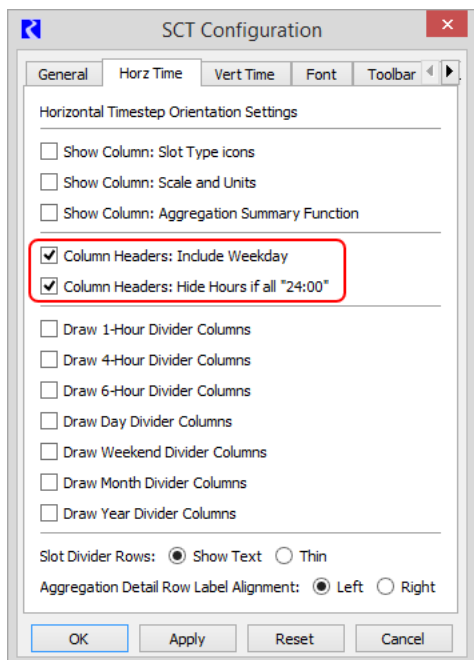
Using the SCT

- **Aggregation Summary Function**—identifies the summary function assigned to each slot, and the information is displayed only when aggregated view is enabled.



5. Specify the information you want to include in the timestep column header. Select a check box to include the information, or clear the check box to hide it.

- **Include Weekday**—identifies the day of the week.
- **Hide Hours if all "24:00"**—uses 24-hour time.



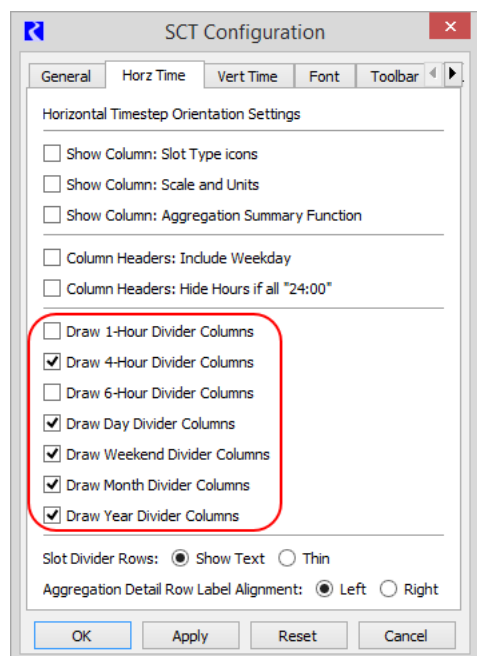
6. Specify the divider columns you want to include in the display. Each divider appears as a thick line in a user-defined color.

You can select *only one* of the following check boxes:

- **Draw 1-Hour Divider Columns**
- **Draw 4-Hour Divider Columns**
- **Draw 6-Hour Divider Columns**

You can select any number of the following check boxes:

- **Draw Day Divider Columns**
- **Draw Weekend Divider Columns**
- **Draw Month Divider Columns**
- **Draw Year Divider Columns**

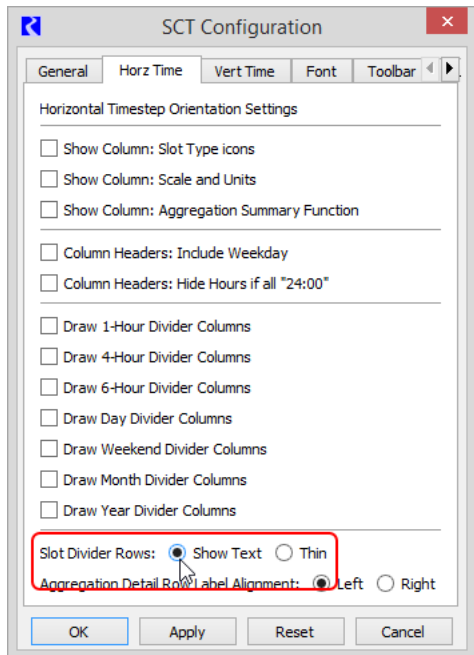


7. Select one of the following Slot Divider Rows options:
 - **Show Text**—includes the user-defined label in the divider row. The height of the divider row is adjusted so there is enough room to fit the text.

Chapter 3

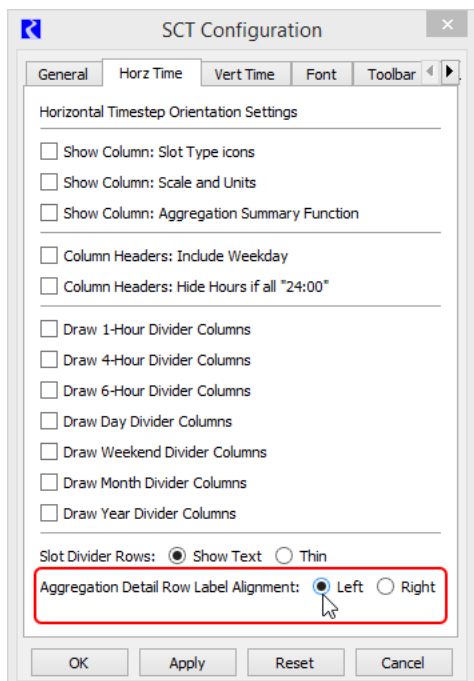
Using the SCT

- **Thin**—excludes the user-defined label from the divider row. The divider appears as a thin line with no text. This option is useful if you need to conserve vertical space.



8. Select one of the following Aggregation Detail Row Label Alignment options. This setting applies only to aggregated view using either Show/Hide Details or Show All Details.

- **Left**—makes the label text left-aligned.
- **Right**—makes the label text right-aligned.



9. Select **Apply** or **OK** to apply the settings to the SCT.

The SCT display is immediately updated.

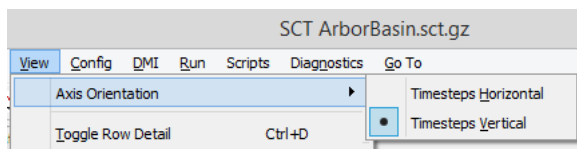
Related topics

- [Series Slots tab](#), page 7
- [Timestep orientation](#), page 16
- [Customize divider and cell colors](#), page 190

Configure vertical timestep display

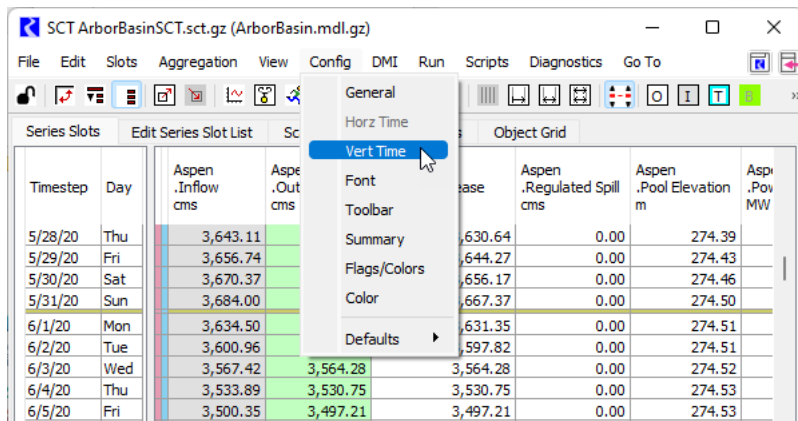
Use this procedure to configure the vertical timestep orientation on the Series Slots tab. See “[Vert Time tab](#)” on page 291 for details on these settings.

1. On the SCT menu, select **View**, then **Axis Orientation**, then **Timesteps Vertical**.



The SCT is displayed in vertical timestep orientation.

2. Go to the **Series Slots** tab.
3. On the SCT menu, select **Config**, then **Vert Time**.



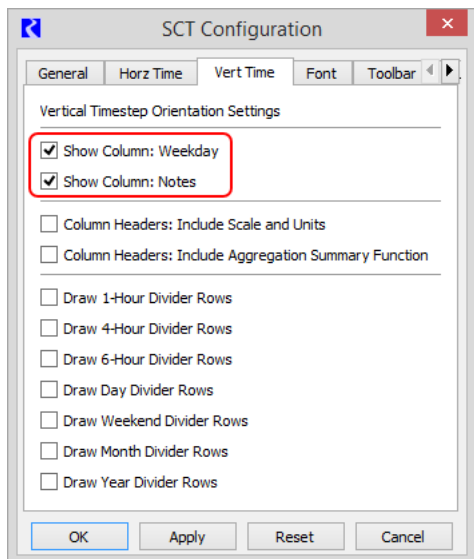
The SCT Configuration window opens at the Vertical Timestep Orientation Settings tab.

4. Specify the columns to include in the timestep row header. Select a check box to include the column, or clear the check box to hide it.
 - **Show Column: Weekday**—identifies the day of the week.

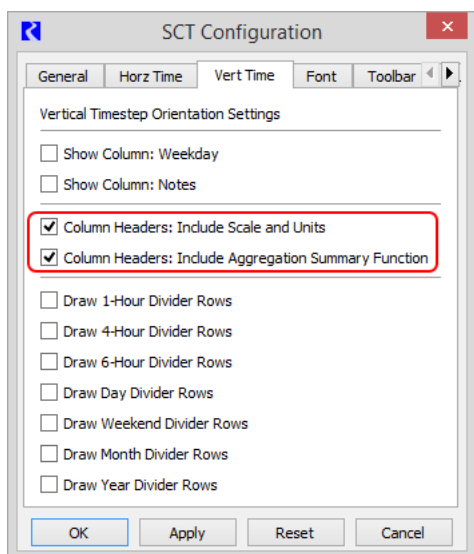
Chapter 3

Using the SCT

- **Show Column: Notes**—displays text of all notes assigned to slots at that timestep.



5. Specify the information you want to include in the slot column header. Select a check box to include the information, or clear the check box to hide it.
 - **Column Headers: Include Scale and Units**—identifies the unit type for each slot.
 - **Column Headers: Include Aggregation Summary Function**—identifies the summary function assigned to each slot, and the information is displayed only when aggregated view is enabled.



6. Specify the divider rows you want to include in the display. Each divider appears as a thick line in a user-defined color.

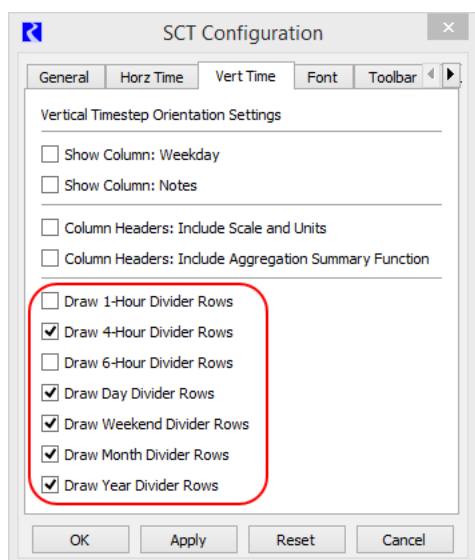
You can select *only one* of the following check boxes:

- **Draw 1-Hour Divider Rows**
- **Draw 4-Hour Divider Rows**

- **Draw 6-Hour Divider Rows**

You can select any number of the following check boxes:

- **Draw Day Divider Rows**
- **Draw Weekend Divider Rows**
- **Draw Month Divider Rows**
- **Draw Year Divider Rows**



7. Select **Apply** or **OK** to apply the settings to the SCT.

The SCT display is immediately updated.

Related topics

- [Series Slots tab](#), page 7
- [Timestep orientation](#), page 16
- [Customize divider and cell colors](#), page 190

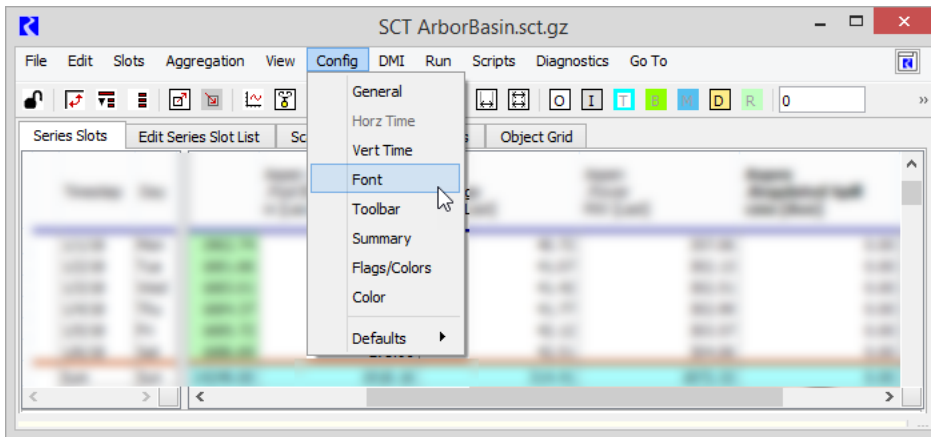
Configure and select the SCT font

Use this procedure to select the font to use for SCT screen display and printing. You can choose to use either the default SCT font or a user-defined font; you can configure up to two user-defined fonts (Font A and Font B).

Chapter 3

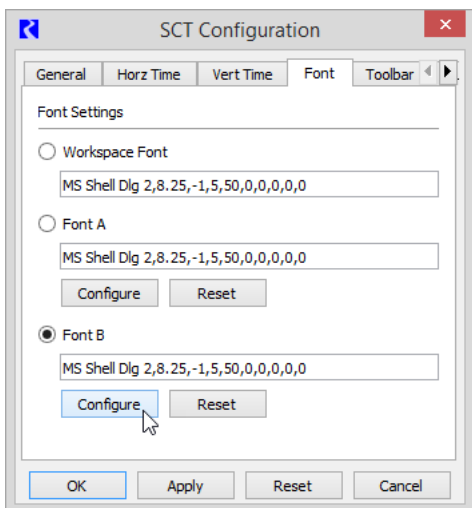
Using the SCT

1. On the SCT menu, select **Config**, then **Font**.



The SCT Configuration window opens at the Font Settings tab.

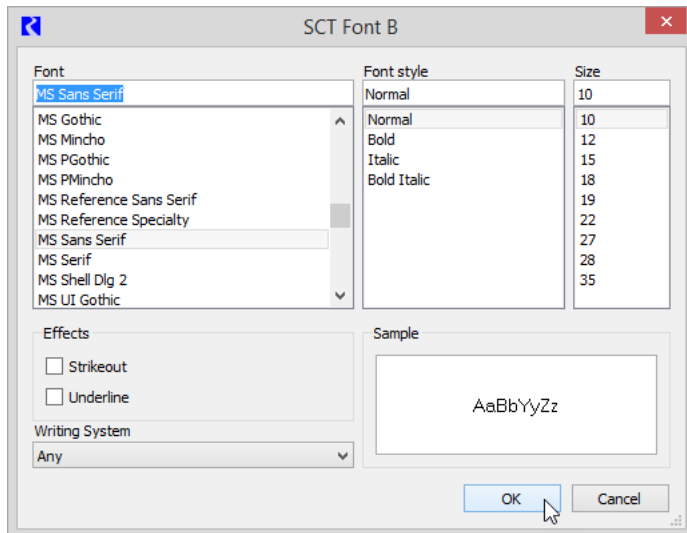
2. Use the following steps to configure a user-defined font (Font A or Font B).
 - a. Select either **Font A** or **Font B**, then select **Configure**.



The SCT Font dialog box opens.

- b. Use the font selector to specify the font characteristics, including family, style, size, and strikeout or underline effects.

- c. Select **OK** to apply the settings.



The dialog box is closed and the font specification is updated on the Fonts tab.

- d. Optionally, repeat [Step a.](#) through [Step c.](#) to configure the other user-defined font.
3. On the **Font** tab, specify the font to use in the SCT by selecting one of the following options, then select **Apply** or **OK** to apply the font settings.
- **Workspace Font**
 - **Font A**
 - **Font B**

The SCT display is updated; this may take a moment to complete since the SCT geometry needs to be readjusted.

Related topics

- [SCT Layout](#), page 6

Show or hide buttons on the SCT toolbar

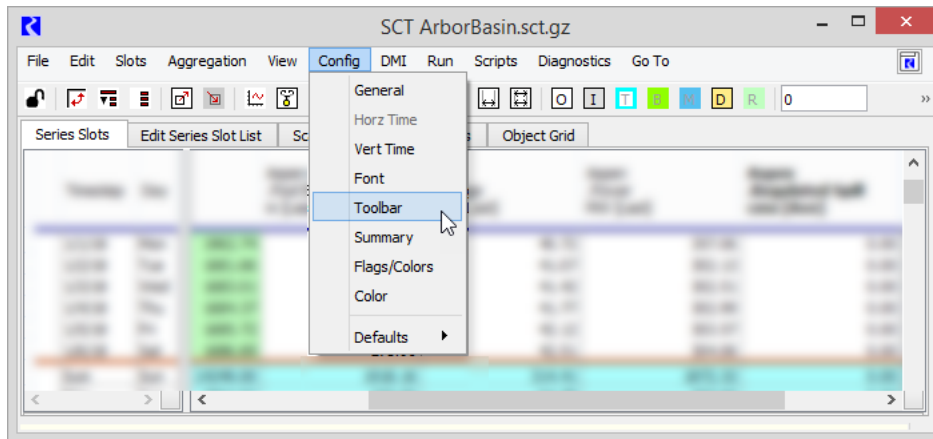
Use this procedure to specify which buttons to include in the SCT toolbar. You can include the buttons you use frequently and hide the ones you do not need. See [“SCT Toolbar”](#) on page 273 for details about the available buttons.

Note: Even if a button is hidden, you can still access the function from the SCT menus.

Chapter 3

Using the SCT

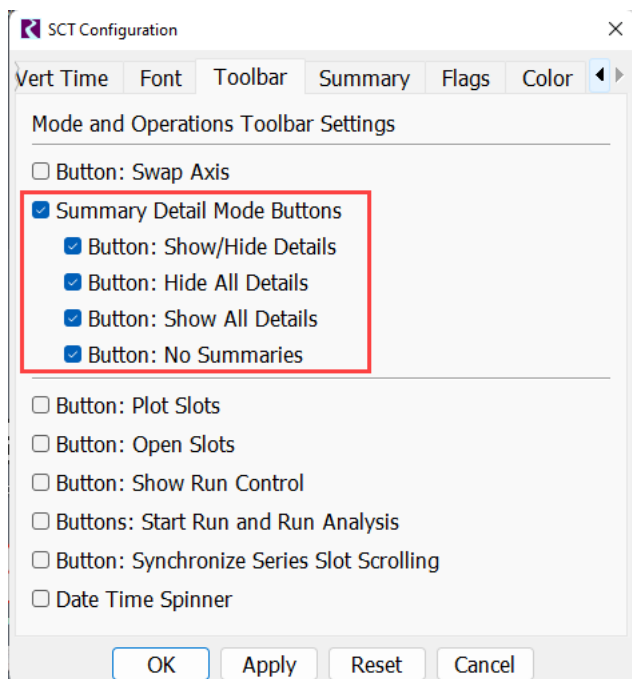
1. On the SCT menu, select **Config**, then **Toolbar**.



The SCT Configuration window opens at the Mode and Operations Toolbar Settings tab.

2. Specify whether to include detail mode buttons, as follows. These buttons control the display of aggregated slot values.
 - To hide all detail mode buttons, clear the **Summary Detail Mode Buttons** check box.
 - To show one or more detail mode buttons, select the **Summary Detail Mode Buttons** check box. Select the check boxes of the buttons you want to include, and clear the check boxes of the buttons you want to hide.
 - **Button: Show/Hide Details**—this button enables aggregated view, with summary and detail values in a dynamic expand/collapse format.
 - **Button: Hide All Details**—this button enables aggregated view locked in summary view.
 - **Button: Show All Details**—this button enables aggregated view locked in summary and detail view.

- **Button: No Summaries**—this button disables aggregated view.

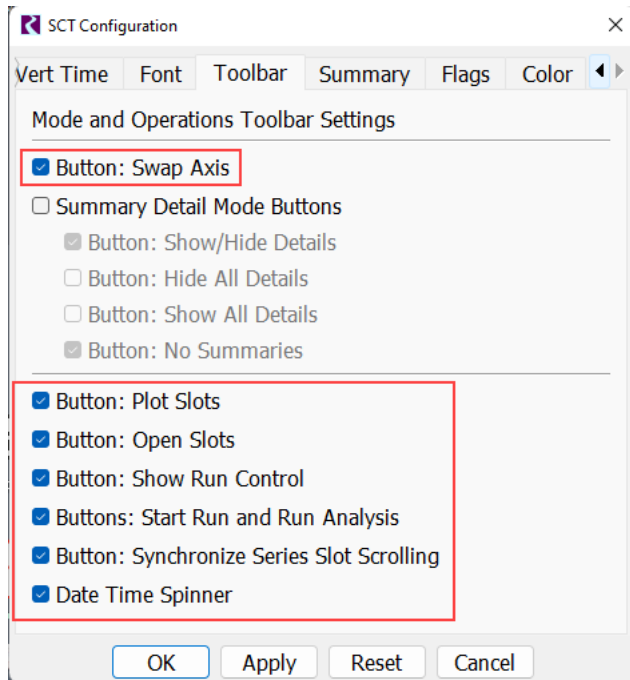


3. Select any of the following check boxes to include the specified item, or clear the check box to hide it.
 - **Button: Swap Axis**—this button switches between horizontal and vertical timestep orientation.
 - **Button: Plot Slots**—this button opens the selected slots in the Plot Page Editor.
 - **Button: Open Slots**—this button opens the selected slots in the Slot Viewer window.
 - **Button: Show Run Control**—this button opens the Run Control window.
 - **Button: Start Run and Run Analysis**—the Start Run button initiates a model simulation run; the Run Analysis button opens the Model Run Analysis tool.
 - **Button: Synchronize Series Slot Scrolling**—this button toggles the synchronized scrolling on and off. See also [“Synchronized Scrolling”](#) on page 18

Chapter 3

Using the SCT

- **Button: Date Time Spinner**—this field allows you to enter a timestep date, or step forward and backward through timesteps, one at a time. The Date Time Spinner shows the last selected cell's timestep.



4. Select **Apply** or **OK** to apply the settings to the SCT.

The SCT display is immediately updated.

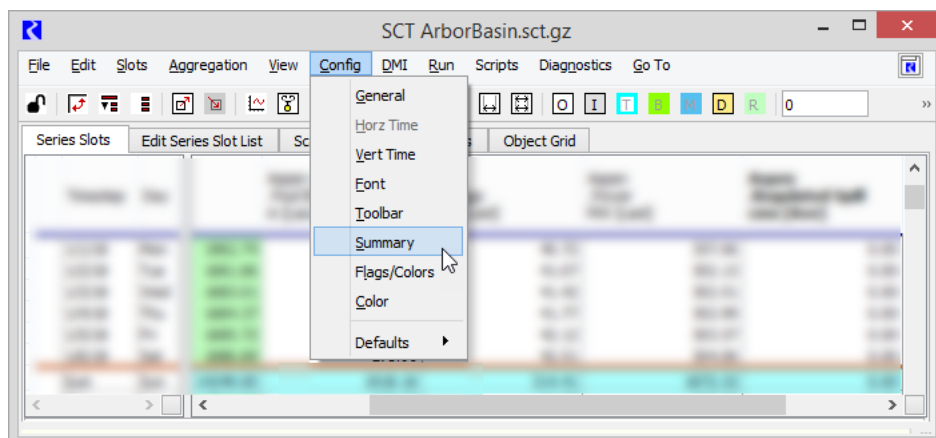
Related topics

- [SCT Layout](#), page 6

Configure display of summary cells

Use this procedure to configure the background colors used for aggregation summary rows, specify whether to display summary statistics in the SCT Status area, and define the colors used in the SCT Status area. See [“Summary tab”](#) on page 296 for details on these settings.

1. On the SCT menu, select **Config**, then **Summary**.



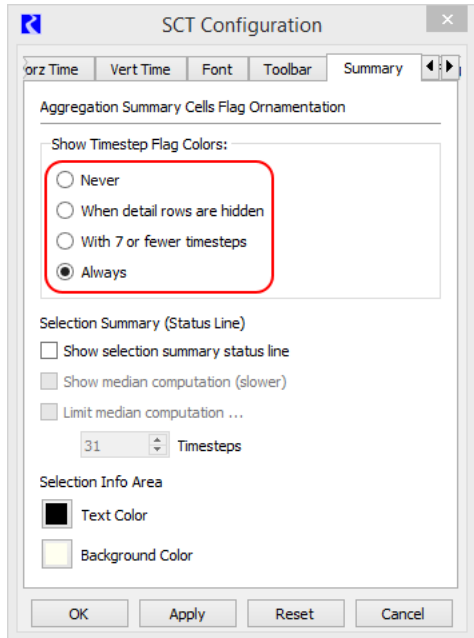
The SCT Configuration window opens at the Aggregation Summary Cells Flag Ornamentation tab.

2. In the **Show Timestep Flag Colors** setting, select one of the following options to indicate whether the background color for summary rows should be the cell value flag color or the Summary Cell Background color. These options allow you to experiment with ways of distinguishing summary rows from detail rows. They also support a faster screen refresh rate with large aggregations.
 - **Never**—summary rows never use flag colors in the background; they always use the Summary Cell Background color.
 - **When detail rows are hidden**—summary rows uses the appropriate flag color in the background when detailed timesteps are hidden; they use the Summary Cell Background color when details are shown.
 - **With 7 or fewer timesteps**—summary rows with seven or fewer timesteps use the appropriate flag color in the background; rows with more than seven timesteps use the Summary Cell Background color.

Chapter 3

Using the SCT

- **Always**—summary rows always use the appropriate flag color in the background. This is a good option in most circumstances.



3. Use the following steps to configure the SCT Status area.
 - a. Select or clear the **Show selection summary status line** check box, as follows.
 - Select to enable the display of summary statistics in the SCT Status area when one or more cells are selected in the same slot.

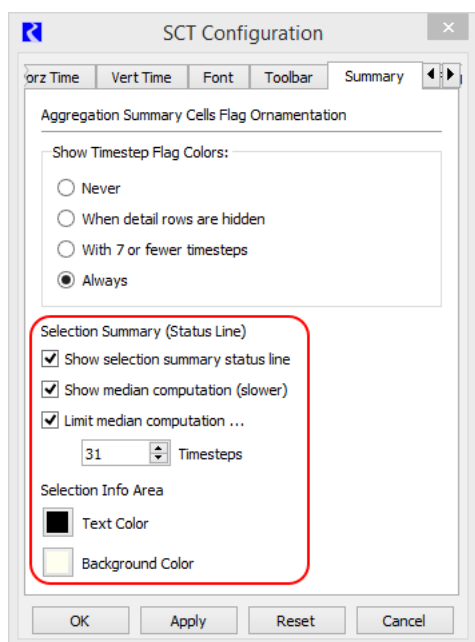
Note: If cells are selected in multiple slots, summary statistics are *not* displayed.

 - Clear to disable the display of summary statistics when cells are selected; proceed to [Step 4](#).
 - b. Configure the display of median values in the summary statistics, as follows.
 - Select the **Show median computation (slower)** checkbox to display the median value for selected cells.
 - Clear the check box to exclude median values; proceed to [Step 4](#).

Note: The median value requires the selected values to be sorted; therefore, enabling this option may lead to slower response times in the SCT display. You can mitigate this by disabling the median value display when more than a specified number of cells have been selected.
 - c. Optionally disable the display of median values when more than a specified number of cells have been selected, as follows.
 - Select the **Limit median computation** check box to include median values only for a specified number of selected cells. In the **Timesteps** field, enter the maximum number of selected cells for which median values will be computed. The default is the number of timesteps in the aggregation interval.
 - Clear the check box to display the median regardless of the number of cells selected; proceed to [Step 4](#).

The median is displayed only if the number of cells selected is less than or equal to this number. For example, if this field is “5” and you select five or fewer cells, the median is included; if you select six or more cells, the median is *not* included.

4. Configure the appearance of the SCT Status area, as follows.
 - Select the **Text Color** button to customize the text color used in the SCT Status area. This selection opens a color selector dialog box that allows you to define or select a custom color.
 - Select the **Background Color** button to customize the background color used in the SCT Status area. This selection opens a color selector dialog box that allows you to define or select a custom color.



5. Select **Apply** or **OK** to apply the settings to the SCT.

The SCT display is immediately updated.

Related topics

- [Displaying Aggregated Values](#), page 123
- [Enable or disable aggregated view](#), page 127

Show or hide flag buttons on the SCT toolbar

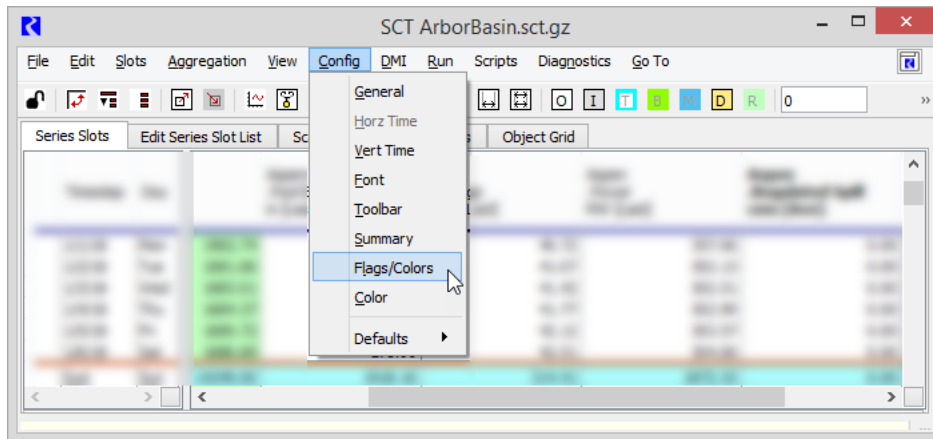
Use this procedure to specify which value flag buttons to include on the SCT toolbar. You can include the flags you use frequently and hide the others. See “[Flags tab](#)” on page 298 for details.

Note: Even if a flag button is hidden, you can still set the flag from the SCT menus.

Chapter 3

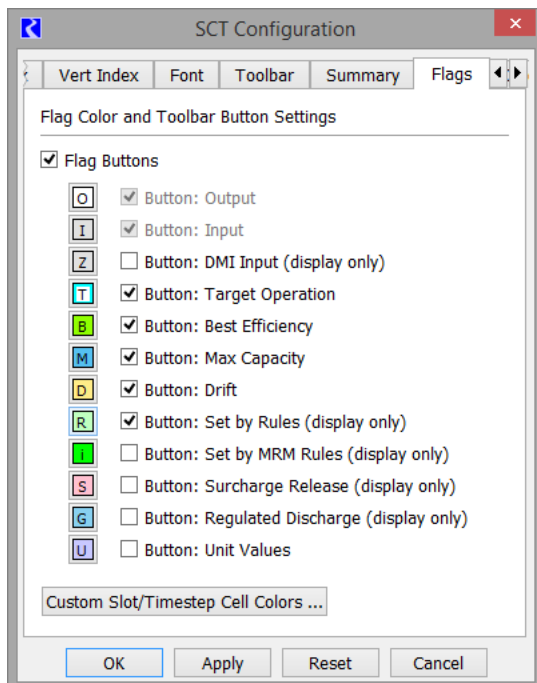
Using the SCT

1. On the SCT menu, select **Config**, then **Flags/Colors**.



The SCT Configuration window opens at the Flag Color and Toolbar Button Settings tab.

2. Specify whether to include one or more flag buttons, as follows.
 - To hide all flag buttons, clear the **Flag Buttons** check box.
 - To show one or more flag buttons, select the **Flag Buttons** check box. Select the check boxes of the buttons you want to include, and clear the check boxes of the buttons you want to hide. See “[Flags tab](#)” on page 298 for descriptions of the flags.



3. Select **Apply** or **OK** to apply the settings to the SCT.

The SCT display is immediately updated.

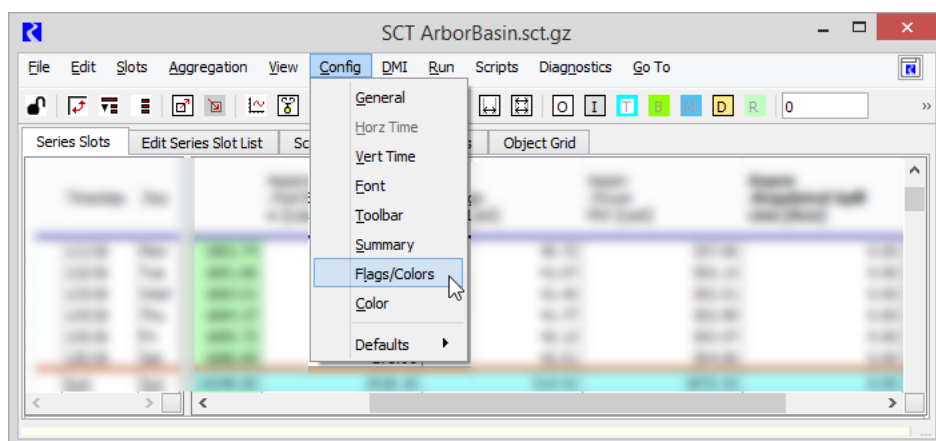
Related topics

- [Flags and cell colors](#), page 26
- [Customize flag colors](#), page 189

Customize flag colors

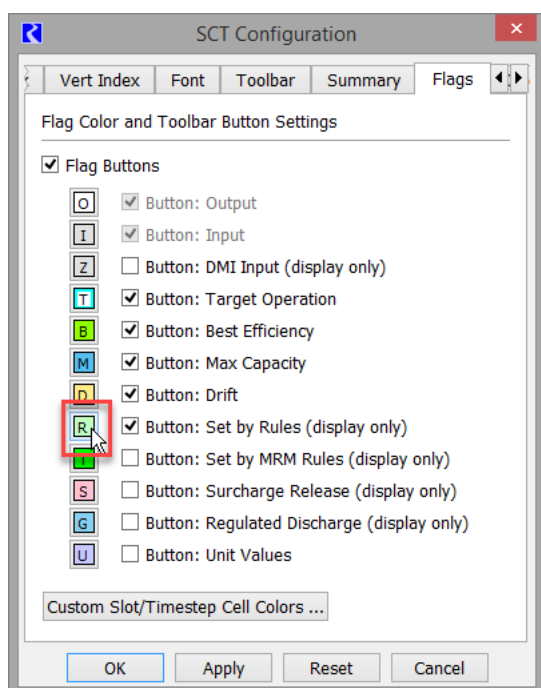
Use this procedure to assign user-defined colors to any of the cell value flags used in the SCT.

1. On the SCT Menu, select **Config**, then **Flags/Colors**.



The SCT Configuration window opens at the Flag Color and Toolbar Button Settings tab.

2. Select the color button next to the flag you want to modify; see [“Flags tab”](#) on page 298 for descriptions of the flags.



Chapter 3

Using the SCT

A Color Chooser dialog box opens.

3. Use one of the color selector tools in the dialog box to define the color to use.
4. Select **OK** to apply the change.

The flag button is updated with the new color.

5. Optionally repeat [Step 2.](#) through [Step 4.](#) for additional flags.
6. Select **Apply** to apply the changes to the SCT.

The window closes and all cells using the flag are updated.

Related topics

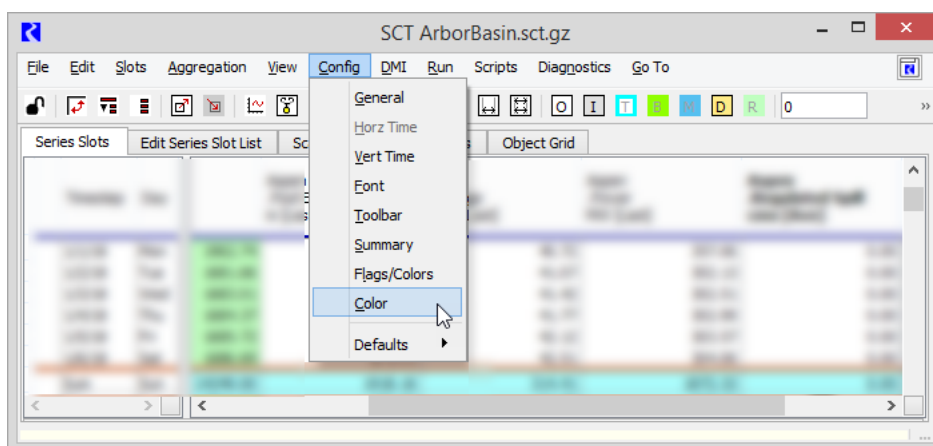
- [Flags and priorities](#) , page 26
- [Show or hide flag buttons on the SCT toolbar](#), page 187

Customize divider and cell colors

Use this procedure to assign user-defined colors to any of the following items used throughout the SCT:

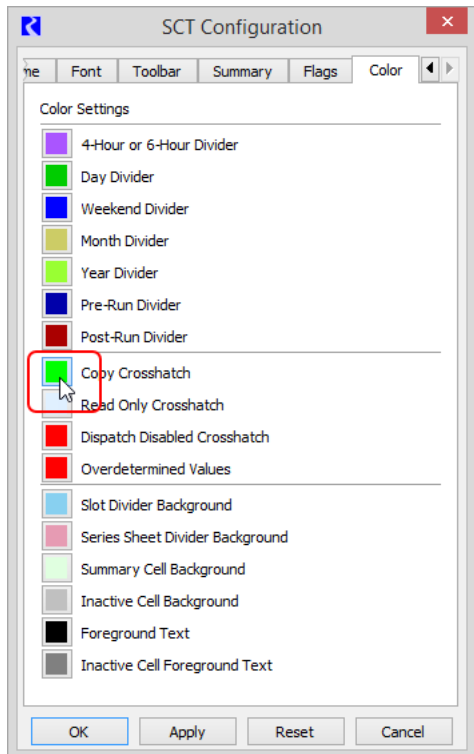
- Slot group and sheet dividers
- Summary cell and inactive cell background
- Time-period dividers (day, month, year, for example)
- Pre-run and post-run dividers
- Cell crosshatching
- Cell text

1. On the SCT Menu, select **Config**, then **Color**.



The SCT Configuration window opens with the Color tab active.

2. Select the color button next to the item you want to modify; see “Color tab” on page 300 for descriptions of the items.



A Color Chooser dialog box opens.

3. Use one of the color selector tools in the dialog box to define the color to use.
4. Select **OK** to apply the change.

The color button is updated with the new color.

5. Optionally repeat [Step 2.](#) through [Step 4.](#) for additional items.
6. Select **Apply** to apply the changes to the SCT.

The window closes and the affected items in the SCT are updated with the new colors.

Related topics

- [Add slot dividers](#), page 54
- [Add sheet dividers](#), page 56
- [Enable or disable aggregated view](#), page 127

Define custom colors for highlights and color alerts

Use the following procedure to define custom color entries, which specify a custom text color, custom cell background color, or combination of both.

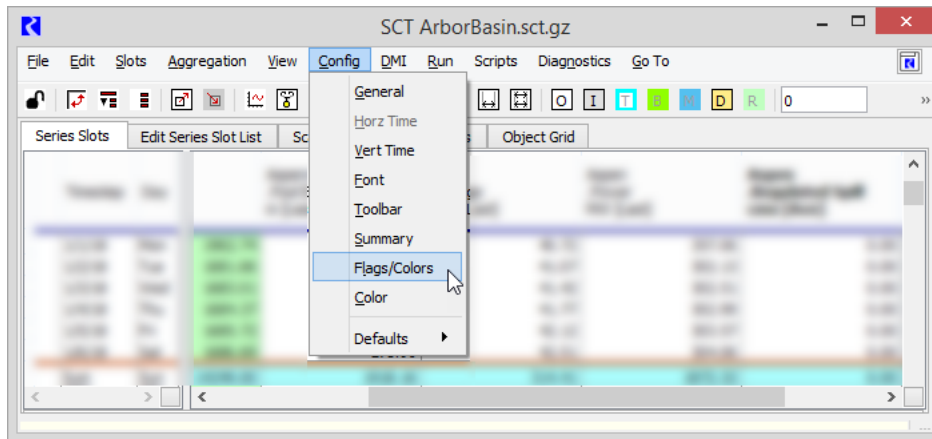
Chapter 3

Using the SCT

Each defined color entry is identified by a unique user-defined label. You can later assign the custom colors, by label, to selected SCT cells or Color Alerts; see [“Apply custom colors to timestep values”](#) on page 117 and [“Configure Color Alerts”](#) on page 150 for details.

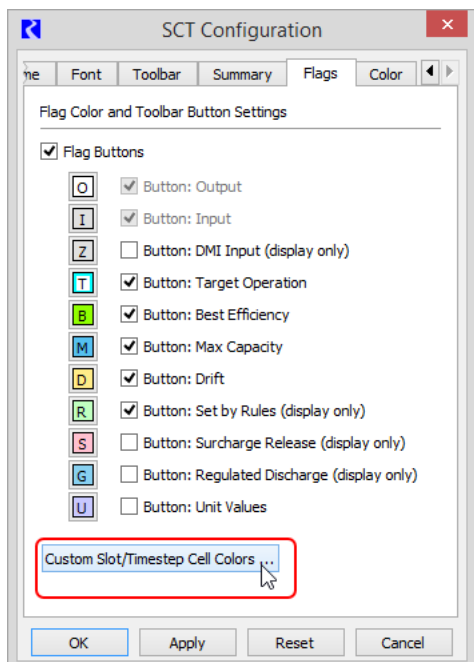
Note: These colors differ from custom colors assigned to dividers and cell value flags. See [“Customize divider and cell colors”](#) on page 190 for details about these colors.

1. On the SCT menu, select **Config**, then **Flags/Colors**.



The SCT Configuration dialog box opens at the Flags tab.

2. Select **Custom Slot/Timestep Cell Colors**.

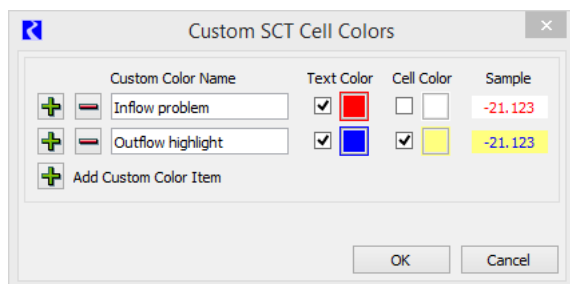


The Custom SCT Colors dialog box opens. Each row represents a custom color entry.

3. Add and delete rows in the dialog box, as follows:

- Add a row by selecting **Plus**  or delete a row by selecting **Minus** .
 - Select **Add Custom Color Item** to add a new row at the end of the list.
4. Use the following steps to define each custom color entry.
- a. In the **Custom Color Name** field, enter a unique name for the entry; an entry in this field is required.
 - b. Define a text color, background cell color, or combination of both, as follows:
 - Select the **Text Color** check box to apply the entry to cell text.
 - Select the **Cell Color** check box to apply the entry to cell backgrounds.

Note: If a check box is not selected, the default color for that attribute (text or cell background) is used.
 - c. For each selected check box—**Text Color** or **Cell Color**—select the associated color button.
A Color Chooser dialog box opens.
 - d. Use one of the color selector tools in the dialog box to define the color to apply, then select **OK**.
- The **Sample** field in the Custom SCT Cell Colors window shows the resulting text and background color combination.



5. When you are done defining colors, select **OK** to confirm the updates.
The Custom SCT Cell Colors dialog box closes.
6. In the SCT Configuration window, select **OK** or **Apply** to apply the definitions to the SCT.
7. Save the SCT to save the custom colors with the SCT.

Related topics

- [Apply custom colors to timestep values](#), page 117
- [Configure Color Alerts](#), page 150

Using an SCT in Operations

The procedures in this section allow you to perform RiverWare operations from the SCT. You can use the SCT as your single point of system control to modify values, run simulations, and analyze results.

Chapter 3

Using the SCT

Topics

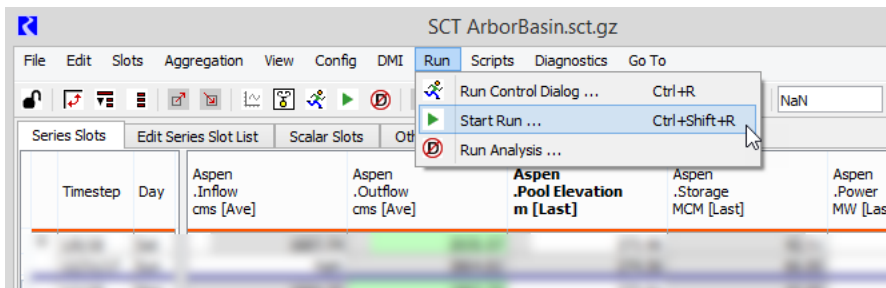
- [Run a model simulation from the SCT](#), page 194
- [Enable or disable dispatching](#), page 197
- [Set flags on series slot values](#), page 198
- [Set a target operation](#), page 201
- [Clear a target operation](#), page 203
- [Show or hide simulation diagnostic messages](#), page 204

Run a model simulation from the SCT

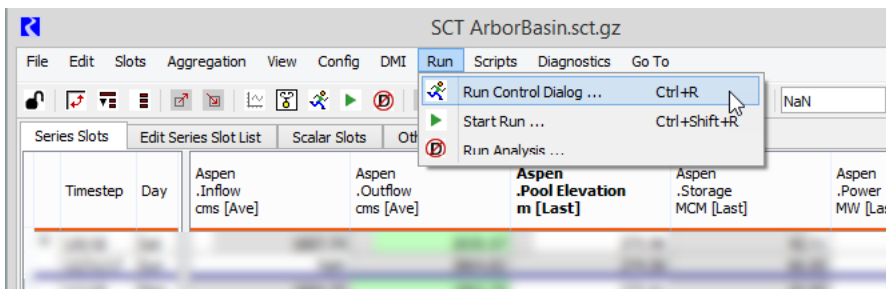
Use this procedure to run model simulations. You can do this directly from the SCT menu or from the Run Control window.

Note: You can perform this procedure from any SCT tab, but you must go to the Series Slots tab to see the run status display, if it is enabled.

1. Go to any tab in the SCT.
2. Proceed using one of the following methods.
 - To run the simulation using the existing parameters, select **Run**, then **Start Run** on the SCT menu. Alternatively, press Ctrl+Shift+R.



- To modify one or more run parameters before starting the run, use the following steps:
 1. Select **Run**, then **Run Control Dialog** on the SCT menu. Alternatively, press Ctrl+R.



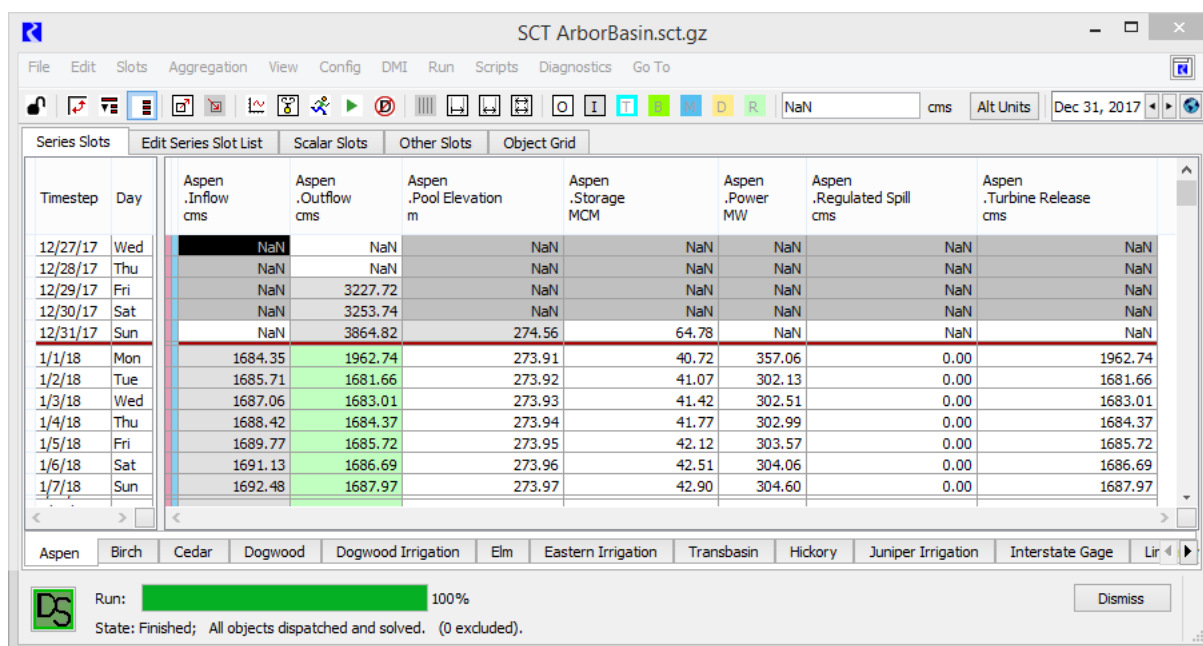
The Run Control window opens.

2. Set the run parameters to your specifications. See [“Run Control” in User Interface](#) for details.

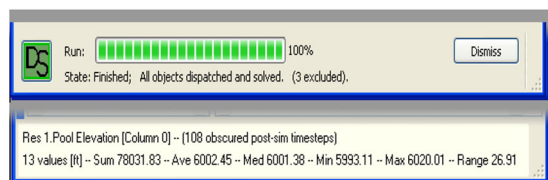
3. In the Run Control window, select **Start** to initiate the run.
3. Optionally, go to the **Series Slots** tab to view the run progress and results.

If the Show Run Status setting is enabled in the SCT configuration, the Status area on the Series Slots tab displays the status of the run as it progresses. You can select **Dismiss** or any cell in the SCT to dismiss the run status display and return to the default Status area display. See [“Enable or disable the run status display”](#) on page 166 for details.

When the run finishes, the updated timestep Output values are displayed on the Series Slots tab.



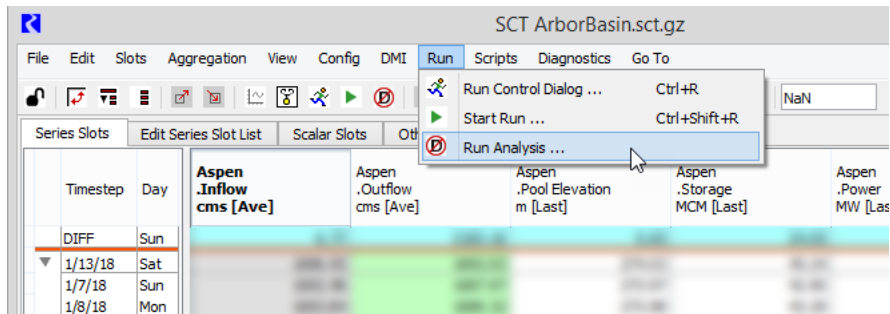
Note: Unlike the Diagnostic Output, the Run Status is simultaneously displayed in multiple SCTs and as a standalone dialog. However, the standalone window is not automatically raised if the status can be displayed in an open SCT—that is, by showing and raising any SCT that is already open.



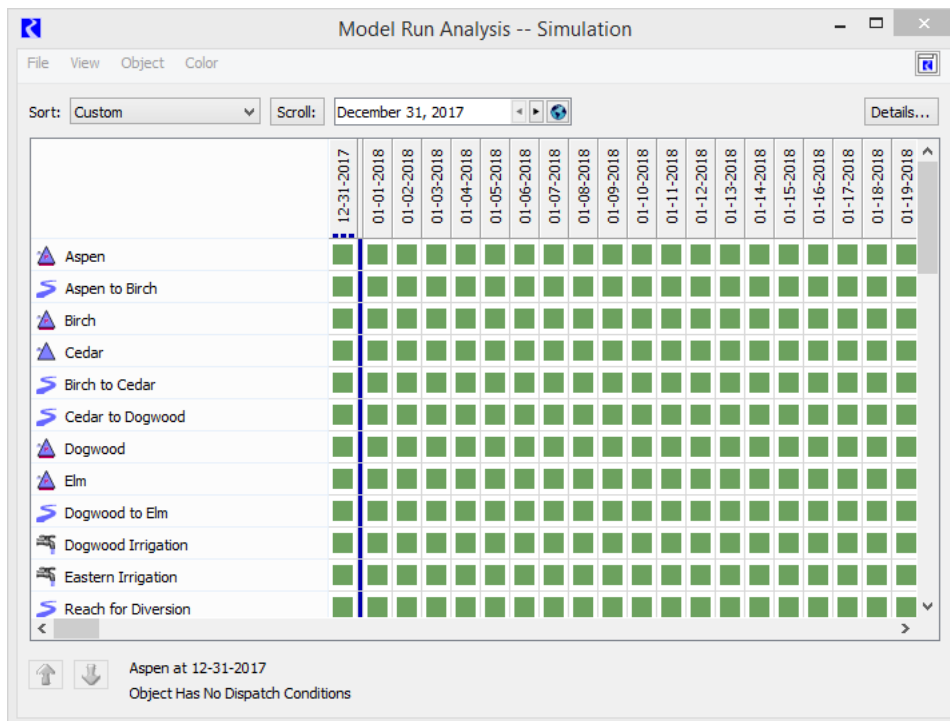
Chapter 3

Using the SCT

- Optionally, to analyze the run results, select **Run**, then **Run Analysis** on the SCT menu.



The Model Run Analysis tool opens, and the first relevant object at the earliest timestep is highlighted. See “[Model Run Analysis](#)” in *Debugging and Analysis* for details on using the tool.



Related topics

- [Enable or disable the run status display](#), page 166
- “[Model Run Analysis](#)” in *Debugging and Analysis*

Alternate methods

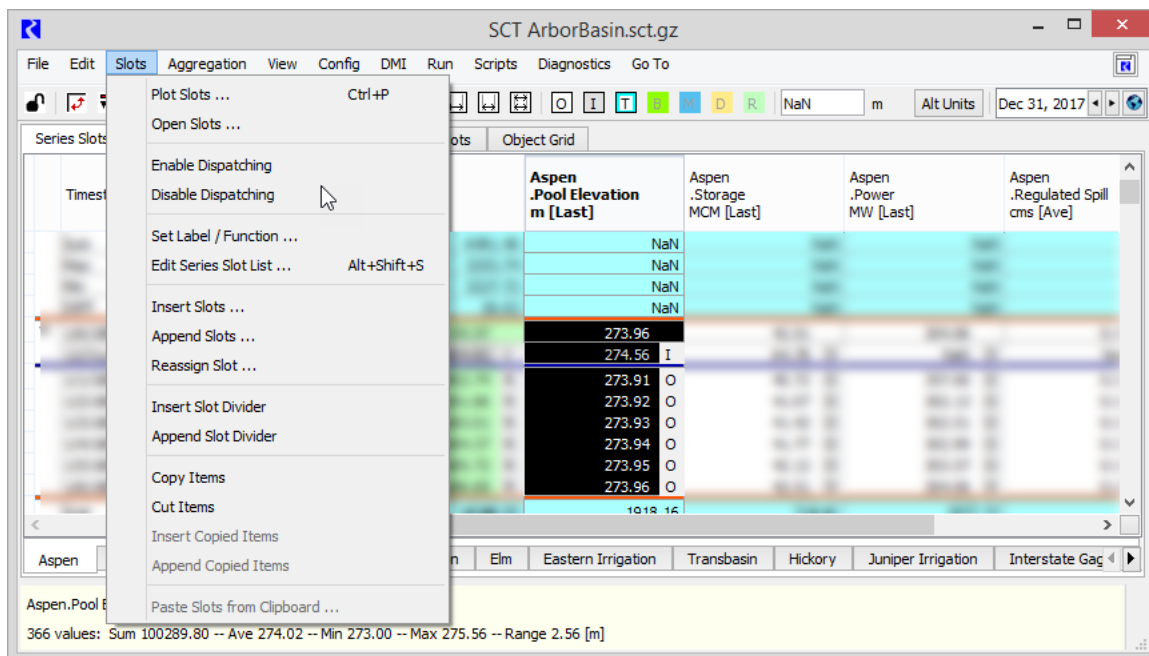
You can use one of the following methods to start the run.

- Select **Start Run**  on the SCT toolbar.
- Press Ctrl+Shft+R in the SCT window.

Enable or disable dispatching

Use this procedure to enable or disable dispatching on all series slots for a selected object.

1. Select any slot on the object you want to modify.
2. Make one of the following selections:
 - To enable dispatching, select **Slots**, then **Enable Dispatching**.
 - To disable dispatching, select **Slots**, then **Disable Dispatching**.

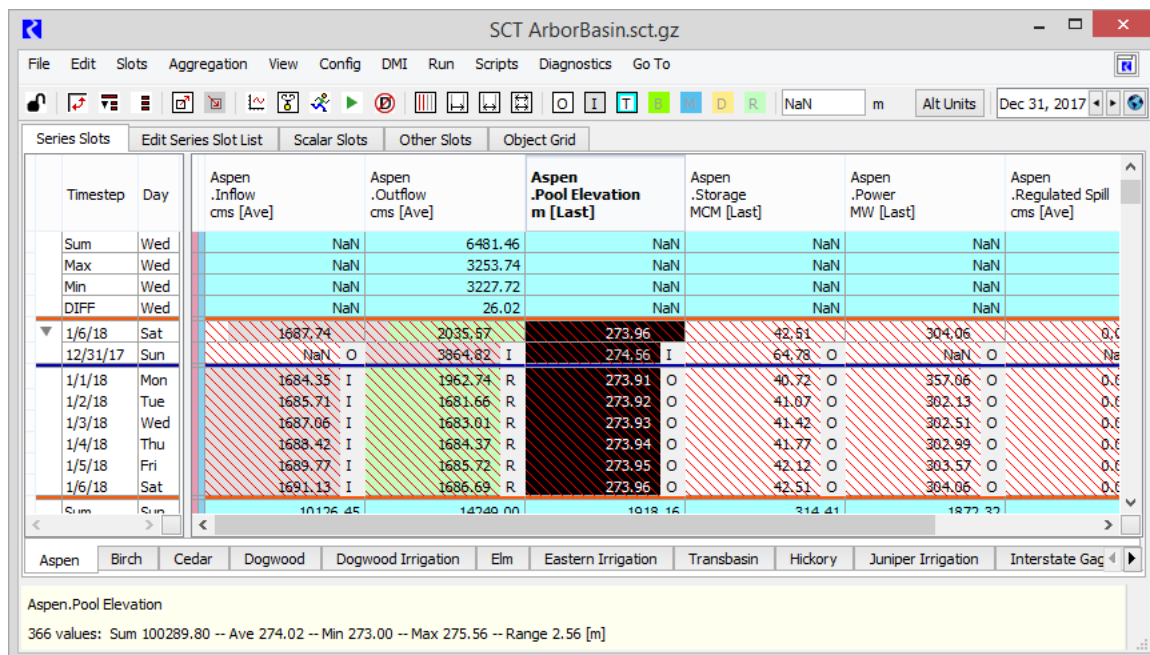


The dispatching mode is applied to all slots on the object.

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Using the SCT

If you have disabled dispatching, all slots on the object are displayed with crosshatching in a user-defined Dispatch Disabled color.



Related topics

- [Customize divider and cell colors](#), page 190

Set flags on series slot values

Use this procedure to change the value type flags on one or more selected cells. Flags are generally depicted in the SCT using background shading in user-defined colors; see [Customize flag colors](#), page 189 for details on defining the colors.

Following are some restrictions on flags:

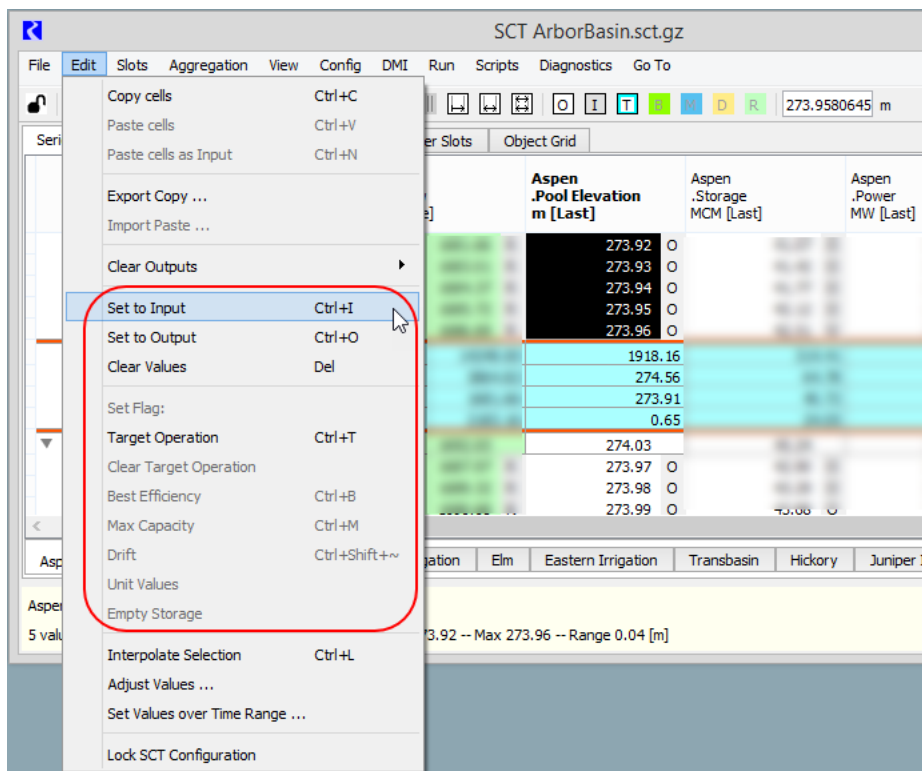
- You can set the Input (I) and Output (O) flags on editable slot timesteps.
- The Input flag can be set only on timesteps with a defined (non-NaN) value.
- Other flags can be set only on certain slots; see [“Flags on the Timestep I/O Menu” in User Interface](#) for details.
- The Target Operation flag may involve coordinating pairs of internal flags on different timesteps.

1. Go to the **Series Slots** tab.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool Elevation m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. Select the cells whose flags you want to modify. You can select any combination of cells, including one cell, multiple non-contiguous cells, a range of cells, one or more entire slots, or entire sheet.
3. In the SCT menu, select **Edit**, then one of the following flag options. Alternatively, use the designated keyboard shortcut.
 - Set to Input (Ctrl+I)
 - Set to Output (Ctrl+O)
 - Target Operation (Ctrl+T)
 - Clear Target Operation
 - Best Efficiency (Ctrl+B)
 - Max Capacity (Ctrl+M)
 - Drift (Ctrl+Shift+~)
 - Unit Values
 - Empty Storage

Note: Only flags that apply to *all* currently selected cells are active in the menu; if a flag is not applicable to one or more cells in the selection, it is not available in the menu.



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Using the SCT

The selected cells are set to the flag you have selected, and the cell background is updated to the user-defined color for that flag.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Birch .Inflow cms [Ave]	Birch .Pool Elevation m [Last]	Birch .Outflow cms [Ave]	Birch .Storage MCM [Last]
1/1/18	Mon			3274.55 O	
1/2/18	Tue			3389.30 M	
1/3/18	Wed			1892.47 M	
1/4/18	Thu			1700.00 M	
1/5/18	Fri			1700.00 O	
1/6/18	Sat			1700.00 O	
Sum	Sun			16740.05	
Max	Sun			3389.30	
Min	Sun			1700.00	
DIFF	Sun			1689.30	

Related topics

- [Customize flag colors](#), page 189
- [Set a target operation](#), page 201
- “Flags on the Timestep I/O Menu” in *User Interface*

Alternate methods

- In the SCT toolbar, select the button for the flag you want to assign. Active flags are indicated with a black border.

Note: The available buttons are user-defined; depending on the SCT configuration, some buttons may not be available. See “[Show or hide buttons on the SCT toolbar](#)” on page 181 for details.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]
1/1/18	Mon			273.91 O	
1/2/18	Tue			273.92 O	
1/3/18	Wed			273.93 O	

- Use the following keyboard accelerators:
 - Ctrl+I—Input
 - Ctrl+O—Output
 - Ctrl+T—Target (T)
 - Ctrl+B—Best Efficiency (B)
 - Ctrl+M—Max Capacity (M)
 - Ctrl+Shft+~ —Drift (D)
 - none—Unit Values

Set a target operation

Use this procedure to set a target operation for a simulation run. You can apply a target operation to a single timestep or a range of timesteps on the following slots:

- Reservoir.Storage
- Reservoir.Pool Elevation

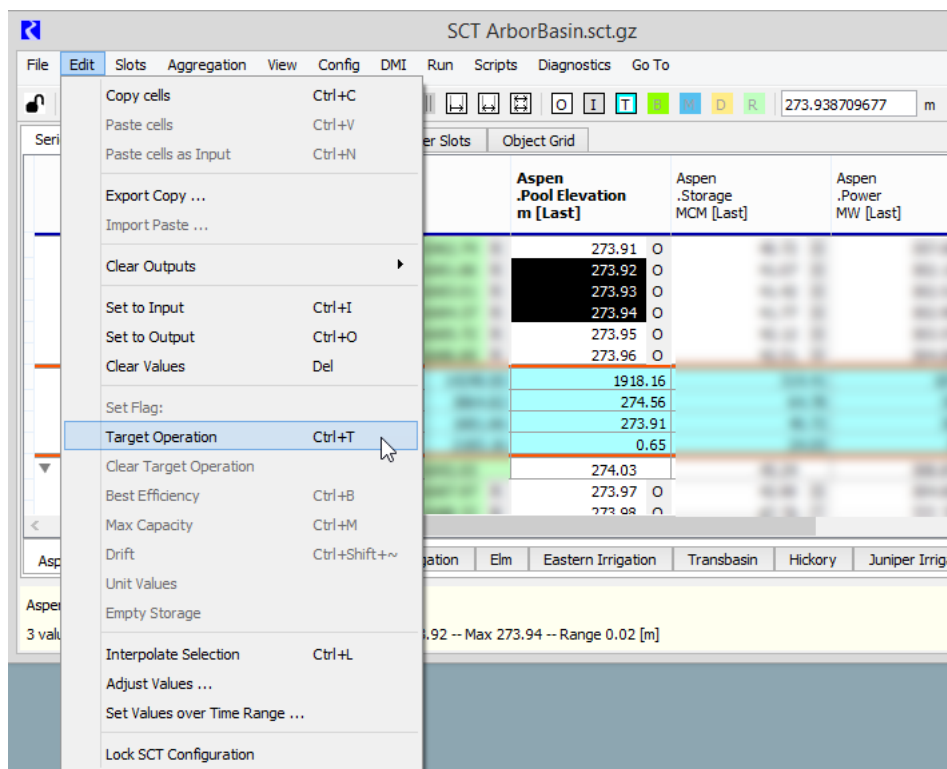
1. Go to the **Series Slots** tab.

Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool Elevation m
5/27/20	Wed	49	3,617.01	0.00	273.91
5/28/20	Thu	11	3,630.64	0.00	273.92

2. In a Reservoir.Storage or Reservoir.Pool Elevation slot, select the timesteps you want to include in the target range. You can select a single timestep or a range.

Note: While it is recommended that you mark the beginning of a target range with an explicit Target Begin flag, there are situations in which this may not be possible.

3. Select **Edit**, then **Target Operation** to set the target range.



The SCT display is updated as follows:

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- If you have selected a single timestep, that timestep is assigned the Target flag, which is indicated by a solid background in the user-defined Target flag color. The Target Begin timestep is implied by the flag values in the preceding timesteps, as determined during the model run.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Outflow cms [Ave]	Aspen .Storage MCM [Last]
1/1/18	Mon		273.91	O	
1/2/18	Tue		273.92	O	
1/3/18	Wed		273.93	O	
1/4/18	Thu		273.94	T	
1/5/18	Fri		273.95	O	
1/6/18	Sat		273.96	O	
Sum	Sun		1918.16		

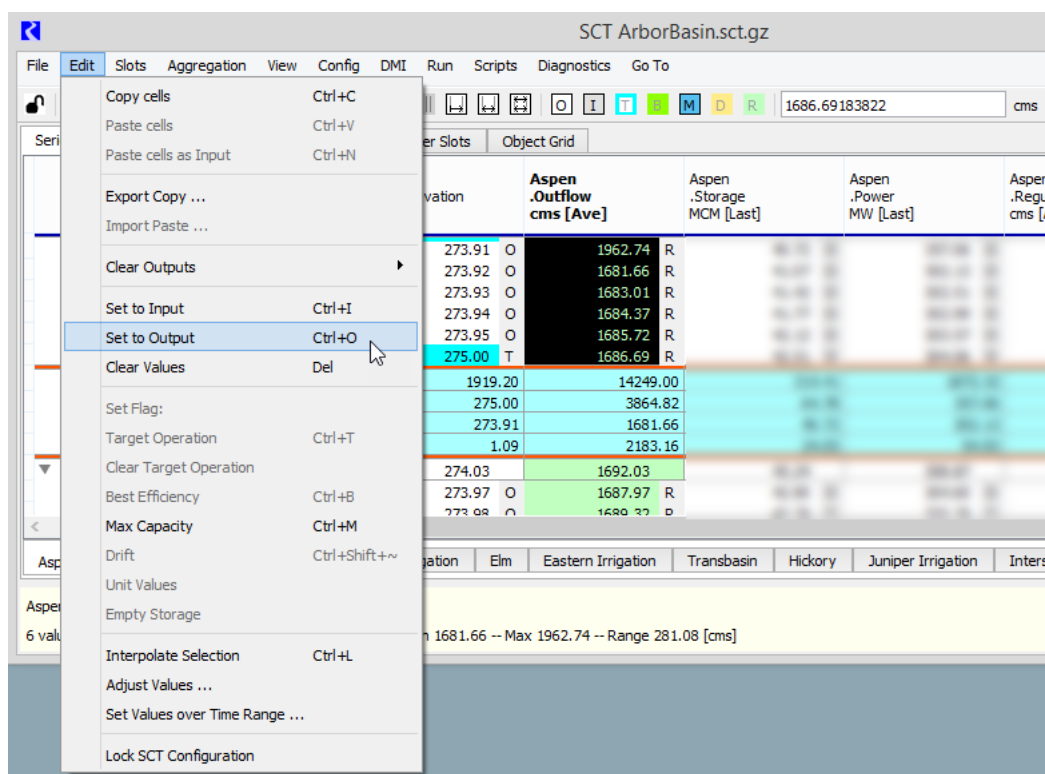
- If you have selected a range of timesteps, a box is drawn around the cells. The first timestep in the selected range is assigned the Target Begin flag. The last timestep in the range is assigned the Target flag, which is indicated by a solid background in the user-defined Target flag color.

Note: Target values are implicitly Input.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Inflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Outflow cms [Ave]	
1/1/18	Mon		273.91	O	
1/2/18	Tue		273.92	O	
1/3/18	Wed		273.93	O	
1/4/18	Thu		273.94	O	
1/5/18	Fri		273.95	O	
1/6/18	Sat		273.96	T	
Sum	Sun		1918.16		
Max	Sun		274.56		

4. In the Target cell, enter the target value.
5. Select the Reservoir.Outflow cells for the same time span as the target range.

6. Select **Edit**, then **Set to Output**.



The selected cells are assigned the Output flag.

7. Execute a simulation run; see [“Run a model simulation from the SCT”](#) on page 194 for details.

After the run completes, you can view the solution values in the Outflow target range.

Related topics

- [Run a model simulation from the SCT](#), page 194

Clear a target operation

Use this procedure to clear a target operation that has been set.

1. Go to the **Series Slots** tab.

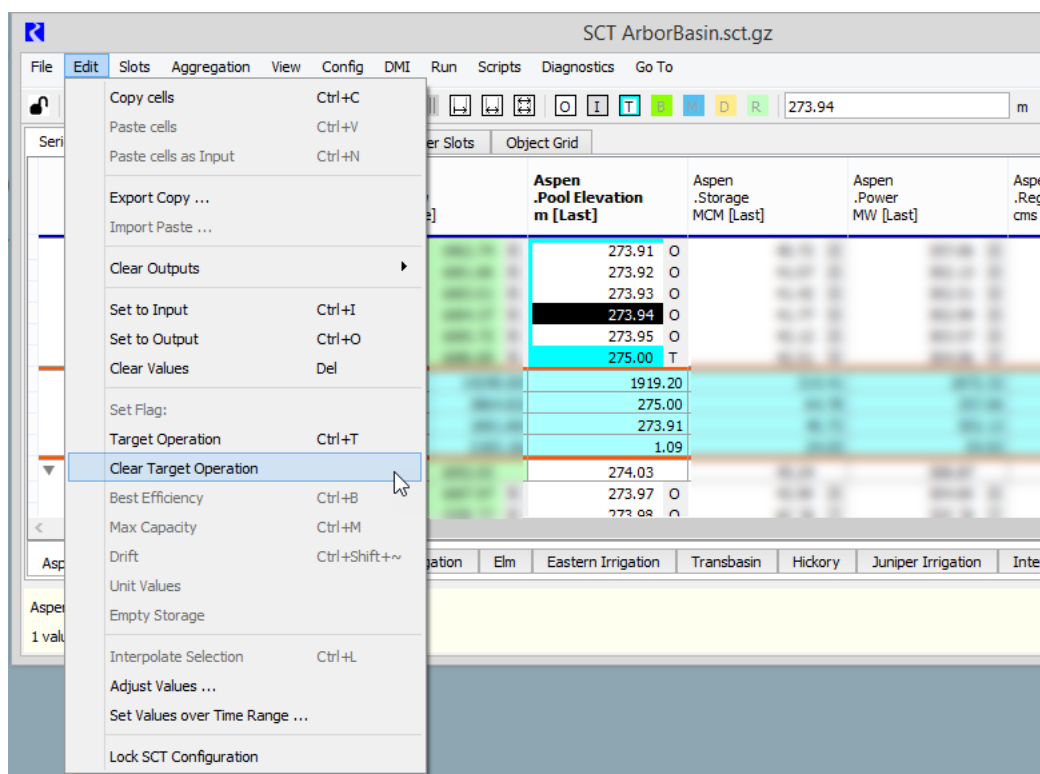
Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. Select a cell in the target range.

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3. Select **Edit**, then **Clear Target Operation**.



The target range is cleared and the cell flags are restored to their original types.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
TimeStep	Day	Aspen .Inflow cms [Ave]	Aspen .Outflow cms [Ave]	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]
1/1/18	Mon			273.91	O
1/2/18	Tue			273.92	O
1/3/18	Wed			273.93	O
1/4/18	Thu			273.94	O
1/5/18	Fri			273.95	O
1/6/18	Sat			275.00	I
Sum	Sun			1919.20	
Max	Sun			275.00	
Min	Sun			273.91	
DIFF	Sun			1.09	
1/13/18	Sat			274.03	
1/7/18	Sun			273.97	O

Related topics

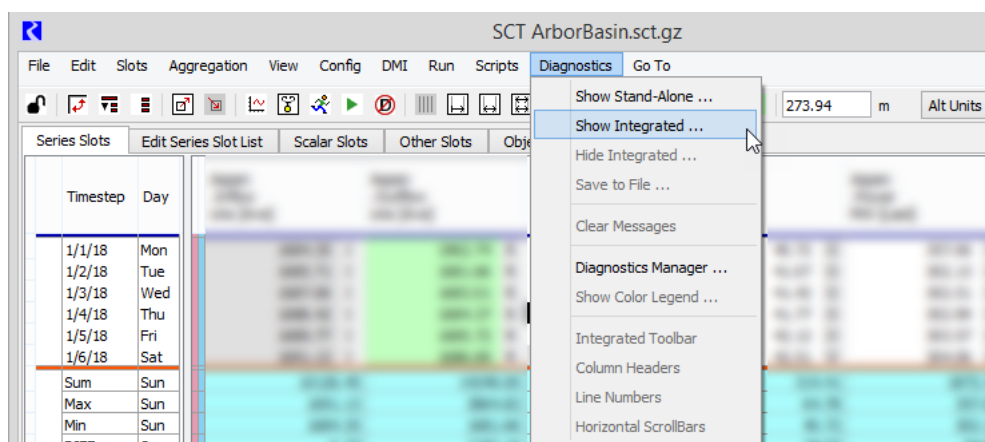
- [Run a model simulation from the SCT](#), page 194

Show or hide simulation diagnostic messages

Use this procedure to display diagnostics messages for a simulation run.

The Diagnostic Output window can be shown as an integrated panel at the bottom of the SCT or as a standalone window; it can appear in only one place at a time. When the panel or window is open, diagnostics messages for all simulation runs are posted there.

1. To show the Diagnostics Output, on the SCT menu, select **Diagnostics**, then one of the following options:
 - **Show Integrated**—displays an integrated Diagnostics Output panel, directly below the sheet tabs and above the Status area. You can resize the height of the panel by dragging its upper border.
 - **Show Stand-Alone**—opens a separate Diagnostics Output window.

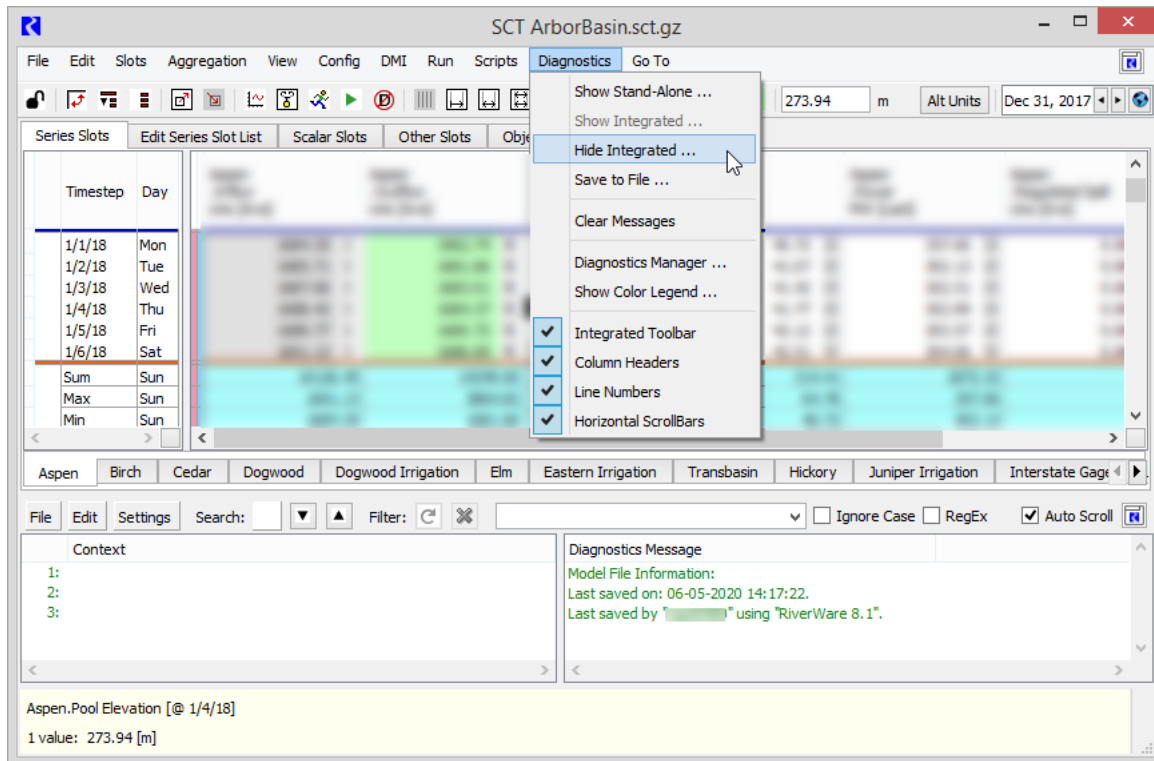


2. To hide the Diagnostics Output, do one of the following:

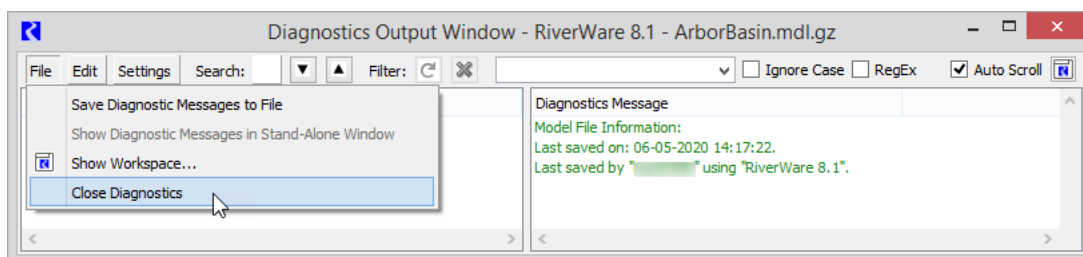
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- To close the integrated Diagnostics Output panel, select **Diagnostics**, then **Hide Integrated** on the SCT menu.



- To close the standalone window, select **File**, then **Close Diagnostics** in the Diagnostics Output window menu.



Related topics

- [SCT Layout](#), page 6
- [“Diagnostics” in Debugging and Analysis](#)

Alternate methods

- Select the **Diagnostics** button in the SCT toolbar.

Plotting and Printing

These procedures allow you to view and print series slot data.

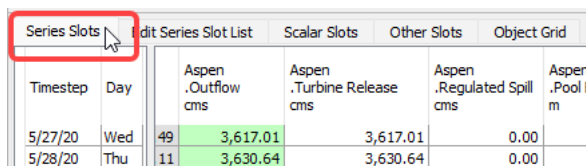
Topics

- [Plot series slots](#), page 207
- [Print the Series Slots tab](#), page 208

Plot series slots

Use this procedure to display a plot of one or more slots directly from the SCT. If you select multiple slots, they are plotted together on the same plot. The slots are plotted over the full timestep range of the SCT.

1. Go to the **Series Slots** tab.

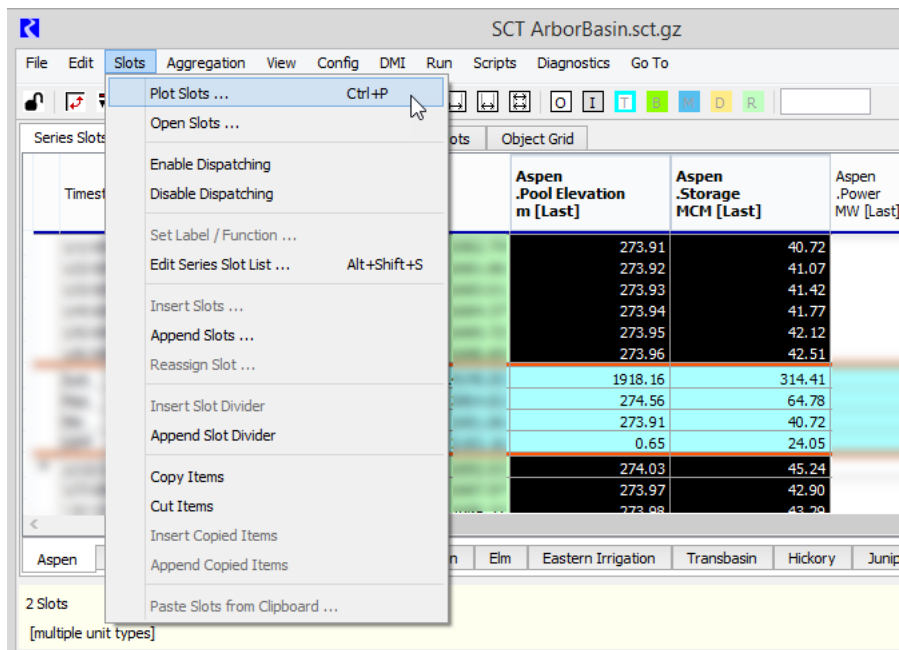


Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

2. Select the slots to plot.

Note: You can select any number of slots, as long as no more than two unit types are represented within your selection.

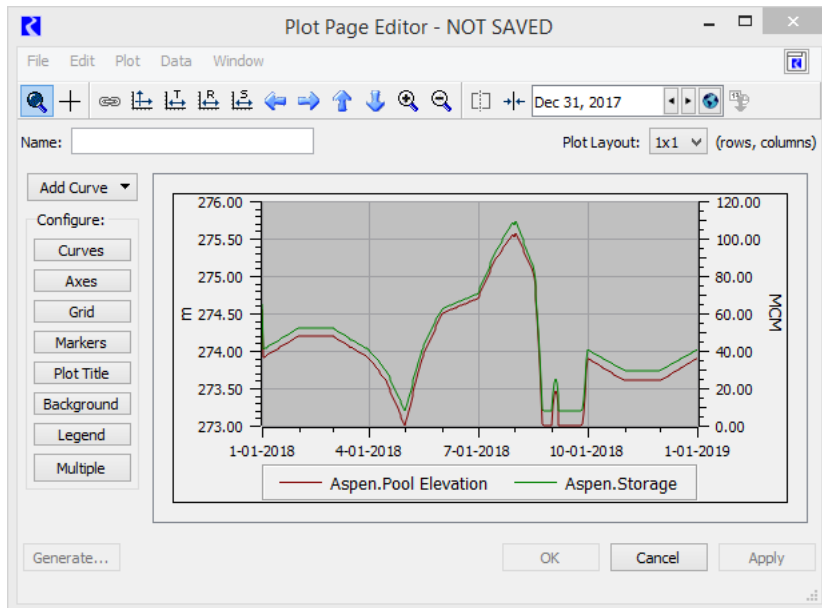
3. On the SCT menu, select **Slots**, then **Plot Slots**; alternatively, press Ctrl+P.



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The slots are plotted in the Plot Page Editor window.



Related topics

- [“Plotting” in Output Utilities and Data Visualization](#)

Print the Series Slots tab

Use this procedure to print all or part of the Series Slots tab to a selected printer. If the necessary drivers are installed on your computer, you can also print to an Adobe PDF file.

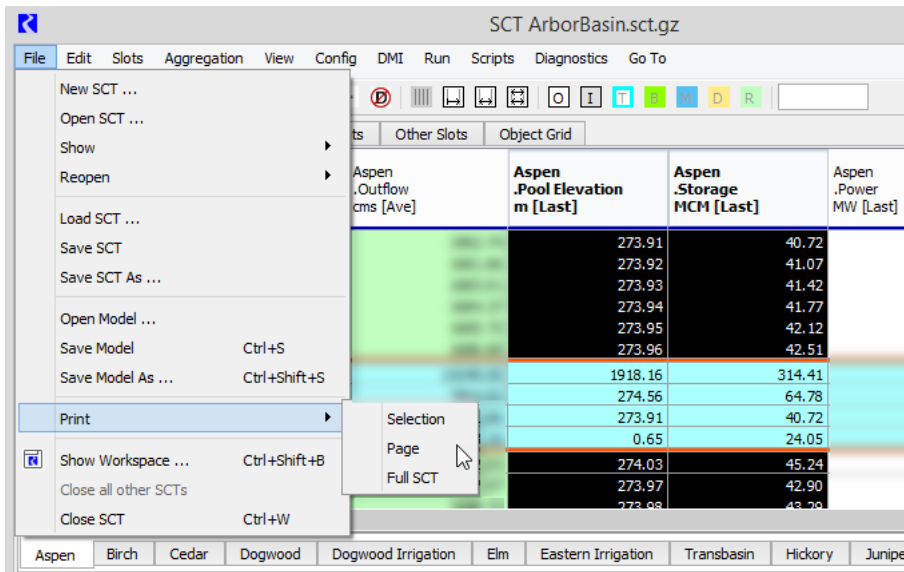
The printout includes the SCT row and column headers and the data table. The page headers and footers include the RiverWare version, model name, date, and page number.

1. Go to the **Series Slots** tab.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool l m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. Optionally, select a range of slots or cells to print.
3. Select **File**, then **Print**, then one of the following options.
 - **Selection**—prints the selection in its entirety. If you selected entire slots, all cells are printed. In aggregated view, summary rows are included only if all their detail rows are included.
 - **Page**—prints one page of the current sheet, starting with the top row currently displayed in the SCT window.

- **Full SCT**—prints the current series slot sheet in its entirety.



The Windows Print dialog box opens.

4. In the Windows Print dialog box, select the printer and other settings, then select **Print**.

The printout is printed on the selected printer.

Related topics

- [Plot series slots](#), page 207

Accessing Slot Dialogs From the SCT

This section includes procedures for opening general slot dialogs from the SCT. These procedures allow you to perform common slot editing tasks directly from the SCT.

Topics

- [Open a slot dialog](#), page 209
- [View and manage a slot description](#), page 212

Open a slot dialog

Use these procedures to open a slot in a separate Slot Viewer or Slot Dialog window.

Open series slots

Use this procedure to open one or more series slots in the Slot Viewer.

Chapter 3

Using the SCT

1. Go to the **Series Slots** tab.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49	3,617.01	3,617.01	0.00
5/28/20	Thu	11	3,630.64	3,630.64	0.00

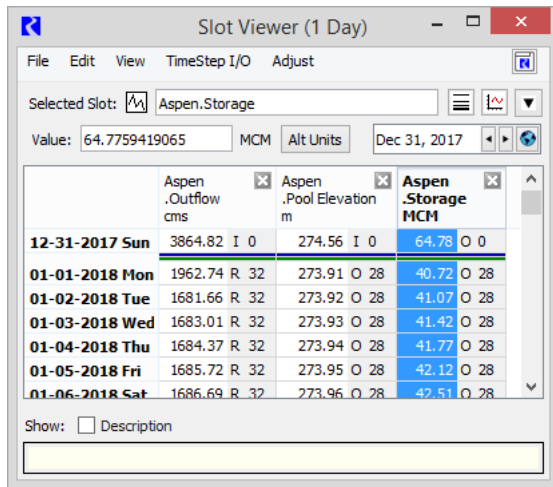
2. Do one of the following to open the slots.
 - To open one slot at a time, select the slot; then right-click and select **Open Slot** in the shortcut menu.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Inflow cms [Ave]	Aspen Elevation [st]	Aspen .Storage MCM [Last]	Aspen .Pool I m
		168			
		1684.35			
		1685.71			
		1687.06			
		1688.42			
		1689.77			
		1691.13			
		10126.45			
		1691.13			
		1684.35			

- To open multiple slots at once, select the slots; then in the SCT menu, select **Slots**, then **Open Slots**.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Pool Elevation m [Last]	Aspen .Storage MCM [Last]	Aspen .Pool I m	Aspen .Pool I m
		3240.73	NaN	NaN	NaN
		NaN	NaN	NaN	NaN
		NaN	NaN	NaN	NaN
		3227.72	NaN	NaN	NaN
		3253.74	NaN	NaN	NaN
		6481.46	NaN	NaN	NaN
		3253.74	NaN	NaN	NaN
		3227.72	NaN	NaN	NaN
		26.02	NaN	NaN	NaN
		NaN	NaN	NaN	NaN
		2035.57	273.96	42.51	42.51
		3864.82	274.56	64.78	64.78
		1962.74	273.91	40.72	40.72
		1681.66	273.92	41.07	41.07
		1683.01	273.93	41.42	41.42
		1684.37	273.94	41.77	41.77

The slots open in the Slot Viewer. Multiple slots appear as separate tabs.

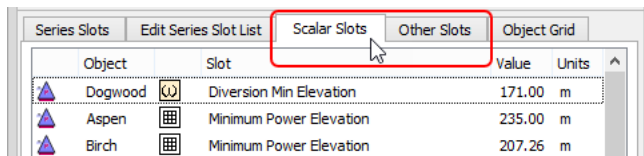


- Optionally, repeat [Step 2](#). to open additional slots. They will open as additional tabs in the Slot Viewer.

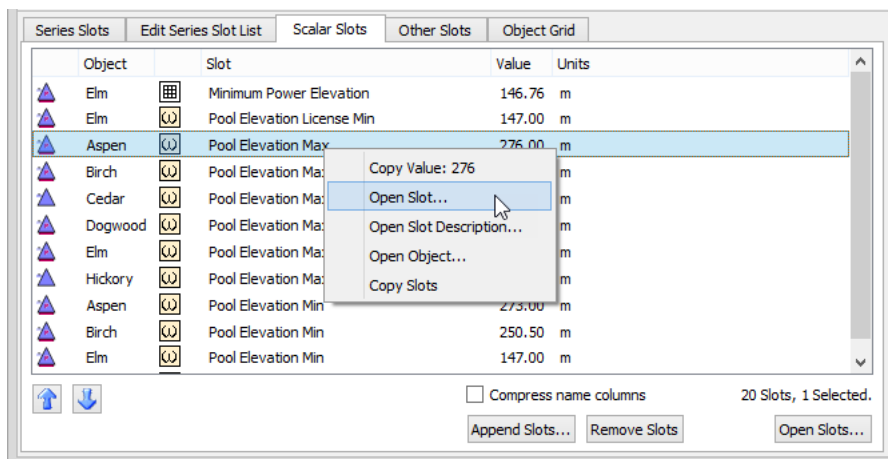
Open other slots

Use this procedure to open other slots, such as scalar, table, and periodic slots, in the Slot Dialog window. You can select only one slot at a time, and each slot opens in a separate Slot Dialog window.

- Go to the **Scalar Slots** tab or the **Other Slots** tab.



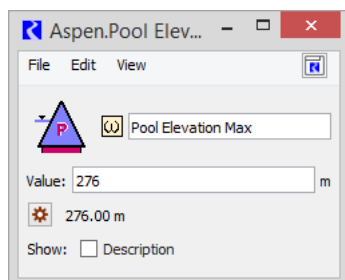
- Select the slot you want to open.
- Right-click and select **Open Slot** in the shortcut menu.



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Using the SCT

The slot opens in the Slot Dialog.



4. Optionally, repeat [Step 2.](#) and [Step 3.](#) to open multiple slots. They will open in separate Slot Dialog windows.

Related topics

- “[Slot Windows](#)” in *User Interface*

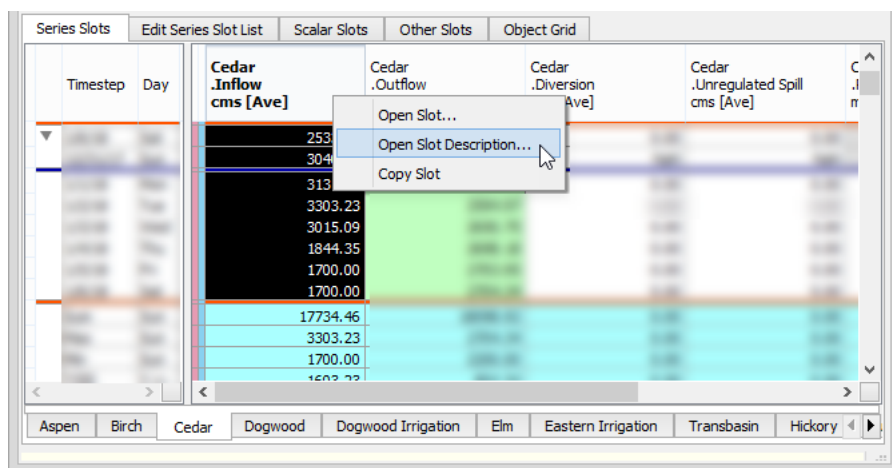
View and manage a slot description

Use this procedure to add, update, or view the slot description. Slot descriptions are optional multi-line descriptions assigned to individual slots. Using descriptions, you can assign searchable keywords or other descriptive information to slots. See “[Slot Descriptions](#)” in *User Interface* for details.

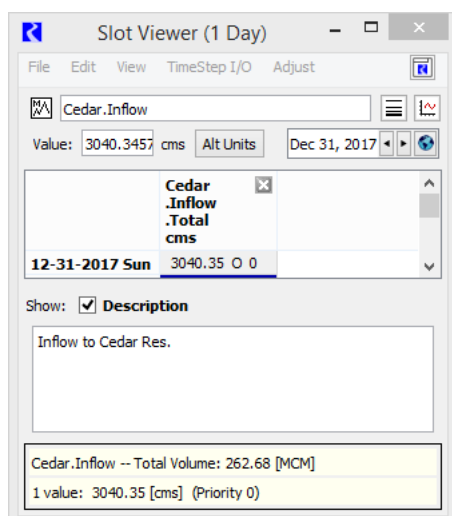
1. Go to the **Series Slots** tab.

Series Slots		Edit Series Slot List	Scalar Slots	Other Slots	Object Grid
Timestep	Day	Aspen .Outflow cms	Aspen .Turbine Release cms	Aspen .Regulated Spill cms	Aspen .Pool I m
5/27/20	Wed	49 3,617.01	3,617.01	0.00	
5/28/20	Thu	11 3,630.64	3,630.64	0.00	

2. Select the header of the slot you want to modify or view.
3. Right-click and select **Open Slot Description** in the shortcut menu.



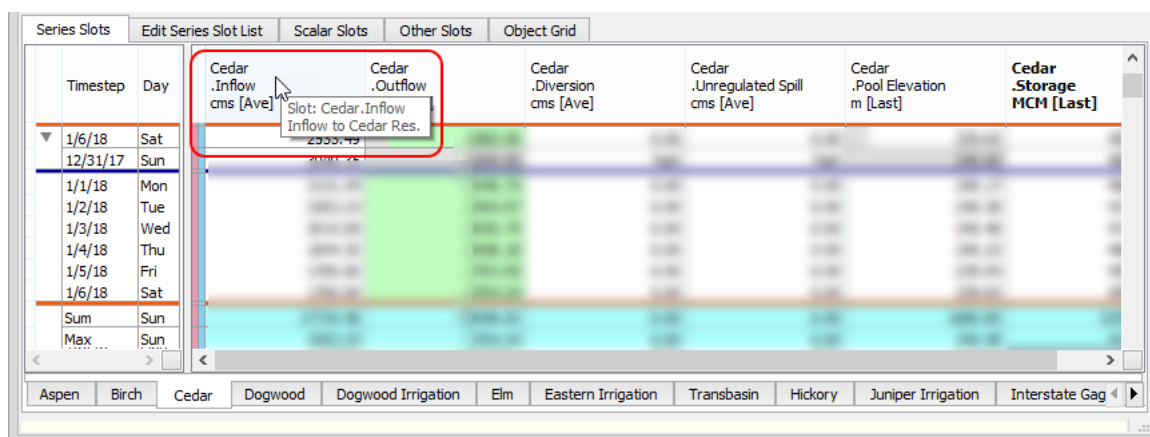
The Slot Viewer dialog box opens, displaying the current slot description, if one has been assigned.



4. If you want to update the description, enter the new description in the **Description** field.
5. Close the Slot Viewer dialog box.

Any description updates are saved to the slot and will be displayed the next time you display the slot description.

Note: Slot descriptions are also displayed in a tooltip when you hover over the slot header.



Related topics

- [Custom slot labels](#), page 26

Organizing Scalar and Table Slots

This section includes procedures for organizing slots on the Scalar Slots and Other Slots tabs.

The Scalar Slots tab displays a list of scalar slots and 1x1 table slots. You can add, rearrange, and remove slots on this list. You can also edit the values of these slots.

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Note: 1x1 table slots were used for scalar values before scalar slots were implemented.

The Other Slots tab displays a list of table slots, periodic slots, statistical table slots, and list slots. You can add, order, and remove slots on this list. You *cannot* edit the values of these slots directly from the SCT, but you can open them in the Slot Viewer.

Topics

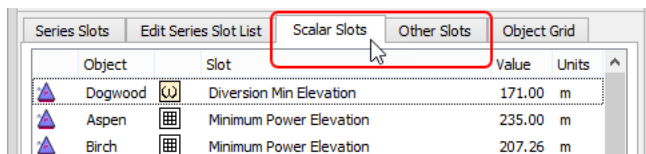
- [Add scalar and table slots to an SCT](#), page 214
- [Compress object and slot name display](#), page 216
- [Save a custom order for scalar and table slots](#), page 217
- [Remove scalar and table slots from an SCT](#), page 218
- [Edit a scalar slot value](#), page 219
- [Set multiple scalar slots to the same value](#), page 219

Add scalar and table slots to an SCT

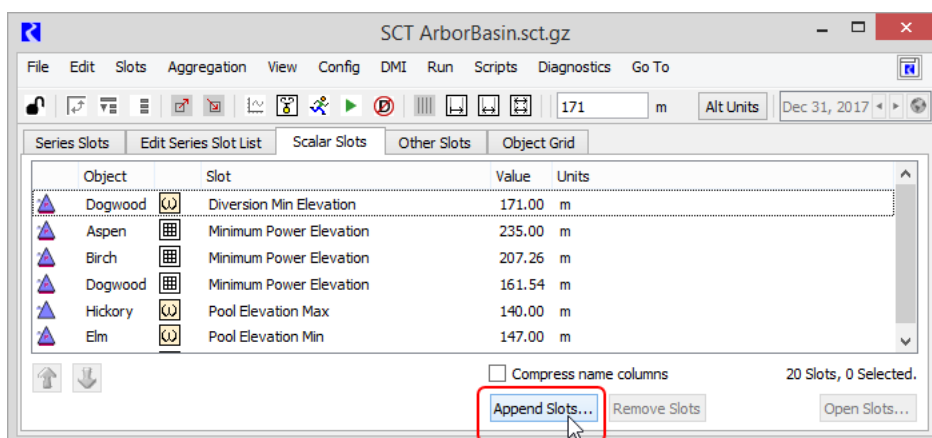
Use this procedure to add non-series related slots to the SCT.

Note: You can only add slots that exist in the currently loaded RiverWare model.

1. Go to the **Scalar Slots** tab or the **Other Slots** tab.



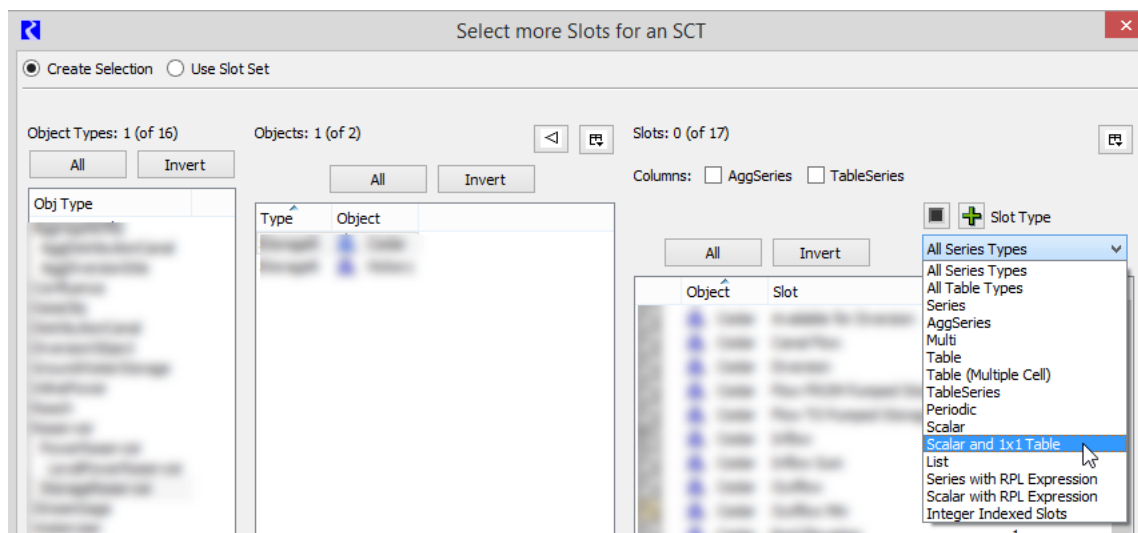
2. Select **Append Slots**.



The Select more Slots for an SCT dialog box opens.

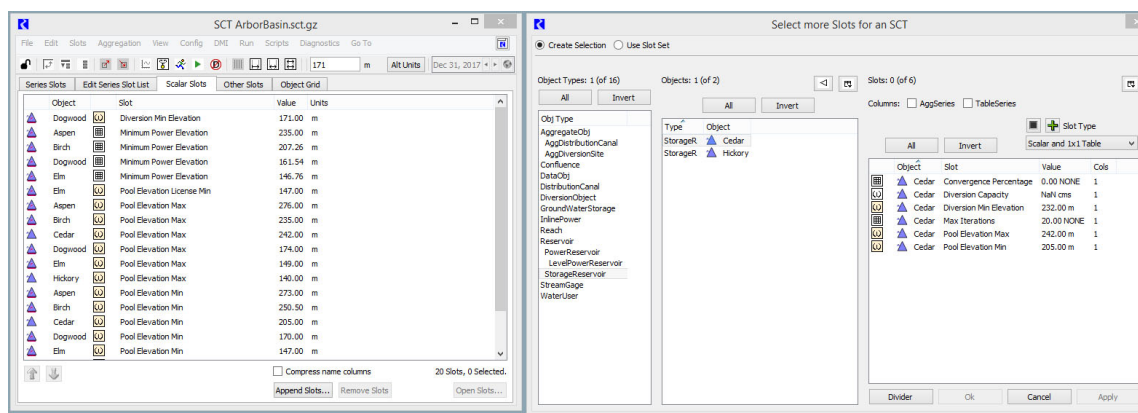
Note: By default, the slot selector dialog box shows series slots only.

3. In the **Slot Type** menu, select **Scalar and 1x1 Tables**.



The slot list is updated to show available scalar slots and 1x1 table slots.

4. Optionally, position the dialog box so you can see it and the SCT together, side-by-side.



5. In the dialog box, select the slots you want to add.

Note: You can include the same slot multiple times within the same SCT.

- a. Select an object type.
- b. Select an object.
- c. Select one or more slots. Press Ctrl+Click or Shift+Click to select multiple slots in the slot list.
- d. Select **Apply** or **Ok**.

The slots are added to the end of the slot list in the selected SCT tab.

6. When you are done adding slots, select **Ok** or **Cancel** in the Select more Slots dialog box to close it.

The dialog box closes, and the selected SCT tab becomes active.

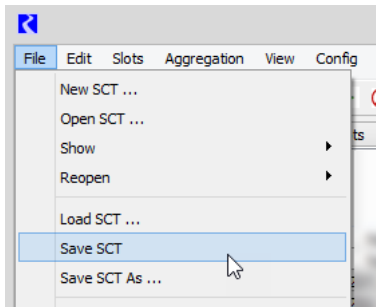
Note: You can add more slots, rearrange them, and make other modifications at a later time.

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7. Select **File**, then **Save SCT** or **Save SCT As** to save the new slots to the SCT. Alternatively, press Ctrl+S or Ctrl+Shift+S.

Note: If you close the SCT without performing this step, the new slots will not be saved to the SCT.



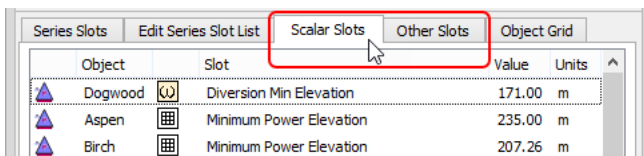
Related topics

- [Add series slots to an SCT](#), page 46

Compress object and slot name display

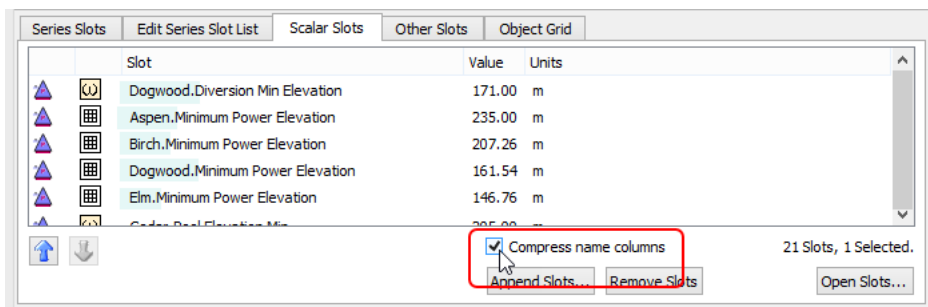
Use this procedure to change the display of the object and slot names for non-series slots. Instead of showing the object and slot names in separate columns, you can compress the names into one column. This procedure is useful when horizontal space is limited.

1. Go to the **Scalar Slots** tab or the **Other Slots** tab.



2. Select or clear the **Compress name columns** check box as follows:

- Clear the check box to expand the object and slot names into separate Object and Slot columns; this is the default setting.
- Select the check box to compress the object and slot names into the Slot column, with the format Object.Slot.



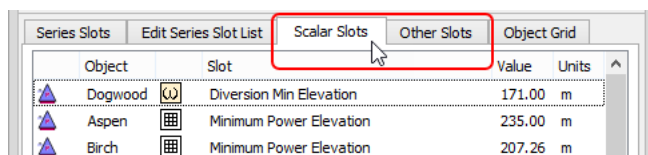
Related topics

- [Add scalar and table slots to an SCT](#), page 214

Save a custom order for scalar and table slots

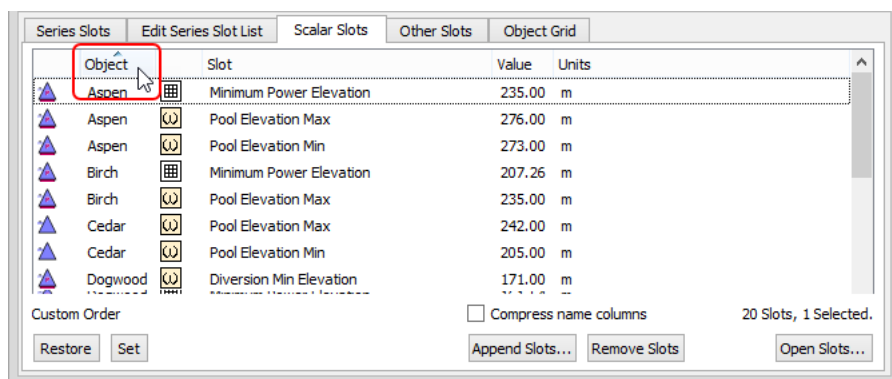
Use this procedure to define and save a custom order for slots on the Scalar Slots tab or the Other Slots tab.

1. Go to the **Scalar Slots** tab or the **Other Slots** tab.



Slots are listed in the current default order for the tab.

2. Select a column header to sort the display by that attribute.



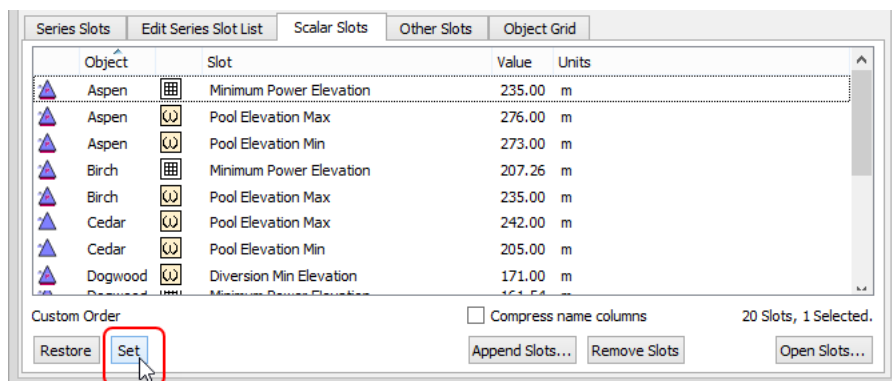
The **Set** and **Restore** buttons replace the arrow buttons.

3. Select **Restore** to restore the default order.

The default order is restored and the arrow buttons replace the **Restore** and **Set** buttons.

4. Use the following steps to set the new order as the default.

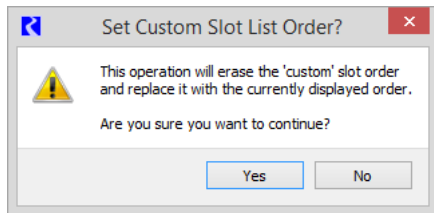
- a. Select **Set**.



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The Set Custom Slot List Order confirmation dialog box opens.



- b. Select **Yes** to confirm the update.
- c. Save the SCT to save the updates; see [“Save an SCT”](#) on page 42 for details.

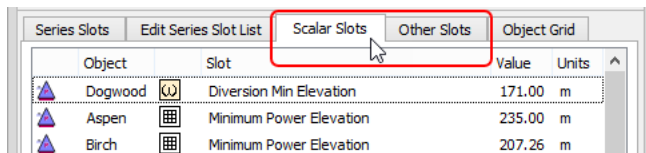
Related topics

- [Add scalar and table slots to an SCT](#), page 214

Remove scalar and table slots from an SCT

Use this procedure to remove non-series-related slots from the SCT.

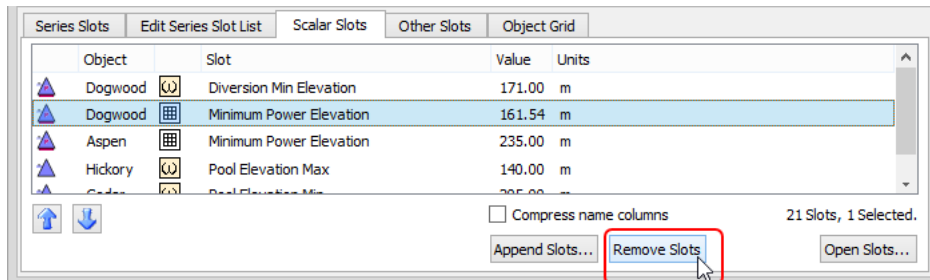
1. Go to the **Scalar Slots** tab or the **Other Slots** tab.



2. Select one or more slots you want to remove.

Caution: There is no confirmation message when you perform the next step; be sure you want to proceed.

3. Select **Remove Slots**.



The slots are immediately removed from the SCT.

Related topics

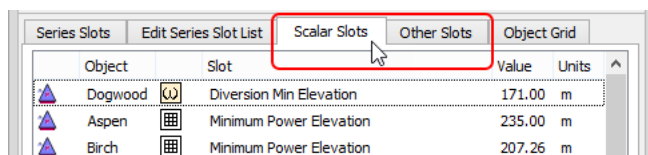
- [Add scalar and table slots to an SCT](#), page 214

Edit a scalar slot value

Use this procedure to edit the value of a single scalar slot or 1x1 table slots

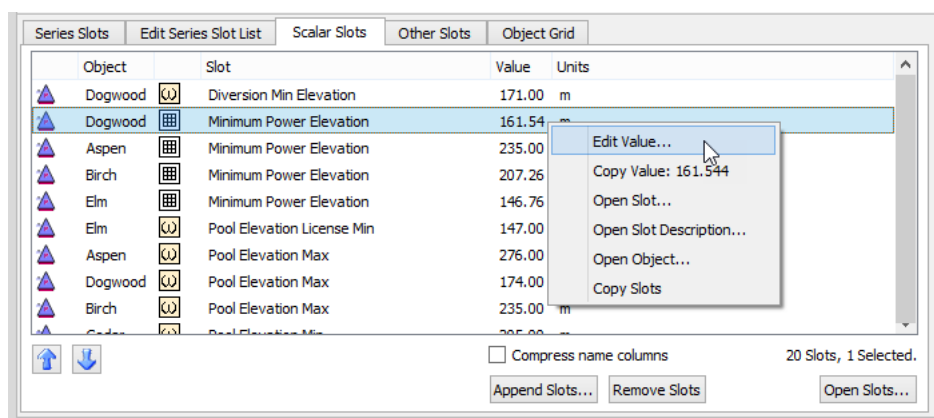
Note: This procedure does not apply to slots on the Other Slots tab.

1. Go to the **Scalar Slots** tab.



2. Select a slot to edit, and use any of the following methods to change the value:

- Double-click the value and enter a new one.
- Double-click the slot and enter a new value in the Slot Viewer.
- Right-click the slot value and select **Edit Value** in the shortcut menu. Enter a new value in the field.



Undo and Redo are available from the Edit menu.

Related topics

- [Display units](#), page 25
- [Manipulating Slot Values](#), page 67
- [Set multiple scalar slots to the same value](#), page 219

Set multiple scalar slots to the same value

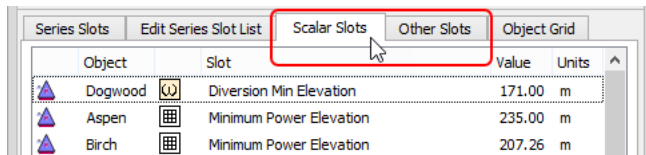
Use this procedure to assign the same value to multiple scalar slots. This is the only method for changing the values of multiple scalar slots at once.

Note: This procedure does not apply to slots on the Other tab.

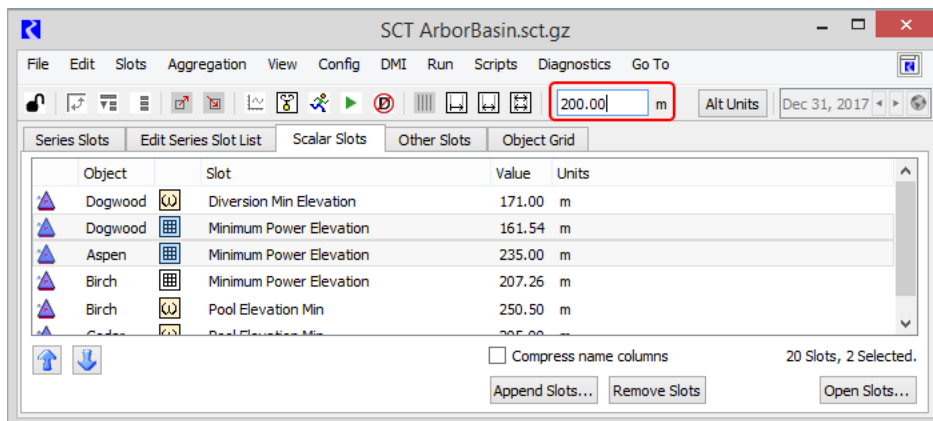
Chapter 3

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1. Go to the **Scalar Slots** tab.



2. Select the slots you want to update; you can select multiple contiguous or non-contiguous slots.
3. Enter the new value in the **Value** field in the SCT toolbar and press Enter.



The new value is applied to all selected slots.

Undo and Redo are available from the Edit menu.

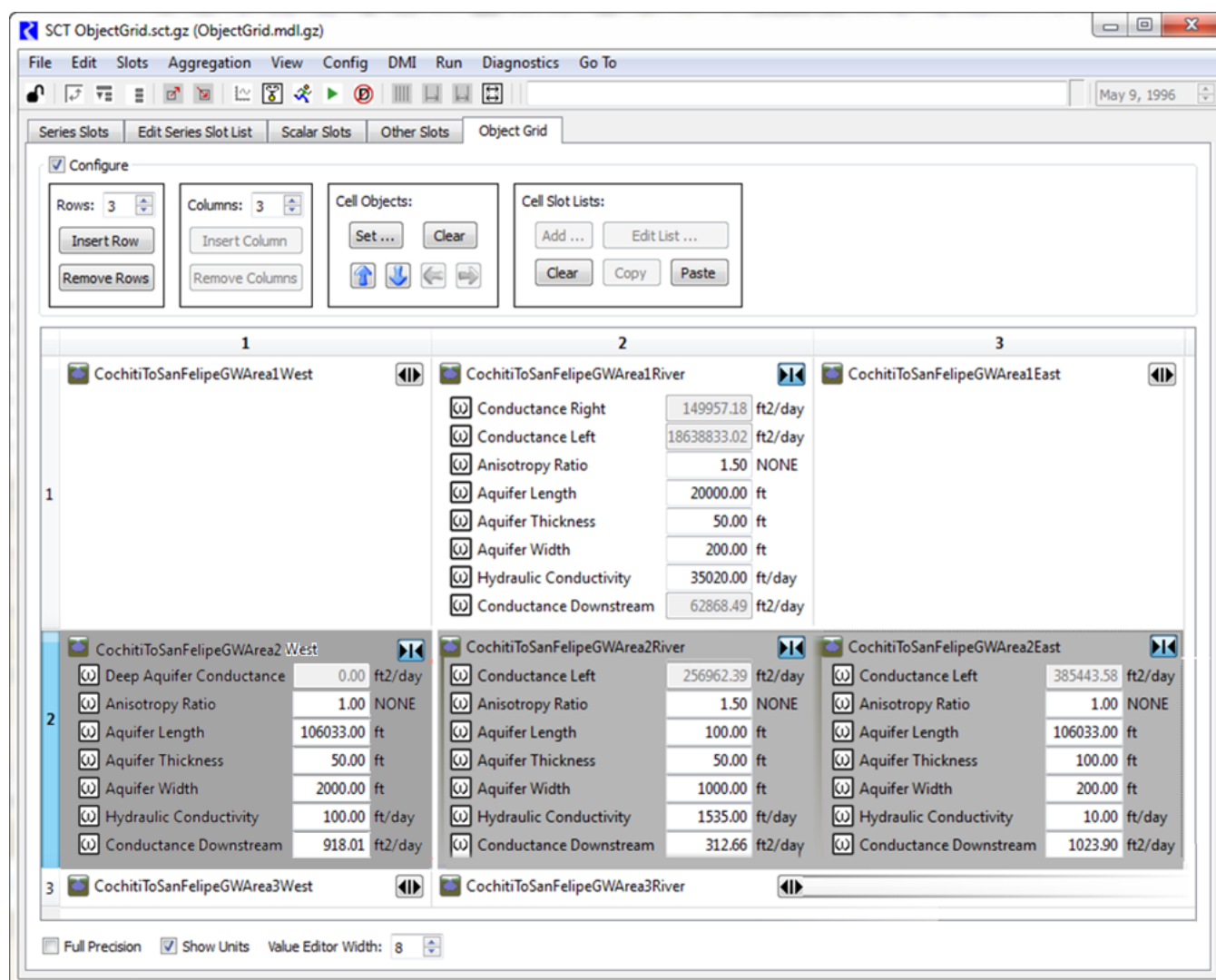
Related topics

- [Display units](#), page 25
- [Manipulating Slot Values](#), page 67
- [Edit a scalar slot value](#), page 219

Organizing Scalar Data in the Object Grid

This section includes procedures for configuring and using the Object Grid tab. This tab displays a user-configured grid of objects and their included scalar and table slots. It provides a convenient location for you to organize and specify parameters on objects that interact with one another in a gridded pattern, such as with groundwater networks.

For example, in the following figure, the Aquifer Widths must have the same value for continuity. The Object Grid tab provides this information in one place.



Topics

- [Display the object grid configuration controls](#), page 222
- [Create a new object grid](#), page 223
- [Add rows and columns to an object grid](#), page 224
- [Assign objects to cells](#), page 226
- [Assign a slot list to a cell](#), page 229
- [Copy and paste slot lists](#), page 231
- [Configure the slot list in a cell](#), page 233
- [Show or hide the slot list in a cell](#), page 235

Chapter 3

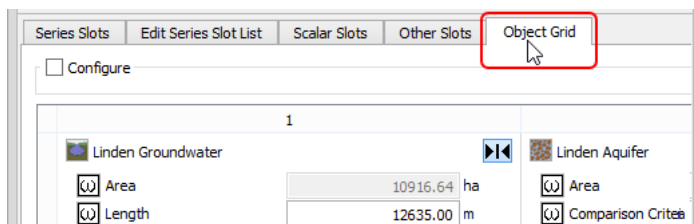
Using the SCT

- [Arrange cells in the object grid](#), page 237
- [Remove slot lists from cells](#), page 238
- [Remove objects from cells](#), page 240
- [Remove rows and columns from the object grid](#), page 241
- [Hide and restore rows and columns in the object grid](#), page 243
- [Configure the object grid display settings](#), page 245
- [Edit a slot value in the object grid](#), page 247
- [Open an object or slot from the object grid](#), page 247
- [Export an object grid to an XML file](#), page 249
- [Import an object grid to an SCT](#), page 250

Display the object grid configuration controls

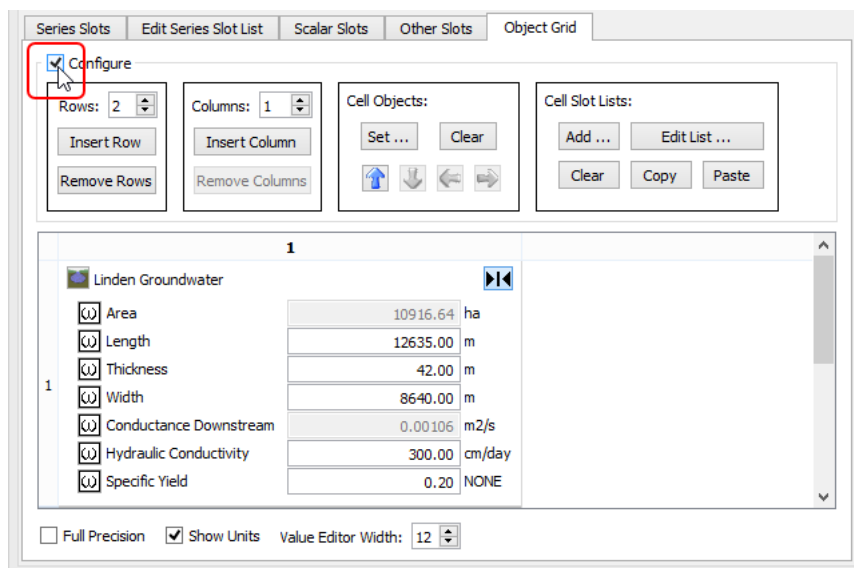
Use this procedure to display the configuration controls at the top of the Object Grid tab. These controls provide an easy way to perform all object grid configuration activities.

1. Go to the **Object Grid** tab.



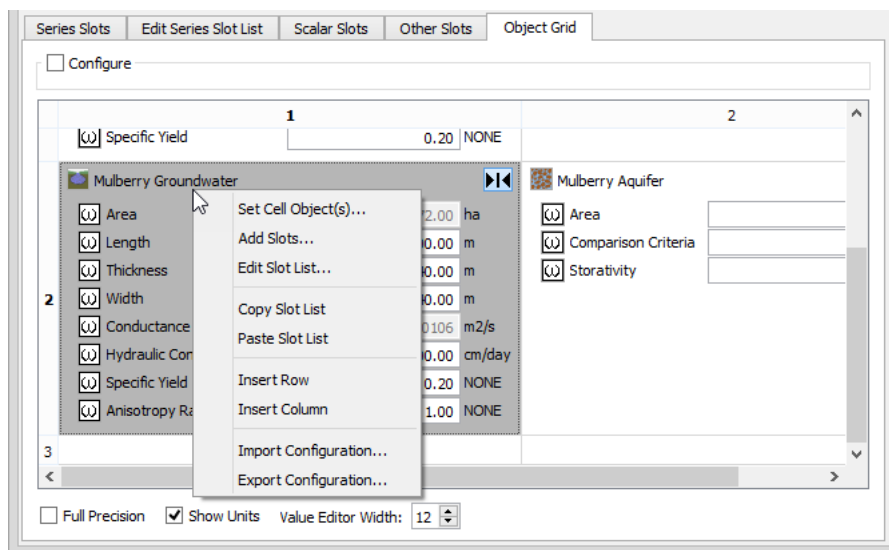
2. Select the **Configure** check box.

The configuration controls are displayed at the top of the page.



- Optionally, when you are finished configuring the object grid, clear the **Configure** check box to hide the controls and free up vertical space on the grid.

Note: Many configuration functions are available from the right-click shortcut menu when the configuration controls are hidden.



Related topics

- [Display the object grid configuration controls](#), page 222

Create a new object grid

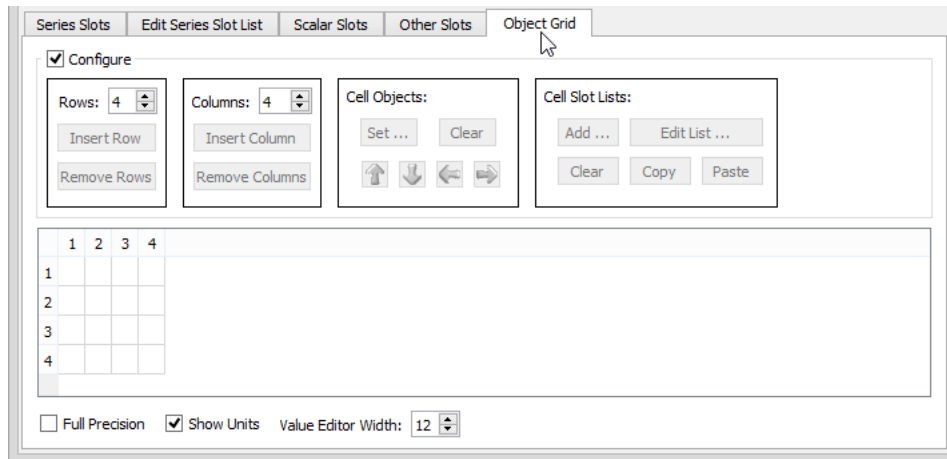
Use this procedure to create a new object grid in the SCT.

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Using the SCT

1. Open a new SCT or an existing SCT that has no object grid.
2. Go to the **Object Grid** tab.

The object grid is empty. By default, a new object grid is configured for four rows and four columns.



3. If necessary, display the configuration controls; see [Display the object grid configuration controls](#), page 222 for details.

The configuration controls are displayed at the top of the page.

4. Optionally, in the **Rows** and **Columns** text boxes, enter the number of rows and columns you want to assign to the grid.

Note: You can configure a grid size to get started, and then later, add or delete rows and columns, as needed.

The grid size is adjusted according to your selections.

5. Use the following procedures to add objects and slot lists to the grid.
 - a. [Assign objects to cells](#), page 226
 - b. [Assign a slot list to a cell](#), page 229
 - c. [Add rows and columns to an object grid](#), page 224

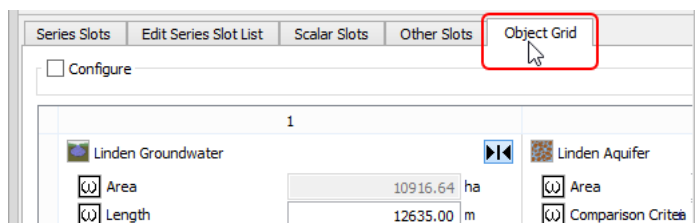
Related topics

- [Display the object grid configuration controls](#), page 222

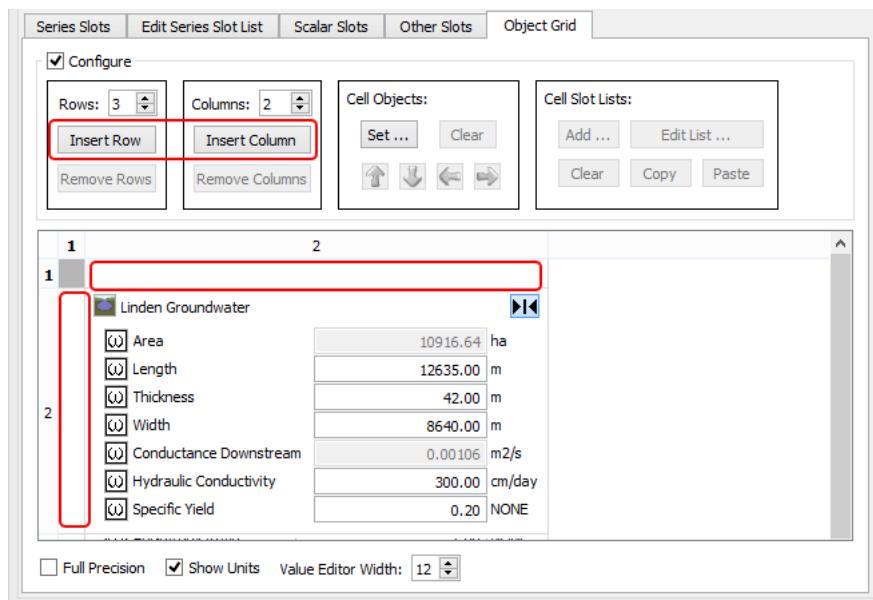
Add rows and columns to an object grid

Use this procedure to add empty rows and columns to the middle or the end of an existing object grid.

1. Go to the **Object Grid** tab.



2. To insert a row or column in the middle of the grid, select a cell at the location where you want to add the item, then perform the following actions as applicable.
 - In the Rows area, select **Insert Row**. An empty row is added above the current one.
 - In the Columns area, select **Insert Column**. An empty column is added to the left of the current one.

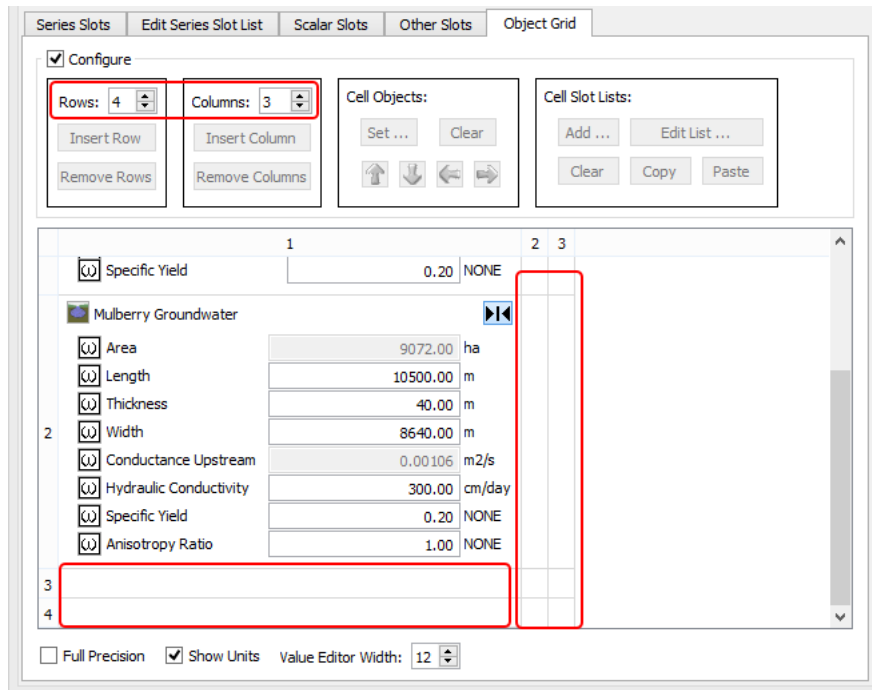


3. To add one or more rows or columns to the end of the grid, perform the following actions, as applicable.
 - In the **Rows** text box, enter an increased number of rows. For example, to add three rows to a five-row grid, enter 8. The empty rows are added to the bottom of the grid.

Chapter 3

Using the SCT

- In the **Columns** text box, enter an increased number of columns. For example, to add two columns to a three-column grid, enter 5. The empty columns are added to the end (right side) of the grid.



Related topics

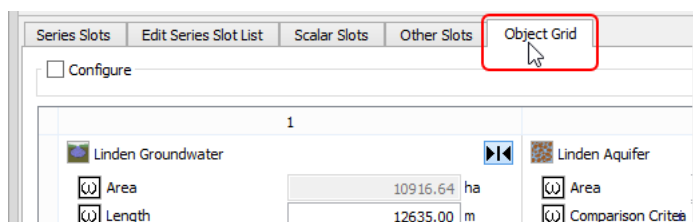
- [Display the object grid configuration controls](#), page 222
- [Assign objects to cells](#), page 226
- [Remove rows and columns from the object grid](#), page 241
- [Hide and restore rows and columns in the object grid](#), page 243

Assign objects to cells

Use this procedure to populate the object grid with one or more objects. Each object is assigned to one cell.

Note: It is possible to assign the same object to multiple cells. This is useful when you want to track different slot views for the same object. You can assign a different slot list to each cell the object is assigned to.

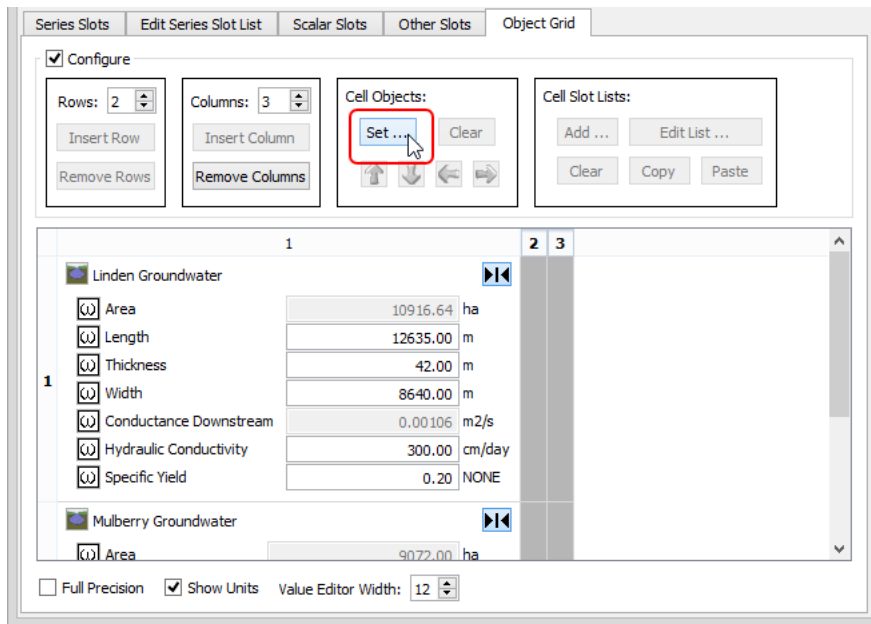
1. Go to the **Object Grid** tab.



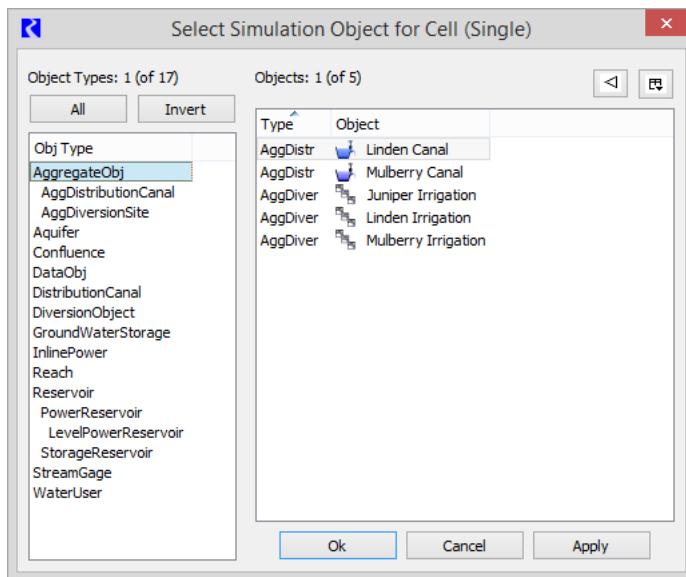
2. Select all the cells you want to populate.

Note: If a selected cell already has an assigned object, the existing object will be replaced.

3. In Cell Objects area, select **Set**.



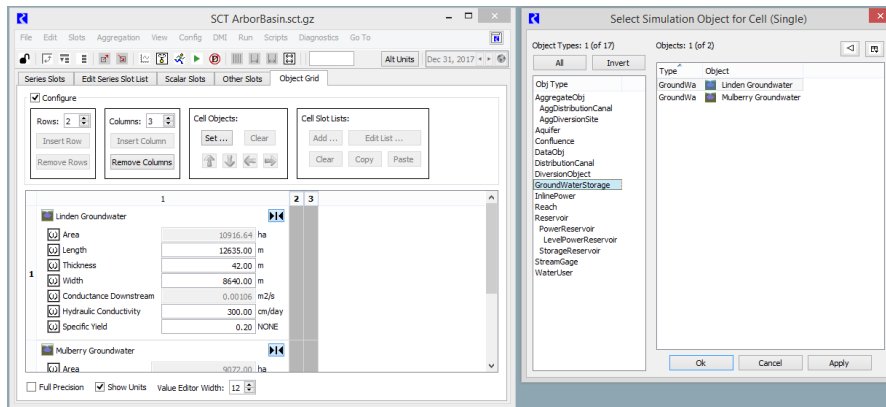
The Select Simulation Object for Cell dialog box opens.



Chapter 3

Using the SCT

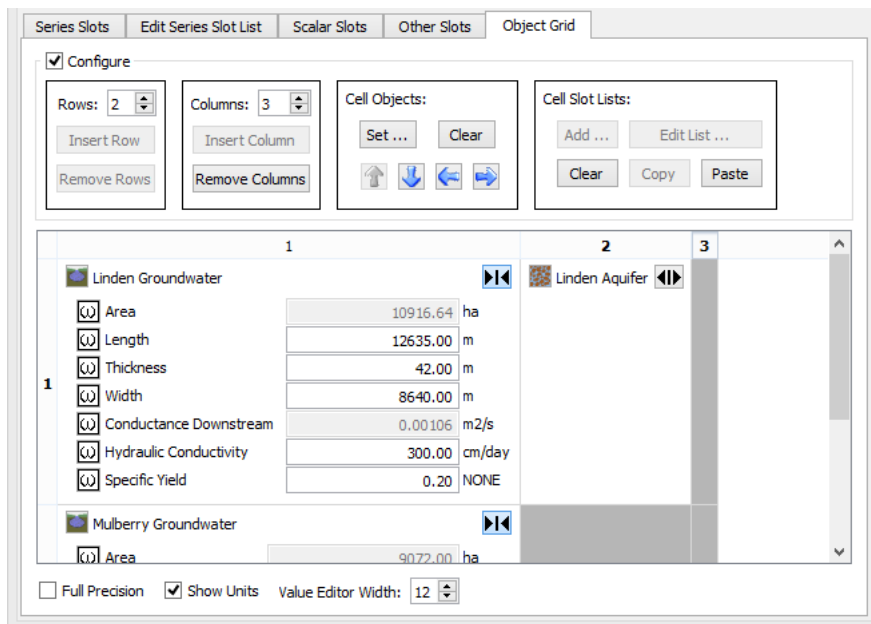
- Optionally, position the two windows so you can see both side-by-side.



- In the Object Selector dialog box, select an object, then select **Apply**.

The object is assigned to a selected cell, as follows:

- If you have selected only one cell, the object is assigned to that cell. You must close the Object Selector dialog box before assigning any additional objects to cells; proceed to Step [Step 7](#).
- If multiple cells are selected, the object is assigned to the first (top-left) cell in the selection.



- Repeat Step [Step 5](#). to assign additional objects, one-by-one, to the remaining selected cells.

Objects are assigned to the remaining selected cells. The first selected column is filled from top to bottom, then the next column to the right is filled, and so on, until all selected cells are filled or you stop assigning objects.

- Select **Ok** in the Object Selector dialog box when you are done assigning objects.

The Object Grid tab becomes active.

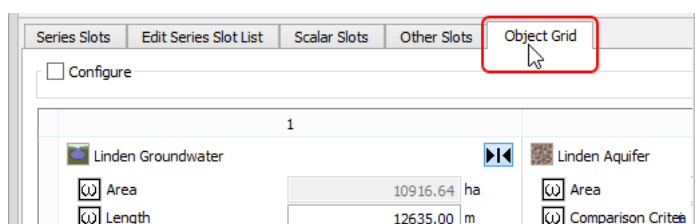
Related topics

- [Display the object grid configuration controls](#), page 222
- [Assign a slot list to a cell](#), page 229

Assign a slot list to a cell

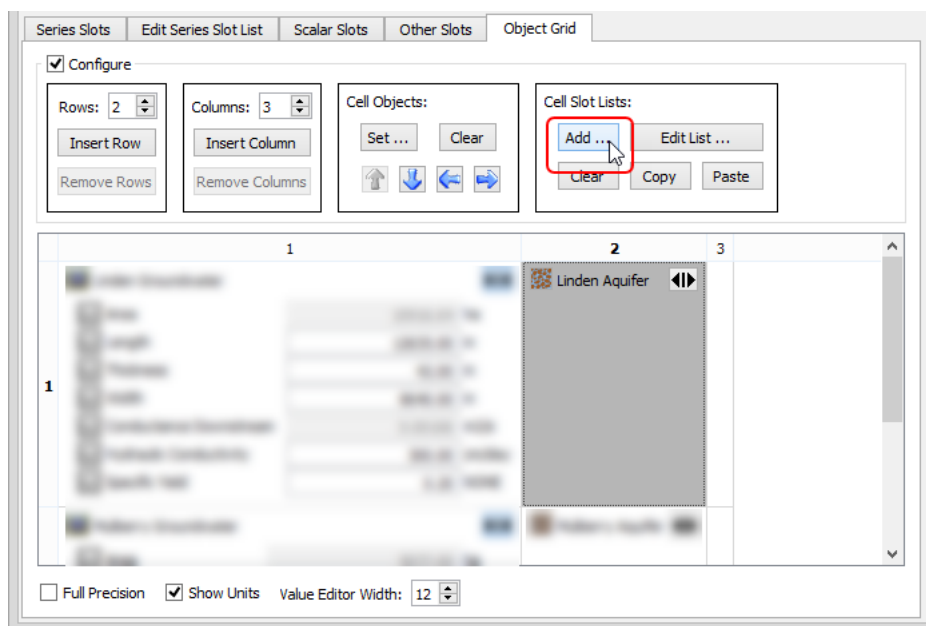
Use this procedure to assign a slot list to a cell. Before performing this procedure, you must assign an object to the cell. A slot list can include any scalar or table slots assigned to the object in the model.

1. Go to the **Object Grid** tab.



2. Select a single cell that has an assigned object.

3. In the Cell Slot Lists area, select **Add**.



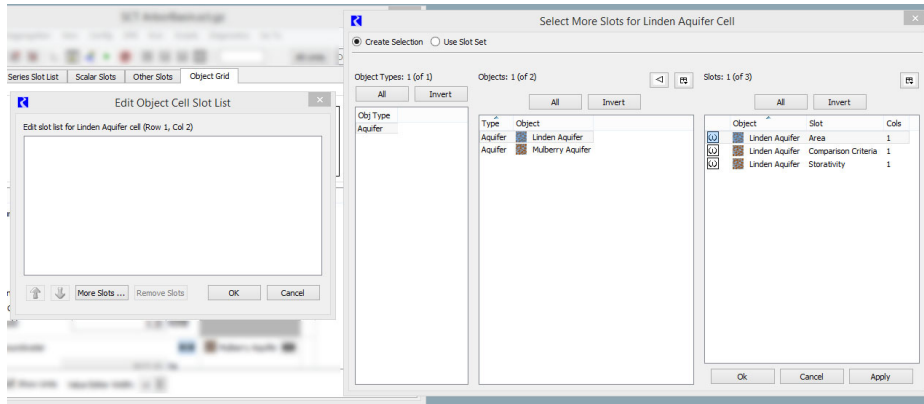
The following dialog boxes open:

- Edit Object Cell Slot List—displays all objects included in the model.
- Select More Slots for object Cell—displays all slots for the object.

Chapter 3

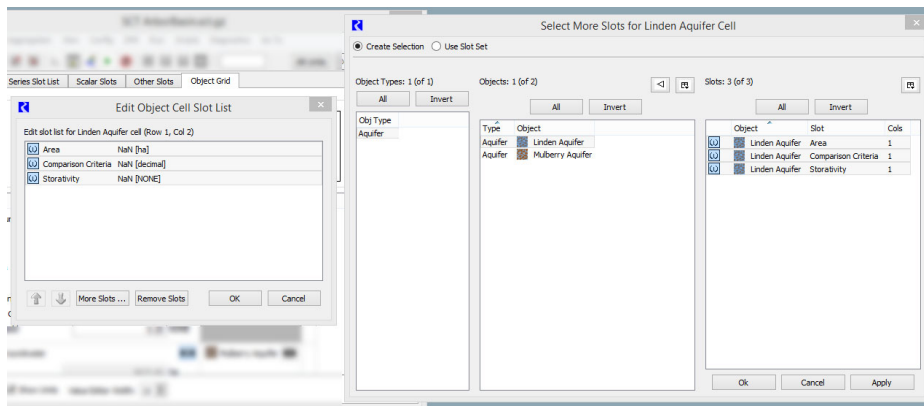
Using the SCT

4. Optionally, position the Select More Slots dialog box so it is side-by-side with the Edit Object Cell Slot List dialog box.



5. Use the following steps to complete the Select More Slots dialog box.
 - a. Select one or more slots to add to the object grid cell.
 - b. Select **Apply**.

The slots are added to the Edit Object Cell Slot List dialog box.

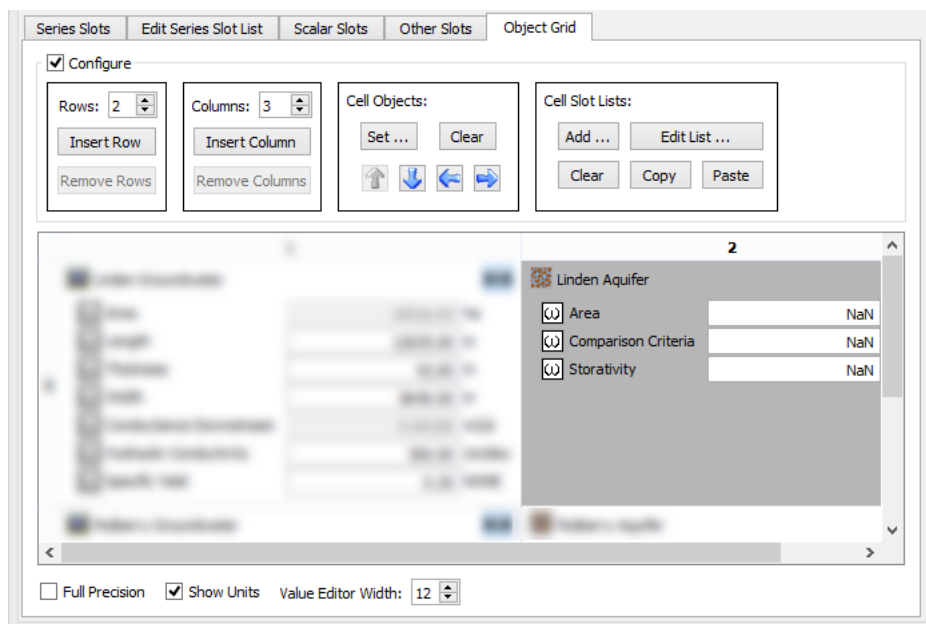


- c. When you are finished adding slots, select **OK** in the Select More Slots dialog box.

The Select More Slots dialog box is closed and the Edit Object Cell Slot List dialog box becomes active.

6. Use the following steps to complete the Edit Object Cell Slot List dialog box.
 - a. Optionally, add, delete, or rearrange slots in the list; see [“Configure the slot list in a cell”](#) on page 233 for details.
 - b. When you are finished making changes to the slot list, select **OK**.

The Slot List is closed and the object cell is updated with the changes.



Related topics

- [Display the object grid configuration controls](#), page 222
- [Assign objects to cells](#), page 226
- [Configure the slot list in a cell](#), page 233

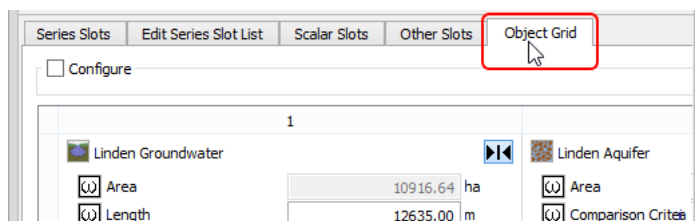
Copy and paste slot lists

Use this procedure to copy a slot list from one cell to another. Slots that do not exist on the target object are displayed as unavailable (dimmed).

Note: If the target cell already has a slot list, it will be replaced when you perform the paste.

This procedure allows you to easily create slot lists for objects that have similar properties. You can copy the same list to multiple objects, then add or delete selected slots to refine the lists as necessary.

1. Go to the **Object Grid** tab.

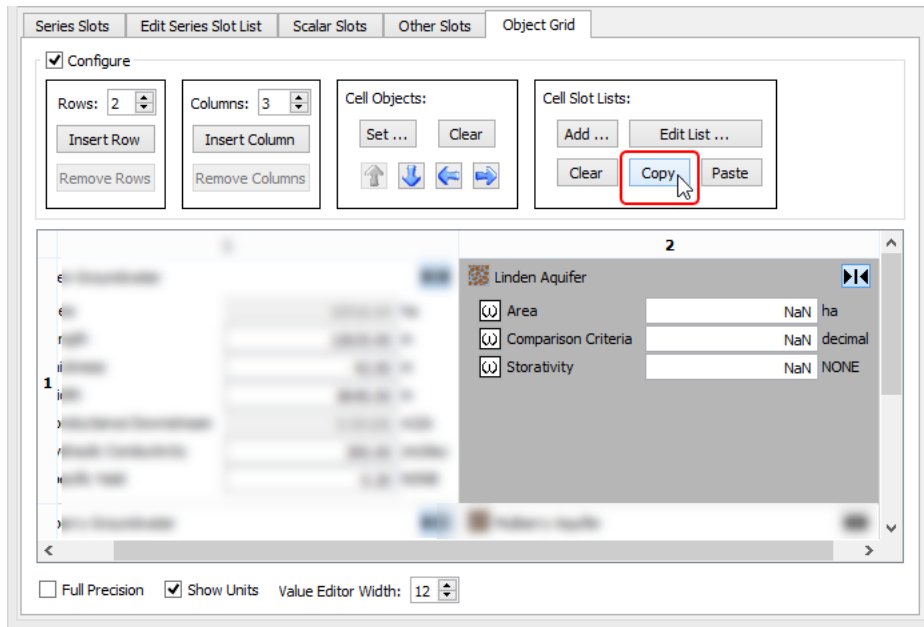


2. Select the cell whose slot list you want to copy.

Chapter 3

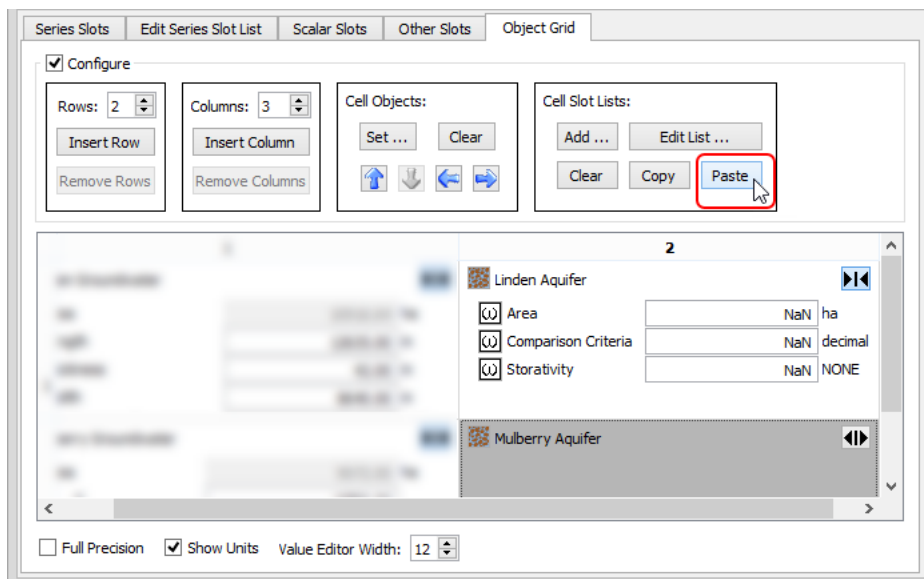
Using the SCT

3. In the Cell Slot Lists area, select **Copy**.

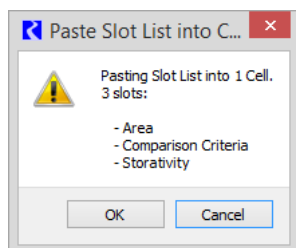


The slot list is copied exactly as is.

4. Select the cell you want to populate. The cell must have an assigned object, and it may or may not have an existing slot list.
5. In the Cell Slot Lists area, select **Paste**.

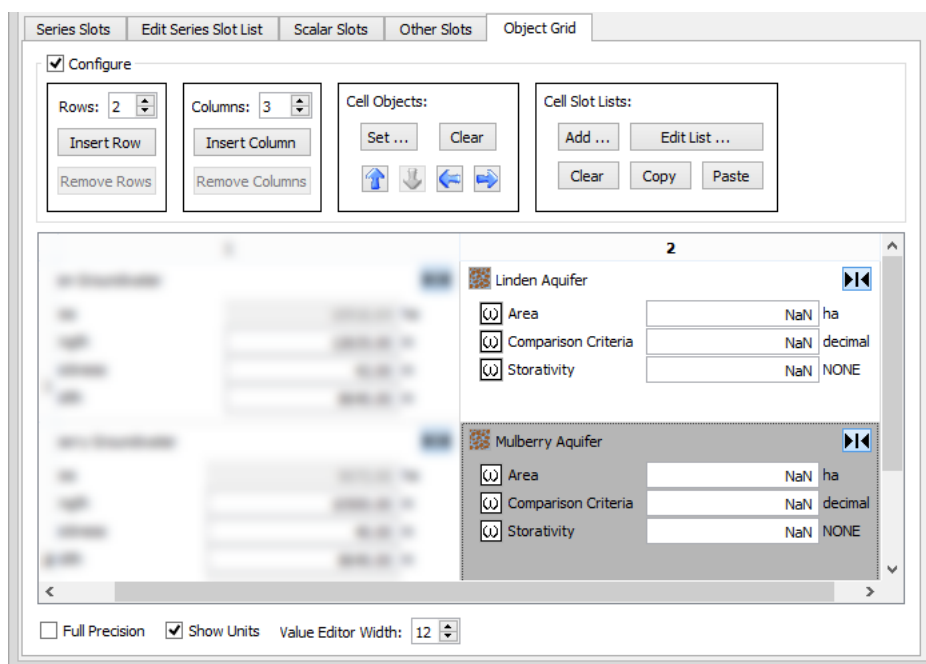


A confirmation dialog box appears, listing the slots that will be pasted.



6. Select **OK** in the dialog box.

The slot list is added to the cell. Any existing slot list is replaced.



7. Optionally, repeat [Step 5.](#) and [Step 6.](#) to paste the same slot list to other cells.

Alternate methods

- Right-click and select **Copy Slot List** in the shortcut menu.

Related topics

- [Display the object grid configuration controls](#), page 222
- [Assign a slot list to a cell](#), page 229

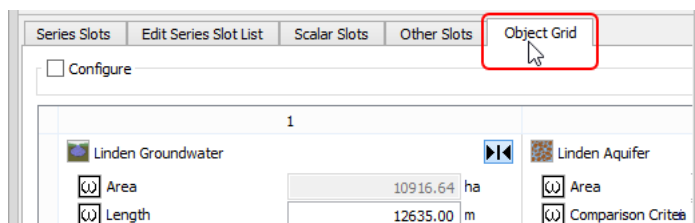
Configure the slot list in a cell

Use this procedure to customize the composition and order of the slots in a slot list. You can add or delete slots from the list, and you can arrange the slots in any order.

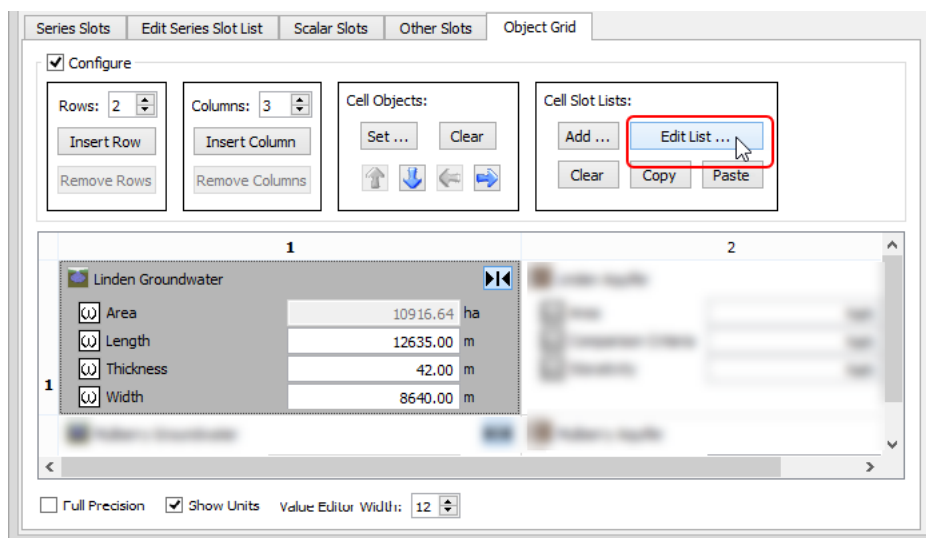
Chapter 3

Using the SCT

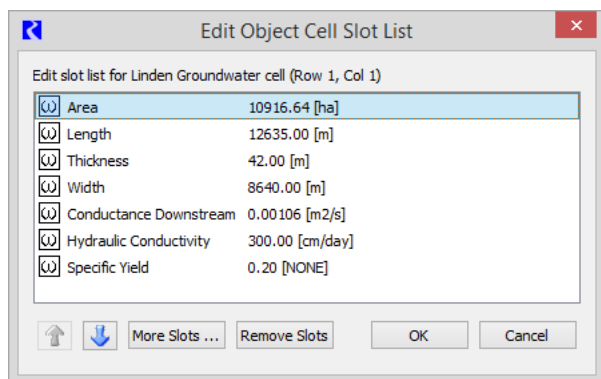
1. Go to the **Object Grid** tab.



2. Select the cell you want to modify.
3. In the Cell Slot Lists area, select **Edit List**.



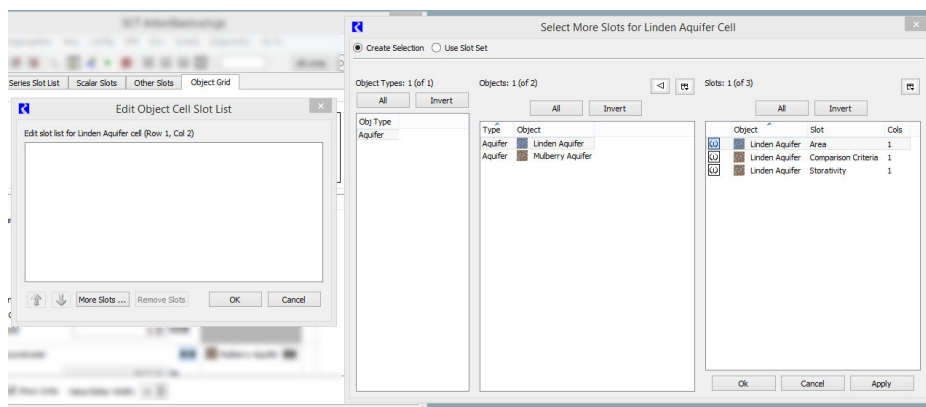
The Edit Object Cell Slot List dialog box appears.



4. Use the following steps to add slots to the slot list.
 - a. In the Slot List, select **More Slots**.

The Select More Slots dialog box opens.

- b. Optionally, arrange the windows so you can see them all side-by-side.



- c. In the Select More Slots dialog box, select the slots you want to add.

- d. Select **Apply**.

The slots are added to the Slot List dialog box.

- e. When you are finished adding slots, select **Ok** in the Select More Slots dialog box.

The dialog box is closed, and the Slot List dialog box is active.

5. Use the following steps to delete slots from the slot list.

- a. In the Slot List dialog box, select the slots you want to delete. You can select one or more contiguous or non-contiguous slots.

- b. Select **Remove Slots**.

The slots are immediately removed from the Slot List.

6. Use the following steps to reorder slots in the slot list.

- a. In the Slot List dialog box, select the slots you want to move. You can select one or more contiguous or non-contiguous slots.

- b. Select the **Up** or **Down** arrow buttons to move the slots in the indicated direction.

7. When the Slot List is arranged the way you want, select **OK**.

The Slot List dialog box is closed, and the selected cell in the Object Grid is updated according to your configuration selections.

Related topics

- [Display the object grid configuration controls](#), page 222
- [Assign a slot list to a cell](#), page 229

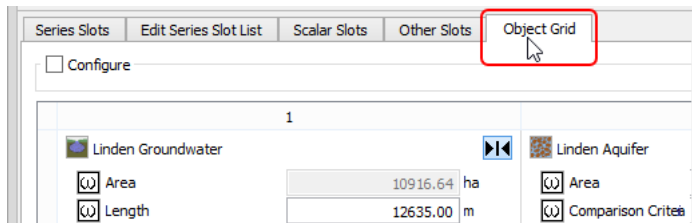
Show or hide the slot list in a cell

Use this procedure to show or hide the slot list in one or more selected cells. This is useful when you want to hide some slot lists so you can focus on others in a large object grid.

Chapter 3

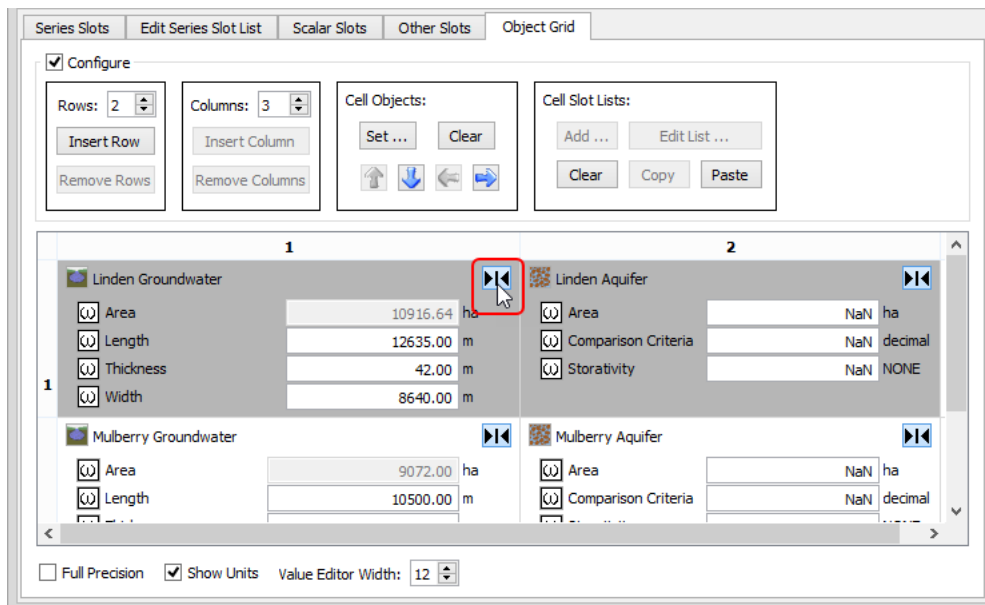
Using the SCT

1. Go to the **Object Grid** tab.



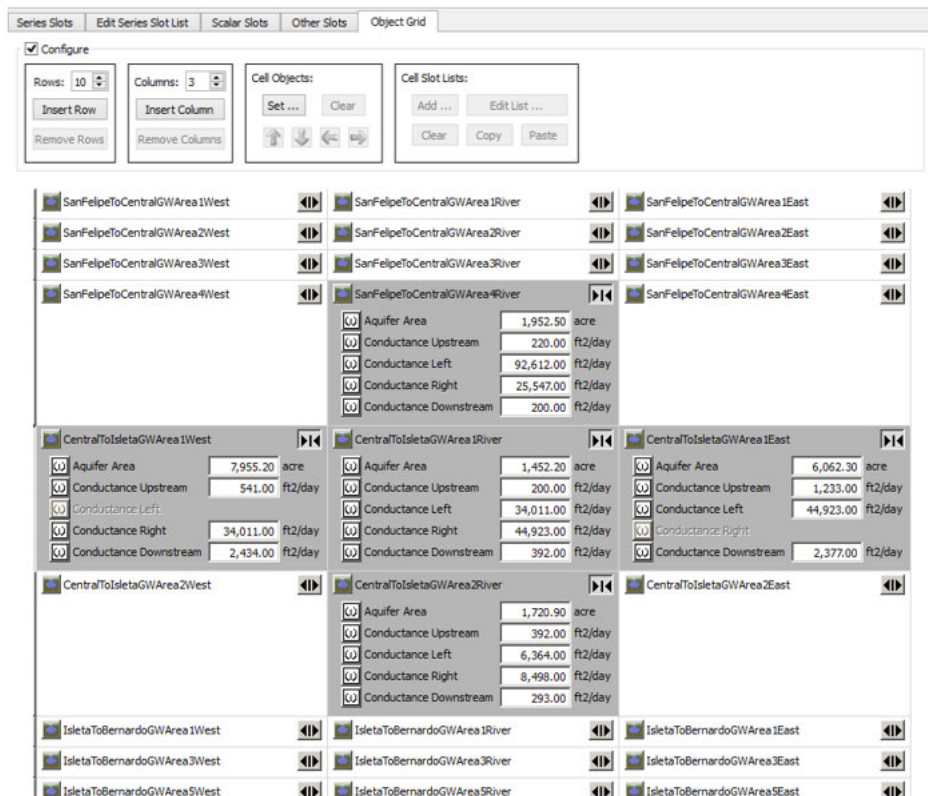
2. Perform one or more of the following actions, as applicable.

- To open or close a single cell, select the **Open/Close Selected Cells** button in that cell.
- To open or close multiple contiguous or non-contiguous cells, select the cells, then select the **Open/Close Selected Cells** button in one of the selected cells.
- To open or close an entire row or column, select the row or column header, then select the **Open/Close Selected Cells** button in one of the selected cells.



- To open a cell and the ones immediately adjacent to it, do one of the following:
 - Shift+select the **Open/Close Selected Cells** button in a cell. The selected cell and the ones immediately adjacent to it are simultaneously selected and opened, while the remaining cells on the grid are *left unchanged*. The selected cells remain selected so you can dynamically open and close them as a group by selecting one of their Open/Close Selected Cells buttons.
 - Ctrl+select the **Open/Close Selected Cells** button in a cell. The selected cell and the ones immediately adjacent to it are simultaneously selected and opened, while the remaining cells on the grid are *closed*.

The selected cells remain selected so you can dynamically open and close them as a group by selecting one of their Open/Close Selected Cells buttons.



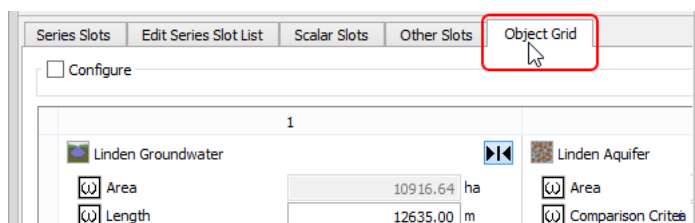
Related topics

- [Display the object grid configuration controls](#), page 222
- [Assign a slot list to a cell](#), page 229

Arrange cells in the object grid

Use this procedure to move cells in the object grid to create any layout that meets your needs.

1. Go to the **Object Grid** tab.



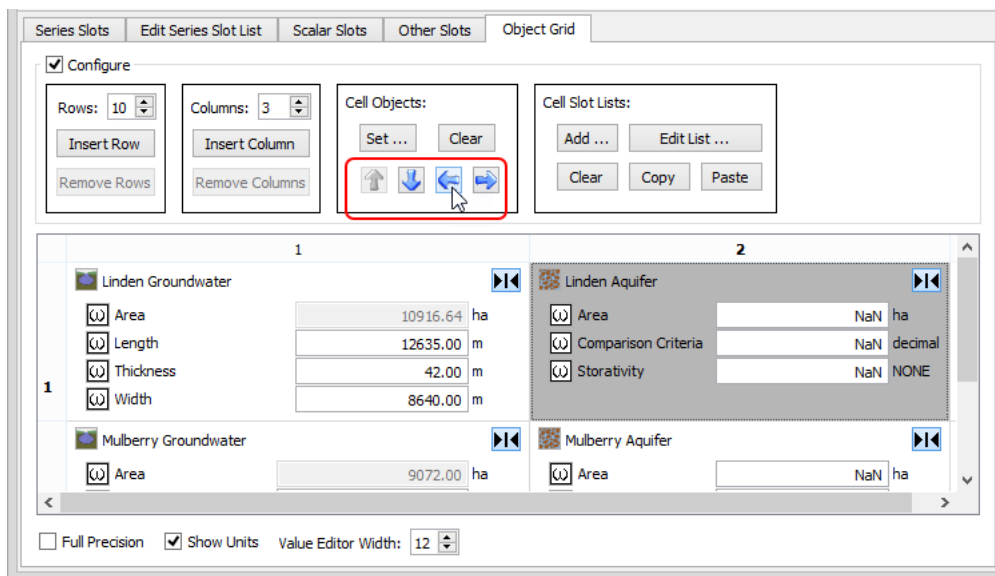
2. Use the following steps to move individual cells.
 - a. Select the cells you want to move. You can select one or more contiguous or non-contiguous cells.

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Using the SCT

- b. In the Cell Objects area, select an arrow button to move the cells one step in the indicated direction. If the adjacent cells in that direction are populated, the contents are swapped, not overwritten.
3. Use the following steps to move an entire row or column.
 - a. Select the row or column header.

The entire row or column is selected.
 - b. In the Cell Objects area, select an arrow button to move the rows or columns one step in the indicated direction. If the adjacent rows or columns in that direction are populated, the contents are swapped, not overwritten.



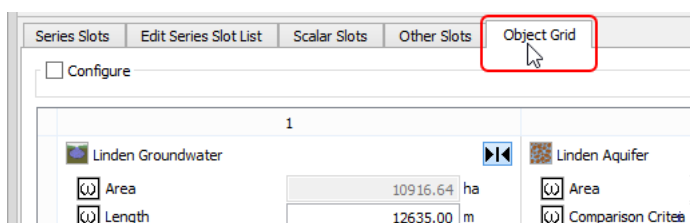
Related topics

- [Display the object grid configuration controls](#), page 222

Remove slot lists from cells

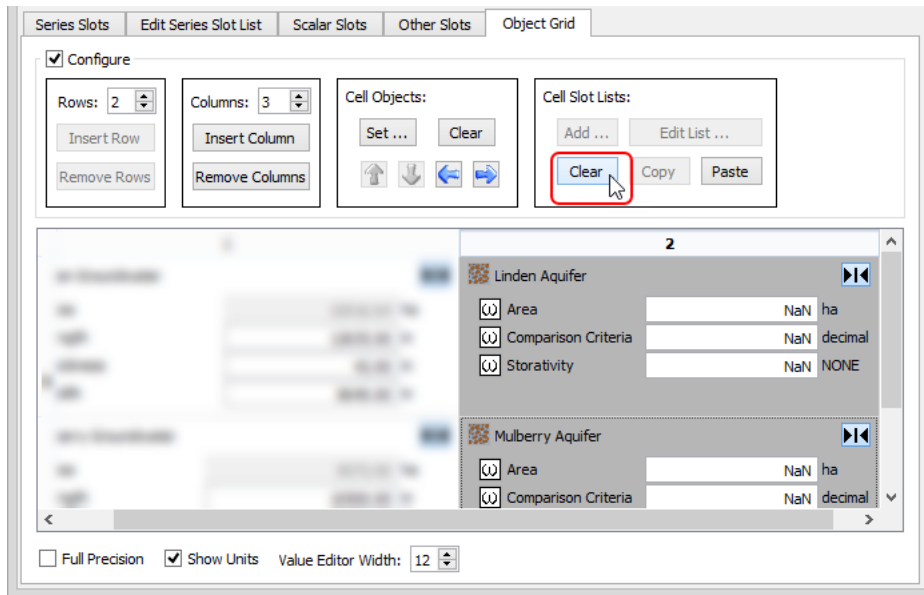
Use this procedure to clear slot lists from selected object grid cells, leaving the objects unchanged.

1. Go to the **Object Grid** tab.

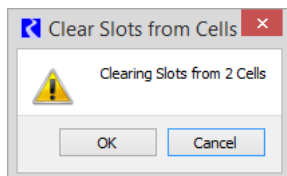


2. Select the cells you want to clear. You can select one or multiple contiguous or non-contiguous cells. You can also select an entire row or column by selecting the header.

3. In the Cell Slot Lists area, select **Clear**.

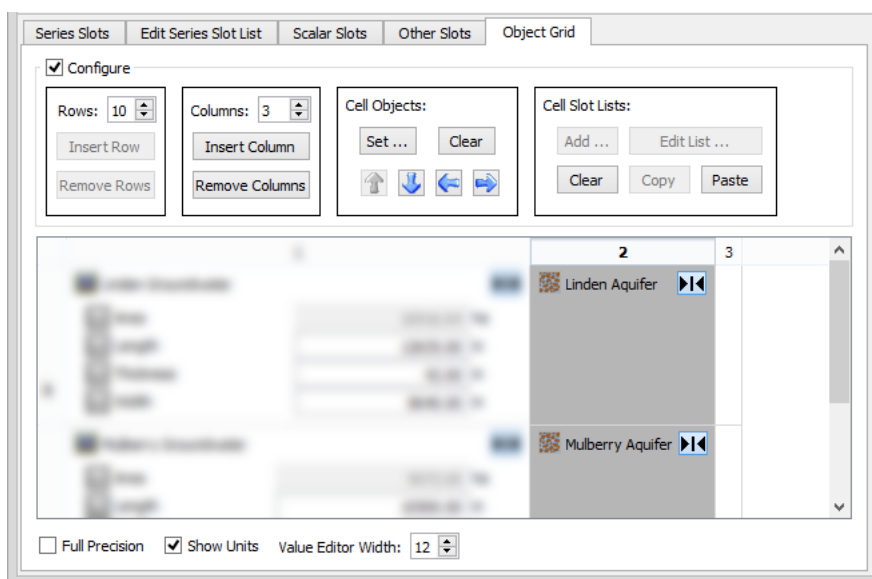


If you selected multiple cells, a confirmation message appears specifying the number of cells from which slot lists will be removed.



4. Select **OK** to confirm the update.

The slot lists are removed from the selected cells.



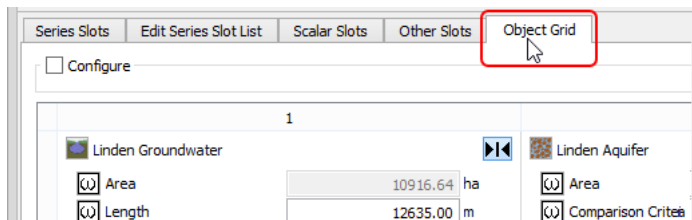
Related topics

- [Display the object grid configuration controls](#), page 222
- [Assign a slot list to a cell](#), page 229

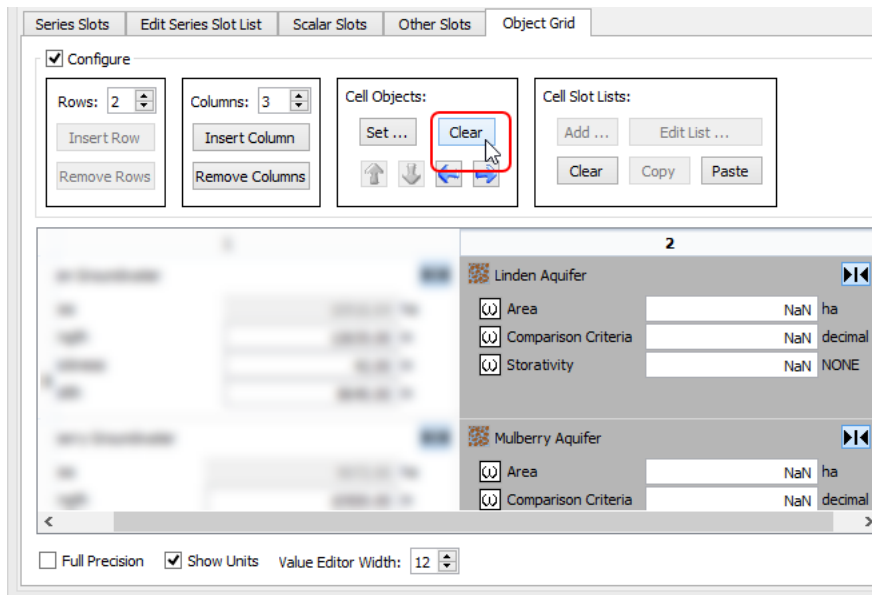
Remove objects from cells

Use this procedure to clear all objects and their assigned slot lists from selected object grid cells.

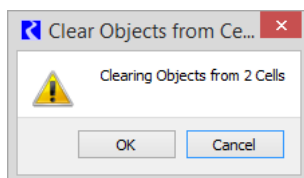
1. Go to the **Object Grid** tab.



2. Select the cells you want to clear. You can select one or multiple contiguous or non-contiguous cells. You can also select an entire row or column by selecting the header.
3. In the Cell Objects area, select **Clear**.

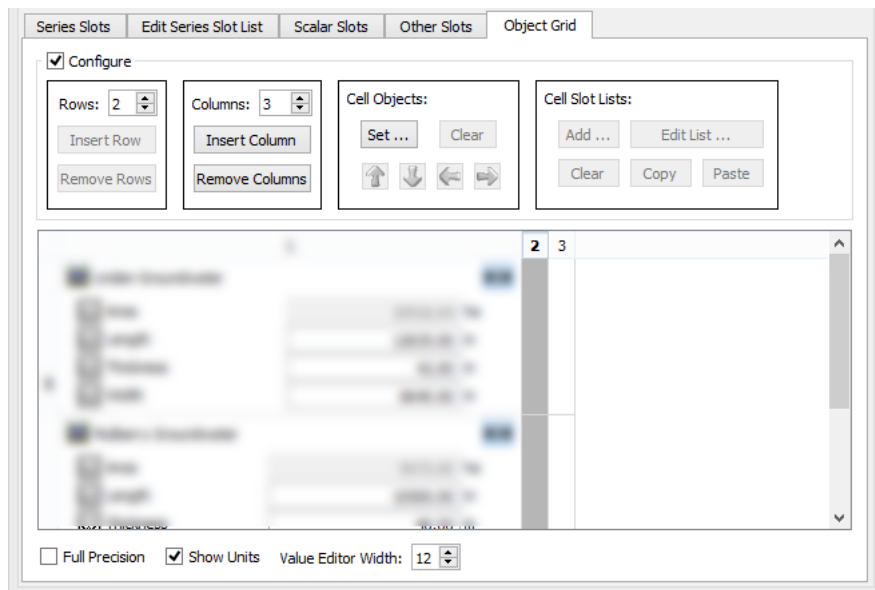


If you selected multiple cells, a confirmation message appears, specifying the number of cells from which objects will be removed.



4. Select **OK** to confirm the update.

The objects and their slot lists are removed from the selected cells.



Related topics

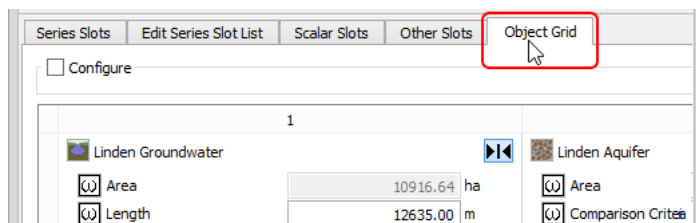
- [Display the object grid configuration controls](#), page 222
- [Assign objects to cells](#), page 226

Remove rows and columns from the object grid

Use this procedure to remove rows or columns from the object grid.

Note: You cannot restore rows and columns once they are removed with this procedure, except by closing the SCT without saving it.

1. Go to the **Object Grid** tab.

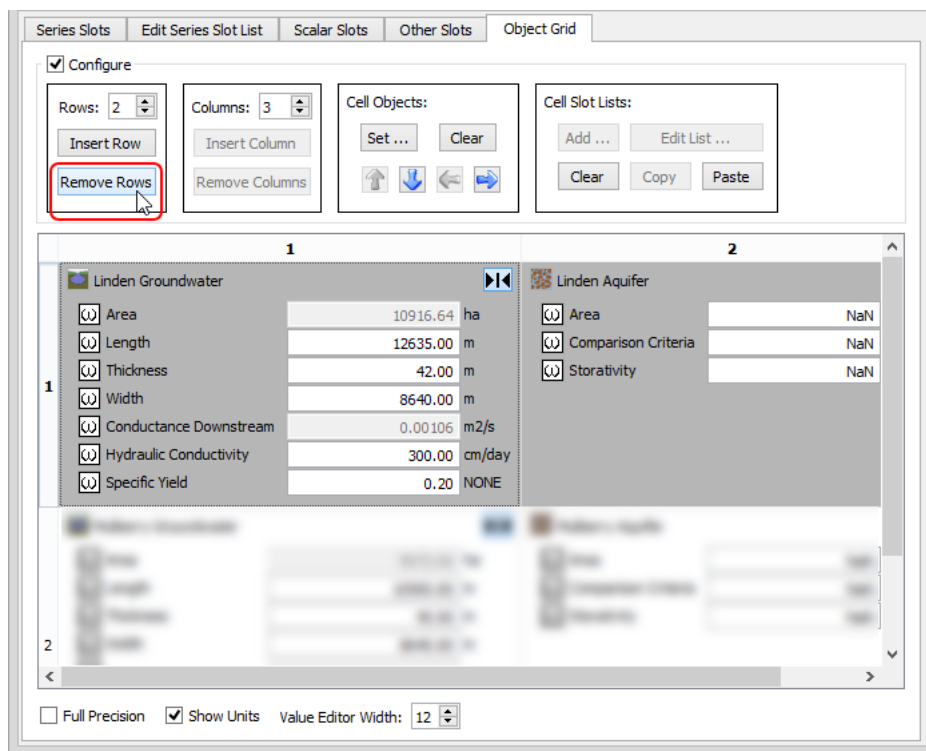


2. Use the following steps to remove rows.
 - a. Select the headers of the rows you want to remove. You can select one or more contiguous or non-contiguous rows.

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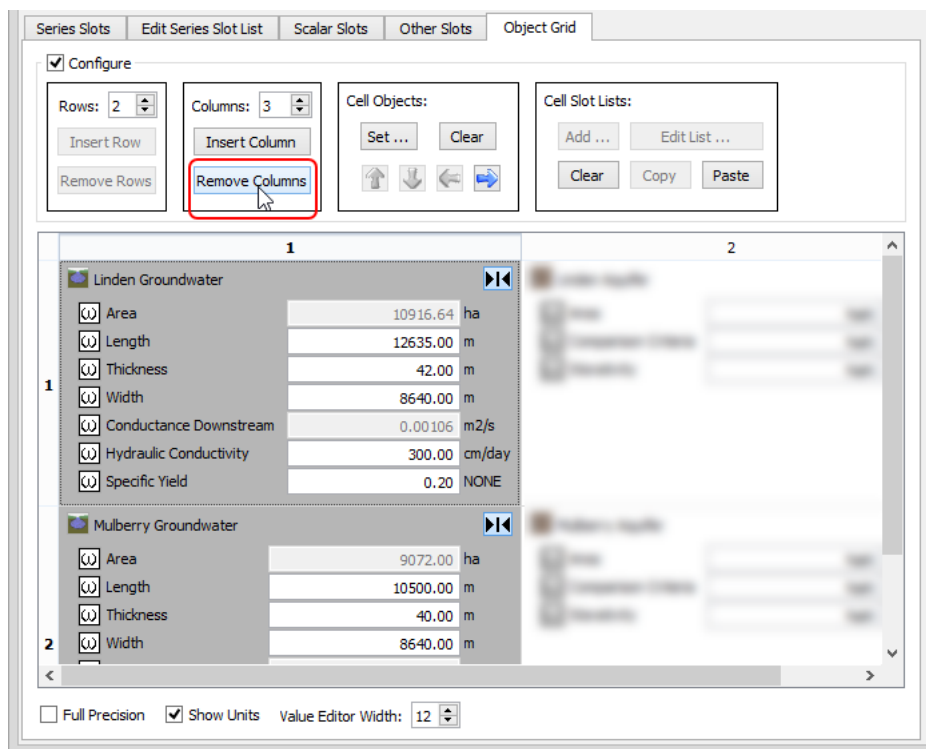
- b. In the Rows area, select **Remove Rows**.



The selected rows are removed from the grid.

3. Use the following steps to remove columns.
- a. Select the headers of the columns you want to remove. You can select one or more contiguous or non-contiguous columns.

b. In the Columns area, select **Remove Columns**.



The selected columns are removed from the grid.

4. Review the changes and proceed as follows.

- To cancel your changes and recover all rows and columns, close the SCT without saving it.
- To delete the rows and columns permanently, save the SCT under its current name.

Related topics

- [Display the object grid configuration controls](#), page 222
- [Add rows and columns to an object grid](#), page 224

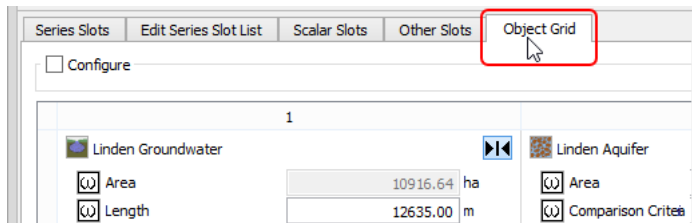
Hide and restore rows and columns in the object grid

Use this procedure to hide rows or columns from the end of the object grid display. Rows are hidden from the bottom of the grid, and columns are hidden from the right side. You can optionally restore the rows and columns hidden by this procedure, making it useful for trying out different layouts when configuring an object grid.

Chapter 3

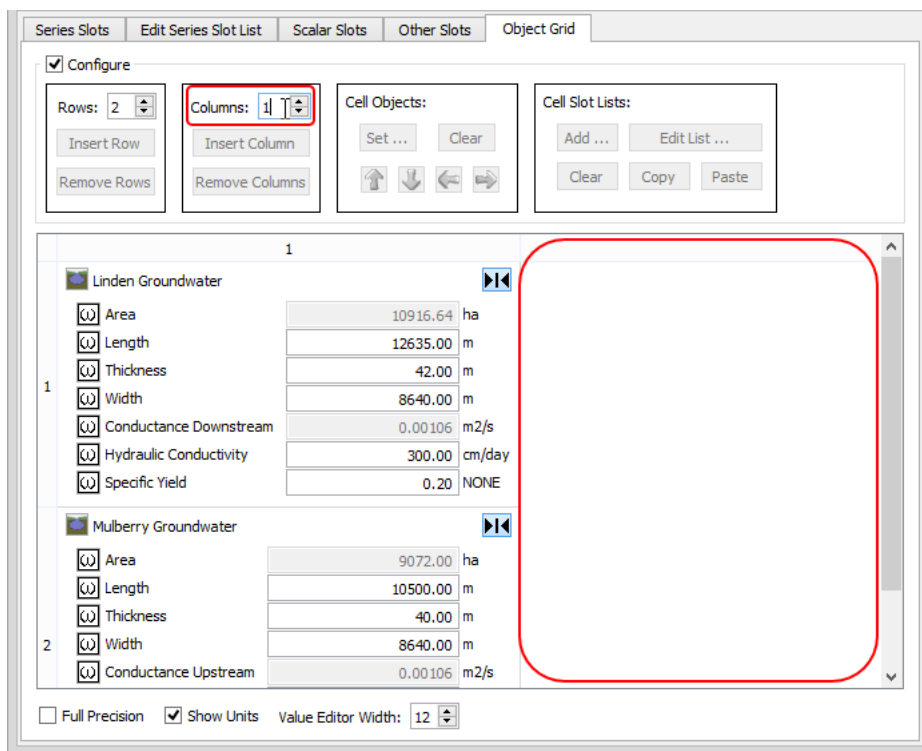
Using the SCT

1. Go to the **Object Grid** tab.



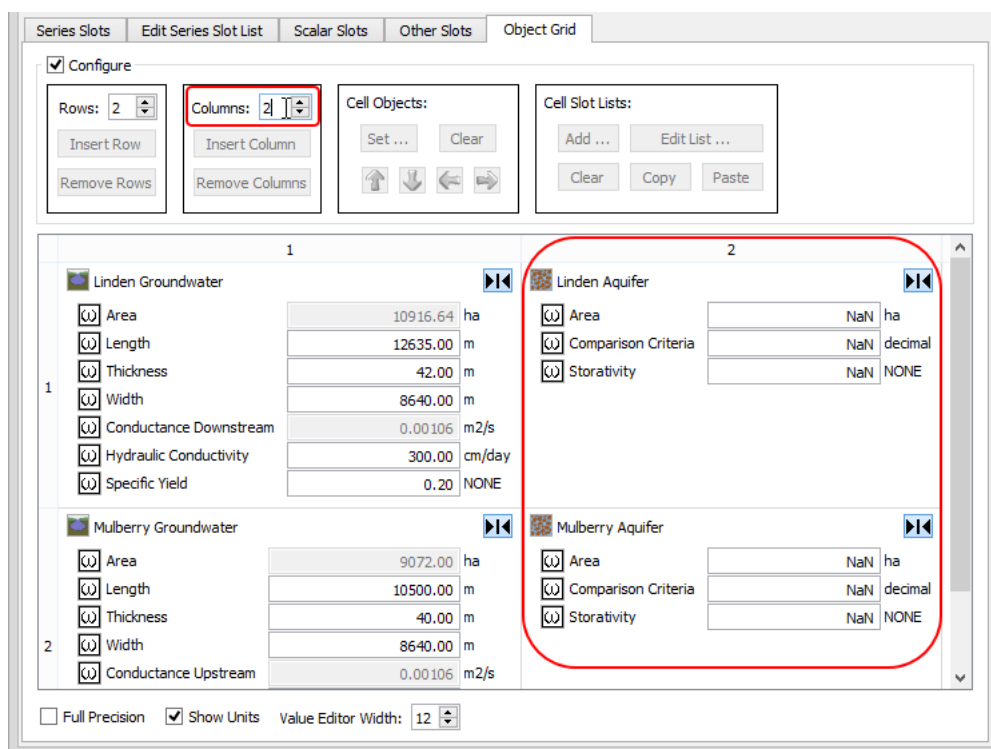
2. To hide rows or columns, perform the following actions, as applicable.

- In the **Rows** text box, enter a reduced number of rows. For example to remove three rows from an eight-row grid, enter 5. The rows are hidden at the bottom of the grid.
- In the **Columns** text box, enter a reduced number of columns. For example to remove two columns from a five-column grid, enter 3. The columns are hidden at the end (right side) of the grid.



3. To restore hidden rows or columns, increase the count in the **Rows** or **Columns** text box, as applicable.

The specified number of rows or columns are restored to the end of the display.



4. Review the changes and proceed as follows.
 - To cancel your changes and recover all hidden rows and columns, close the SCT without saving it.
 - To delete the hidden rows and columns permanently, save the SCT under its current name.

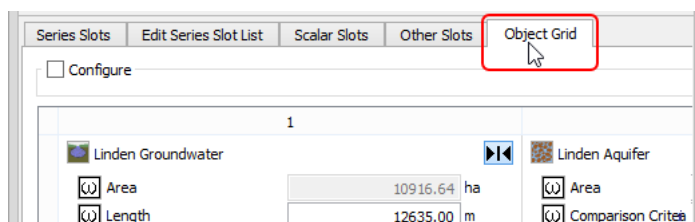
Related topics

- [Display the object grid configuration controls](#), page 222
- [Add rows and columns to an object grid](#), page 224

Configure the object grid display settings

Use this procedure to configure the display attributes of the object grid.

1. Go to the **Object Grid** tab.



2. Configure slot display settings as follows.

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- Select the **Full Precision** check box to display slot values at full precision. Full precision is the precision necessary to precisely represent the internal floating-point numeric value. Clear the check box to display slot values at the display precision only.

Note: This is just a display setting; you can always edit slot values at full precision, regardless of the precision displayed.

- Select the **Show Units** check box to display units next to each slot value. Clear the check box to hide the units.

Note: When units are hidden, you can hover over the slot value to display a tooltip with the unit.

- In the **Value Editor Width** text field, enter the total number of characters you want to display in the slot value fields. This is useful if horizontal space is limited. The supported range is 3 to 22, and the default is 12.

Note: This is just a display setting; it does not affect the number of characters saved.

The screenshot shows the 'Object Grid' tab in the SCT interface. The 'Configure' checkbox is unchecked. The 'Value Editor Width' is set to 12. The 'Full Precision' and 'Show Units' checkboxes are both checked. The 'Linden Groundwater' slot is selected, and its properties are displayed in a table. The 'Linden Aquifer' slot is also visible.

Property	Value	Unit
Area	10916.64	ha
Length	12635	m
Thickness	42	m
Width	8640	m
Conductance Downstream	0.00106332396801383	m2/s
Hydraulic Conductivity	300	cm/day
Specific Yield	0.2	NONE

The screenshot shows the 'Object Grid' tab in the SCT interface. The 'Configure' checkbox is unchecked. The 'Value Editor Width' is set to 7. The 'Full Precision' and 'Show Units' checkboxes are both unchecked. The 'Linden Groundwater' slot is selected, and its properties are displayed in a table. The 'Linden Aquifer' slot is also visible.

Property	Value	Unit
Area	10916.64	ha
Length	12635.00	m
Thickness	42.00	m
Width	8640.00	m
Conductance Downstream	0.00106	m2/s
Hydraulic Conductivity	300.00	cm/day
Specific Yield	0.20	NONE

Related topics

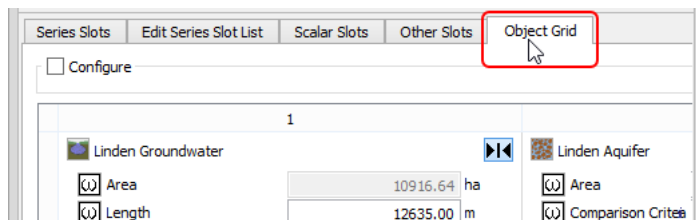
- [Edit a slot value in the object grid](#), page 247

Edit a slot value in the object grid

Use this procedure to modify the values of scalar slots and single-cell table slots directly from the SCT object grid.

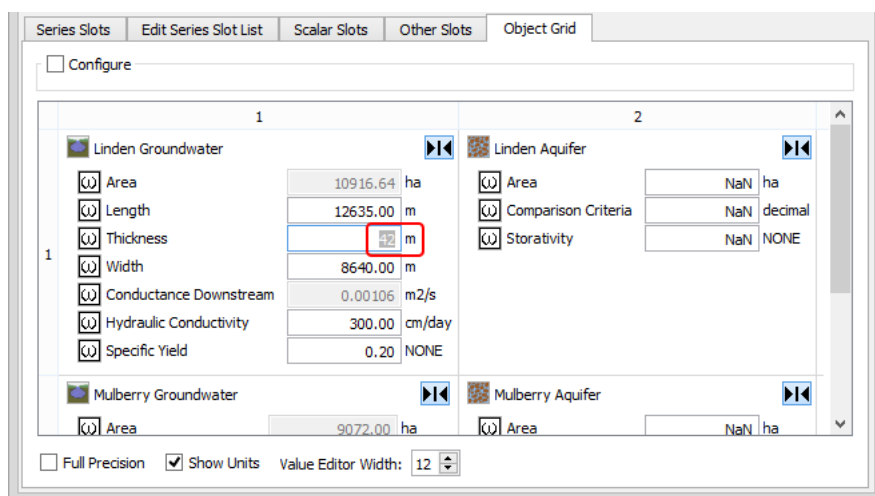
Note: The SCT is just a *window* into the model and all values are stored in the model itself. You must save the model to preserve the new values.

1. Go to the **Object Grid** tab.



2. Use one of the following methods to edit the value.

- In the object grid, select the slot value you want to modify, and enter the new value.



- Open the slot in the Slot dialog, and enter the new value in the **Value** field.

The value is updated in the object grid.

Undo and Redo are available from the Edit menu.

Related topics

- [Configure the object grid display settings](#), page 245
- [Open an object or slot from the object grid](#), page 247

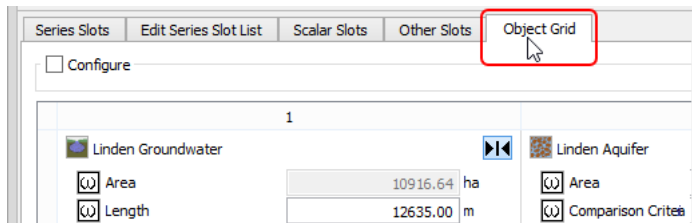
Open an object or slot from the object grid

Use this procedure to open a selected object in the Object Viewer or a selected slot in the Slot Viewer.

Chapter 3

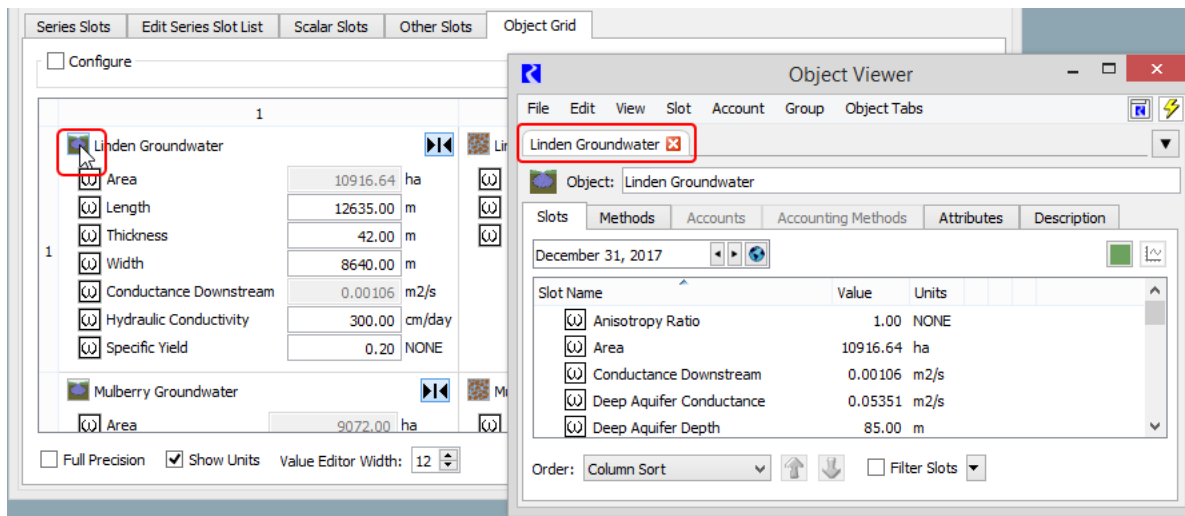
Using the SCT

1. Go to the **Object Grid** tab.



2. To open an object, select the object type button next to the object name.

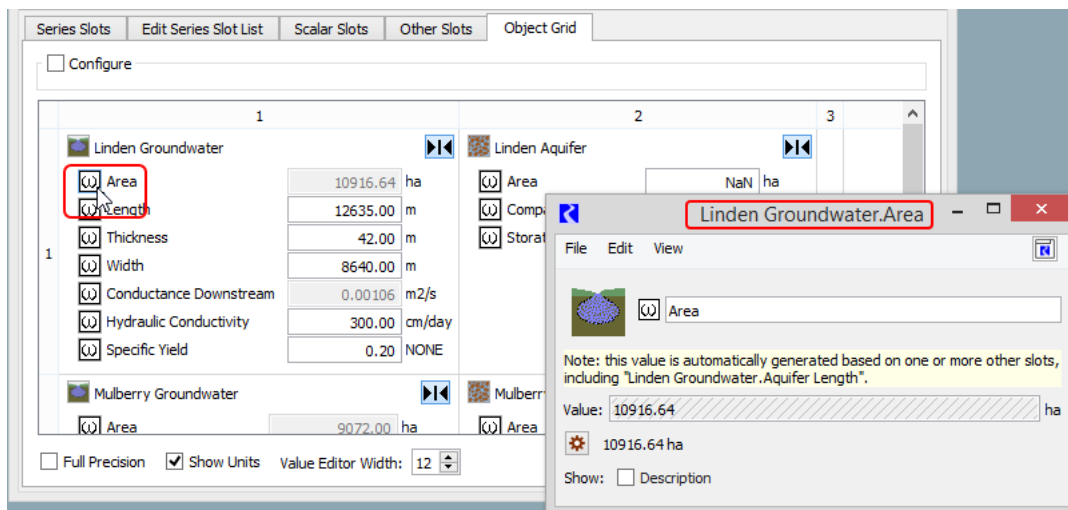
The slot opens in the Object Viewer; you can perform any actions available in the Object Viewer.



3. To open a slot, select the slot type button next to the slot name.

The slot opens in the Slot Viewer; you can perform any actions available from the Slot Viewer.

Note: Changing the number of rows or columns in a 1x1 table slot can produce unexpected results when that slot is edited from the SCT. A previously displayed 1x1 table slot that currently has more rows or columns is not displayed.



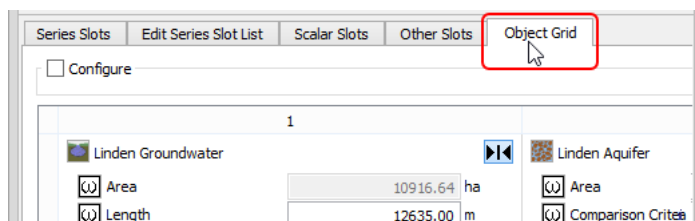
Related topics

- [Edit a slot value in the object grid](#), page 247

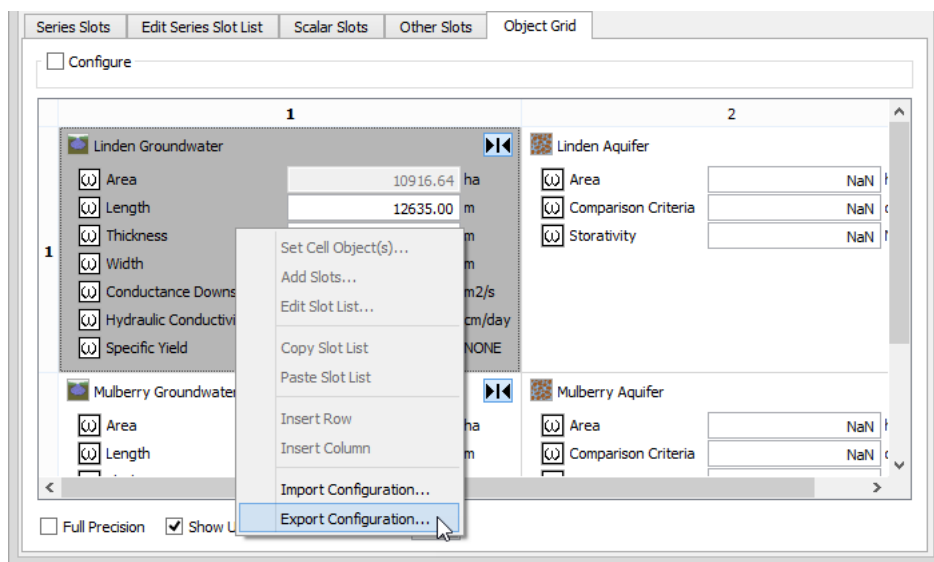
Export an object grid to an XML file

Use this procedure to export the entire object grid configuration of an SCT to an XML file. The configuration includes the cell layout and assigned objects and slot lists. This is useful for transferring an object grid from one SCT to others.

1. Go to the **Object Grid** tab.



2. Right-click any cell in the object grid and select **Export Configuration** in the shortcut menu.



The Windows File Chooser opens.

3. In the File Chooser, navigate to the folder where you want to save the object grid configuration. In the **File name** field, enter a name with a .xml extension.

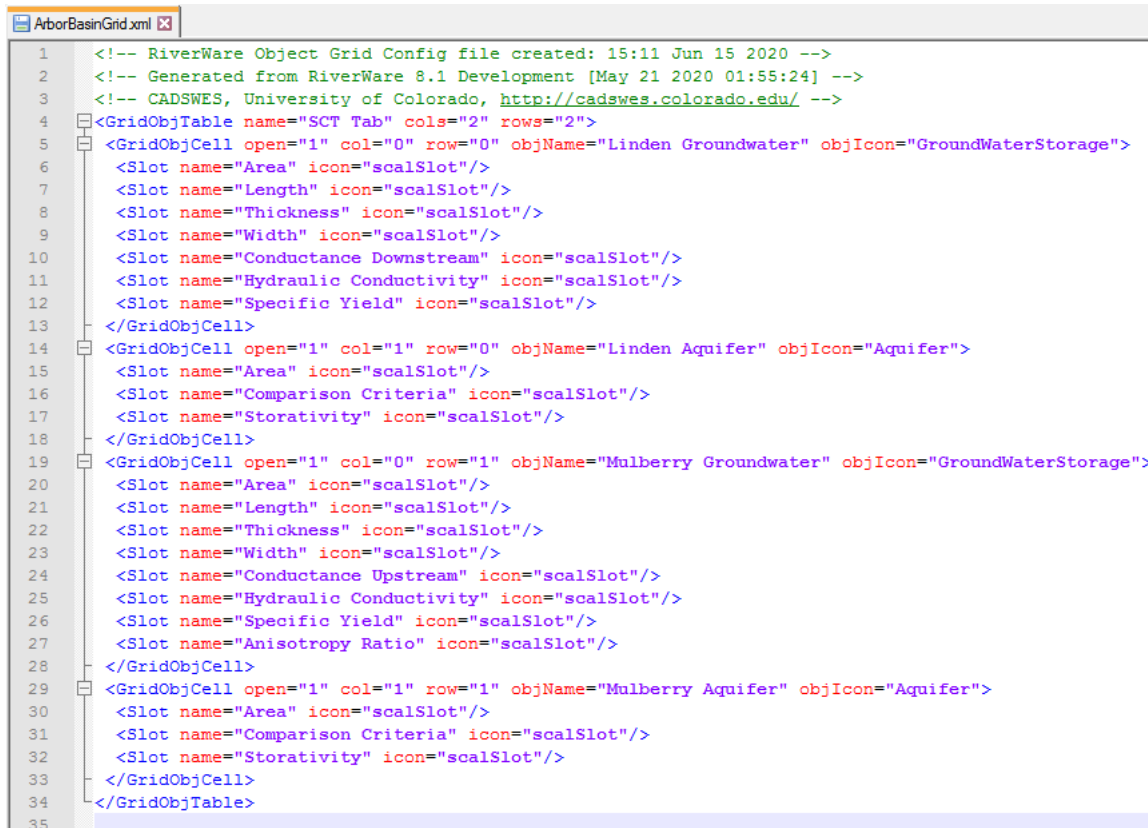
Note: The extension is not assigned automatically; you must enter it explicitly.

4. Select **Save**.

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The file is saved to the specified location. Optionally, you can view the contents of the file by opening it in a text editor, such as Notepad.



```
1 <!-- RiverWare Object Grid Config file created: 15:11 Jun 15 2020 -->
2 <!-- Generated from RiverWare 8.1 Development [May 21 2020 01:55:24] -->
3 <!-- CADSWES, University of Colorado, http://cadswes.colorado.edu/ -->
4 <GridObjTable name="SCT Tab" cols="2" rows="2">
5   <GridObjCell open="1" col="0" row="0" objName="Linden Groundwater" objIcon="GroundWaterStorage">
6     <Slot name="Area" icon="scalSlot"/>
7     <Slot name="Length" icon="scalSlot"/>
8     <Slot name="Thickness" icon="scalSlot"/>
9     <Slot name="Width" icon="scalSlot"/>
10    <Slot name="Conductance Downstream" icon="scalSlot"/>
11    <Slot name="Hydraulic Conductivity" icon="scalSlot"/>
12    <Slot name="Specific Yield" icon="scalSlot"/>
13  </GridObjCell>
14  <GridObjCell open="1" col="1" row="0" objName="Linden Aquifer" objIcon="Aquifer">
15    <Slot name="Area" icon="scalSlot"/>
16    <Slot name="Comparison Criteria" icon="scalSlot"/>
17    <Slot name="Storativity" icon="scalSlot"/>
18  </GridObjCell>
19  <GridObjCell open="1" col="0" row="1" objName="Mulberry Groundwater" objIcon="GroundWaterStorage">
20    <Slot name="Area" icon="scalSlot"/>
21    <Slot name="Length" icon="scalSlot"/>
22    <Slot name="Thickness" icon="scalSlot"/>
23    <Slot name="Width" icon="scalSlot"/>
24    <Slot name="Conductance Upstream" icon="scalSlot"/>
25    <Slot name="Hydraulic Conductivity" icon="scalSlot"/>
26    <Slot name="Specific Yield" icon="scalSlot"/>
27    <Slot name="Anisotropy Ratio" icon="scalSlot"/>
28  </GridObjCell>
29  <GridObjCell open="1" col="1" row="1" objName="Mulberry Aquifer" objIcon="Aquifer">
30    <Slot name="Area" icon="scalSlot"/>
31    <Slot name="Comparison Criteria" icon="scalSlot"/>
32    <Slot name="Storativity" icon="scalSlot"/>
33  </GridObjCell>
34 </GridObjTable>
35
```

Related topics

- [Create a new object grid](#), page 223
- [Import an object grid to an SCT](#), page 250

Import an object grid to an SCT

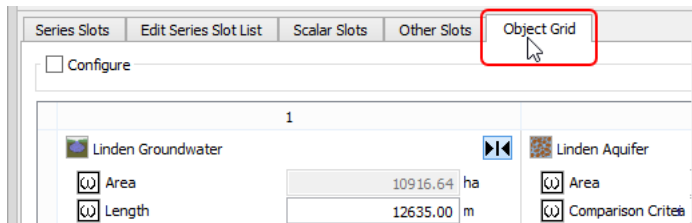
Use this procedure to import an object grid into a new or existing SCT. This procedure allows you to easily create a new SCT object grid using a similar one as a basis. After importing the object grid, you can modify the layout and assigned objects and slot lists, as necessary.

Note: Before using this procedure, you must export an existing object grid to an XML file.

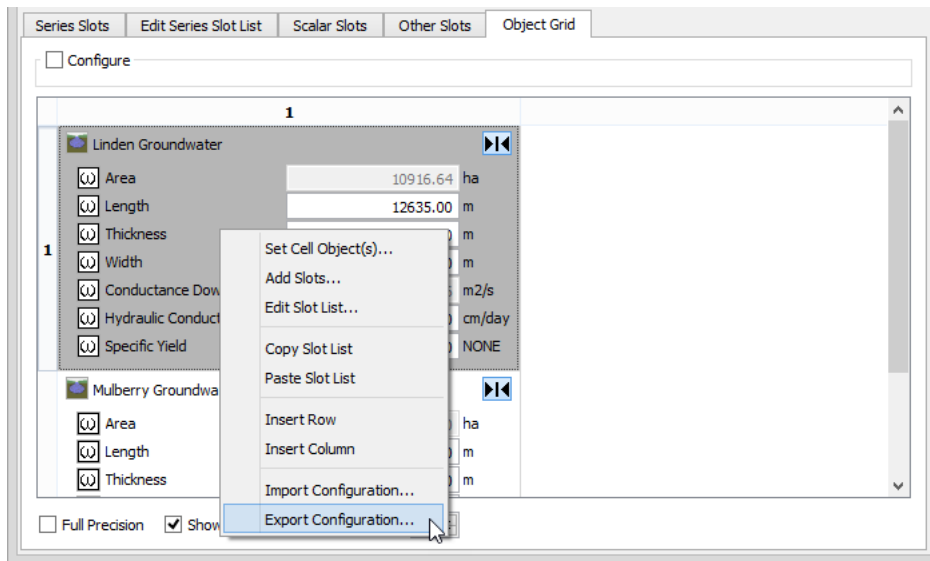
Note: If you import to an existing SCT, the contents of the existing object grid are replaced by the imported configuration.

1. Open a new or existing SCT.

2. Go to the **Object Grid** tab.



3. Right-click any cell in the object grid and select **Import Configuration** in the shortcut menu.



The Windows File Chooser opens.

4. In the File Chooser, navigate to the location where the exported SCT is saved, and select the XML file.
Note: If the file you select is not a valid object grid, the import will fail with an error message.
5. Select **Open**.

Chapter 3

Using the SCT

The imported configuration, including the assigned objects and slot lists, is loaded into the SCT, overwriting any existing configuration.

	1	2
Linden Groundwater		Linden Aquifer
Area	10916.64 ha	NaN ha
Length	12635.00 m	NaN decimal
Thickness	42.00 m	NaN NONE
Width	8640.00 m	
Conductance Downstream	0.00106 m2/s	
Hydraulic Conductivity	300.00 cm/day	
Specific Yield	0.20 NONE	
Mulberry Groundwater		Mulberry Aquifer
Area	9072.00 ha	NaN ha
Length	10500.00 m	NaN decimal
Thickness	40.00 m	NaN NONE

☐ Full Precision ☒ Show Units Value Editor Width: 12

Related topics

- [Create a new object grid](#), page 223
- [Import an object grid to an SCT](#), page 250

Using the Default SCT Configurations

This section includes procedures for managing the user-defined and factory-default SCT configurations.

Topics

- [Create a user-defined default SCT configuration](#), page 252
- [Revert an SCT to the current default configuration](#), page 253
- [Restore the factory default SCT configuration](#), page 254

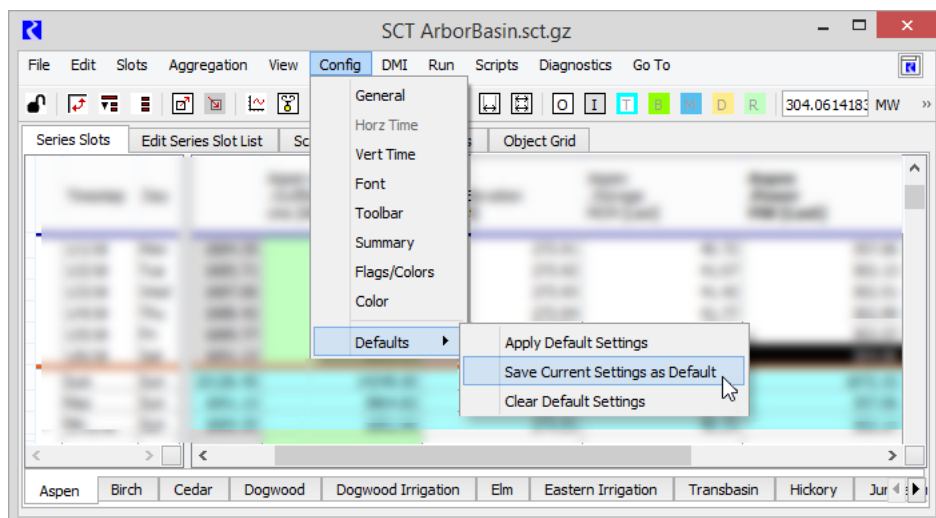
Create a user-defined default SCT configuration

Use this procedure to save the current SCT configuration settings as the default. These settings will be applied to all new SCTs when they are created. This procedure does not affect existing SCTs.

Note: The default SCT configuration does not include the slot item list.

1. Create or open the SCT you want to use for the new default configuration.

2. On the SCT menu, select **Config**, then **Defaults**, then **Save Current Settings as Default**.



The current configuration is saved as the default in the location specified in the settings manager. See [“Defaults \(Settings File\)”](#) in *User Interface* for more information.

Related topics

- [Create a new SCT](#), page 38
- [Load an existing SCT](#), page 39
- [Restore the factory default SCT configuration](#), page 254
- [SCT Configuration](#), page 33
- [“Settings Reference”](#) in *User Interface*

Revert an SCT to the current default configuration

Use this procedure to revert the current SCT to the default SCT configuration. If a user-defined default configuration has been created, those settings are used; if there is no user-defined default, the factory-default settings are used. The default SCT settings file’s location is specified in the settings manager. See [“Defaults \(Settings File\)”](#) in *User Interface* for more information.

This procedure removes any saved or unsaved customizations made to this SCT. This procedure does not affect other SCTs.

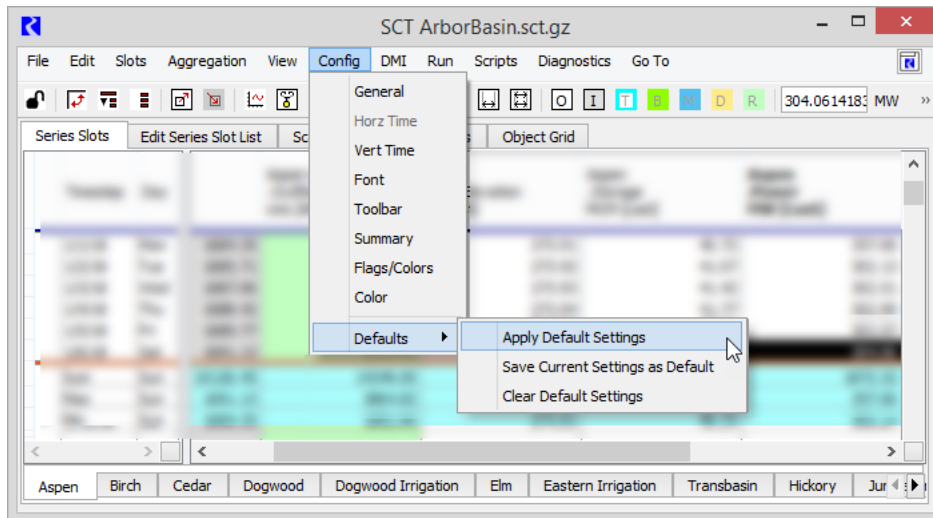
Note: Any custom settings previously saved with the SCT file will be removed if you save the SCT after performing this procedure; therefore, you may want to make a copy of the SCT before proceeding.

1. Open the SCT whose configuration you want to revert.

Chapter 3

Using the SCT

2. On the SCT menu, select **Config**, then **Defaults**, then **Apply Default Settings**.



The default SCT configuration settings are applied to the SCT.

Note: This does not affect the slot item list.

3. Proceed as follows:
 - To save these settings with the SCT, save the SCT.
 - To retain previously saved custom settings, close the SCT without saving it.

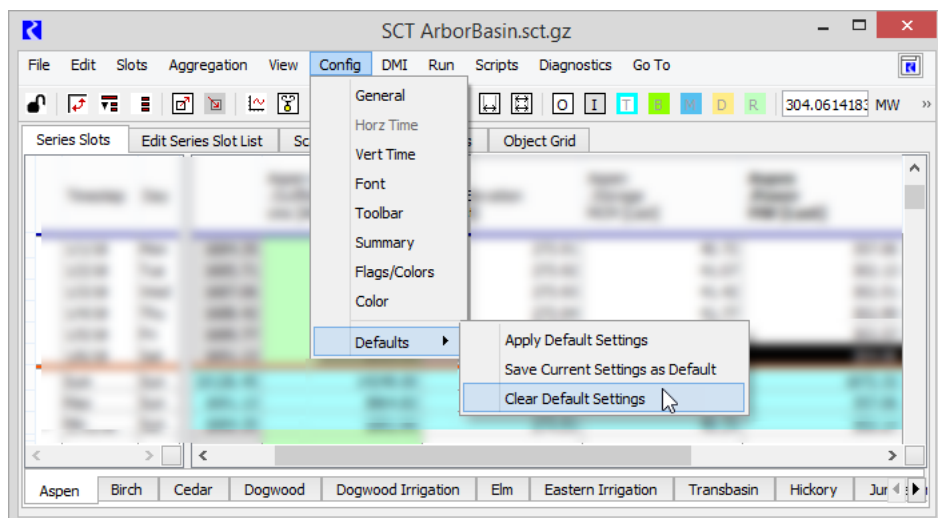
Related topics

- [Create a user-defined default SCT configuration](#), page 252
- [SCT Configuration](#), page 33
- “Settings Reference” in *User Interface*

Restore the factory default SCT configuration

Use this procedure to restore the *factory default* SCT configuration—that is, the configuration that is delivered with the RiverWare installation. This replaces the user-defined default configuration, if one exists. The default SCT settings file’s location is specified in the settings manager. See “[Defaults \(Settings File\)](#)” in *User Interface* for more information. The factory-default settings will be applied to all new SCTs when they are created. This procedure does not affect existing SCTs.

1. On the SCT menu, select **Config**, then **Defaults**, then **Clear Default Settings**.



The default SCT settings are restored to the factory default configuration.

Related topics

- [Create a user-defined default SCT configuration](#), page 252
- [SCT Configuration](#), page 33
- “Settings Reference” in *User Interface*

Chapter 3

Using the SCT

Chapter 4

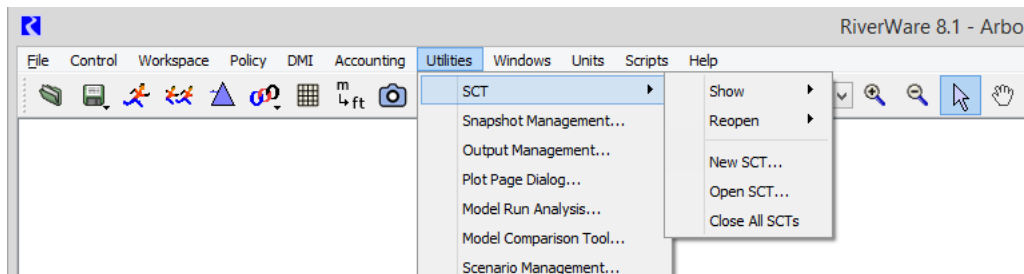
SCT Reference

Topics

- [Workspace Utilities Menu, page 257](#)
- [SCT Menus, page 258](#)
- [SCT Toolbar, page 273](#)
- [SCT Tab Details, page 278](#)
- [SCT Configuration Window, page 284](#)
- [Integer-indexed SCTs, page 302](#)
- [Keyboard Shortcuts, page 311](#)

Workspace Utilities Menu

To access the SCT, select **Utilities**, then **SCT** on the RiverWare workspace.



Show

Bring an open SCT to the front.

Reopen

Reopen a recently accessed SCT file. Up to six SCT files are remembered from one RiverWare session to the next. Provides quick access to all recently opened SCTs.

Currently open SCTs are excluded from this list and appear in the **Show** menu instead.

New SCT

Create an empty SCT in a new SCT window.

Open SCT

Open an existing SCT file in a separate SCT window. Use the File Chooser to select the SCT.

Close All SCTs

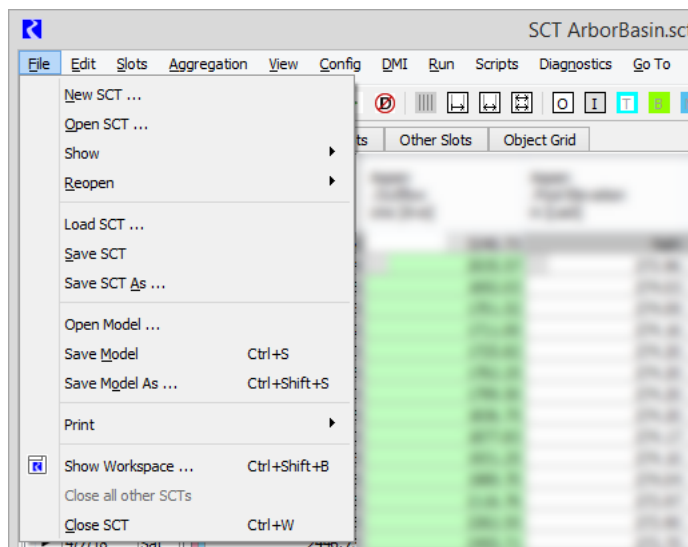
Close all open SCTs without saving them. A confirmation message is displayed for each one, preventing you from inadvertently closing an SCT without saving it first.

SCT Menus

Topics

- [SCT File menu](#), page 259
- [SCT Edit menu](#), page 261
- [SCT Slots menu](#), page 264
- [SCT Aggregation menu](#), page 265
- [SCT View menu](#), page 267
- [SCT Config menu](#), page 269
- [SCT DMI menu](#), page 270
- [SCT Run menu](#), page 270
- [SCT Scripts menu](#), page 271
- [SCT Diagnostics menu](#), page 271
- [SCT Go To menu](#), page 273

SCT File menu



New SCT

Create an empty SCT in a new SCT window.

Open SCT

Open another SCT for this model in a separate SCT window. Allows you to have more than one SCT open at a time.

Load SCT

Open another SCT for this model in a separate SCT window. Allows you to have more than one SCT open at a time.

Show

Bring an open SCT to the front.

Reopen

Open another recently accessed SCT for this model in a separate SCT window. Up to six SCT files are remembered from one RiverWare session to the next. Provides quick access to all recently opened SCTs.

Currently open SCTs are excluded from this list and appear in the **Show** menu instead.

Load SCT

Open another SCT for this model and load it in the current SCT window, replacing the current SCT. Use the File Chooser to select the SCT.

Save SCT

Save the current SCT to the same name and location. No confirmation message is displayed.

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SCT Reference

Save SCT As

Save the current SCT to a new location or name. Use the File Chooser to assign a location and name to the new SCT.

Open Model

Open another model in the workspace, replacing the current one. Use the File Chooser to select the model.

Note: This selection has no effect on the currently open SCTs.

Save Model

Save the current model to the same name and location.

Save Model As

Save the current model to a new location or name. Use the File Chooser to assign a location and name to the new model.

Print

Available only for the Series Slots tab. Options are:

- Selection—Print the currently selected cells.
- Page—Print the currently displayed cells.
- Full SCT—Print all series slots and data in the SCT.

Show Workspace

Bring the RiverWare workspace to the front.

Close all other SCTs

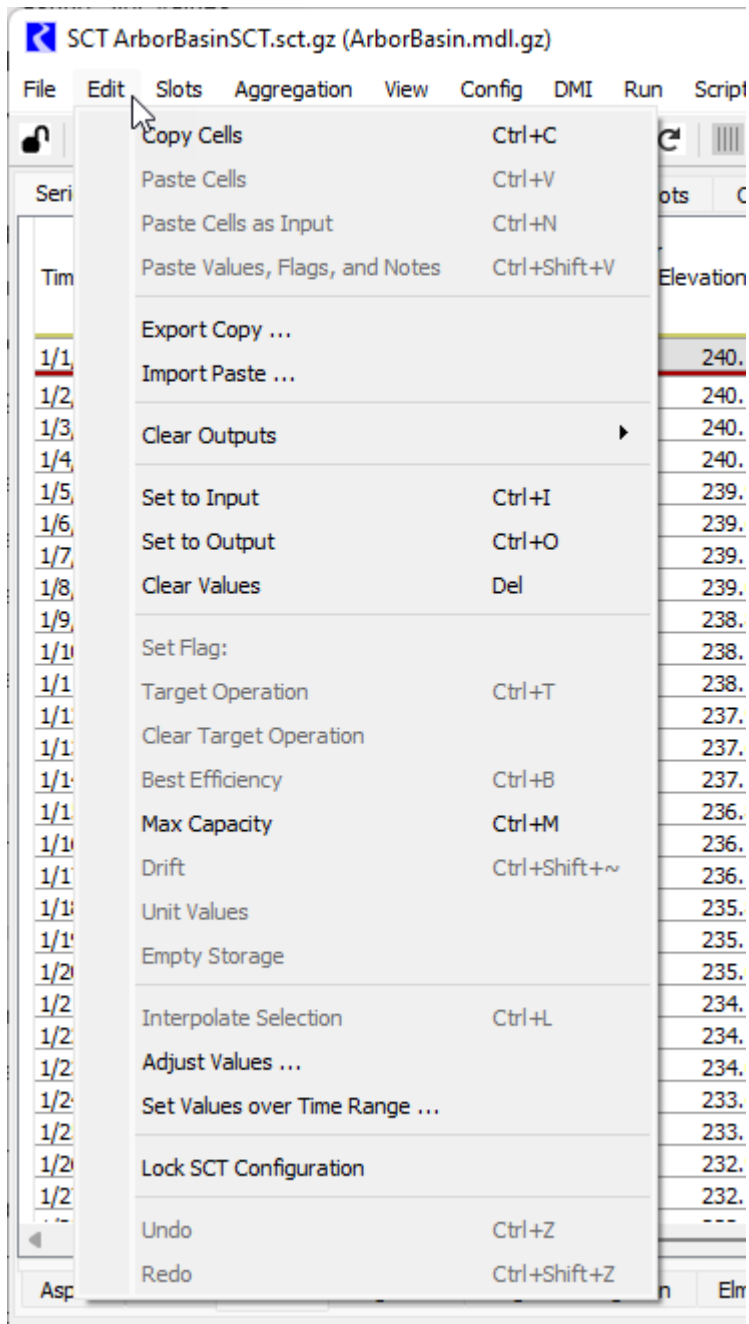
Available only if more than one SCT is open.

Close all SCTs except the current one without saving them. A confirmation message is displayed for each one, preventing you from inadvertently closing an SCT without saving it first.

Close SCT

Close the current SCT without saving it. A confirmation message is displayed, preventing you from inadvertently closing the SCT without saving it first.

SCT Edit menu



Copy cells

Copy the selected cell values to the RiverWare clipboard.

Paste cells

Paste cell values from the RiverWare clipboard, starting at the current location. Only copied values that had the Input (I) flag will be pasted.

Paste cells as Input

Paste cell values from the RiverWare clipboard to cells starting at the current location. All values are pasted, and all pasted values are assigned the Input (I) flag.

Paste Values, Flags, and Notes

Paste all copied values just as they were copied, including the copied flags and any notes on the slot values (see [Notes on series slots](#), page 29).

Export Copy

Copy the selected cell values to the Windows clipboard. They can be pasted from the Windows clipboard to external applications.

Import Paste

Paste values from the Windows clipboard to SCT cells starting at the current location.

Clear Outputs

Select one of the following options to clear output values in the SCT:

- within Selection—clear all output values in the selected cells.
- whole SCT—clear all output values in the SCT.

Set to Input

Assign the Input (I) flag to the selected cells.

Set to Output

Assign the Output (O) flag to the selected cells.

Clear Values

Clear all values in the selected cells.

Target Operation

Assign the Target (T) flag to the last timestep in the selected cells.

Clear Target Operation

Clears the target operation for the selected cells, if applicable.

Best Efficiency

Assign the Best Efficiency (B) flag to the selected cells.

Max Capacity

Assign the Max Capacity (M) flag to the selected cells.

Drift

Assign the Drift (D) flag to the selected cells.

Unit Values

Assign the Unit Values (U) flag to the selected cells.

Interpolate Selection

Perform linear interpolation on the values included in the selection.

Adjust Values

Adjust the values of the selected cells by a fixed value or a percentage. Enter the adjustment value in the Adjust Selected Slot Values dialog box.

Set Values over Time Range

Set a single value over a range of timesteps, where the first timestep is defined by the cell selection, and you specify either the last timestep or the number of timesteps and the value to set in a Set Slot Values over Time Range dialog that opens.

Lock SCT Configuration

Lock or unlock the SCT configuration to prevent any changes to the configuration. You can still edit values.

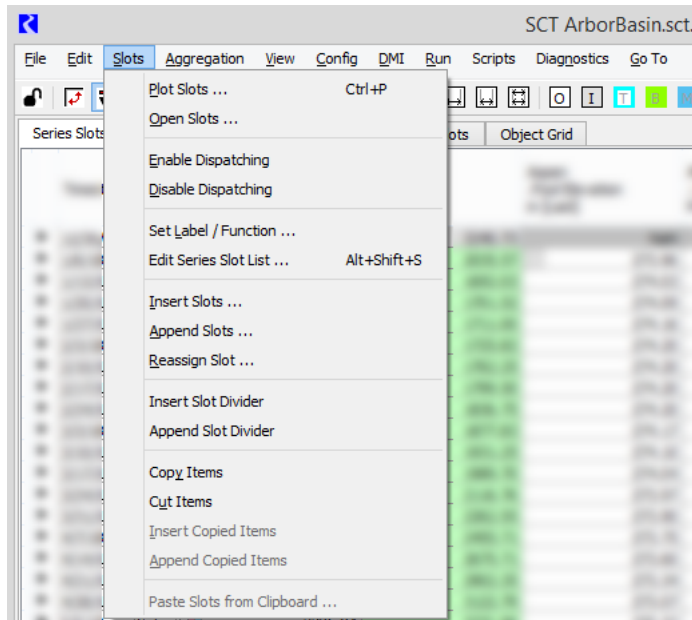
Undo and Redo

Undo or Redo the previous slot editing operation. Undo returns all affected slot data (values and flags) to the values before the most recent tracked edit. Redo returns all affected slot data (values and flags) to the values after the latest tracked edit.

Note: Structural changes, such as adding or removing a slot from the SCT are not tracked and cannot be undone or redone. Configuration changes (colors, showing flags, etc) are also not tracked and cannot be undone or redone.

Note: External changes, those changes to the slots that are initiated outside of the SCT (for example from a slot dialog or a script action), are not tracked. If you Undo or Redo an operation and there has been an intervening untracked change to any of the values affected by the edit, you are presented with a confirmation dialog and given the opportunity to cancel the operation. If you proceed, it will undo the value that was last tracked by the SCT dialog, not the external change.

SCT Slots menu



Plot Slots

Plot one or more selected slots in the Plot Page Editor.

Open Slots

Open one or more selected slots in the Slot Viewer.

Enable Dispatching

Enable dispatching for all objects containing one or more selected slots.

Disable Dispatching

Disable dispatching for all objects containing one or more selected slots.

Slots on a simulation object with dispatching disabled are indicated with crosshatching in a user-defined Dispatch Disabled Crosshatch color. In horizontal timestep orientation, the crosshatching is on the slot labels only. See

Set Label/Function

Assign a custom label and a summary function to the selected slot; active only if just one slot is selected.

Edit Series Slot List

Go to the **Edit Series Slot List** tab.

Insert

Add one or more slots before the current selection. Select the slots from the Select more Slots for an SCT dialog box.

Append

Add one or more slots after the current selection. Select the slots from the Select more Slots for an SCT dialog box.

Reassign

Replace the selected slot with a different one. Select the replacement slot from the Select more Slots for an SCT dialog box. The SCT displays the timestep values for the new slot.

Copy Items

Copies the selected cell values to the RiverWare clipboard. Same as Ctrl+C.

Cut Items

Cuts the entire selected slot and puts it on the RiverWare clipboard. You can paste the slot to a different location.

Note: The entire slot is cut, even if only part it is selected.

Insert Copied Items

Pastes copied or cut slots from the RiverWare clipboard directly before the current selection. No checking is done to determine whether the slot already exists in the SCT.

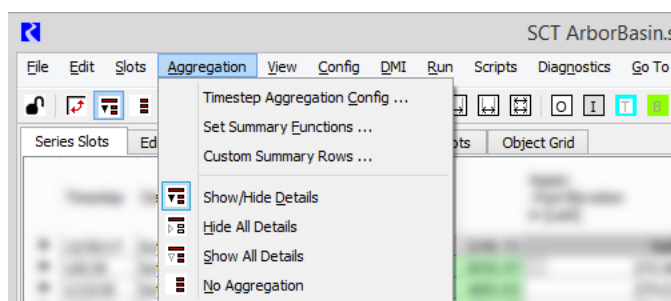
Append Copied Items

Pastes copied or cut slots from the RiverWare clipboard directly after the current selection. No checking is done to determine whether the slot already exists in the SCT.

Paste Slots from Clipboard

Pastes copied or cut slots from the RiverWare clipboard to the current location. A confirmation message is displayed if the slot already exists in the SCT.

SCT Aggregation menu



Timestep Aggregation Config

Opens the SCT Timestep Aggregation Dialog, which allows you to define the aggregation interval used in the SCT. This can be a fixed number of timesteps or a regular interval, such as daily, weekly, monthly, and so on. See [“Define the aggregation interval for the SCT”](#) on page 125 for details.

Set Summary Functions

Opens the Set SCT Aggregation Summary Functions dialog, which allows you to define the summary functions used when aggregating values for specified unit types or slots. See [“Define summary functions by unit type”](#) on page 129 and [“Define summary functions by slot name”](#) on page 131 for details.

Custom Summary Rows

Opens the SCT Custom Aggregation Summary Rows dialog box, which allows you to one or more custom summary rows for all slots in the SCT. Each row can have a different summary function, and the rows are added to end of each aggregation interval. See [“Create custom summary rows for all slots”](#) on page 137 for details.

Note: Custom summary rows are displayed in vertical timestep orientation only, and aggregated view must be enabled using either Show/Hide Details or Show All Details. Custom summary rows are not visible when timestep detail is hidden.

Show/Hide Details

Enables aggregated view. You can selectively expand and collapse aggregations to show or hide detail.

Note: If the Double Click Data Cell Toggles Detail Rows option is enabled, you can expand and collapse an aggregation interval by double-clicking any cell in the interval; see [“General tab”](#) on page 285.

Hide All Details

Enables aggregated view locked in summary mode for all aggregations. You cannot show detail.

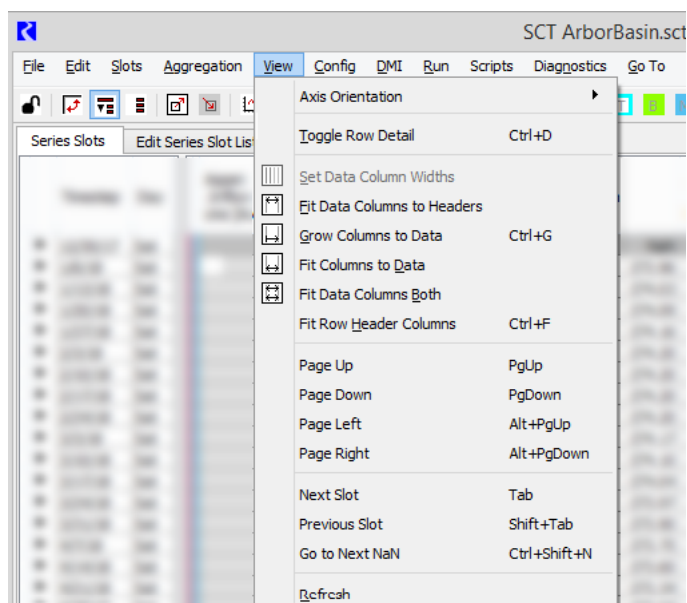
Show All Details

Enables aggregated view locked in full detail mode for all aggregations. You cannot hide detail.

No Aggregation

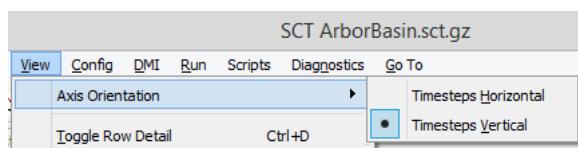
Disables aggregated view. Aggregation data is not visible.

SCT View menu



Axis Orientation

Switch between horizontal and vertical timestep orientation.



Synchronize Series Slot Scrolling

Toggle the Synchronize Series Slot Scrolling setting on or off. See [“Synchronized Scrolling”](#) on page 18 for more information

Toggle Row Detail

Active only when aggregated view is enabled using Show/Hide Details.

Select to switch between showing and hiding all detail for all timestep aggregations at once.

Set Data Column Widths

Note: See also [“Automatically Resize Column Widths to Fit Data”](#) on page 287 for information on optionally automatic resizing of column width.

Available only when a column is selected. Sets all columns to the width of the selected column.

Fit Data Columns to Headers

Sets the width of all columns to fit the text in the column headers. Grows and shrinks columns as needed.

Grow Columns to Data

Expands the column widths to fit the cell data values, as needed. Grows columns only.

Fit Columns to Data

Sets the column widths to fit the cell data values. Grows and shrinks columns as needed.

Fit Data Columns Both

Sets the column widths to fit the column headers and the cell data values. Grows and shrinks columns as needed.

Fit Row Header Columns

Sets the column widths in the row header table to fit the column headers. Also adjusts the splitter between the row header table and the SCT data table to fit the row header table exactly.

Note: Many, though not all, of the column width adjustments are preserved when you save and reload an SCT.

Page Up, Page Down, Page Left, Page Right

Scroll the SCT window display in the indicated direction.

Next Slot, Previous Slot

Move the active cell to the next or previous slot.

Go To Next NaN

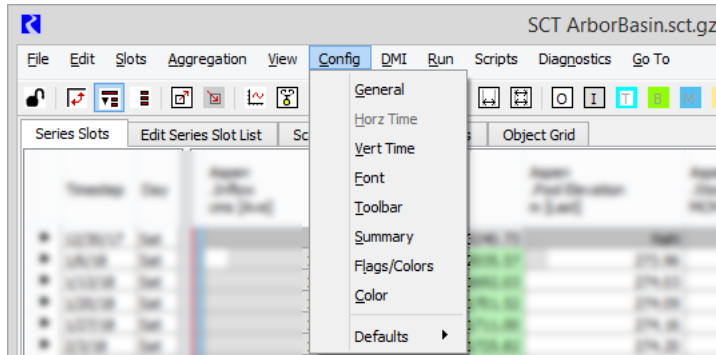
Go to the next cell with a NaN value. The search always goes through timesteps first, then through subsequent slots.

Refresh

Refresh the SCT display to clear any selections and update any Color Alerts that use condition functions (see [“Configuring Color Alerts”](#) on page 150).

SCT Config menu

Opens the SCT Configuration window at the selected tab.



General

Allows you to specify options applicable to the overall SCT including pre- and post-simulation timesteps and whether to display NaNs or the SCT background grid.

Horz Time

Allows you to add weekend dividers and change whether to display weekdays, user units and scale, and the summary function.

Vert Time

Allows you to add weekend dividers and change whether to display weekdays, user units and scale, and the summary function.

Font

Allows you to customize the fonts to use in the SCT. You can specify up to three options.

Toolbar

Allows you to select the buttons to display on the user-configurable portion of the SCT toolbar.

Summary

Allows you to change attributes of the Selection Summary to improve performance of the SCT in very large models.

Flags/Colors

Allows you to select the flag buttons to include on the SCT toolbar. Also allows you to customize the flag colors.

Color

Allows you to customize the colors of slot dividers, weekend dividers, and simulation start boundary.

Chapter 4

SCT Reference

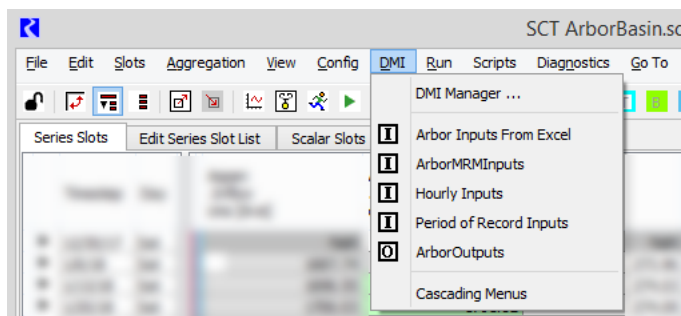
Defaults

These selections apply to the default SCT configuration.

- Apply Default Settings
- Save Current Settings as Default
- Clear Default Settings

SCT DMI menu

Allows you to access DMIs and the DMI Manager from the SCT.



DMI Manager

Opens the DMI Manager window.

DMI name

Lists all defined DMI items with their type icon: Input DMIs (I), Output DMIs (O), and DMI Groups (G).

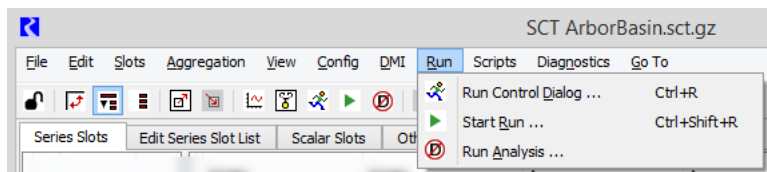
Note: Selecting a DMI or Group from this menu invokes that DMI.

Cascading Menus

Switch between listing all DMI items in the top-level DMI menu or cascaded into sub-menus. The top-level menu option is useful if there are not too many items.

SCT Run menu

Allows you to run a model simulation directly from the SCT.



Run Control Dialog

Opens the Run Control dialog box.

Start Run

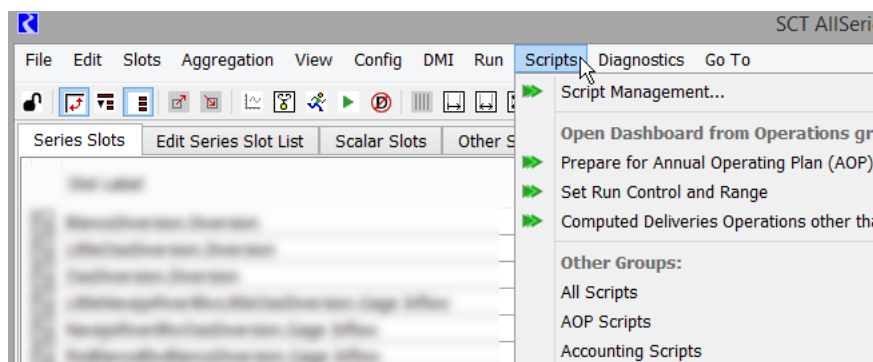
Initiates a run using the controller and run times selected in the Run Control dialog box.

Run Analysis

Opens the Model Run Analysis tool.

SCT Scripts menu

Allows you to access scripts directly from the SCT.



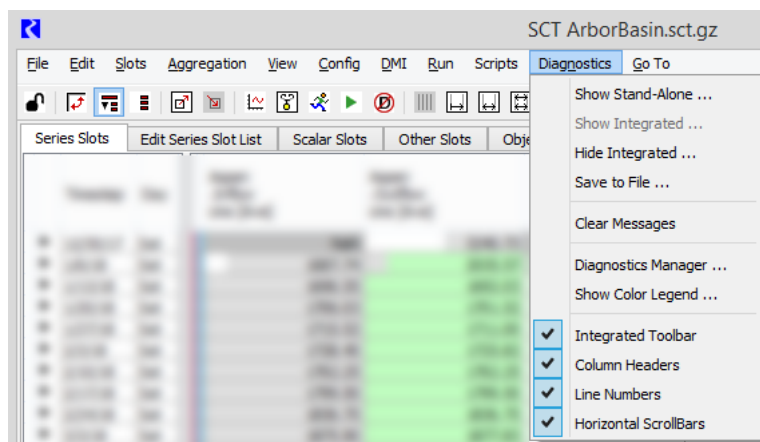
Script Management

Opens the Script Manager window.

Open Dashboard

Lists the available scripts. Select a script to open its Dashboard window.

SCT Diagnostics menu



Chapter 4

SCT Reference

The Diagnostic Output window can be shown as a panel at the bottom of a single SCT or as a stand-alone window. It appears in only one place at a time. You can enable diagnostics in or out of the SCT using the Diagnostics icon in the SCT toolbar or the SCT Diagnostics menu.

Show Stand-Alone

Opens the Diagnostics Output window in a separate window.

Show Integrated

Opens the integrated Diagnostics Output panel at the bottom of the SCT window.

Hide Integrated

Active only when the integrated Diagnostics Output panel is displayed.

Closes the integrated Diagnostics Output panel.

Save to File

Allows you to save the Diagnostics Output messages to a text file.

Clear Messages

Clears all Diagnostics Output messages.

Diagnostics Manager

Opens the Diagnostics Manager window.

Show Color Legend

Displays the Diagnostics Output messages color legend in a separate window. You can leave this window open while you run model simulations and evaluate diagnostics messages.

Integrated Toolbar

Active only when the integrated Diagnostics Output panel is displayed.

Allows you to display or hide the Diagnostics toolbar in the panel. Hiding the toolbar is useful if you need to save space; the toolbar operations are also available in the SCT **Diagnostics** menu.

Column Headers

Active only when the integrated Diagnostics Output panel is displayed. Hiding the column headers is useful if you need to save space.

Allows you to display or hide the column headers in the panel.

Line Numbers

Active only when the integrated Diagnostics Output panel is displayed. Hiding the line numbers is useful if you need to save space.

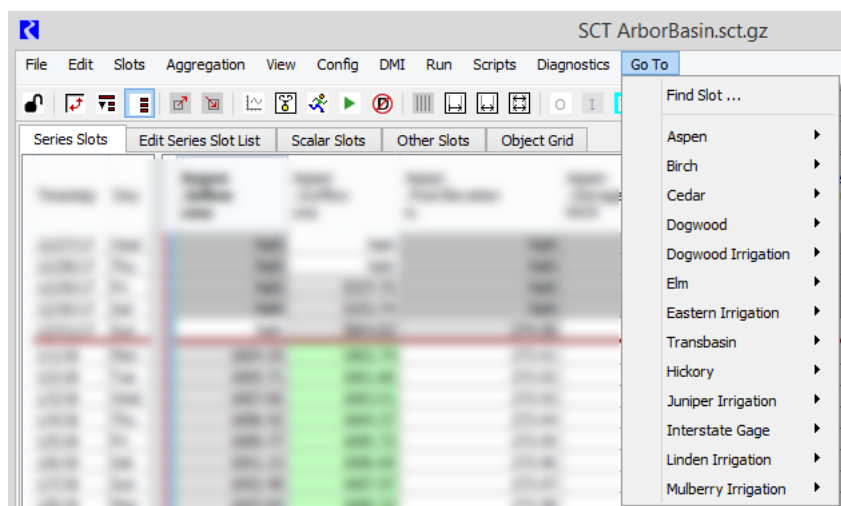
Allows you to display or hide the line numbers in the Context column of the integrated panel.

Horizontal Scrollbars

Active only when the integrated Diagnostics Output panel is displayed. Hiding the scrollbars is useful if you need to save space.

Allows you to display or hide the horizontal scrollbars at the bottom of the integrated panel.

SCT Go To menu



Find Slot

Displays the Find Slot dialog box, which allows you to search for slots by name.

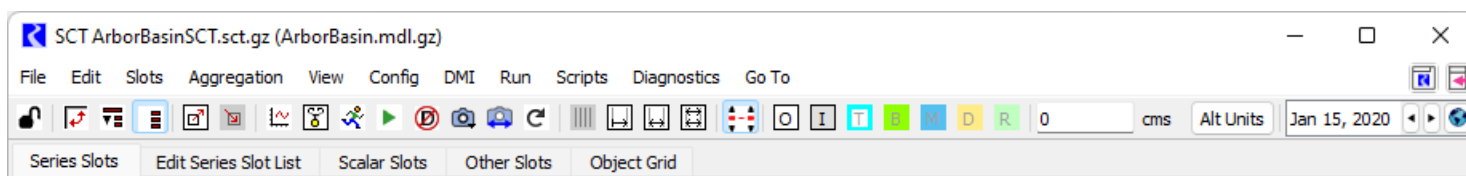
Sheet divider name

If the SCT has sheet dividers, the menu lists each sheet. If a sheet has at least one slot group divider, the following secondary menu options are also available:

- **Start of Sheet**—available if the sheet has at least one slot group divider. Goes to the beginning of the sheet.
- **Slot group divider name**—lists named slot group dividers. Goes to the selected divider.
- **Object.Slot name**—lists the first slot in an unnamed slot group. Goes to the selected slot.

SCT Toolbar

Following is a list of all possible items on the SCT toolbar. The toolbar is customizable; therefore, some items may not be included on a given SCT.





Lock/Unlock

Always appears.



Swap Axis

Can be hidden.



User Controlled Details

Can be hidden. Allows you to switch between detail and summary views.



Summaries Only

Can be hidden. Allows you to display summary info only on the entire tab.



All Details Open

Can be hidden. Allows you to display all details on the entire tab.



Details Only

Can be hidden. Disables time aggregation.



Copy Data

Always appears.



Paste Data

Always appears. This pastes copied values with the Input (I) flag only, the equivalent of the [Paste cells](#) operation from the Edit menu.



Plot Slots

Can be hidden.



Open Slots

Can be hidden.



Show/Hide Diagnostics

Always appears.



Run Control

Can be hidden.



Start Run

Can be hidden.



Model Run Analysis

Can be hidden.



Snapshot

Always appears. Selecting this button presents a menu with options for creating and viewing snapshots. (See [“Snapshots” in Output Utilities and Data Visualization](#) for more information on snapshots.)

- **Snapshot Manager**—opens the Snapshot Manager (see [“Snapshot Manager” in Output Utilities and Data Visualization](#)).
- **Create Snapshot**—creates a new snapshot object using the Snapshot Template (slot selection and name) currently configured in the Snapshot Manager.
- **Apply Snapshot**—presents a list of all snapshot objects in the model. Selecting a snapshot to apply copies all data, flags, and series slot notes from all slots on the snapshot and pastes them into the corresponding source slots.
- **Show No Snapshots**—this is the default selection. When selected, snapshot slots are not shown in the SCT (unless they have been added manually by using Insert Slots or Append Slots).
- **Show Selected Snapshots**—this selection opens the Select Snapshots dialog, where you can select which snapshots in the model should have their slots displayed in the SCT together with the source slots. (See [Displaying Snapshot Slots](#), page 145.)
- **Show N Most Recent Snapshots**—shows slots from the most recent snapshots in the SCT together with the source slots. (See [Displaying Snapshot Slots](#), page 145.)



Show All Snapshots

Always appears. This operation overrides the per slot setting for whether a source slot should have its corresponding snapshot slots displayed when one of the Show Recent Snapshot options is selected. Selecting this button shows snapshot slots for all source slots in the SCT that have slots in the selected snapshots. Selecting the button again returns to using the per slot settings to determine which snapshot slots to show. (See [Configure which snapshot slots to show](#), page 146.)



Refresh

Always appears. This operation updates the data displayed in the SCT, and it updates the evaluation of condition functions for color alerts (see [Configuring Color Alerts](#), page 150 for more information).



Resize Columns: Set All to Selected Column Width

Can be hidden. Sets all columns to the same width as the selected column.



Grow Columns: Fit Data

Can be hidden. Increases the width of any columns that are too narrow; does not modify columns that are too wide



Resize Columns: Fit Data

Can be hidden. Decreases the width of any columns that are too wide; does not modify columns that are not wide enough



Resize Columns: Fit All

Can be hidden. Does both; fits text by increasing and decreasing widths, as required



Synchronize Series Slot Scrolling

Can be hidden. Turn on or off synchronized series slot scrolling. See [“Synchronized Scrolling”](#) on page 18 for more information.

User-configurable flags area

Appears on the **Series Slots** tab only. Select a button to assign the flag to the selected cells. Depending on how the SCT is configured, any or all of the following flags may be available:



Output (O)



Input (I)



DMI Input (Z)—display only



Target Operation (T)



Best Efficiency (B)



Max Capacity (M)



Drift (D)



Set by Rules (R)



Surcharge Release (S)



Regulated discharge (G)



Unit Values (U)

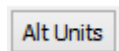
 cms

Value

Appears on all tabs.

Displays the value of the selected cell. If multiple cells are selected, displays the value of the last selected cell. The value is displayed to full precision from the model.

You can enter a value to assign to all selected data cells.



Flow/Volume Display Mode

Appears on all tabs.

Menu allows you to select the display mode.

- Standard - Display flows and volumes without conversion
- Flow - Display volumes as flows (as appropriate)
- Volume - Display Flows as Volumes

 ◀ ▶

Timestep Navigator

Appears on all tabs. Displays the current timestep and provides a variety of navigation options:

- Enter a date in the text field to go to the specified timestep.
- Select the **Previous** and **Next** arrows to move one timestep at a time.



Global Time Scroll

Appears on all tabs.

Menu provides options to go directly to the first or last timestep in the model.

- Run Start

- [Run Finish](#)



Show Workspace

Select to bring the RiverWare workspace window to the front.

SCT Tab Details

Topics

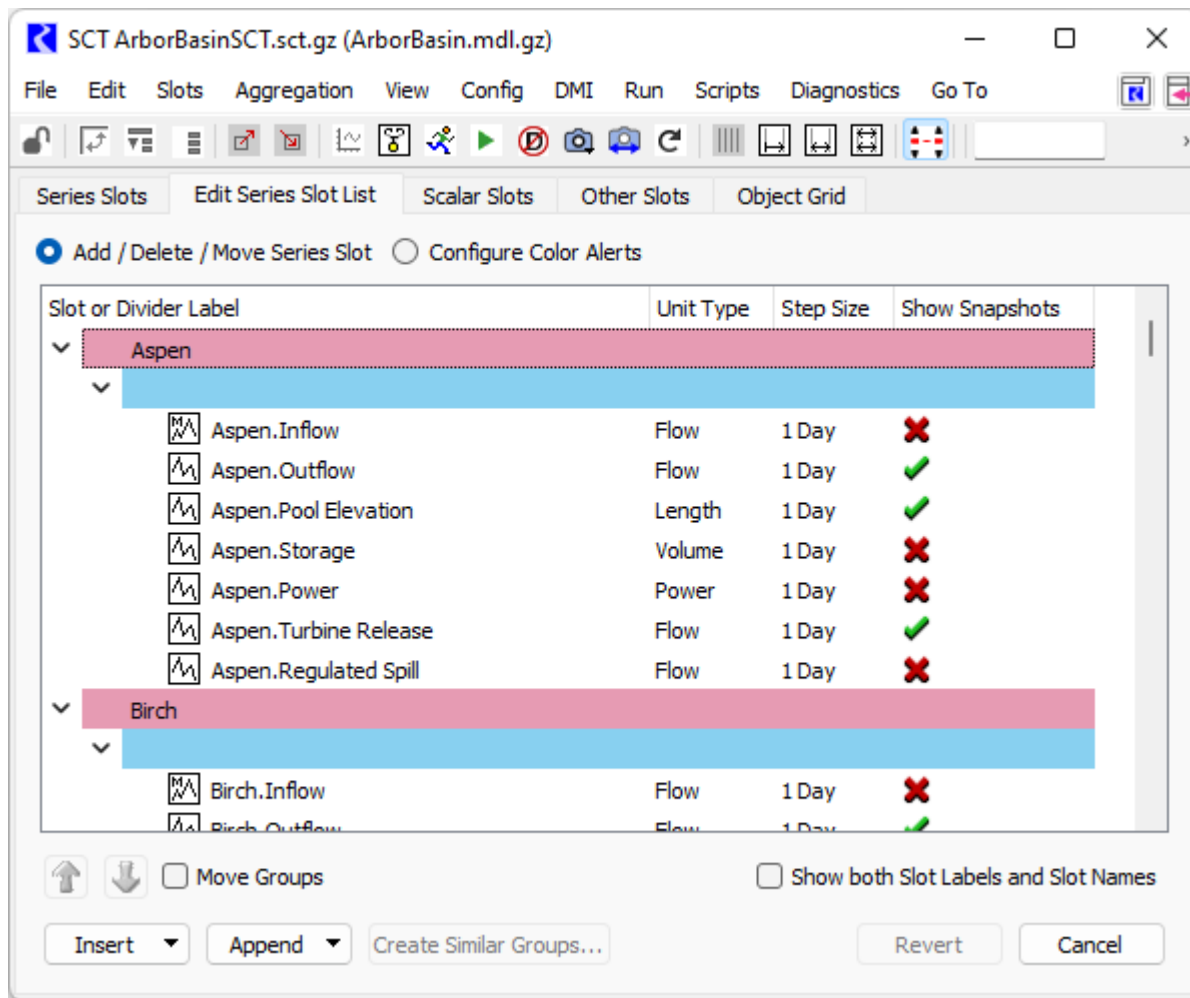
- [Edit Series Slot List tab details](#), page 278
- [Scalar Slots tab details](#), page 281
- [Other Slots tab details](#), page 283

Edit Series Slot List tab details

This tab lists the series slots, slot group dividers, and sheet dividers included in the Series Slots tab, and it allows you to organize the Series Slots tab display. The screen fields vary depending on the selected mode, as follows:

- [Add / Delete / Move Series Slot mode](#), page 279

- [Configure Color Alerts mode](#), page 281



Add / Delete / Move Series Slot mode

Add / Delete Move Series Slots

Select to organize series slots on the SCT. You can add, remove, and arrange the order of slots. You can also add group dividers to group the slots into logical groups, and sheet dividers to display the slots on separate sheets.

Up/Down arrows

Available only if one or more rows are selected. Moves the selected items in the indicated direction.

Move Groups

Select to move selected slot groups or sheets as a whole. Allows you to reorganize entire groups and sheets.

Clear to move selected slot group dividers, sheet dividers, and individual slots only. Allows you to reorganize slots within groups and sheets.

Show both Slot Labels and Slot Names

Select to add the “Slot Name” column to the display. This displays the default slot name in Object.Slot format.

Clear to hide the “Slot Name” column from the display.

Insert

Available only if just one row is selected. Select to insert slots, slot group dividers, or sheet dividers *directly above* the selected row. Menu options are as follows:

- **Insert Slots**—opens the Select more Slots for an SCT window, allowing you to select slots to add.
- **Insert Slot Divider**—inserts a slot group divider.
- **Insert Sheet Divider**—inserts a sheet divider.

Append

Select to add slots, slot group dividers, or sheet dividers *to the end of the list*. Menu options are as follows:

- **Append Slots**—opens the Select more Slots for an SCT window, allowing you to select slots to add.
- **Append Slot Divider**—adds a slot group divider.
- **Append Sheet Divider**—adds a sheet divider.

Create Similar Groups

Available only if just one slot group divider or sheet divider is selected. Select to replicate the slot structure in the selected group or sheet to other objects in the model. Opens the Select *object_type* for new SCT slot groups window, allowing you to select the other objects.

Show Snapshots

In this column, specify for each slot whether corresponding snapshot slots should be displayed when one of the Show Recent Snapshot slots options is selected. (See [Displaying Snapshot Slots](#), page 145 for more information.)

Revert

Available only if you have made changes to the slot organization. Discards all changes made during this editing session.

Accept

Available only if you have made changes to the slot organization. Applies all changes made during this editing session to the series slots data table and opens the Series Slots tab.

Cancel

Available only if no changes have been made to the slot organization. Opens the Series Slots tab.

Configure Color Alerts mode

Configure Color Alerts

Select to define custom, labeled background and text colors to be used on the Series Slots tab. These colors will be applied automatically to slot cells based on user-defined conditions. They can be used to highlight selected cells for any purpose. (See [“Configuring Color Alerts”](#) on page 150.)

Custom Cell Colors

Opens the Custom SCT Cell Colors dialog box, allowing you to define custom, labeled background and text colors. (See [“Define custom colors for highlights and color alerts”](#) on page 191.)

Color Alerts

Opens the SCT Color Alert Configuration dialog box, allowing you to define custom Color Alerts. Each alert is based on either a value exceeding a specified threshold or meeting a condition defined in a function. If a slot value meets the condition, the cell is automatically highlighted with the defined custom color. (See [“Configuring Color Alerts”](#) on page 150.)

Cancel

Opens the Series Slots tab.

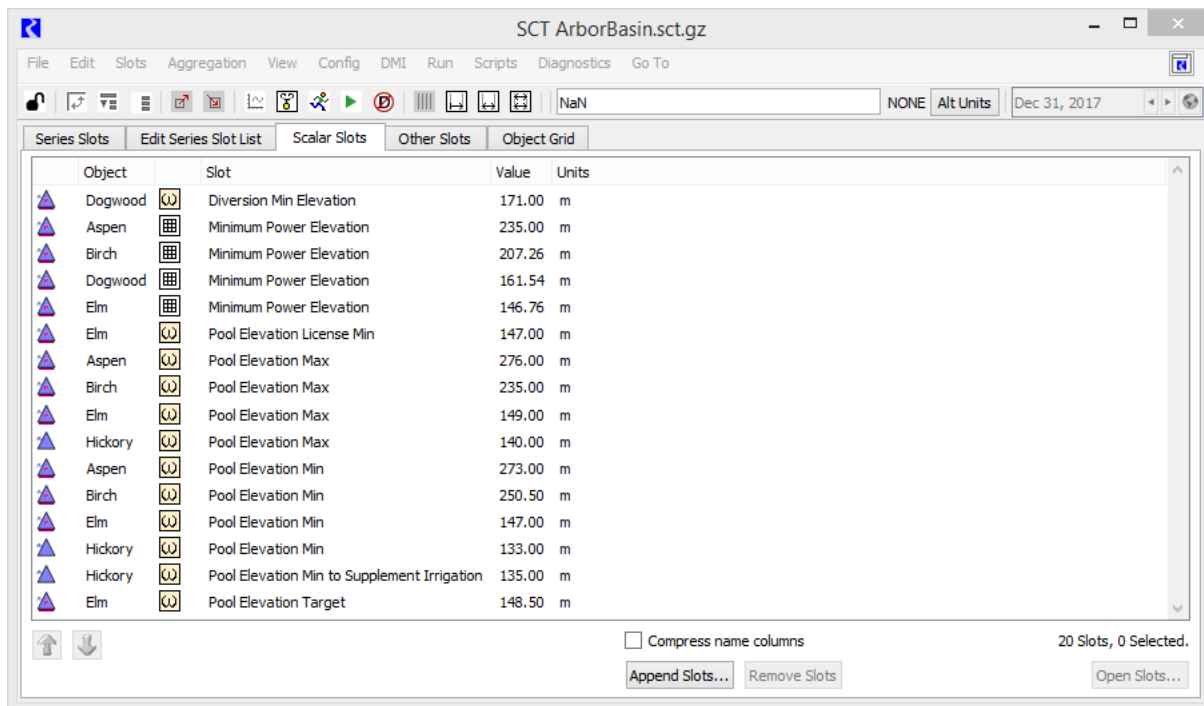
Scalar Slots tab details

This tab shows a list of scalar and 1x1 table slots. It allows you to display, organize, and set values on non-series data in one convenient location.

Chapter 4

SCT Reference

Note: 1x1 table slots were used for scalar values before Scalar Slots were implemented.



Up/Down arrows

Available only if one or more slots are selected. Moves the selected items in the indicated direction.

Compress name columns

Select to compress the object and slot names into one “Slot” column, using the Object.Slot format.

Clear to display the object and slot names in separate “Object” and “Slot” columns.

x Slots, y Selected

Display only. Displays the total number of slots in the list and the number currently selected.

Append Slots

Select to open the Select more Slots for an SCT dialog box, allowing you to add slots *to the end of the list*.

Remove Slots

Available only if one or more slots are selected. Select to *immediately* remove the selected slots from the list.

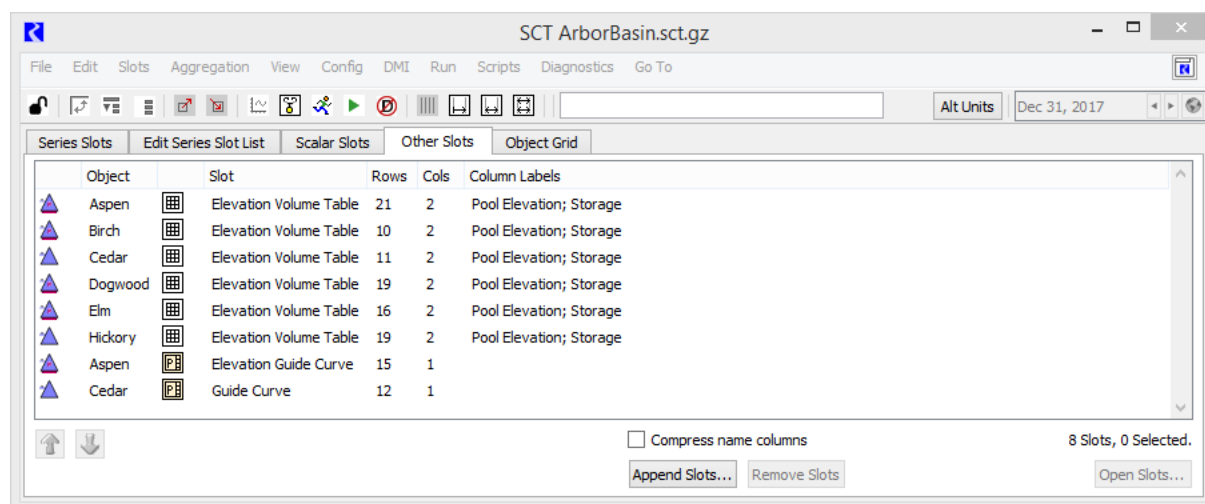
Note: There is no confirmation message before the slots are removed.

Open Slots

Available only if one or more slots are selected. Opens the selected slots in separate Slot Viewer windows.

Other Slots tab details

This tab displays a list of table slots, periodic slots, and statistical table slots. It allows you to display and organize these slots in one convenient location.



Up/Down arrows

Available only if one or more slots are selected. Moves the selected slots in the indicated direction.

Compress name columns

Select to compress the object and slot names into one “Slot” column, using the Object.Slot format.

Clear to display the object and slot names in separate “Object” and “Slot” columns.

x Slots, y Selected

Display only. Displays the total number of slots in the list and the number currently selected.

Append Slots

Select to open the Select more Slots for an SCT dialog box, allowing you to add slots *to the end of the list*.

Remove Slots

Available only if one or more slots are selected. Select to *immediately* remove the selected slots from the list.

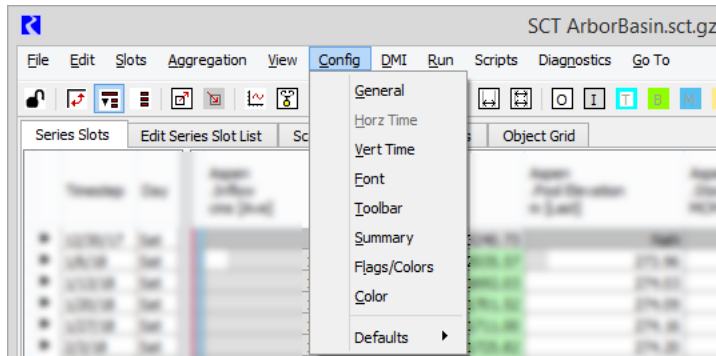
Note: There is no confirmation message before the slots are removed.

Open Slots

Available only if one or more slots are selected. Opens the selected slots in separate Slot Viewer windows.

SCT Configuration Window

To open this window, select **Config** on the SCT menu, then select any of the options.



Most SCT configuration settings are controlled through this window. These settings are saved to the SCT when you save it.

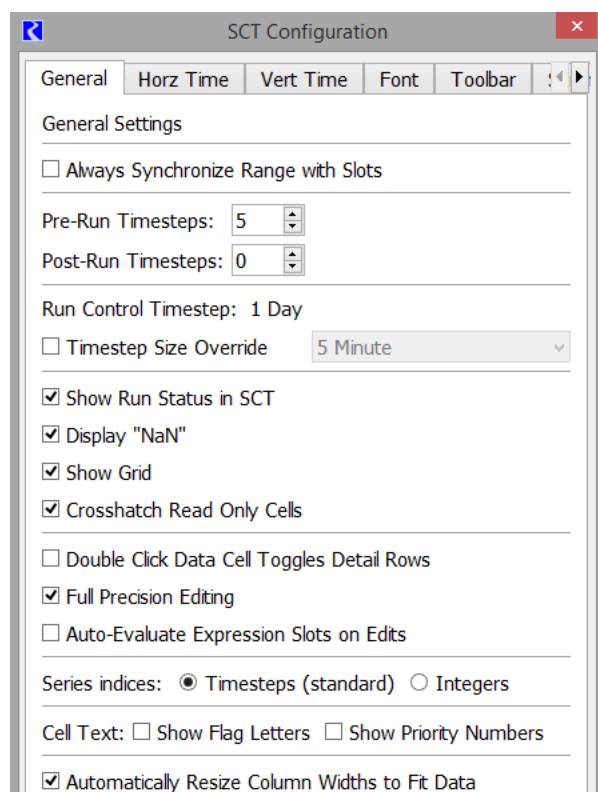
Note: Modifications to these settings require the SCT configuration to be unlocked. See [“Lock or unlock the SCT Configuration”](#) on page 164 for details.

Topics

- [General tab](#), page 285
- [Horz Time tab](#), page 288
- [Vert Time tab](#), page 291
- [Font tab](#), page 292
- [Toolbar tab](#), page 294
- [Summary tab](#), page 296
- [Flags tab](#), page 298
- [Color tab](#), page 300

General tab

This tab provides a variety of global SCT settings, including pre- and post-simulation timesteps, whether to display simulation run status and NaNs in the SCT, and the how slot values are edited.



Always Synchronize Range with Slots

Select to configure the SCT to always use the same timesteps as the included slots. Selecting this check box disables the remaining time range settings on this dialog box.

Clear this check box if you want to configure the SCT to use a different run control range from the model.

Pre-Run Timesteps

This field is active only if Always Synchronize Range with Slots is selected.

Specify the number of pre-run timesteps to include in the SCT. The default is 0, which indicates the SCT should start at the same timestep as the model.

The specified number of timesteps are added to the beginning of the model run control range on the SCT. See [“Timestep size”](#) on page 15 for details.

Post-Run Timesteps

Specify the number of post-run timesteps to include in the SCT. The default is 0, which indicates the SCT should end at the same timestep as the model.

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SCT Reference

The specified number of timesteps are added to the end of the model run control range on the SCT. See [“Timestep size”](#) on page 15 for details.

Run Control Timestep

Display only. Displays the timestep size defined for the model.

Timestep Size Override

Select to have the SCT use a different timestep size than the model. Select the SCT timestep size in the menu.

Clear to have the SCT use the same timestep size as the model.

Show Run Status in SCT

Select if you want the Status area at the bottom of the SCT to display status information for the current model simulation run. The run status replaces the default status display during and upon completion of a simulation run.

Clear if you want the status area at the bottom of the SCT to always display cell summary information, even during a simulation run.

Display “NaN”

Select if you want the SCT to display undefined numeric values with the string “NaN”.

Clear if you want the SCT to display undefined numeric values as empty, or null, cells.

Show Grid

Select to display a thin border around each cell on the SCT.

Clear to hide the borders around each cell.

Crosshatch Read Only Cells

Select to highlight read-only cells with crosshatching in a user-defined color. See [“Color tab”](#) on page 300 for the crosshatch color settings.

Clear to have read-only cells display with no special markings.

Read-only cells contain non-editable values. These include cells for expression slots and accounting system slots.

Double-click Data Cell Toggles Detail Rows

This setting allows you to configure what happens when you double-click an SCT cell. By default, double-clicking a cell opens it so you can edit the value directly.

Select if you want to prevent cells from being edited directly when they are double-clicked. Double-clicking a cell does the following:

- The cell remains locked, but you can change the value in the Value field in the SCT toolbar. This applies to all aggregated and non-aggregated views.

- In aggregated view with Show/Hide Details, the aggregation interval display switches between show and hide details. See [“Enable or disable aggregated view”](#) on page 127 for details.

Clear to use the SCT default behavior in which double-clicking a cell opens it for editing. See [“Edit slot values manually”](#) on page 68 for details.

Full Precision Editing

Note: Enabling or disabling this setting applies to all SCT cells and the Value field on the SCT toolbar.

Select to enable full-precision data value editing in all SCT cells and the Value field on the SCT toolbar.

Clear to disable full-precision data value editing in cells and the Value field on the SCT toolbar. You can only edit values to the display precision.

Auto-Evaluate Expression Slots on Edits

Select to cause expression slots included in the SCT to fully evaluate across all timesteps whenever a single timestep value is edited. This causes the SCT to behave like a spreadsheet, where expression slot calculations are performed immediately after editing a value. For example, you might have an expression slot that sums the Outflows from multiple reservoirs. This checkbox allows you to see this sum each time you edit an Outflow.

Note: This may cause refresh delays in the SCT if there are many expression slots or if the slots do not evaluate quickly.

Series indices

Select one of the following options to indicate the type of slots that can be included in the SCT. An SCT can include either timeseries slots or integer-indexed slots, not both.

- **Timesteps (standard)**—select to use the SCT to display and manage timeseries data only. All slots included in the SCT have the same timestep size. This is the default option.
- **Integers**—select to use the SCT to display and manage integer-indexed slots. An integer-indexed SCT has different properties than a timeseries SCT; see [“Integer-indexed SCTs”](#) on page 302 for details.

Cell Text

In the Cell Text area, select one or both of the following selections:

- **Show Flag Letters:** show the flag letter(s) for the cell value (I, O, R, for example). This is useful when threshold alert colors override the flag colors.
- **Show Priority Numbers:** show the rule priority for the cell value. This is useful for rulebased simulation models.

Automatically Resize Column Widths to Fit Data

Use this option to specify if series slot columns should resize automatically. When this option is enabled/checked, column widths will resize to fit the data during certain operations, like stopping a run. In most cases, the columns will only grow, but some operations cause the width to shrink.

If deselected, column widths will be preserved. See the resize operations under the View menu. See [“SCT View menu”](#) on page 267

Chapter 4

SCT Reference

Buttons

OK

Select to apply the changes and close the SCT Configuration window.

Apply

Select to apply the changes and keep the SCT Configuration window open.

Reset

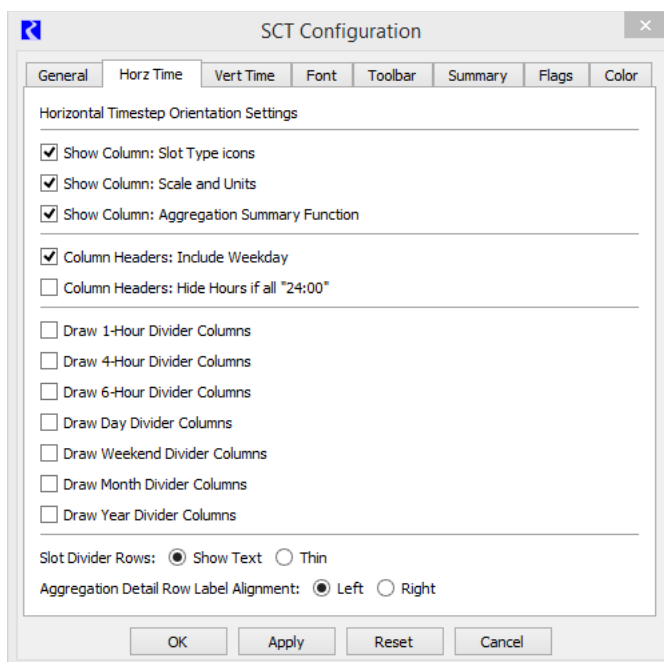
Select to cancel the changes and keep the SCT Configuration window open.

Cancel

Select to cancel the changes and close the SCT Configuration window.

Horz Time tab

This tab provides settings for configuring the horizontal timestep orientation.



Show Column: Slot Type icons

Select to include the slot type icon in the slot header. The icon identifies the slot type of each slot.

Clear to hide this column.

Show Column: Scale and Units

Select to include the scale and unit type column in the slot header. This column identifies the unit type for each slot.

Clear to hide this column.

Show Column: Aggregation Summary Function

Select to include the summary function column in the slot header. This column identifies the summary function assigned to each slot. This information is displayed only when aggregated view is enabled.

Clear to hide this column.

Column Headers: Include Weekday

Select to include the day of the week in the timestep header.

Clear to hide this information.

Column Headers: Hide Hours if all “24:00”

Select to hide the hour of the day in the timestep header for timesteps greater than 12 hours.

Clear to show this information.

Draw time period Divider Columns

Select to display a divider column to mark the specified time period. The divider appears as a thick line in a user-defined color. See “[Color tab](#)” on page 300 for the divider color settings.

The dividers are visible in all aggregated and non-aggregated views.

Note: You can select only one hourly divider column (1-Hour, 4-Hour, or 6-Hour). However, you can select multiple divider columns for the remaining time periods (Day, Weekend, Month, and Year).

Options are:

- **Draw 1-Hour Divider Columns**—show divider columns at the end of each hour. If you select this check box, you cannot select the 4-Hour or 6-Hour dividers.
- **Draw 4-Hour Divider Columns**—show divider columns at the end of each four-hour period. If you select this check box, you cannot select the 1-Hour or 6-Hour dividers.
- **Draw 6-Hour Divider Columns**—show divider columns at the end of each six-hour period. If you select this check box, you cannot select the 1-Hour or 4-Hour dividers.
- **Draw Day Divider Columns**—show divider columns at the end of each day.
- **Draw Weekend Divider Columns**—show divider columns before Saturday and after Sunday.
- **Draw Month Divider Columns**—show divider columns at the end of each month.
- **Draw Year Divider Columns**—show divider columns at the end of each year.

Slot Divider Rows

Select one of the following options to indicate how you want to display slot divider rows, which mark the beginning of a new SCT series slot sheet.

- **Show Text**—select to include the user-defined label in the divider row. The height of the divider row is adjusted so there is enough room to fit the text.
- **Thin**—select to exclude the user-defined label from the divider row. The divider appears as a thin line with no text. This option is useful if you need to conserve vertical space.

Aggregation Detail Row Label Alignment

Select one of the following options to indicate how you want the timestep label text to be aligned within the timestep column. This applies only to aggregated view using either Show/Hide Details or Show All Details.

- **Left**—select to make the label text left-aligned.
- **Right**—select to make the label text right-aligned.

Buttons

OK

Select to apply the changes and close the SCT Configuration window.

Apply

Select to apply the changes and keep the SCT Configuration window open.

Reset

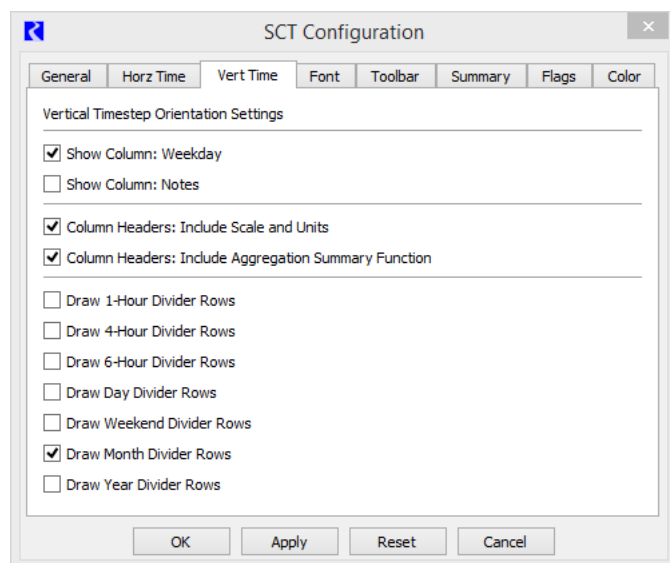
Select to cancel the changes and keep the SCT Configuration window open.

Cancel

Select to cancel the changes and close the SCT Configuration window.

Vert Time tab

This tab provides settings for configuring the vertical timestep orientation.



Show Column: Weekday

Select to include the Day column in the timestep header. This column displays the day of the week of each timestep.

Clear to hide this column.

Show Column: Notes

Select to include the Notes column in the timestep header. This column displays the text of all notes that have been assigned to slots at that timestep. You can also add and edit notes when the column is displayed.

Clear to hide this column.

See [“Display and edit series slot notes”](#) on page 109 for details.

Column Headers: Include Scale and Units

Select to include the scale and unit type for each slot in the slot header.

Clear to hide this information.

Column Headers: Include Aggregation Summary Function

Select to include the summary function assigned to each slot in the slot header. This information is displayed only when aggregated view is enabled.

Clear to hide this information.

Draw time period Divider Rows

Select to display a divider row to mark the specified time period. The divider row appears as a thick line in a user-defined color. See “[Color tab](#)” on page 300 for the divider color settings.

The dividers are visible in non-aggregated view and in aggregated view when details are shown. The dividers are not visible in aggregated view when details are hidden.

Note: You can select only one hourly divider row (1-Hour, 4-Hour, or 6-Hour). However, you can select multiple divider rows for the remaining time periods (Day, Weekend, Month, and Year).

Options are:

- **Draw 1-Hour Divider Rows**—show dividers rows at the end of each hour. If you select this check box, you cannot select the 4-Hour or 6-Hour dividers.
- **Draw 4-Hour Divider Rows**—show dividers rows at the end of each four-hour period. If you select this check box, you cannot select the 1-Hour or 6-Hour dividers.
- **Draw 6-Hour Divider Rows**—show dividers rows at the end of each six-hour period. If you select this check box, you cannot select the 1-Hour or 4-Hour dividers.
- **Draw Day Divider Rows**—show divider rows at the end of each day.
- **Draw Weekend Divider Rows**—show divider rows before Saturday and after Sunday.
- **Draw Month Divider Rows**—show dividers rows at the end of each month.
- **Draw Year Divider Rows**—show divider rows at the end of each year.

Buttons

OK

Select to apply the changes and close the SCT Configuration window.

Apply

Select to apply the changes and keep the SCT Configuration window open.

Reset

Select to cancel the changes and keep the SCT Configuration window open.

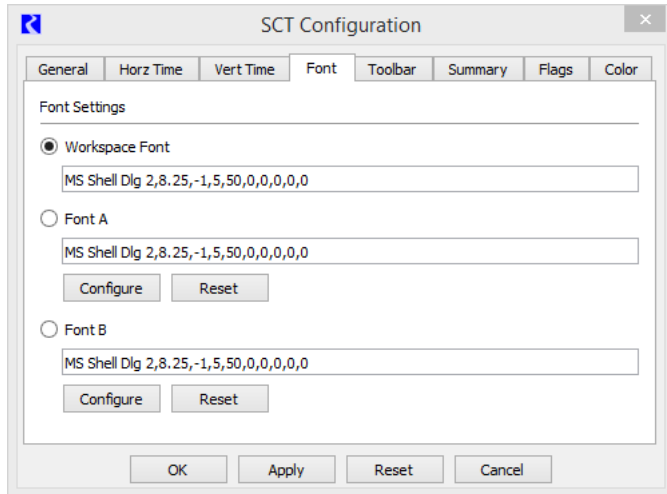
Cancel

Select to cancel the changes and close the SCT Configuration window.

Font tab

This tab allows you to specify the font to use for the SCT. The selected font is used for screen display and printing.

You can configure up to three possible fonts and switch between them depending on your needs. Only one font can be selected at a time.



Default Font

Select this option to use the default RiverWare workspace font in the SCT.

You cannot change the properties of this font.

Font A

Select this option to use Font A in the SCT.

Select **Configure** to configure Font A. This opens the font selector, which allows you to specify the font properties, such as family, size, and weight. You can choose any font installed on your computer.

Select **Reset** to restore the Font A specification to the default font.

Font B

Select this option to use Font B in the SCT.

Select **Configure** to configure Font B. This opens the font selector, which allows you to specify the font properties, such as family, size, and weight. You can choose any font installed on your computer.

Select **Reset** to restore the Font B specification to the default font.

Buttons

OK

Select to apply the changes and close the SCT Configuration window.

Apply

Select to apply the changes and keep the SCT Configuration window open.

Reset

Select to cancel the changes and keep the SCT Configuration window open.

Cancel

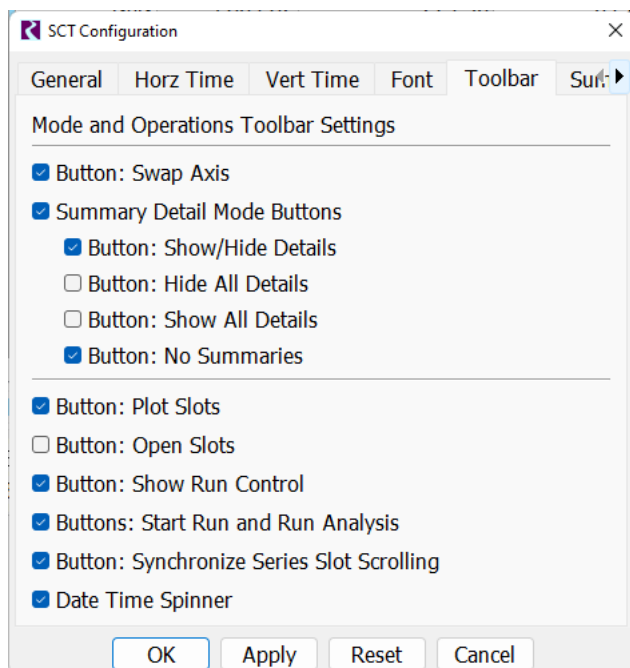
Select to cancel the changes and close the SCT Configuration window.

Toolbar tab

This tab allows you to specify which buttons to include in the SCT toolbar. You can include the buttons you use frequently and hide the ones you do not need. See [“SCT Toolbar”](#) on page 273 for details.

Note: Even if a button is hidden, you can still access the function from the SCT menus.

Note: Use the **Flags** tab to select which cell value flag buttons to include in the SCT toolbar. See [“Flags tab”](#) on page 298 for details.



Button: Swap Axis

Select to include the Swap Axis button on the SCT toolbar. Clear to hide the button.

This button is equivalent to one of the following selections on the SCT main menu:

- **View**, then **Axis Orientation**, then **Timesteps Horizontal**
- **View**, then **Axis Orientation**, then **Timesteps Vertical**

Summary Detail Mode Buttons

Select to include one or more detail mode buttons on the SCT toolbar. Clear to hide all the detail mode buttons.

Button: Show/Hide Details

Applies only if **Summary Detail Mode Buttons** is selected.

Select to include the Show/Hide Details button on the SCT toolbar. Clear to hide the button.

Button: Hide All Details

Applies only if **Summary Detail Mode Buttons** is selected.

Select to include the Hide All Details button on the SCT toolbar. Clear to hide the button.

Button: Show All Details

Applies only if **Summary Detail Mode Buttons** is selected.

Select to include the Show All Details button on the SCT toolbar. Clear to hide the button.

Button: No Summaries

Applies only if **Summary Detail Mode Buttons** is selected.

Select to include the No Summaries button on the SCT toolbar. Clear to hide the button.

Button: Plot Slots

Select to include the Plot Slots button on the SCT toolbar. Clear to hide the button.

This button is equivalent to the following selection on the SCT menu:

- **Slots**, then **Plot Slots**

Button: Open Slots

Select to include the Open Slots button on the SCT toolbar. Clear to hide the button.

This button is equivalent to the following selection on the SCT menu:

- **Slots**, then **Open Slots**

Button: Show Run Control

Select to include the Show Run Control button on the SCT toolbar. Clear to hide the button.

This button is equivalent to the following selection on the SCT main menu:

- **Run**, then **Run Control Dialog**

Buttons: Start Run and Run Analysis

Select to include the Start Run and Run Analysis buttons on the SCT toolbar. Clear to hide the buttons.

These buttons are equivalent to the following selections on the SCT main menu:

- **Run**, then **Start Run**
- **Run**, then **Run Analysis**

Chapter 4

SCT Reference

Button: Synchronize Series Slot Scrolling

Select to include the Synchronize Series Slot Scrolling button on the tool bar. Clear to hide the button. This button toggles the synchronized scrolling on and off. See also [“Synchronized Scrolling”](#) on page 18

Date Time Spinner

Select to include the Date Time selection field on the SCT toolbar. Clear to hide the field.

Buttons

OK

Select to apply the changes and close the SCT Configuration window.

Apply

Select to apply the changes and keep the SCT Configuration window open.

Reset

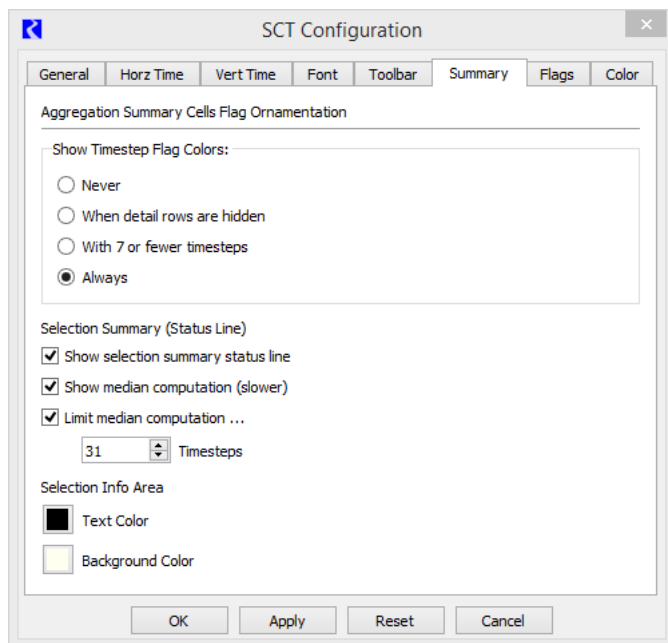
Select to cancel the changes and keep the SCT Configuration window open.

Cancel

Select to cancel the changes and close the SCT Configuration window.

Summary tab

This tab allows you to configure the background colors used for aggregation summary rows, specify whether to display summary statistics in the SCT Status area, and define the colors used in the SCT Status area.



Show Timestep Flag Colors

Indicate whether the background color for summary rows should be the appropriate cell value flag color or the Summary Cell Background color. See “[Flags tab](#)” on page 298 for flag color options. See “[Color tab](#)” on page 300 for the Summary Cell Background color.

Options are as follows:

- **Never**—summary rows never use flag colors in the background; they always use the Summary Cell Background color.
- **When detail rows are hidden**—summary rows use the appropriate flag color in the background when detailed timesteps are hidden; they use the Summary Cell Background color when details are shown.
- **With 7 or fewer timesteps**—summary rows with seven or fewer timesteps use the appropriate flag color in the background; rows with more than seven timesteps use the Summary Cell Background color.
- **Always**—summary rows always use the appropriate flag color in the background.

Show selection summary status line

Select if you want to enable the display of summary statistics in the SCT Status area when you select one or more cells in the same slot. The summary statistics are as follows:

- When you select one cell, the cell value is displayed.
- When you select multiple cells in the same slot, the sum, average, median (optional), minimum, maximum, and range of all selected cell values is displayed.

Note: If you select cells in multiple slots, summary statistics are *not* displayed.

Clear if you want to disable the display of summary statistics when cells are selected.

Show median computation (slower)

Select if you want the summary statistics to include the median value for the selected cells.

Note: The median value requires the selected values to be sorted; therefore, this option may lead to slower response times in the SCT display.

Clear if you do not want the summary statistics to include the median value.

Limit median computation

Applies only if **Show selection summary status line** is selected.

Select to include the median value only when the selected number of cells is less than or equal to the number specified in the Timesteps field.

Clear to display the median regardless of the number of cells selected.

Timesteps

Applies only if **Limit median computation** is selected.

Enter the maximum number of timesteps that can be selected for the median to be included in the summary statistics. The default is the number of timesteps in the aggregation interval.

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The median is included only if the number of cells selected is less than or equal to this number. For example, if this field is “5” and you select five or fewer cells, the median is included; if you select six or more cells, the median is *not* included.

Text Color

Select the color button to customize the text color used in the SCT Status area. This selection opens a color selector dialog box that allows you to define or select a custom color.

Background Color

Select the color button to customize the background color used in the SCT Status area. This selection opens a color selector dialog box that allows you to define or select a custom color.

Buttons

OK

Select to apply the changes and close the SCT Configuration window.

Apply

Select to apply the changes and keep the SCT Configuration window open.

Reset

Select to cancel the changes and keep the SCT Configuration window open.

Cancel

Select to cancel the changes and close the SCT Configuration window.

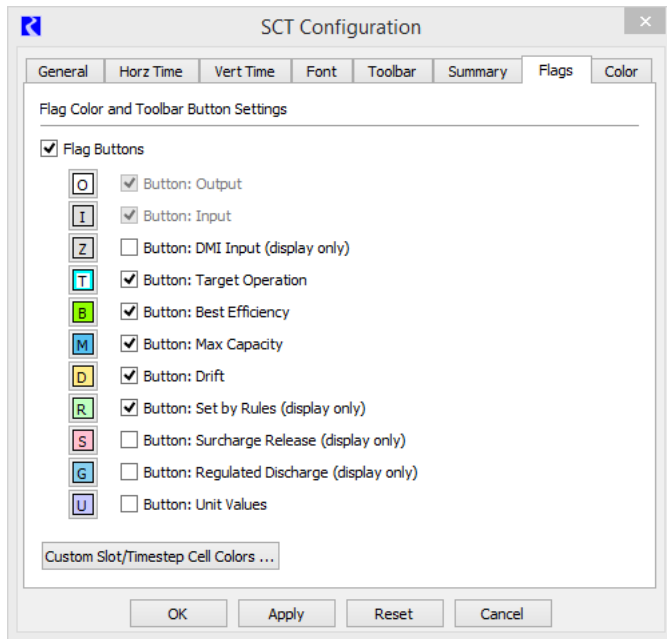
Flags tab

Allows you to show or hide the specified flag buttons on the SCT toolbar and define custom background colors for each flag.

Note: Even if a flag button is hidden, you can still set the flag from the SCT menus.

Note: See “[Color tab](#)” on page 300 for custom color options for other SCT elements.

Note: See “Toolbar tab” on page 294 for details on other toolbar buttons.



Flag Buttons

Select to enable flag buttons on the SCT toolbar. Use the remaining check boxes to select the buttons you want to show.

Clear to disable all flag buttons, removing the flag button section from the SCT toolbar.

Flag options are:

- Output (O)—indicates the value was set by simulation. This button is always displayed when flag buttons are enabled.
- Input (I)—indicates user-input data which may not be overwritten by simulation results. This button is always displayed when flag buttons are enabled.
- DMI Input (Z)—value was set by an input DMI or initialization rule.
- Target Operation (T)—available for Storage and Pool Elevation slots.
- Best Efficiency (B)—power calculation flags available for Energy slots.
- Max Capacity (M)—available to Outflow slots.
- Drift (D)—available to Regulated Spill and Bypass slots.
- Set by Rules (R)—indicates the value was set directly by a rule.
- Surcharge Release (S)—available to the Surcharge Release slot.
- Regulated Discharge (G)—available to the Regulation Discharge slot.
- Unit Values (U)—available to Turbine Release and Energy slots on power reservoirs with the Unit Power Table method selected.

Color button

Select the color button to customize the background color used for the specified cell value flag. This selection opens a color selector dialog box that allows you to define or select a custom color.

Button: flag name

Select the check box to include the specified button in the flag portion of the SCT toolbar.

Buttons

Custom Slot/Timestep Cell Colors

Select to open the Custom SCT Cell Colors dialog box, which allows you to customize text and background colors for cell highlights and threshold alerts. See [“Define custom colors for highlights and color alerts”](#) on page 191 for details.

OK

Select to apply the changes and close the SCT Configuration window.

Apply

Select to apply the changes and keep the SCT Configuration window open.

Reset

Select to cancel the changes and keep the SCT Configuration window open.

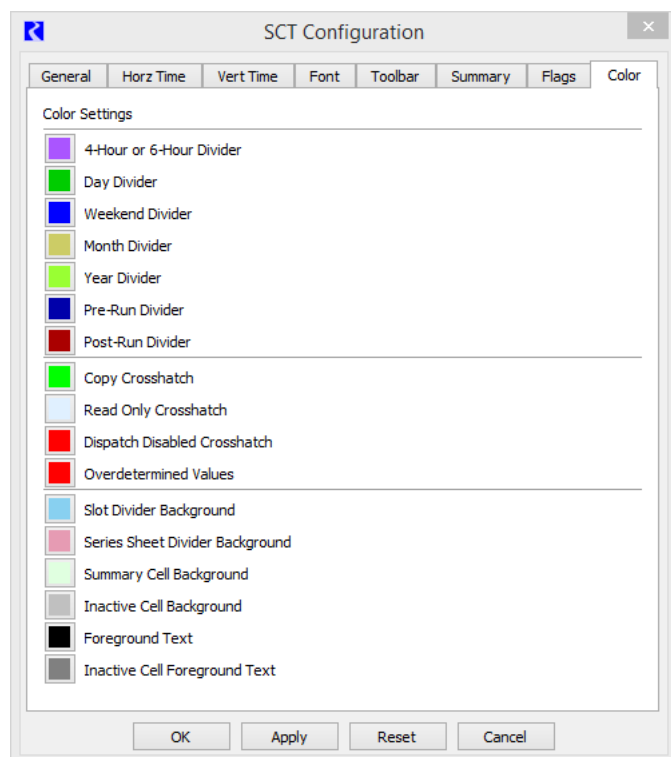
Cancel

Select to cancel the changes and close the SCT Configuration window.

Color tab

Allows you to define custom colors for each of the specified SCT elements.

Note: This tab is not used to define cell value type flag colors; see “Flags tab” on page 298 for details about those options.



Time divider colors

Color of the divider used to mark the end of the specified time period.

- 4-Hour or 6-Hour Divider
- Day Divider
- Weekend Divider
- Month Divider
- Year Divider
- Pre-Run Divider
- Post-Run Divider

Crosshatch colors

Note: Use crosshatch and background colors that contrast well with text colors so cell values are not obscured.

Color of crosshatching applied to cells that meet the specified criteria.

- Copy Crosshatch—the cell value has been copied and will be pasted in the next paste operation.
- Read Only Crosshatch—the cell value is read only.

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- Dispatch Disabled Crosshatch—dispatching has been disabled for this slot timestep.
- Overdetermined Values—the cell value is overdetermined.

Background colors

Note: Use crosshatch and background colors that contrast well with text colors so cell values are not obscured.

Color of the background for the specified divider or cell.

- Slot Divider Background
- Series Sheet Divider Background
- Summary Cell Background
- Inactive Cell Background

Text colors

Note: Use crosshatch and background colors that contrast well with text colors so cell values are not obscured.

Color of the text used in the specified cells.

- Foreground Text—used for cells in the model run range
- Inactive Cell Foreground Text—used for cells in pre- and post-run timesteps

Buttons

OK

Select to apply the changes and close the SCT Configuration window.

Apply

Select to apply the changes and keep the SCT Configuration window open.

Reset

Select to cancel the changes and keep the SCT Configuration window open.

Cancel

Select to cancel the changes and close the SCT Configuration window.

Integer-indexed SCTs

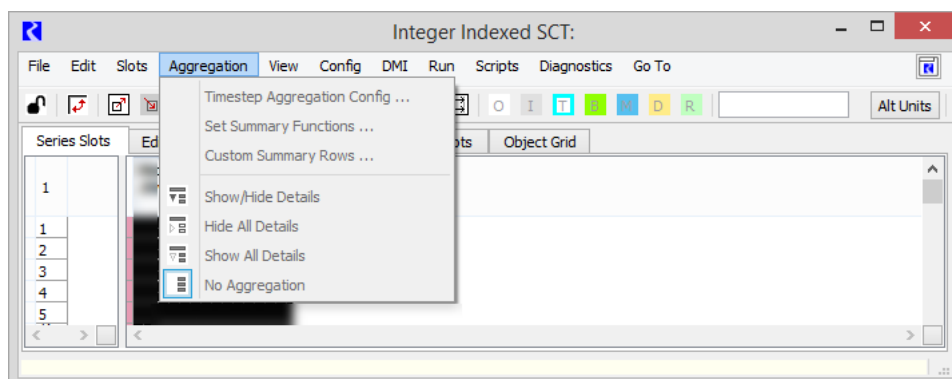
The SCT is most often used to display timeseries data, but you can use it to display integer-indexed slots instead. A new SCT is set up for timeseries data by default. If you want to include integer-indexed data, you must explicitly specify that before adding any slots; see [“Specify the type of slot indexing for the SCT”](#) on page 171. Once you start adding slots of one type, you cannot switch to the other.

Some SCT functionality is not available for integer-indexed SCTs. Following are the ways in which the menus and dialog boxes in an integer-indexed SCT differ from those in a timeseries SCT.

Topics

- [SCT Aggregation menu](#), page 303
- [SCT Config menu](#), page 304
- [SCT View menu](#), page 304
- [SCT Configuration dialog box, General tab](#), page 305
- [SCT Configuration dialog box, Horz Time tab](#), page 306
- [SCT Configuration dialog box, Vert Time tab](#), page 307
- [SCT Configuration dialog box, Toolbar tab](#), page 308
- [SCT Configuration dialog box, Summary tab](#), page 309
- [SCT Configuration dialog box, Color tab](#), page 310

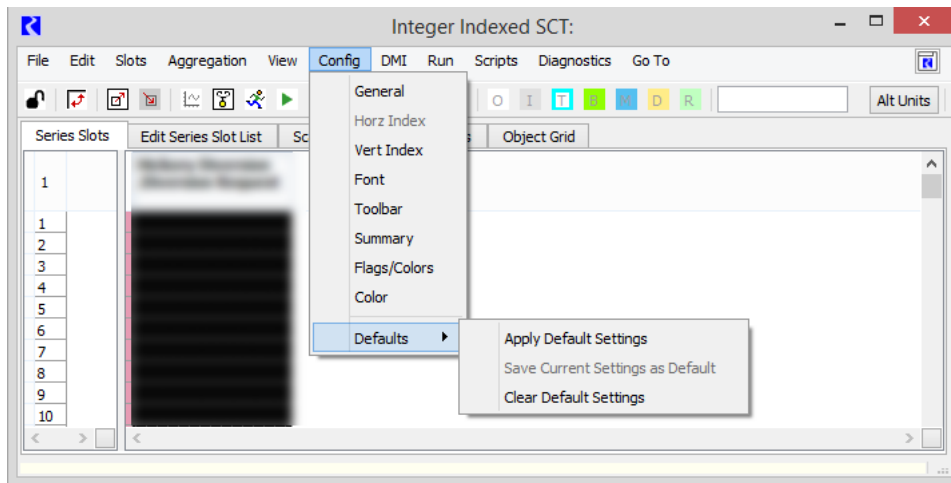
SCT Aggregation menu



The following menu options on the Aggregation menu are not available because they are not valid.

- **Aggregation**, then **Aggregation Config**
- **Aggregation**, then **Set Summary Functions**
- **Aggregation**, then **Custom Summary Rows**
- **Aggregation**, then **Show/Hide Details**
- **Aggregation**, then **Hide All Details**
- **Aggregation**, then **Show All Details**
- **Aggregation**, then **No Aggregation**

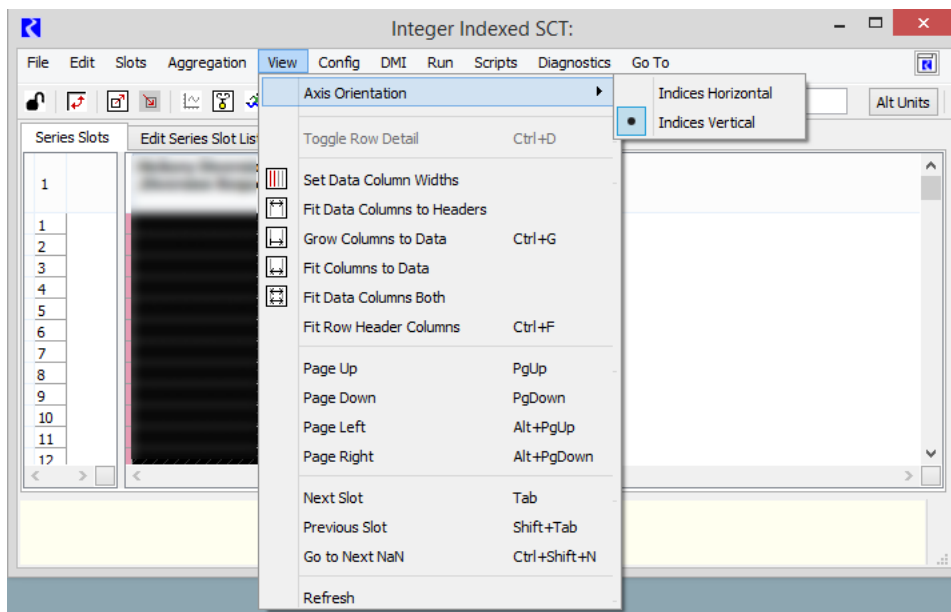
SCT Config menu



The following option on the Config menu is not available because it is not valid.

- **Config**, then **Defaults**, then **Save Current Settings as Default**

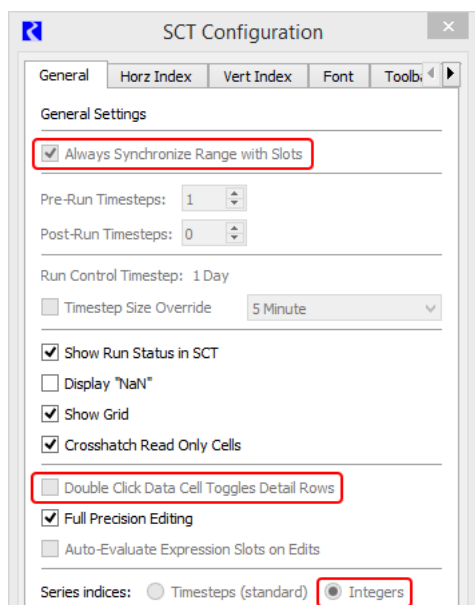
SCT View menu



The following options on the View menu are relabeled to better indicate the operations.

- **View**, then **Axis Orientation**, then **Indices Vertical**
- **View**, then **Axis Orientation**, then **Indices Horizontal**

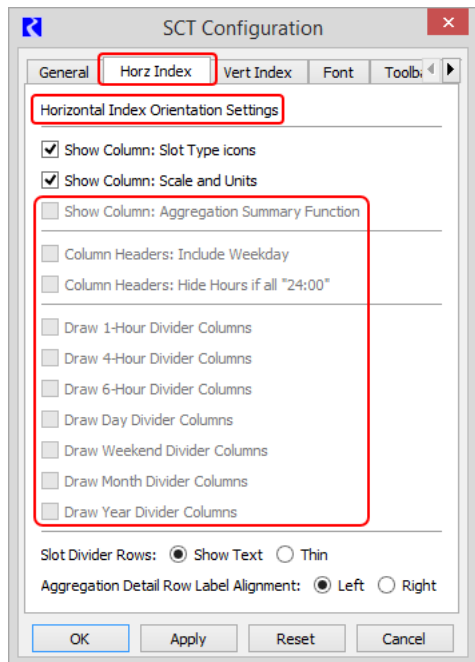
SCT Configuration dialog box, General tab



The following options are set to the specified state and made not available.

- **Always Synchronize Range to Slots**, selected
Note: When this option is selected, alternative range controls are not available.
- **Double-click Data Cell Toggles Detail Rows**, cleared

SCT Configuration dialog box, Horz Time tab



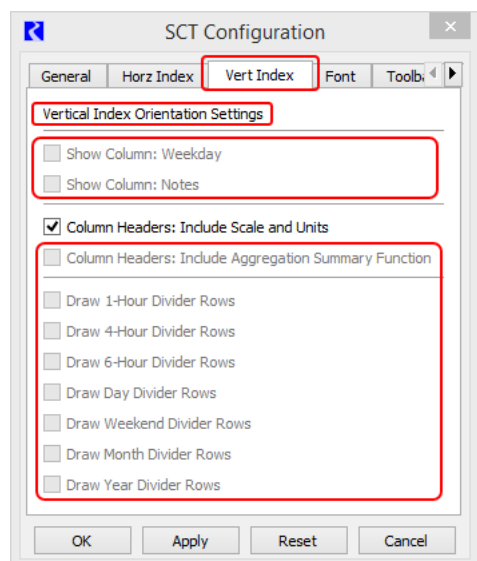
The following items are renamed.

- The tab is changed to **Horz Index**.
- The description is changed to “Horizontal Index Orientation Settings”.

The following options are set to the specified state and made not available.

- **Column Headers: Include Weekday**, cleared
- **Column Headers: Hide Hours if all “24:00”**, cleared
- **Show Column: Agg. Summary Function**, cleared
- **Draw 1-Hour Divider Columns**, cleared
- **Draw 4-Hour Divider Columns**, cleared
- **Draw 6-Hour Divider Columns**, cleared
- **Draw Day Divider Columns**, cleared
- **Draw Weekend Divider Columns**, cleared
- **Draw Month Divider Columns**, cleared
- **Draw Year Divider Columns**, cleared

SCT Configuration dialog box, Vert Time tab



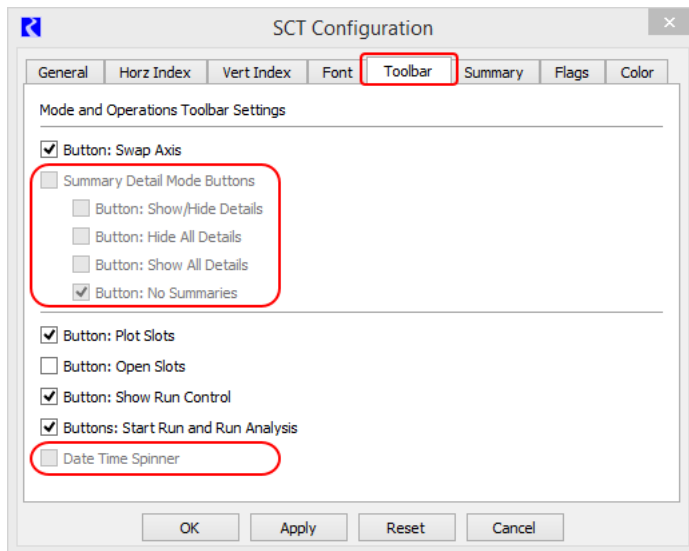
The following items are renamed.

- The tab is changed to **Vert Index**.
- The description is changed to “Vertical Index Orientation Settings”.

The following options are set to the specified state and made not available.

- **Show Column: Weekday**, cleared
- **Show Column: Notes**, cleared
- **Column Headers: Include Aggregation Summary Function**, cleared
- **Draw 1-Hour Divider Rows**, cleared
- **Draw 4-Hour Divider Rows**, cleared
- **Draw 6-Hour Divider Rows**, cleared
- **Draw Day Divider Rows**, cleared
- **Draw Weekend Divider Rows**, cleared
- **Draw Month Divider Rows**, cleared
- **Draw Year Divider Rows**, cleared

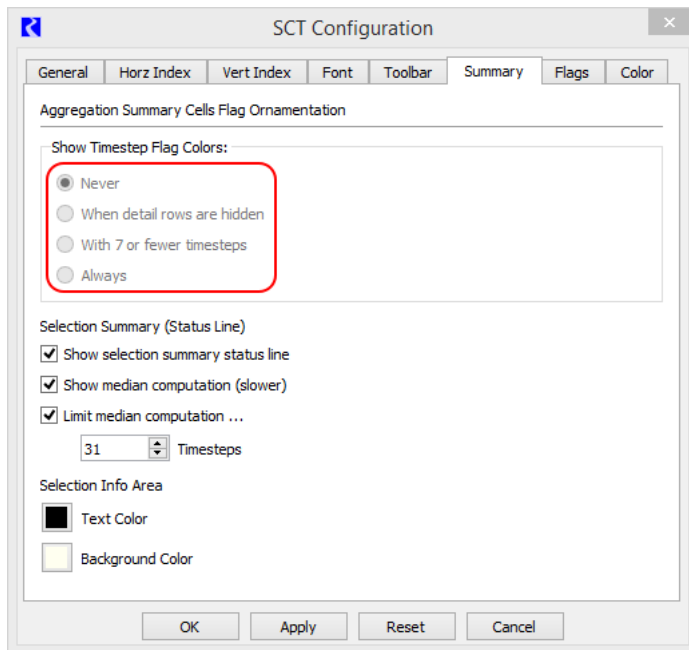
SCT Configuration dialog box, Toolbar tab



The following options are set to the specified state and made not available.

- **Summary Detail Mode Buttons**, cleared
- **Button: Show/Hide Details**, cleared
- **Button: Hide All Details**, cleared
- **Button: Show All Details**, cleared
- **Button: No Summaries**, selected
- **Date Time Spinner**, cleared

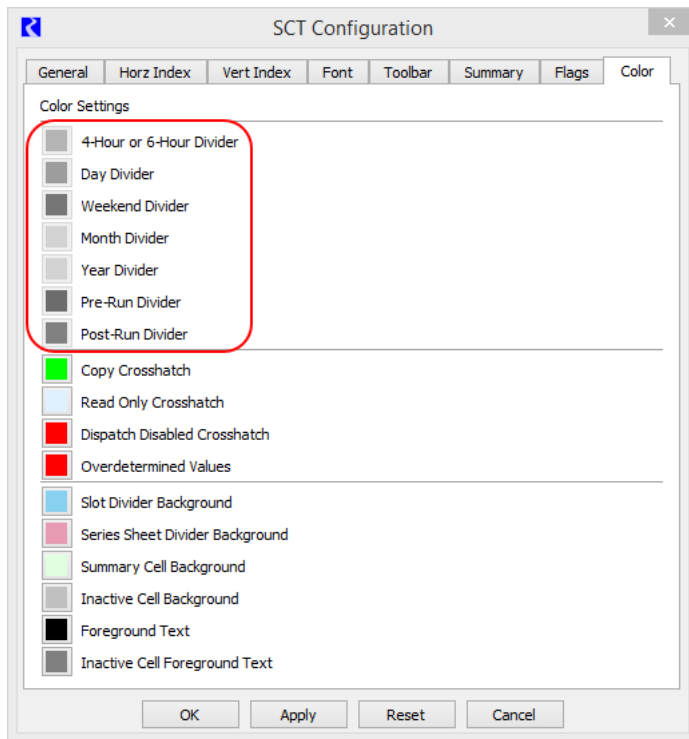
SCT Configuration dialog box, Summary tab



The following options in the “Show Timestep Flag Colors” area are set to the specified state and made not available.

- **Never**, selected
- **When detail rows are hidden**, cleared
- **With 7 or fewer timesteps**, cleared
- **Always**, cleared

SCT Configuration dialog box, Color tab



The following color setting options are made not available.

- **4-Hour or 6-Hour Divider** button
- **Day Divider** button
- **Weekend Divider** button
- **Month Divider** button
- **Year Divider** button
- **Pre-Run Divider** button
- **Post-Run Divider** button

Related topics

- [Specify the type of slot indexing for the SCT](#), page 171

Keyboard Shortcuts

Category	Action	Shortcut
Navigating and selecting cells	Go to the Edit Series Slot List tab	Alt+Shft+S
	Next cell	arrow
	Next NaN	Ctrl+Shft+N
	Next Slot	Tab
	Previous Slot	Shft+Tab
	Page Down	PgDown
	Page Up	PgUp
	Page Left	Alt+PgUp
	Page Right	Alt+PgDown
	Page through SCT	Ctrl+Shft+arrow
	Extend selection	Shft+arrow
Manipulating slot values	Clear Values	Del
	Copy Cells	Ctrl+C
	Paste Cells	Ctrl+V
	Paste Cells as Input	Ctrl+N
	Paste Values, Flags, and Notes	Ctrl+Shift+V
	Interpolate Selection	Ctrl+L
	Undo	Ctrl+Z
	Redo	Ctrl+Shift+Z
Setting series slot Value flags	Set to Best Efficiency (B)	Ctrl+B
	Set to Drift (D)	Ctrl+Shft+~
	Set to Input (I)	Ctrl+I
	Set to Max Capacity (M)	Ctrl+M
	Set to Output (O)	Ctrl+O
	Set to Target Operation (T)	Ctrl+T
Changing how data is displayed	Fit Row Header Columns	Ctrl+F
	Grow Columns to Data	Ctrl+G
	Toggle Row Detail (available only when Show/Hide Details is enabled)	Ctrl+D
Accessing other utilities	Open Run Control window	Ctl+R
	Close SCT	Ctrl+W
	Save Model	Ctrl+S

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Category	Action	Shortcut
	Save Model As	Ctrl+Shift+S
	Plot Slots	Ctrl+P
	Start Run	Ctrl+Shift+R